Cosmochrony: A Non-Injective Projection Theory of Emergent Physics

Quantum Mechanics, Spacetime Geometry, and Gauge-Matter Structure from a Static Relational Substrate

Jérôme Beau[1*]

[1*]Independent Researcher, Paris, France.

Corresponding author(s). E-mail(s): jerome.beau@cosmochrony.org;

Abstract

Cosmochrony is a pre-geometric relational framework that does not replace quantum mechanics or general relativity, but operates at a deeper structural level from which both emerge as effective descriptions in appropriate regimes. Its aim is to expose the common infra-physical foundations underlying these theories, to clarify how their conceptual structures are mutually consistent, and to show that several phenomena currently treated as independent postulates or open problems follow necessarily from a minimal common origin.

The framework postulates a single static substrate χ, encoding only admissibility constraints. Observable states are obtained through a fixed generically non-injective projection Π, whose non-injectivity is a structural necessity rather than an independent assumption: any framework that distinguishes an infra-physical level from an observable level must admit such a map. No fundamental spacetime, metric, or quantum structure is assumed. From this minimal ontology, temporal ordering, spatial geometry, the causal structure of spacetime, the origin of quantum indeterminacy, and the structure of matter emerge as projective consequences rather than independent postulates. The speed of light and Planck's constant arise as complementary bounds on projectability and spectral resolvability. The three-generation structure of Standard Model fermions, the quantization of electric charge, and the absence of magnetic monopoles follow from admissibility constraints on the projection fiber.

The framework yields falsifiable predictions: a $\sim$5% enhancement of $H(z)$ at $z \sim 1$ relative to ΛCDM, environment-dependent deviations in kinematic inference in cosmic voids testable with current surveys, and a structural upper bound on the capacity exponent governing the fermion mass hierarchy.

The present document serves as the conceptual and structural reference for the Cosmochrony programme. Formal derivations, spectral numerical evidence, and phenomenological applications are developed in a series of companion papers cited throughout.

Keywords: Pre-geometric substrate, emergent spacetime, relational dynamics, Born–Infeld dynamics, spectral geometry, quantum non-injectivity

Contents

Part I
Foundations

This part establishes the conceptual and ontological foundations of the Cosmochrony framework. Its purpose is to state explicitly the minimal assumptions from which all subsequent developments follow, without presupposing spacetime, geometry, or quantization as fundamental primitives.

We introduce the pre-geometric relational substrate χ, clarify its ontological status, and define the structural principles governing its admissible configurations. Central notions such as non-injective projection, bounded admissibility constraints, projective ordering, and effective projectability are defined in a canonical manner and will be used throughout the document without repetition.

Effective geometric, dynamical, and temporal descriptions are shown to arise only within restricted projective regimes, as consequences of the non-injective structure of Π rather than properties of χ itself. This part therefore serves as the reference layer for all later physical interpretations, formal developments, and phenomenological applications.

1 Introduction and Motivation

Modern fundamental physics is built upon two highly successful yet conceptually distinct frameworks: quantum mechanics and general relativity [1, 2]. Quantum theory accurately describes microscopic phenomena, while general relativity provides a geometric account of gravitation and spacetime dynamics at macroscopic and cosmological scales. Despite their empirical success, these theories rely on incompatible foundational assumptions and resist unification within a single coherent conceptual framework [3–5].

A central difficulty underlying this tension concerns the status of spacetime itself. In general relativity, spacetime geometry is dynamical and responds to matter, yet it remains the arena within which all physical processes are formulated. In quantum theory, time is treated as an external parameter, while spatial locality is assumed as a primitive input. These roles are difficult to reconcile in regimes where spacetime itself is expected to emerge or lose operational meaning, such as near cosmological origins or in strongly quantum gravitational settings.

It is important to emphasize that this tension does not, by itself, require spacetime to have a temporal origin. Scenarios in which spacetime is eternal, cyclic, or undergoes conformal transitions remain logically and physically consistent. The difficulty addressed here is therefore not the existence of spacetime, but the explanatory sufficiency of spacetime-based descriptions for accounting for their own structural properties.

Any attempt to account for the emergence, persistence, or breakdown of spacetime descriptions faces a conceptual constraint: spacetime cannot, on its own, provide a complete account of the conditions under which its notions of time, causality, locality, and dynamics become meaningful. Descriptions that presuppose temporal evolution, spatial propagation, or geometric structure are necessarily limited when used to explain how such notions arise or cease to apply. This suggests that spacetime, even if taken as fundamental or eternal, may not be explanatorily self-sufficient.

A closely related motivation concerns a broad class of phenomena commonly described as emerging from the vacuum. In contemporary quantum field theories, effects such as zero-point energies, vacuum fluctuations, virtual processes, and renormalized self-energies are routinely invoked to account for observable phenomena. While these constructs are operationally successful, they raise a conceptual tension similar to that encountered for spacetime itself.

If physical observables are defined entirely within spacetime-based descriptions, then entities attributed to the vacuum cannot originate from observable degrees of freedom alone. Their explanatory role therefore implicitly points toward a deeper level of description, one that is not itself accessible as a physical observable.

From this standpoint, the vacuum should not be regarded as a physically populated medium, but as a manifestation of structural limitations of the effective description. Apparent vacuum-related phenomena then signal the presence of unresolved degrees of freedom associated with a non-observable, infra-physical, and pre-geometric level of description.

This observation reinforces the need for a framework in which both spacetime structure and vacuum-related effects arise as effective concepts, emerging from the same underlying constraints. The question is therefore not what fills the vacuum, but what structural conditions render such effective descriptions necessary in the first place.

From this perspective, the problem of unification is not merely one of combining dynamical laws within spacetime, but of identifying at least one additional level of description from which spacetime, causality, and dynamical evolution can appear as effective concepts. Such a level need not provide an ultimate origin of physical reality. Its role is more modest: to supply the minimal structural conditions required for spacetime-based descriptions to be applicable at all.

The framework developed in this work, referred to as *Cosmochrony*[1], is motivated by this requirement. It explores whether the emergence of spacetime geometry, gravitation, and quantum phenomena can be understood as consequences of deeper structural constraints operative at an infra-physical level, without assuming spacetime itself to be explanatorily complete.

The present introduction is therefore concerned exclusively with the *necessity* of a pre-geometric level of description. The definition of the underlying entity, its ontological status, and the minimal assumptions governing its admissible descriptions are introduced systematically in Section 2.

Cosmochrony does not aim to replace the Standard Model or general relativity in their empirically validated domains. It offers a coherent infra-physical framework designed to expose the common structural foundations from which time, geometry, gravitation, and quantum correlations emerge, and to show that several phenomena currently treated as independent postulates admit a common projective origin.

1.1 The Unification Problem

Quantum mechanics and general relativity differ not only in their mathematical formalisms but also in their foundational concepts. Quantum theory is intrinsically probabilistic, relies on a fixed causal structure, and treats time as an external parameter [1, 6]. General relativity describes gravitation as the dynamics of spacetime geometry itself, with time acquiring a coordinate-dependent and observer-relative status [2, 3].

This conceptual mismatch becomes acute in regimes where both quantum effects and strong gravitational fields are relevant, such as near spacetime singularities or in the early universe [7, 8]. Direct attempts to quantize gravity encounter persistent difficulties, including the problem of time, non-renormalizability, and the absence of a preferred background structure. These difficulties suggest that the tension may reflect a deeper incompatibility in the assumed ontological status of time and geometry.

Several major research programs have sought to address these challenges. Quantum field theory in curved spacetime accounts for particle creation and vacuum effects but retains a classical spacetime background [4]. Canonical and covariant approaches to quantum gravity attempt to quantize spacetime geometry itself, often at the cost of substantial mathematical complexity and interpretational ambiguity. String theory introduces extended fundamental objects and higher-dimensional structures, offering deep mathematical unification but leading to a large space of possible low-energy realizations [5]. While internally rich, these approaches face ongoing challenges concerning empirical testability and the physical interpretation of their fundamental degrees of freedom.

[1] From $\kappa\acute{o}\sigma\mu o\varsigma$ and $\chi\rho\acute{o}\nu o\varsigma$, denoting a framework in which cosmic structure and temporal ordering are treated as emergent.

These limitations motivate the exploration of alternative perspectives in which spacetime geometry, matter, and quantum behavior emerge from a common underlying mechanism operating at a pre-geometric level.

1.2 Minimalism as a Guiding Principle

The framework developed in this work adopts minimalism as a guiding principle. Rather than introducing multiple fundamental fields, additional dimensions, or independent quantization rules, Cosmochrony rests on two structural ingredients: a single static pre-geometric substrate χ, encoding only admissibility constraints, and a fixed generically non-injective projection Π mapping substrate configurations onto observable states.

The substrate χ is not interpreted as a conventional matter field, nor as a component of spacetime geometry, nor as an entity endowed with intrinsic dynamics. It encodes relational structure and constrains what projections are admissible; it does not itself evolve. Time and space are not independent primitives but emerge as effective properties of the projective description: temporal ordering from the monotone accumulation of projected spectral structure, spatial relations from correlations within admissible projected configurations. Effective geometric and quantum descriptions arise only through coarse-grained, generally non-injective projections of χ-configurations admissible under these constraints.

1.3 Temporal Ordering, Irreversibility, and Cosmological Expansion

A central motivation for the Cosmochrony framework is the close connection between time, irreversibility, and cosmological expansion. In standard cosmology, expansion is described kinematically through the scale factor, while the arrow of time is typically attributed to boundary conditions or entropy growth [7–9]. Neither account addresses the structural origin of temporal ordering itself.

In Cosmochrony, temporal ordering is not postulated as a property of the substrate χ, nor derived from an underlying dynamics. It emerges instead as a consequence of the projective structure: the cumulative projected spectral capacity $I(n)$, measuring the spectral structure stably realised at each step of the admissibility cascade, is monotone non-decreasing along all tested admissible trajectories [10]. This monotonicity defines an intrinsic ordering on effective states without requiring any evolution of χ itself. One-way activation is the condition of possibility for this ordering: because spectral modes, once stably projected, do not deactivate, $I(n)$ is well-defined as a globally consistent monotone quantity, and irreversibility follows. One-way activation is a property of *spectral modes*, not of the coherence of projected configurations; the structural reorganization of unstable configurations into more stable ones is separately accounted for by the admissible factorization mechanism (Section 4.4).

Three levels of status attach to this picture and should be kept distinct. The monotonicity of $I(n)$ is an *established numerical result*, confirmed across 169 conjugate pairs and 310 post-burn-in BFS steps for $q \in \{29, 61, 101, 151\}$, with zero physically meaningful regression [10]. The one-way activation property that underpins it — formally, that $\sigma_{\mathrm{pair}}(n) \leq \sigma_{\mathrm{pair}}(n-1)$ for all admissible q, all conjugate pairs, and all

$n \geq 1$ — is an *open conjecture*: its analytical proof from the Gram–Schmidt structure, the Born–Infeld admissibility constraints, and the non-injectivity of Π remains to be established (see Section 2.6). The identification of this ordering with physical time is an *interpretation*: temporal ordering is the relational structure induced by admissible projections, not an evolution of χ itself.

Cosmological expansion admits an analogous interpretation. The progressive growth of $I(n)$ at the global scale corresponds to an ever broader range of mutually distinguishable projected configurations becoming stably realised. Expansion is therefore not driven by an external energy component but reflects the monotone accumulation of admissible projected structure. In the effective geometric description, this global growth manifests as the large-scale expansion of the observable universe.

1.4 Conceptual Context and Related Approaches

The idea that spacetime geometry and gravitation may be emergent has been explored in a variety of contemporary theoretical frameworks. Several approaches interpret the spacetime metric as an effective description arising from deeper geometric, informational, or dynamical structures, and recast gravitation as a collective phenomenon [11, 12]. Cosmochrony belongs to this broad conceptual lineage while adopting a deliberately minimalist ontological stance: a single static pre-geometric substrate χ, together with a fixed non-injective projection Π, from which physical observables emerge as admissible projected descriptions.

Like Loop Quantum Gravity (LQG), Cosmochrony holds that spacetime geometry is not fundamental [5]. However, the two frameworks operate at distinct explanatory levels. LQG provides a quantized description of geometry once a spacetime structure is already assumed, encoding areas and volumes through spin networks and holonomies. Related ideas have also appeared in modern holographic approaches formulated in asymptotically flat spacetimes, such as celestial holography, where scattering amplitudes are reorganized as conformal correlators on the celestial sphere [13]. These approaches reformulate or quantize geometric degrees of freedom, but do not address the origin of spacetime geometry itself.

Several approaches have proposed that gravitational dynamics can be encoded in spectral data, notably through the spectral action principle in which geometry and its quantum corrections are derived from the spectrum of a Dirac-type operator [14, 15].

While these constructions demonstrate that a rich geometric structure can be recovered from spectral information, the spectral operator itself is typically postulated as fundamental. In contrast, Cosmochrony addresses the origin of spectral structure, deriving it from the relational Laplacian of the χ-substrate under bounded admissibility constraints and finite projectability.

Beyond frameworks centered on the quantization or reformulation of geometry, several recent proposals have emphasized that specific physical entities or processes may not admit a fully spacetime-embedded description. In particular, arguments based on null worldlines in relativistic kinematics have motivated interpretations in which photons are described as experiencing no proper time between emission and absorption, leading to conceptual models in which light is treated as effectively atemporal or as residing outside the standard spacetime description [16]. Other approaches have similarly explored descriptions in which physical objects are interpreted as stable modal or vibrational structures of an underlying substrate, with spacetime notions emerging only at an effective level.

While these perspectives share with Cosmochrony the motivation to question the fundamental status of spacetime and time, they differ at a structural level. In Cosmochrony, the substrate χ is static and carries no intrinsic dynamics. Temporal ordering and all dynamical notions arise solely at the level of effective projected descriptions, as consequences of the non-injective projective structure rather than properties of the substrate itself. The emphasis is therefore not placed on the atemporality of specific entities, nor on the postulation of an underlying field with prescribed dynamics, but on the admissibility constraints and finite projectability of the relational structure.

Cosmochrony thus addresses an earlier and more primitive explanatory level. It neither quantizes geometry nor posits timeless physical entities as fundamental. Rather, it seeks to explain how geometric and temporal notions themselves arise as effective, coarse-grained descriptions of underlying χ-configurations. The emergence of spacetime is mediated by a non-injective projection from the pre-geometric substrate to effective observables, allowing geometric, dynamical, and quantum features to appear only once specific relational and spectral conditions are met. From this perspective, Cosmochrony does not compete with LQG or related frameworks but conceptually precedes them, aiming to account for the physical origin of the geometric and temporal degrees of freedom that may subsequently be treated within spacetime-based or quantized descriptions.

1.5 Scope and Limitations

The aim of this work is exploratory rather than definitive. Cosmochrony does not seek to replace established theories within their empirically validated domains, but to offer a coherent reinterpretation that may clarify persistent conceptual difficulties concerning time, geometry, and quantization. Emphasis is placed on internal consistency, conceptual clarity, and qualitative contact with observable phenomena, while openly acknowledging open questions and limitations.

1.6 Structure of the Paper

The paper is organized in five parts. Part I introduces the χ substrate, its minimal structural properties, and its ontological status (Section 2). Part II develops the effective dynamics of χ (Section 3), the emergence of particle-like excitations (Section 4), and the projection fiber with gauge emergence (Section 5). Part III addresses gravity as a collective effect (Section 7) and cosmological implications (Section 8). Part IV presents the unified treatment of quantum phenomena, entanglement, and the relation to quantum formalism (Section 9). Part V collects radiation and quantization (Section 10), the spectral mass spectrum (Section 11), testable predictions (Section 12), and the discussion with conclusion (Section 13). Appendices A–F provide mathematical foundations, conceptual extensions, cosmological and numerical supplements, and a glossary of notation.

2 The χ Substrate: Definition, Properties, and Ontology

This section introduces the single fundamental entity at the core of Cosmochrony: the pre-geometric relational substrate χ. We identify the minimal structural properties required for effective descriptions of spacetime, dynamics, and physical observables to arise in appropriate regimes, and we clarify the ontological status of χ and its relation to emergent physical quantities.

No spacetime manifold, metric structure, or background geometry is assumed. Geometric and dynamical notions are recovered only through non-injective, coarse-grained projections of χ configurations once suitable stability conditions are met. This minimality principle, referred to as a regime of *ontological poverty*, is adopted as a foundational constraint.

2.1 Definition of the χ Field

We define the existence of a single pre-geometric relational substrate, denoted χ, which constitutes the primitive ontological basis of physical reality. The quantity χ is not defined on a pre-existing spacetime manifold and does not presuppose any metric, causal, or geometric structure. Spacetime notions arise only as effective descriptions of the relational and spectral properties of χ configurations, accessed through non-injective projection.

Ontologically, χ is not a scalar order parameter and does not possess local values. Scalar order parameters arise only at the effective level, as coarse-grained descriptors of projected χ configurations once a geometric regime is established. Dimensional quantities associated with length, duration, or mass arise only when χ configurations admit a stable geometric interpretation.

Temporal ordering and spatial separation are not fundamental primitives. Temporal ordering arises from the monotone accumulation of projected spectral structure along the admissibility cascade (Section 2.6); spatial separation arises from relational correlations within admissible projected configurations once a quasi-stable geometric regime is reached.

The analogy with thermodynamic order parameters applies only at the effective level: χ itself is not an order parameter but gives rise to effective order parameters once projected. The substrate χ therefore provides the minimal ontological basis from which time, space, inertial mass, gravitation, and quantum phenomena jointly emerge as consequences of a single non-injective projective structure.

Further interpretative clarifications and the fully relational formulation of χ are provided in Appendix B.1 and Appendix E.1.

2.2 Effective Descriptions from Projective Structure

Effective Observables from χ Correlations

Quantities conventionally described in geometric terms—such as time intervals, spatial separation, and causal ordering—arise as effective summaries of relational patterns within the χ substrate, accessed only through projected, coarse-grained representations.

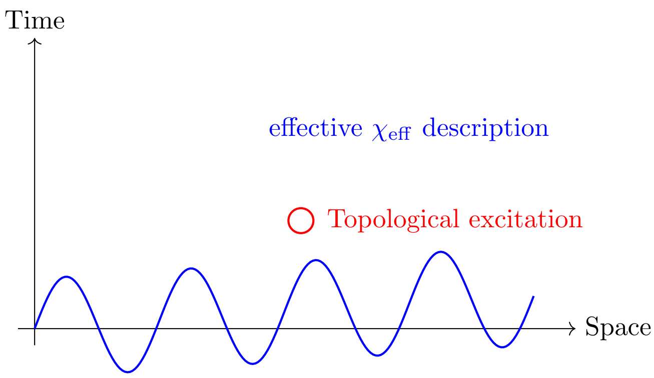

Fig. 1 Conceptual representation of Cosmochrony. An effective spacetime depiction of the projected scalar description of χ, used for visualization purposes only. The monotone accumulation of projected spectral structure gives rise to an effective temporal ordering, while localized topological excitations correspond to particle-like configurations in the emergent geometric regime.

The configurations σ represent internal relational states of χ and are defined without reference to external spacetime coordinates. The measure $d\mu(\sigma)$ denotes an invariant integration over configuration space, defined intrinsically from the correlation structure associated with χ.

Effective scalar descriptor.

In regimes where projected χ configurations admit a stable geometric interpretation, we introduce an *effective scalar descriptor* $\chi_{\rm eff}$. This quantity is a coarse-grained, projected representation of relational and spectral features of χ, defined only within the emergent spacetime description. It does not correspond to the fundamental substrate itself.

Operational time intervals are defined from the accumulated ordering of projected configurations along admissible paths:

$$\tau_{AB} \;\propto\; \int_{\gamma_{AB}} \mathcal{D}_\lambda \chi_{\rm eff}\, d\lambda, \tag{1}$$

where $\mathcal{D}_\lambda \chi_{\rm eff}$ is an effective ordering functional of the projected description and λ an admissible ordering parameter.

Operational spatial separation is quantified by the decay of correlations between effective descriptors:

$$d(x,y) \;\propto\; -\log\!\left(\frac{\langle \chi_{\rm eff}(x)\, \chi_{\rm eff}(y)\rangle}{\langle \chi_{\rm eff}^2\rangle}\right), \tag{2}$$

where $\langle \chi_{\rm eff}(x)\, \chi_{\rm eff}(y)\rangle$ denotes an effective correlation functional encoding relational proximity between projected configurations x and y.

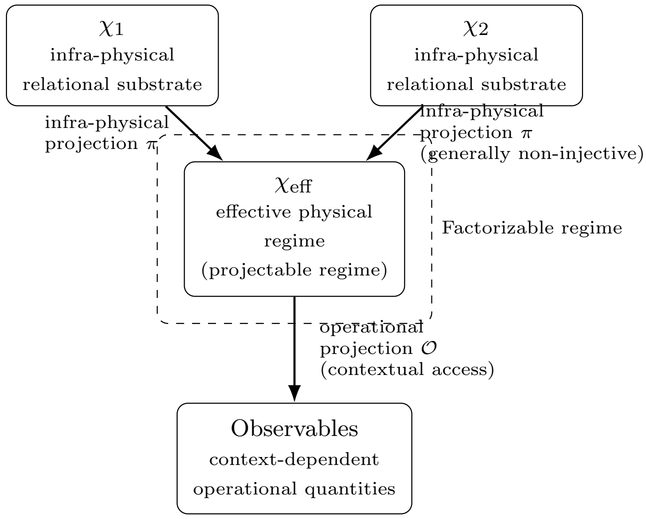

Fig. 2 Ontological pipeline in Cosmochrony. The infra-physical projection π maps the fundamental χ substrate to an effective physical regime χ_{eff} in projectable regimes, generally in a non-injective manner (multiple infra-physical configurations can correspond to the same projected description). Physical observables arise from an operational projection $\mathcal{O}$ that specifies how χ_{eff} is accessed under measurement contexts.

Effective Metric as a Descriptive Tool

In regimes where projected configurations exhibit smooth and stable correlation patterns, the relational observables may be summarized by an operational tensor $g_{\mu\nu}[\chi_{\text{eff}}]$. Such smooth regimes arise when the projected spectrum is dominated by low-eigenvalue modes of the relational operator, ensuring slowly varying correlations and enabling a differentiable metric description to emerge. This effective metric is a derived descriptor summarizing relational correlations, not a fundamental geometric structure. No background $\eta_{\mu\nu}$ is assumed; Minkowski space appears only as an effective approximation in weak-gradient regimes.

In projectable regimes admitting a low-dimensional embedding, one may write locally

$$d(i,j)^2 \;\approx\; g_{\mu\nu}(x)\,\Delta x^{\mu}\Delta x^{\nu}.$$

The continuum line element ds^2 is therefore a derived descriptive construct.

Consistency with General Relativity

The effective metric reproduces the phenomenology of general relativity in appropriate regimes: the weak-field limit yields Einstein-like dynamics; near localized χ excitations, the metric encodes time dilation and spatial curvature as emergent consequences of locally constrained projective ordering; homogeneous projected expansion yields an effective Hubble-like expansion law. All predictive content resides in the underlying projective structure.

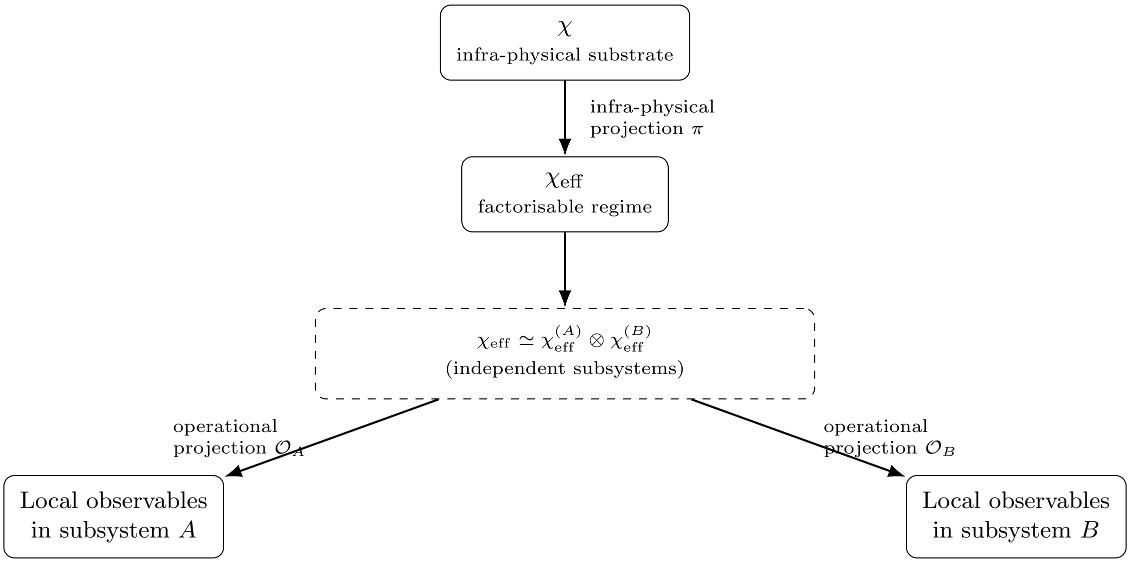

Fig. 3 Classical (factorisable) regime. After the infra-physical projection π, the effective description admits an approximate decomposition into independent subsystems. Operational projections then yield compatible local observables, recovering standard classical descriptions. This contrasts with the non-factorisable regimes discussed in Section 9.

Operational origin of the effective metric.

The explicit construction of $g_{\mu\nu}$ from operational distances is given in Appendix E, where geometric quantities are shown to arise only in projectable regimes admitting a smooth continuum approximation.

2.3 Physical Interpretation

In regimes where the infra-physical projection from χ to an effective description χ_{eff} yields a factorisable structure, an approximate decomposition into subsystems and the definition of operational observables become possible. This factorisation underlies the emergence of classical locality, compatibility of measurements, and standard relativistic descriptions, as illustrated in Fig. 3.

Because the projection from χ to χ_{eff} is generically non-injective, distinct χ configurations may correspond to identical effective descriptions, while a single underlying configuration may admit multiple correlated operational realizations. This structural asymmetry is central to the emergence of both classical and quantum phenomenology. The physical content of the theory resides entirely in the projective structure relating χ configurations to effective descriptions, while spacetime notions function as emergent, context-dependent descriptive tools.

2.4 Relational Projection and Admissibility Principles

Admissible physical descriptions arise through a projection from the relational substrate χ onto effective observable configurations. This projection does not operate directly on individual configurations but on their relational structure.

Relational operator.

Let $\mathcal{H}_\chi$ denote the configuration space of admissible χ states. We introduce a relational operator

$$L_\chi : \mathcal{H}_\chi \to \mathcal{H}_\chi, \tag{3}$$

defined purely in terms of correlations within the χ substrate. In discrete realizations (Appendix D) L_χ reduces to a graph Laplacian. In the continuum limit it defines an effective self-adjoint operator encoding relational connectivity.

Spectral structure.

The operator L_χ admits a spectral decomposition

$$L_\chi \psi_n = \lambda_n \psi_n, \tag{4}$$

where $\{\lambda_n\}$ are non-negative eigenvalues and $\{\psi_n\}$ the associated eigenmodes. These modes provide a natural basis for describing relational configurations of the substrate.

Admissibility principle.

Only a restricted subset of relational configurations can give rise to stable effective descriptions. This restriction can be expressed as a spectral filtering acting on the relational operator L_χ. Formally one may represent this selection through an admissibility filter

$$F_{\lambda_*} : \mathcal{H}_\chi \to \mathcal{H}_{\text{eff}}, \tag{5}$$

which associates relational configurations with admissible effective descriptions. The scale λ_* represents a spectral threshold separating projectively admissible from non-admissible relational modes.

Modes with $\lambda_n \leq \lambda_*$ contribute directly to effective descriptions, while higher modes remain encoded in the fibre of the projection and appear only indirectly through fluctuations or renormalized parameters. Such modes constitute infra-projectable invariants of the relational description.

Non-injectivity and equivalence classes.

The admissibility filter is generally non-injective. Distinct relational configurations may share identical spectral content below the admissibility threshold and therefore correspond to the same effective observable description. Effective states therefore arise as equivalence classes of relational configurations compatible with a common observable structure.

Admissibility is relational, not absolute.

Admissibility is never an intrinsic predicate of an observable state taken in isolation. It is a relational constraint on projective transitions: a configuration O_n is admissible only relative to a preceding configuration O_{n-1}, whose accumulated spectral and fibre constraints determine the local set of accessible successors. We write $O_n \in \mathcal{A}(O_{n-1})$ for the *admissible neighbourhood from* O_{n-1}, defined as the set of configurations reachable from O_{n-1} under the admissibility filter and the Born–Infeld flux bound. The phrasing "O_n is admissible" is always shorthand for $O_n \in \mathcal{A}(O_{n-1})$: admissibility is always *from-here* admissibility, conditioned on the already-realised projective position. There is no fixed ambient space of possible observable states; there is a field of local admissible neighbourhoods, each determined by the point already reached.

Born–Infeld admissibility and its two structural roles.

The admissibility filter introduced above is not arbitrary: among all possible spectral thresholds, the Born–Infeld (BI) structure is selected as the unique analytic completion that preserves gauge invariance, reduces to Maxwell electrodynamics at low field strength, and bounds the field invariants [17, 18]. The Born–Infeld constraint plays two structurally distinct roles in the framework, which should not be conflated.

Its *first role* is the selection of admissible fibres: the parity of the BI action $S_{\mathrm{BI}}[\chi] = S_{\mathrm{BI}}[-\chi]$ forces every projective fibre $\Pi^{-1}(y)$ to contain the involution $\chi \mapsto -\chi$, establishing the conjugate-pair structure $\{c, q-c\}$ of the Weil representation as a structural necessity rather than a modelling choice [19]. This is a *geometric* role: it constrains the topology of the fibres of Π.

Its *second role* is to bound the effective projectability rate: the saturation scale c_χ sets the maximal rate at which relational ordering can be projected from χ onto effective observables. This is a *kinematic* role: it governs how fast the admissibility cascade can advance, and underlies the emergence of relativistic bounds (Section 2.6) and the finite cutoff replacing ultraviolet renormalisation (Section 10.4). The two roles are complementary and not reducible to each other: fibre geometry is a property of Π at fixed λ, while the rate bound operates along the cascade as λ advances.

Programme of spectral admissibility.

The structural consequences of this admissibility principle are developed in detail in Chapter 6. In particular, the admissibility constraints on χ impose quantitative conditions on admissible spectral modes through an admissibility envelope that limits their effective amplitudes.

This constraint induces a hierarchy among spectral sectors and leads to a classification of relational structures capable of supporting stable projected descriptions. In later sections this mechanism will be shown to select specific finite symmetry groups compatible with admissible relational dynamics.

2.5 Structural Principles and Projective Regimes

All effective physical observables arise through a relational projection Π from the underlying configuration space Ω of the χ substrate. Effective descriptions are constrained by three structural principles that govern the admissibility of projected regimes.

Principle I: Substratic Locality and Bounded Relaxation.

The admissibility constraints encoded in χ are strictly local and subject to universal bounds. The transport of relational structure within the effective reprojection sequence admits a maximal admissible flux, constraining the rate at which structural information can be redistributed. This bound arises from the intrinsic stability conditions of χ itself.

Principle II: Non-Injective Projective Realization.

The projection $\Pi : \Omega \to O$ from substratic configurations to effective observables is generically non-injective. Distinct configurations of χ may be structurally identified at the level of observable descriptions. This implies that effective descriptions need not admit a factorisable or locally complete representation, even when the underlying dynamics remains strictly local.

Remark (Non-injectivity as structural necessity). Principle II is not a modelling hypothesis but a provable structural necessity. Any surjective projection $\Pi : \Omega \to O$ whose observables are entirely defined by Π and whose effective level is not isomorphic to Ω must be non-injective, with $S_\Pi > 0$. A fully injective emergent description would reduce emergence to a relabelling of the fundamental level — a contradiction. Within this projective-emergent setting, the following equivalences hold:

$$\text{genuine emergence} \iff \Pi \text{ non-injective} \iff S_\Pi > 0. \tag{6}$$

A complete proof, independent of any specific physical framework, is given in [20]; the structural consequences for gauge structure and Bell correlations are developed in Appendix B.23 and [21].

Principle III: Projective Compensation.

Whenever Π fails to resolve the full relational complexity of χ, effective descriptions compensate through the inflation of effective parameters—temperature, curvature, horizon structure—encoding unresolved relational structure within a reduced descriptive framework. These quantities function as Lagrange multipliers, not as additional fundamental degrees of freedom.

Together, these principles delineate distinct projective regimes. When the projection is approximately injective and the effective reprojection sequence is far from saturation, standard local and geometric descriptions apply. Near structural saturation or strong non-injectivity, effective parameters grow large, signaling the breakdown of spacetime-based descriptions.

2.6 Projective Temporal Ordering and One-Way Activation

Temporal ordering in Cosmochrony is not postulated as a property of the substrate χ, nor derived from an intrinsic dynamics. It emerges as a structural consequence of the admissibility cascade.

Along the breadth-first search cascade on the Cayley graphs of the Heisenberg group $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, define the *cumulative projected capacity*

$$I(n) \;=\; \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n), \tag{7}$$

where $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ is the residual projective capacity of the conjugate pair $(c, q{-}c)$ at BFS depth n [22]. The quantity $I(n)$ measures the spectral structure stably projected up to depth n.

Proposition 2.1 (Projective temporal ordering — numerical form [10])**.** *For $q \in \{29, 61, 101, 151\}$, $I(n)$ is strictly non-decreasing at every BFS step and for every conjugate pair $(c, q{-}c)$, with no physically meaningful regression across* 169 *pairs and* 310 *post-burn-in steps.*

This monotonicity defines an intrinsic ordering on effective states:

$$U_n \prec U_{n+1} \iff I(U_n) \le I(U_{n+1}). \tag{8}$$

The index n is the BFS shell depth, an ordering parameter on projected configurations; it does not represent a physical time variable at the substrate level. The ordering is not a dynamics on χ but a relational structure induced by the non-injective projection Π: it acts entirely within the fixed observable rank established in [22], without enlarging the observable degrees of freedom.

Admissible neighbourhood and recursive structure.

Each state U_n reached by the cascade determines the set of admissible successors at the next step. We denote this set the *admissible neighbourhood from* U_n:

$$\mathcal{A}(U_n) \;:=\; \big\{\, U_{n+1} \mid I(U_{n+1}) \ge I(U_n) \text{ and the transition } U_n \to U_{n+1} \text{ respects the Born–Infeld flux bound} \,\big\}. \tag{9}$$

Admissibility is therefore never an absolute predicate of U_{n+1} taken in isolation, but a relational property of the transition: U_{n+1} is admissible *from* U_n. The cascade proceeds as the recursive chain

$$U_{n-1} \;\longrightarrow\; U_n \;\longrightarrow\; \mathcal{A}(U_n) \;\longrightarrow\; U_{n+1}, \tag{10}$$

in which the reached state immediately becomes the generator of the next set of constraints. Because $\mathcal{A}(U_n) \neq \mathcal{A}(U_{n-1})$ generically, the cascade never evolves from an invariant reference frame: the observable structure at step n is the current projective position within the space of admissible reprojections, not a static section of a fixed ambient space.

One-way activation.

The observed monotonicity is underpinned by an empirical property of the cascade: spectral modes, once stably projected, do not deactivate. This *one-way activation* property is a numerical result established in [10]; its analytical derivation from first principles constitutes the central open conjecture of the temporal ordering programme.

One-way activation is a property of *spectral modes* in the Gram–Schmidt span, not of the coherence of projected configurations. A mode once acquired remains in V_n; this is compatible with the fact that a particular coherent combination of modes — corresponding to an unstable particle — can lose its admissible coherence and reorganize into simpler configurations. The persistence of modes and the instability of configurations are therefore not in conflict: the former is a property of the spectral substrate, the latter a property of the relational coherence of specific projected states. The structural interpretation of particle decay in this framework is discussed in Section 4.4.

Conjecture 2.1 (One-way activation)**.** *For all admissible primes q, all conjugate pairs $(c, q-c)$, and all $n \geq 1$,*

$$\sigma_{\mathrm{pair}}(n) \;\leq\; \sigma_{\mathrm{pair}}(n-1). \tag{11}$$

Three structural facts are immediate from the Gram–Schmidt construction — the cumulative span $V_n = \mathrm{span}\{f_\omega : \omega \in S_0 \cup \cdots \cup S_n\}$ is non-decreasing, $\dim V_n \leq q$, and the rank increment $\delta r_n = \dim V_n - \dim V_{n-1} \geq 0$ at every step — but they do not imply Conjecture 2.1. The conjecture requires, in addition, quantitative control of the shell-wise efficiency with which successive BFS shells generate new independent Weil fingerprint vectors relative to shell size $|S_n|$. A proof would rest on three identified structural ingredients:

(i) the Gram–Schmidt monotonicity of cumulative rank,
(ii) the Born–Infeld admissibility constraints on the amplitude of admissible projective flux — here BI acts as a structural admissibility constraint, not as a fundamental dynamical law — and
(iii) the non-injectivity of Π, which controls how spectral structure accumulates across fibres.

A fourth structural ingredient, identified in [23], is the rigidity theorem of O27: any admissible morphism $\Phi_{q,\rho}$ necessarily factors through $\mathfrak{su}(2)$, confining the Weil fingerprint vectors to a three-dimensional real sector and providing the non-concentration structure required to control shell-wise efficiency. Quantitative control of shell-wise spectral saturation in this $\mathfrak{su}(2)$-confined sector of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ is the analytical task required to close the conjecture.

Arrow of time and irreversibility.

One-way activation is the condition of possibility for the projective arrow of time: because spectral modes persist, $I(n)$ is well-defined as a monotone quantity, and the ordering it induces is globally consistent. Irreversibility then follows: no admissible trajectory can decrease $I(n)$. The arrow of time is identified with the directed growth of $I(n)$, arising prior to any statistical or thermodynamic description [8, 24]. The relation to thermodynamic irreversibility is discussed in Section 8.9.

Energy as residual projective capacity.

Energy is interpreted as the residual projective capacity of effective configurations: it measures the degree to which further accumulation of projected spectral structure remains admissible. Formally, $\sigma_{\text{pair}}(n) = \sigma_{\text{pair}}(0) - I(n)$ is the remaining capacity at depth n, decreasing monotonically as the cascade advances. Admissible ordering paths exclude any effective decrease of $I(n)$, which would correspond to a restoration of projective capacity incompatible with the one-way activation property.

Projectability and kinematic saturation.

A change of velocity corresponds to a modification of the relational coherence constraints maintained by the projection. As velocity increases, the informational demand on the projection grows, progressively saturating its capacity. The bound c_χ is a structural limit beyond which no stable and globally consistent projection can be maintained. Approaching saturation, part of the relational content of χ becomes inaccessible to the effective description, manifesting as time dilation, length contraction, and horizon formation. Relativistic kinematics thus emerges as a consequence of finite projection capacity.[2]

Planck scale and relativistic bounds as projection limits.

The constants c and h are interpreted as complementary manifestations of a finite resolution of the projection from χ to effective observables. The bound c limits the maximal admissible rate at which relational ordering can be projected, while h sets a lower bound on the granularity with which the relational flux can be resolved. Relativistic and quantum constraints thus emerge as complementary facets of a single structural limitation: the finite capacity and resolution of the projection.

2.7 Local Admissible Ordering Rate

The effective local ordering rate associated with projected χ configurations is bounded. In effective geometric descriptions, this constraint takes the form

$$|\mathcal{D}_{\text{loc}}\chi_{\text{eff}}| \leq c, \tag{12}$$

where $\mathcal{D}_{\text{loc}}\chi_{\text{eff}}$ denotes an effective local relaxation functional. This inequality limits the maximal rate at which effective causal connectivity can be established within admissible descriptions. The quantity c characterizes the causal structure of the projected regime.

Local particle propagation, signal transmission, and field interactions are all constrained by this bound in effective spacetime descriptions. Apparent superluminal

[2] This kinematic saturation concerns the ordering and coherence capacity of the projection itself and should not be conflated with thermodynamic entropy production.

recession velocities at cosmological scales arise from cumulative global effects of projected ordering and do not violate this local causal constraint.

2.8 Relation to Conventional Fields

Effective descriptions derived from projected χ configurations may formally resemble scalar or tensor fields used in cosmology and particle physics. This resemblance reflects the emergence of a spacetime-based descriptive language, not the presence of an additional fundamental field.

Energy and quantization are not fundamental attributes of χ but arise at the effective level as consequences of the non-injective projection. Matter, radiation, and interactions correspond to effective degrees of freedom arising from structural constraints, spectral organization, and long-lived relational patterns of projected configurations. Standard Model fields are recovered as accurate effective descriptions within the appropriate coarse-grained regimes.

Cosmochrony does not extend the Standard Model by introducing new fundamental fields. It provides an ontological explanation for the emergence, applicability, and structural properties of effective field descriptions themselves.

2.9 Initial Conditions and Global Structure

The framework does not postulate initial conditions in the conventional temporal sense. It assumes that χ admits a minimal admissible ordering state, denoted χ_0, defining a structural boundary of admissible projected descriptions. This state does not correspond to a distinguished moment in time but to the earliest configurations for which an effective ordering interpretation becomes meaningful.

In effective geometric regimes, the characteristic scale near χ_0 coincides numerically with the Planck scale. This correspondence reflects the breakdown of projectability below this regime, not the presence of a fundamental cutoff or underlying discreteness.

Cosmic history is interpreted as the progressive and irreversible growth of stably projected structure, accumulating away from this minimal admissible boundary. No spacetime singularity is required at the fundamental level. Apparent singular behavior arises only when classical notions are extrapolated beyond the regime in which projected configurations admit a stable geometric interpretation.

Ontological poverty and the growth of admissible structure.

The minimal state χ_0 corresponds to a regime of *ontological poverty*: only a severely restricted class of simple and highly coherent configurations can be projected. As the admissibility cascade advances, the space of stably projected configurations expands, enabling the emergence of increasingly rich, localized, and hierarchical effective structures. The global structure of admissible descriptions is constrained by the ordering properties of the projective structure.

2.10 Ontological Interpretation

The χ Substrate as a Pre-Temporal Relational Reservoir

The substrate χ admits an ontological interpretation as a pre-temporal relational reservoir from which spacetime, matter, and effective physical laws emerge. It is not a dynamical history, nor a prescriptive blueprint of admissible universes, but a relational support supplying a continuous flux of configurational possibilities. Temporal succession is emergent, corresponding to the monotone growth of projected spectral structure along the admissibility cascade, as established in Section 2.6.

The term "pre-temporal" emphasizes that χ does not encode time or causal succession as primitive notions. Rather, time arises as a property of stable projected descriptions that remain mutually compatible under successive updates. No teleology, determinism, or block-universe ontology is implied: multiple effective histories may correspond to the same relational substrate through non-injective projection.

Relational Ontology and Conceptual Lineage

The relational character of χ bears a conceptual affinity with relational approaches in physics, notably those of Rovelli [5, 25], which trace part of their lineage to Aristotelian relational ontology [26, 27]. Cosmochrony shares the rejection of intrinsic, observer-independent properties but extends relationalism to a deeper ontological level: χ configurations are not relations *between* fundamental objects but relational structures that give rise to objects only upon projection.

Relativistic causality and spacetime locality emerge without postulating spacetime as fundamental [28]. Relationality is therefore not a property of spacetime itself, but of the pre-geometric substrate whose projected stabilization yields spacetime as an effective description.

Projection, Reality, and Ontological Asymmetry

The emergence of spacetime and physical law is a *projection* from χ, not a dual or bidirectional description. The projected universe is fully real at the level of physical experience, but ontologically derivative: spacetime entities and dynamical laws do not possess ontological primacy.

While all physical descriptions depend on the projection of χ, the admissibility of a given projection is constrained by the compatibility of the update with the current effective state U. This introduces an ontological asymmetry: χ supplies configurations, whereas U restricts which projected descriptions can remain stable under successive updates.

Apparent fine-tuning is thus reinterpreted as a projective selection effect: only those updates preserving the coherence of an effective state appear as physically realized universes. Cosmochrony does not postulate a multiverse; the universe is unique at the level of physical reality, even though its relational description in terms of χ is non-unique.

Formal developments related to projection, including its fiber-bundle formulation and the emergence of gauge interactions, are developed in Section 5.

Configurational State Structure

The substrate χ defines a configurational space of relational structures and an irreversible flux of candidate configurations. It does not prescribe a fixed set of admissible macroscopic states. Physical reality corresponds instead to projected realizations that remain stable under finite resolution and compatibility constraints imposed by the current effective state U.

What appears as temporal evolution corresponds to the monotone ordering of projected configurations induced by the admissibility constraints of Π, not to any evolution of χ itself. Universal bounds such as c_χ and $\hbar_\chi$ reflect invariant limits of the projection and update process, rather than prescriptive constraints encoded in χ itself.

Projective Origin of Observable Variability

Observable variability and probabilistic behavior arise exclusively at the level of projected descriptions, as a direct consequence of the non-injectivity of the projection Π. The χ substrate does not fluctuate or evolve at the pre-geometric level: temporal ordering, change, and variation are notions defined only within effective projected descriptions. Randomness is therefore projective in origin: it reflects the multiplicity of effective realizations compatible with a given relational configuration of χ and a given effective state U. No intrinsic dynamical indeterminacy of χ is assumed or required.

The role of the substrate is not to provide an infra-physical dynamics of fluctuations, but to supply a structurally rich relational support whose distinct configurations may remain indistinguishable within the projected description. Observable fluctuations arise only when this multiplicity becomes effective in the projected description.

Relation to Holographic Descriptions

Cosmochrony is not a holographic theory in the technical sense. It does not posit a lower-dimensional boundary description or a dual equivalence between bulk and boundary physics. The limitation of physically accessible information within a spacetime region reflects the degeneracy of underlying χ configurations corresponding to the same effective projection, as constrained by finite resolution and compatibility.

Scaling behaviors reminiscent of holography are interpreted as emergent signatures of projection. Similarly, Cosmochrony differs from thermodynamic approaches to gravity in that it does not posit entropy or information as primitive quantities. Thermodynamic descriptions arise only at the effective level, as secondary languages applicable when projected configurations admit coarse-grained statistical interpretations.

2.11 Energy, Mass, and Fundamental Constants

Energy as Residual Projective Capacity

Energy is the residual projective capacity of effective configurations: it measures the degree to which further accumulation of projected spectral structure remains admissible under the current state of the cascade. It is not a fundamental conserved substance but an emergent quantity characterizing the degree to which a given projected configuration retains the ability to undergo further relaxation. Conservation laws arise only at the effective level, as structural regularities of projected dynamics. Without non-injective projection (Section 2.10), the notion of energy at the effective level would become trivial: a fully injective projection would resolve all relational distinctions, leaving no multiplicity of configurations compatible with a given observable state, and therefore no effective capacity for further relaxation.

Mass as Frozen Information

Localized and long-lived configurations of χ correspond to regions in which further relaxation is strongly inhibited. Such configurations trap a fixed amount of unresolved structural information, preventing it from participating in the global relaxation process. Mass represents *frozen projective capacity*: structural information whose contribution to further spectral accumulation has been locally saturated[3]. Particle annihilation, decay, or radiation emission correspond to the partial or complete release of frozen structural information. The operational definition $m = E/c^2$ and its derivation from resistance to relaxation ordering are developed in Section 4.3.

Quarks as Non-Projectable Internal Modes

Quarks are not independent localized excitations but internal structural modes of composite solitonic configurations. They are required to characterize the internal organization of hadronic excitations but do not admit an autonomous projection into spacetime. Confinement reflects a structural constraint: isolated quark-like modes do not correspond to admissible standalone projections of χ_{eff}. Only collective configurations in which such internal modes are topologically and relationally closed admit a stable spacetime manifestation. Quark confinement thus appears as a direct consequence of the non-injective character of the projection.

The Role of the Universal Bound c_χ

The universal invariant bound c_χ characterizes an absolute structural limit on the degree to which relational information can be locally confined within admissible configurations of χ. It is non-metric and non-temporal: it is not associated with distances, durations, or lightcones, since none of these are defined prior to projection. The constant c_χ expresses a maximal admissible rate of structural ordering.

The effective causal constraint c observed in spacetime is the projected manifestation of c_χ:

$$c \equiv \Pi(c_\chi),$$

acquiring operational meaning only once notions of locality and causal ordering become meaningful. In strong-gravity or near-deprojection regimes, c may lose its geometric interpretation while c_χ remains invariant.

The Role of $\hbar_\chi$ and Reprojection

The parameter $\hbar_\chi$ characterizes a fundamental structural scale of χ, determined by its intrinsic relational density and constraint structure. It specifies a minimal

[3]This notion of frozen structural information should be distinguished from projection-induced entropy, which quantifies loss of distinguishability under non-injective projection.

quantum of *reprojection*: intrinsic structural information encoded in χ can enter an effective spacetime description only in discrete units set by this scale. As spacetime structure stabilizes, reprojection events become increasingly localized, manifesting phenomenologically as vacuum fluctuations.

The Origin of Planck's Constant

Within Cosmochrony, $\hbar$ is a **spectral invariant** of the relaxation and projection process. At the substrate level, the quantum of reprojection is fixed by

$$\hbar_\chi \;=\; \frac{c^3}{K_0\,\chi_c}, \tag{13}$$

where K_0 denotes the coupling density of relational constraints and χ_c the characteristic correlation scale. This relation expresses the *spectral rigidity* of the substrate.

In effective spacetime descriptions, the same scale appears as $\hbar_{\rm eff}$, functioning as the quantum of action in Hilbert-space formulations. The apparent universality of $\hbar$ reflects the fact that K_0 and χ_c are global invariants of the current relaxation epoch. The transition from $\hbar_\chi$ to $\hbar_{\rm eff}$ involves a change in representation, not in value.

Spectral Invariance of Planck's Constant and the Fine-Structure Constant

The fine-structure constant α emerges as a dimensionless spectral ratio, determined by the geometry of the projection fiber Π and the spectral rigidity encoded by K_0:

$$\alpha \;=\; \mathcal{F}\!\left(\frac{\text{geometry and topology of }\Pi}{K_0}\right), \tag{14}$$

where $\mathcal{F}$ is a functional fixed by the structure of admissible projections. If the structural parameters K_0 or χ_c were to vary—for example in primordial high-constraint regimes—both $\hbar$ and α would scale accordingly, preserving the internal structural coherence of the framework.

Part II

Dynamics and Particles

This part develops the effective projective dynamics and explains how localized, long-lived excitations emerge as particle-like structures. Starting from the admissibility constraints of the χ substrate, we derive the effective ordering equations governing χ_{eff} and identify the conditions under which stable or metastable projected configurations arise.

Particles are interpreted not as fundamental objects but as structured excitations of the projected description, characterized by spectral, topological, and projective properties. Within this framework, inertial mass, energy–frequency relations, charge, spin, and antiparticles emerge as effective attributes of admissible projected χ configurations.

The projection fiber and its transmittance properties are then introduced, providing a unified interpretation of gauge structures and internal symmetries. This part establishes the physical content of matter and interactions prior to any gravitational or quantum description.

3 Effective Projective Dynamics

3.1 Parameter-Independent Projective Ordering

Physical evolution appears only at the effective level, as an ordered sequence of projected χ configurations $(\chi_{\mathrm{eff},\lambda})$, where λ is a strictly monotonic ordering parameter labeling admissible stages of the projective cascade. The ordering parameter λ does not represent time; it serves as an index ordering projected configurations once a macroscopic spacetime description becomes applicable.

Admissible projected descriptions are constrained by an effective ordering condition:

$$\mathcal{D}_\lambda \chi_{\mathrm{eff}} = \mathcal{R}[\chi_{\mathrm{eff}}], \tag{15}$$

where $\mathcal{R}[\chi_{\mathrm{eff}}]$ denotes an effective projective ordering functional characterizing the admissible progression of projected configurations. This relation is a consistency condition on admissible projections, not a fundamental dynamical law. The functional $\mathcal{R}$ is defined only in regimes admitting a stable geometric interpretation.

Quantities commonly interpreted as temporal derivatives arise exclusively within effective descriptions. Within this framework, the projective ordering does not occur *in* time; the ordering of projected configurations defines what is operationally identified as physical duration.

3.2 Hamiltonian Derivation of the Evolution Equation

Once projected χ configurations admit a smooth, stable coarse-grained description, admissible local projective ordering patterns may be summarized by a constraint formally analogous to a Hamiltonian condition. The effective local ordering rate is bounded by the invariant causal constant c:

$$(\mathcal{D}_{\mathrm{loc}} \chi_{\mathrm{eff}})^2 + \mathcal{V}_{\chi_{\mathrm{eff}}}^2 = c^2, \tag{16}$$

where $\mathcal{V}_{\chi_{\mathrm{eff}}}$ denotes an effective internal variation functional encoding the resistance of projectively stable configurations to further ordering advancement (e.g., solitonic or bound configurations).

Restricting to the monotonic ordering branch yields

$$\mathcal{D}_{\mathrm{loc}} \chi_{\mathrm{eff}} = c\sqrt{1 - \frac{\mathcal{V}_{\chi_{\mathrm{eff}}}^2}{c^2}}, \tag{17}$$

expressing the universal slowdown of the effective projective ordering rate induced by localized structural constraints.

Emergent Gravitational Description

Projectively stable configurations locally reduce the admissible effective ordering rate. In a spacetime interpretation, this reduction is conventionally described as gravitational time dilation. No independent gravitational field is postulated.

In weak-structure regimes, the spatial distribution of ordering slowdown admits the simplified relation

$$\nabla \cdot \left(\frac{\nabla \chi_{\mathrm{eff}}}{\sqrt{1 - |\nabla \chi_{\mathrm{eff}}|^2/c^2}} \right) \simeq \frac{4\pi G_{\mathrm{eff}}}{c^2} \rho, \tag{18}$$

where ρ denotes the effective density of projectively stable configurations and G_{eff} is an emergent coupling parameter. In the Newtonian limit, this yields an effective Poisson equation for a potential defined operationally by

$$\Phi \equiv c^2 \ln\left(\frac{\mathcal{D}_{\mathrm{loc}} \chi_{\mathrm{eff}}}{c} \right). \tag{19}$$

This potential is a compact summary of how projectively stable configurations locally constrain the effective ordering rate, modulating physical clocks and rulers.

3.3 Microscopic Origin of the Coupling Tensor and the Poisson Equation

The effective coupling governing projected χ configurations depends on the internal structural state of the projected description. A convenient phenomenological parametrization is

$$K_{\mathrm{eff}} = K_0 \exp\left(-\frac{(\Delta \chi_{\mathrm{eff}})^2}{\chi_c^2} \right), \tag{20}$$

where $\Delta \chi_{\mathrm{eff}}$ measures effective internal variation, K_0 is the maximal projective throughput in a homogeneous background, and χ_c sets the scale beyond which inhomogeneities suppress the admissible ordering rate.

Projected configurations exhibiting strong internal variation reduce the effective coupling and locally slow the admissible projective ordering, providing the microscopic origin of emergent gravitational phenomenology.

An effective gravitational potential Φ may then be introduced through

$$\frac{\mathcal{D}_{\mathrm{loc}} \chi_{\mathrm{eff}}}{\mathcal{D}_0} \simeq 1 + \frac{\Phi}{c^2}, \tag{21}$$

summarizing the relative slowdown of the effective projective ordering rate. In the weak-structure regime, the spatial distribution of Φ admits a Poisson-type relation:

$$\nabla^2 \Phi \simeq 4\pi G_{\mathrm{eff}} \rho, \tag{22}$$

where ρ is the effective density of projectively stable configurations and G_{eff} an emergent coupling parameter. Gravitation appears as a descriptive manifestation of reduced projective throughput induced by structured projected configurations.

A fully relational formulation is provided in Appendix E.

3.4 Variational Formulation and Born–Infeld Action

In regimes admitting a stable geometric interpretation, the effective projective ordering constraints may be summarized in a compact variational form. Motivated by Born–Infeld-type non-linear actions [17, 29], we consider the effective Lagrangian density

$$\mathcal{L}_{\text{eff}} = -c^2\sqrt{1 - \frac{|\nabla \chi_{\text{eff}}|^2}{c^2}} + \mathcal{D}_{\text{loc}}\chi_{\text{eff}} - \frac{4\pi G_{\text{eff}}}{c^2}\,\rho\,\chi_{\text{eff}}, \tag{23}$$

where $\mathcal{D}_{\text{loc}}\chi_{\text{eff}}$ is the effective local ordering rate (Section 3.1) and ρ the effective density of projectively stable configurations.

The linear dependence on $\mathcal{D}_{\text{loc}}\chi_{\text{eff}}$ enforces monotonicity without additional propagating degrees of freedom. The square-root structure acts as a non-linear regulator ensuring that effective spatial variations remain bounded by c. The Euler–Lagrange equation reproduces the non-linear elliptic relation

$$\nabla \cdot \left(\frac{\nabla \chi_{\text{eff}}}{\sqrt{1 - |\nabla \chi_{\text{eff}}|^2/c^2}} \right) = \frac{4\pi G_{\text{eff}}}{c^2}\,\rho, \tag{24}$$

coinciding with the effective relation obtained in Section 3.3.

This Born–Infeld-like action is an *auxiliary variational representation*. It does not define a fundamental action principle, nor equations of motion for the χ substrate. Its purpose is to regularize the effective description, enforce universal structural bounds, and facilitate comparison with standard gravitational phenomenology.

The physical interpretation is discussed in Appendix A.1, and the mathematical consistency with the relational dynamics is established in Appendix A.16.

Connection to Emergent Geometry.

In regimes where a geometric description is applicable, the Hessian of $\mathcal{L}_{\text{eff}}$ defines an emergent metric tensor $g_{\mu\nu}$, summarizing the effective geometric regularities of projected configurations.

Why This Is Not a Scalar–Tensor Theory.

The effective scalar descriptor χ_{eff} is not a fundamental dynamical field: it does not possess intrinsic values or conjugate momenta, and does not propagate independently. No modification of the gravitational sector is postulated, no scalar–tensor coupling of the form $f(\chi_{\text{eff}})R$ is introduced, and no additional propagating modes arise. Gravitation emerges from the local constraint on the projective ordering rate, not from the exchange of a scalar mediator.

Operational saturation scale and acceleration proxy

The invariant scale b_χ bounds the maximal rate of admissible projective flux transport in the substrate χ. The kinematic bound c appearing in the square-root structure is the effective projection of this invariant limit in weak-field geometric regimes.

In projectable regimes, the effective constitutive response therefore takes a Born–Infeld-like form. Projected gradients cannot grow without bound while remaining within a smooth quasi-injective update.

In the macroscopic limit, one may introduce an operational acceleration proxy $a_\star$ marking the onset of the saturated response. Dimensional consistency and bounded transport jointly imply that this threshold must scale inversely with the local descriptive resolution ℓ of the projected state U. We therefore parametrize

$$a_\star \sim \kappa \frac{b_\chi}{\ell}, \tag{25}$$

where κ is a dimensionless normalization factor encoding the convention adopted for the projective mapping Π. The role of ℓ and its scale dependence are discussed in Section 13.12.

3.5 Schwinger Effect as a Saturation Threshold of Admissible Flux

In standard quantum field theory, the Schwinger effect is interpreted as a vacuum instability under an electric field exceeding a critical threshold. In Cosmochrony, the Born–Infeld-type effective dynamics provides a structurally different interpretation.

The Schwinger effect corresponds to a *threshold of flux saturation*. When the imposed electric field drives the effective admissible projective flux beyond what can be transported smoothly through the projection fiber, the homogeneous projective regime becomes unstable. The system resolves this instability by activating additional admissible modes of the projection, corresponding to the nucleation of conjugate torsional excitations—naturally identified with electron–positron pairs. As discussed in Section 4.7, such pairs preserve global torsional neutrality, ensuring charge conjugation symmetry.

Pair production thus originates from a topological reconfiguration of the projection fiber under maximal admissible flux stress, not from vacuum fluctuations. Matter creation acts as a dissipation channel that restores admissibility by redistributing excess projective flux into stable, projectable vortical modes. This enlarges the space of admissible configurations, in accordance with the principle of monotonic growth of projected spectral structure.

Cosmochrony therefore predicts the existence of a Schwinger-like threshold *ab initio*, without invoking the Dirac sea or field quantization. The effect emerges as a universal transition between smooth projective ordering and topologically mediated dissipation.

Dissipation by structure creation.

This mechanism illustrates a more general principle: when directed projective fluxes approach their maximal transport capacity, admissibility is restored by creating new stable structures that enlarge the space of admissible configurations. This suggests a unified interpretation of pair-loaded astrophysical jets, primordial matter production, and ultra-high-energy excitations in extreme curvature regimes (Section 12).

3.6 Causality and Locality

Within effective descriptions, the projective ordering encoded by χ_{eff} exhibits locality and causality. Effective locality follows from the fact that variations of χ_{eff} at a given effective event depend only on correlated neighboring projected configurations. Causality is enforced through the universal bound

$$|\mathcal{D}_{\text{loc}}\chi_{\text{eff}}| \leq c,$$

constraining the maximal rate at which correlations between effective configurations may be established. No superluminal propagation of signals or causal influence occurs within effective descriptions. Apparent superluminal recession velocities in cosmological contexts arise from cumulative integration of locally constrained projective ordering over extended regions and remain consistent with effective locality.

The effective causal bound c corresponds to the projected manifestation of the invariant structural bound c_χ defined at the level of the pre-temporal substrate (Section 2.11). Effective causality arises as a derived property of bounded projective realizations.

3.7 Homogeneous Cosmological Limit

In a homogeneous and isotropic regime, projected configurations exhibit no effective spatial variations and the admissible projective ordering rate attains its maximal value:

$$\mathcal{D}_{\text{loc}}\chi_{\text{eff}} = c. \tag{26}$$

Expressed in terms of an effective cosmological time parameter t, this yields

$$\chi_{\text{eff}}(t) = \chi_{\text{eff},0} + c\,t, \tag{27}$$

where $\chi_{\text{eff},0}$ is a reference value. This linear regime underlies the emergent Hubble law derived in Section 8.4.

As shown in Appendix A.6, the monotonicity requirement in an expanding regime implies a minimal residual structural inhomogeneity in projected configurations, manifesting as a non-vanishing lower bound on gravitational acceleration. This bound provides a natural route to MOND-like phenomenology without postulating dark matter particles [30, 31].

3.8 Influence of Local Structure

In regions where projected configurations exhibit non-vanishing effective structural variations, the admissible local projective ordering rate is reduced. Projectively stable configurations—describable as particle-like solitonic structures—act as constraints on the admissible ordering of projected configurations. By increasing the effective structural complexity, they locally constrain the projective ordering rate without introducing any additional interaction or mediator.

When a geometric description is applicable, this constraint manifests as gravitational time dilation and spatial curvature. Gravitation arises as a collective consequence of locally constrained effective projective ordering.

3.9 Unified Origin of Geometric and Field Effects

The relation between the χ substrate and the effective spacetime metric $g_{\mu\nu}$ is strictly hierarchical and may be summarized in three levels. First, at the fundamental level, physical reality is described solely in terms of χ and its intrinsic relational structure. Second, in regimes where projected configurations admit a stable and slowly varying description, the spacetime metric arises as an effective descriptor summarizing correlations and projective ordering structure. Third, projectively stable localized configurations—describable as solitonic structures—are identified with matter degrees of freedom, and gravitational phenomena correspond to local modulations of the effective projective ordering rate induced by such structures.

No independent gravitational interaction or fundamental field is postulated. Matter, geometry, and gravitation emerge as complementary aspects of the same constrained projective ordering of configurations.

3.10 Limitations and Scope

Equation (17) is intentionally minimal. It constrains admissible projected descriptions in regimes where a stable geometric interpretation is applicable, but does not provide a first-principles account of quantum fluctuations or correlations at the level of χ itself. Such phenomena arise from non-injective projection and coarse-graining. Higher-order structural effects and strongly non-projectable regimes lie outside the present scope.

Within these limitations, the constrained ordering relation provides a unified kinematic backbone from which gravitational, quantum, and cosmological phenomena can be consistently recovered at the effective level. More refined treatments of effective fluctuations and extended relational structures are left for future work.

4 Particles as Localized Excitations of the χ Field

4.1 Particles as Stable Wave Configurations

Particles are not fundamental point-like entities. They arise at the level of effective descriptions as stable, localized configurations within projected χ descriptions [32]. Such configurations correspond to persistent patterns that locally constrain the admissible projective ordering rate. In effective geometric regimes, they may be described using soliton-like language. Their stability reflects localized regions in which the advancement of projective ordering is strongly inhibited. Apparent particle propagation corresponds to a continuous reorganization of admissible projected descriptions, not to the motion of an object through a fundamental spacetime.

4.2 Topological Stability

The stability of particle-like excitations does not rely on fundamental conserved charges postulated *a priori*. Certain projected configurations exhibit non-trivial internal organization that prevents them from being continuously deformed into homogeneous effective descriptions without violating admissibility conditions.

This stability is topological in character: it reflects the existence of inequivalent classes of admissible projected configurations that cannot be smoothly connected through continuous reconfiguration while preserving the monotone growth of projected spectral structure. The long-lived character of solitonic structures follows from this topological incompatibility, not from a dynamical balance of forces.

Detailed geometric constructions of topological solitons, including vortex, skyrmion, and knotted configurations, are provided in Appendix B.2. The fully relational formulation is developed in Appendix E.8.

The complete classification of admissible topological invariants — including the homotopy groups π_1 and π_2 of the projection fiber and their physical consequences (charge quantisation, absence of magnetic monopoles, projective chirality) — is given in Section 5.3 and [33].

Stability through self-consistency.

The projection Π does not operate independently of the effective description already in place, but is constrained by the current state U through a compatibility condition. Among the relational configurations supplied by the χ substrate, only those transitions $\chi_n \to \chi_{n+1}$ whose projection remains compatible with U admit a stable effective realization. In this sense, physical laws are not imposed by the substrate χ, but emerge as the only update rules supporting projective continuity across successive descriptions [34].

4.3 Mass as Resistance to Projective Ordering Advancement

Building on the ontological interpretation of mass as frozen projective capacity (Section 2.11), we develop the quantitative formulation. Mass emerges at the effective level as a measure of how strongly a localized projected configuration resists the advancement of admissible projective ordering.

The effective structural energy associated with a projected solitonic configuration $\chi_{\mathrm{eff},s}$ is

$$E[\chi_{\mathrm{eff},s}] \;\equiv\; \int_{\Sigma} \left(\frac{1}{\sqrt{1 - |\nabla \chi_{\mathrm{eff},s}|^2/c^2}} - 1 \right) d\Sigma, \tag{28}$$

where Σ denotes a hypersurface of constant effective ordering parameter and $|\nabla \chi_{\mathrm{eff},s}|$ quantifies effective structural deformation. The inertial mass is then defined operationally as

$$m \;\equiv\; \frac{E[\chi_{\mathrm{eff},s}]}{c^2}. \tag{29}$$

For example, the electron mass m_e reflects its topological stability as a 4π-periodic soliton (Section B.4), while the proton mass arises from a composite 3-soliton configuration.

The relation $E = mc^2$ is interpreted as a kinematic identity: mass quantifies the local saturation of projective capacity, while energy expresses the same quantity in units of residual projective capacity. The question of how distinct particle masses arise from different classes of projected configurations is addressed in Appendix B.2.

4.4 Metastability, Projection, and Particle Decay

A stable particle corresponds to a deep basin of admissible projected configurations; an unstable particle occupies a shallow or fragile basin. Particle decay is interpreted as the structural reorganization of a metastable configuration: when the concentration of constraints exceeds what can be sustained by a single projected entity, admissibility is recovered through factorization into less constrained localized configurations, possibly accompanied by weakly structured excitations.

Entanglement and decay represent two regimes of the same projection structure. Entanglement corresponds to non-factorizability without fragmentation, while decay corresponds to non-factorizability that forces fragmentation. The distinction lies in the stability properties of the projected description under admissible fluctuations.

The finite lifetime of unstable particles reflects the probability of crossing a structural reorganization threshold under admissible variations, yielding the observed exponential decay law as a statistical signature of metastability.

A more technical characterization of metastability, admissible factorization channels, and decay widths is provided in Appendix B.19.

4.5 Energy–Frequency Relation

The energy associated with a particle-like excitation is linked to a characteristic internal spectral scale of the corresponding projected configuration. Configurations associated with higher characteristic frequencies correspond to more tightly constrained structures and encode greater effective saturation of local projective capacity. This yields

$$E \propto \nu, \tag{30}$$

where ν characterizes the spectral scale of internal organization. Planck's constant appears as an effective proportionality factor whose universality reflects the robustness

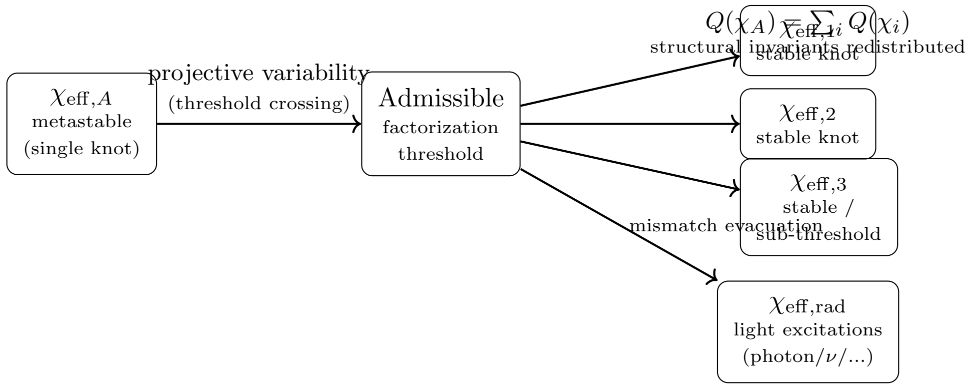

Fig. 4 Structural interpretation of particle decay. A metastable localized projected configuration transitions, via an admissible factorization threshold, into several more stable configurations plus weak excitations that evacuate the residual structural mismatch.

of spectral scales across the current projective cascade epoch. A more explicit realization in the context of radiation is presented in Section 10.3.

4.6 Fermions, Bosons, and Spin

Particle statistics emerge from the topological structure of admissible projected configurations. Configurations requiring a 4π rotation to return to an equivalent projected description give rise to fermion-like behavior, while 2π-periodic configurations correspond to bosons. This distinction reflects a topological obstruction rather than a symmetry principle imposed at the fundamental level.

Spin as a Topological Property

Spin is not an intrinsic kinematic degree of freedom but a purely topological property of admissible projected configurations. Fermionic configurations require a 4π transformation in configuration space to return to an equivalent effective description, implying that the relevant configuration space admits a double covering with fundamental group

$$\pi_1(\mathcal{C}_{\text{eff}}) = \mathbb{Z}_2. \tag{31}$$

When an effective quantum description is applicable, a 2π transformation induces a sign change of the associated wavefunction,

$$\psi \longrightarrow -\psi, \tag{32}$$

while a 4π transformation restores the original state. This 4π-periodicity directly implies fermionic antisymmetry and the Pauli exclusion principle, as detailed in Section E.9.

Exchanging two identical fermionic excitations corresponds topologically to a 2π loop in the combined configuration space and induces a sign change, dynamically excluding symmetric configurations [35]. The spin–statistics connection thus admits a unified topological origin within the effective descriptive framework.

4.7 Charge as a Topological and Projective Property of χ

Unified Projective Budget for Mass and Charge

Both inertial mass and electric charge draw on the same finite projective capacity of χ. A localized excitation mobilizes capacity in two structurally distinct channels: a scalar inhibition channel associated with inertial response (mass), and an oriented or chiral channel associated with a net topological flux (charge). Admissible projected configurations satisfy a single combined bound

$$\mathcal{B}_m[\chi] + \mathcal{B}_e[\chi] \leq \mathcal{B}_{\max}, \tag{33}$$

where $\mathcal{B}_{\max}$ is fixed by the universal projective capacity constraint (Section 3.4). Effective mass and charge are therefore competing manifestations of a common substrate resource, and the observed discreteness of particle masses may be approached as a classification of admissible stable configurations under a single saturation constraint.

Charge as a Directed and Conjugate Projective Mode

Charges arise only after projection, as stable invariants characterizing admissible projected configurations. Charge conjugation is reinterpreted as a transformation between *relationally conjugate* projected configurations: paired topological classes related by an internal reversal of relational organization, reflecting the internal duality of the projection fiber.

At the discrete level, the χ structure induces a bounded admissible flux $\vec{J}_\chi \sim \nabla\chi$. Charge corresponds to the non-integrability of this flux around localized excitations: the resulting winding number defines an integer-valued topological invariant whose sign encodes chirality. Electric attraction and repulsion arise from the geometric compatibility or frustration between torsional flux patterns of neighboring excitations. The bounded nature of $\vec{J}_\chi$ implies a maximal admissible charge density and naturally removes short-distance singularities.

The formal identification of electric charge as a π_1 winding invariant of the $U(1)$ Hopf fiber, and the proof of charge quantisation from $\pi_1(S^1) \cong \mathbb{Z}$, are given in Section 5.3 and [33].

The robustness of this chiral–torsional invariant is tested in Appendix B.22.

CP Symmetry as Projective Chirality.

CP symmetry corresponds to a combined transformation acting on the orientation of the projection itself. When the projection is achiral, conjugate configurations are mapped symmetrically and CP is preserved. When the projection is chiral, CP is violated as a direct consequence of projective asymmetry, without introducing explicit CP-violating terms at the level of χ. The formal definition of projective chirality and its structural mechanisms are given in Section 5.3 and [33].

4.8 Antiparticles, Creation/Destruction, and CPT

Antiparticles as Relationally Conjugate Configurations

A particle and its antiparticle correspond to projected configurations belonging to distinct but conjugate topological classes within the space of admissible projected descriptions, related by an internal reversal of relational organization. Annihilation occurs when a particle-like configuration and its conjugate combine into a composite description that no longer supports localized structural constraints, redistributing relational structure into delocalized radiation-like excitations. This constitutes a process of *structural unknotting*: mass measures the degree of topological obstruction to projective ordering advancement.

Why Antimatter Does Not Require Time Reversal.

The monotone growth of projected spectral structure defines an absolute arrow of admissible projection that cannot be inverted. Antiparticles correspond to topologically conjugate classes, not to reversed temporal trajectories. The apparent association between antimatter and time reversal in standard formalisms is a feature of the effective representation.

Matter–Antimatter Asymmetry without Fundamental CP Violation.

If the projection from χ to effective spacetime is chiral, conjugate classes need not be realized with equal stability or projectability. Matter–antimatter asymmetry emerges as a selection effect imposed by differential projectability, without requiring dynamical CP-breaking interactions.

Particle Creation and Destruction

Particle creation corresponds to a projected configuration acquiring sufficient structural organization to support a stable topological class. Particle destruction occurs when a configuration loses its topological admissibility and admits continuous deformation toward a delocalized effective description.

CPT as a Global Projective Property

C, P, and T are individually effective and representation-dependent. Their combined action corresponds to a full relational conjugation of projected descriptions, mapping any admissible effective configuration to another admissible configuration representing the same underlying χ structure. CPT invariance is therefore not a microscopic symmetry but a global projective consistency condition ensuring that the space of admissible descriptions is closed under full relational conjugation. Violations of CPT would signal a breakdown of projectability itself.

CPT as an Admissibility Consistency Condition

Certain projected configurations carry orientation-sensitive structural invariants (chirality, phase winding, relational orientation). Under admissible factorization, these signed invariants must be redistributed, possibly requiring paired localized excitations carrying opposite orientations—interpreted as particle–antiparticle pairs. CPT symmetry reflects the invariance of admissibility under a combined reversal of signed structural invariants, effective spatial orientation, and the effective ordering parameter. A technical formulation is provided in Appendix B.21.

Why CPT Survives Quantum Gravity.

CPT survives because it expresses the minimal requirement for a coherent physical projection. Quantum gravity may challenge locality, geometry, and the notion of time, but cannot violate CPT without undermining the possibility of a consistent emergent universe.

4.9 Neutrinos as Partially Projectable Modes

Neutrinos correspond to *partially projectable modes* of χ: configurations whose relational structure admits a stable projection in some degrees of freedom while remaining weakly or non-projectable in others. This partial projectability explains their extremely small effective masses, weak interaction strength, and sensitivity to global rather than local structural properties of the projection.

Dirac vs. Majorana Character.

A Dirac neutrino admits distinct conjugate projected realizations; a Majorana neutrino corresponds to a configuration whose partial projectability collapses the distinction between conjugate classes. The Dirac or Majorana character is not a fundamental choice but a manifestation of how fully the projection resolves relational conjugation. Lepton number conservation emerges only in regimes where conjugate configurations are distinguishable; when partial projectability erases this distinction, effective lepton number violation becomes admissible.

Neutrino Oscillations without Fundamental Mass Eigenstates.

Different flavors correspond to closely related partially projectable configurations whose internal relational structures overlap but are not identical. As projected descriptions advance along the admissibility cascade, the relative projectability of these configurations varies, leading to coherent transitions between flavor labels without invoking mass eigenstates as fundamental objects.

Stability of the Projection Boundary.

Neutrinos occupy an intermediate regime between fully localized particles and delocalized radiation. They act as carriers of marginal structural information, redistributing relational organization without inducing strong backreaction, thereby preventing abrupt transitions between projectable and non-projectable regimes. In strong-gravity or near-deprojection regimes, neutrinos remain among the last modes to retain partial projectability.

Why Neutrinos Are the Lightest Fermions.

Neutrino configurations reside closest to the boundary of projectability. Their absence of electromagnetic coupling and partial self-conjugacy prevent the formation of tightly bound projected structures. The smallness of neutrino masses is a structural necessity imposed by their role as marginally projectable modes.

Neutrinos as Probes of Pre-Geometric Structure.

Because neutrinos operate near the projection boundary, they provide a unique observational window into the pre-geometric structure of χ. Their weak localization allows them to sample relational structures inaccessible to fully projectable modes, and deviations in their phenomenology may encode signatures of pre-geometric ordering.

Failure of Absolute Localization.

Neutrinos cannot be fully confined within a bounded spacetime region without loss of admissibility, explaining their absence of sharply defined position operators and extremely small interaction cross-sections.

Limits of Effective Quantum Field Theory.

Neutrinos systematically probe the limits of local QFT assumptions. Their weak localization, extended coherence, and oscillatory behavior signal a breakdown of the strict particle ontology. Standard QFT constructs (mass eigenstates, flavor mixing matrices) are understood as effective bookkeeping devices for marginally projectable configurations.

Experimental Signatures.

Potential signatures include small departures from standard oscillation patterns at ultra-long baselines, coherence and decoherence effects sensitive to global spacetime properties, constraints from neutrinoless double beta decay on the projective resolution of conjugate configurations, and cosmological neutrino backgrounds encoding information about the approach to the projection boundary.

Synthesis.

Neutrinos mark the transition between physics described by local quantum fields on spacetime and the pre-geometric relational dynamics of χ. They constitute the structural frontier of emergent spacetime, where effective geometry, quantum description, and pre-geometric ontology converge.

4.10 Spectral Stability and the Lamb Shift

The stability of charged particle-like excitations is governed not only by topological admissibility but also by the fine spectral structure induced by the coupling between localized modes and the global projective ordering flow. While the linear effective description predicts degenerate energy levels for certain atomic configurations, this degeneracy is lifted once non-linear saturation effects and projectability constraints are taken into account.

In standard QED, the Lamb shift arises from radiative corrections requiring renormalization. In Cosmochrony, no vacuum degrees of freedom are introduced. Atomic bound states correspond to admissible localized projective modes. S-states ($\ell = 0$) probe finer scales of the projection fiber where Born–Infeld saturation and spectral frustration are maximal, whereas P-states ($\ell = 1$) remain less sensitive to these inner-core constraints.

The effective Born–Infeld dynamics induces a finite upward shift of the S-state energy:

$$\Delta E_{\mathrm{Lamb}} \sim \kappa\, \alpha^5 m_e c^2, \tag{34}$$

where α is the dimensionless ratio between the local admissible projective flux and the maximal saturation flux c_χ, and κ is a numerical factor of order unity. Numerically, $\alpha^5 m_e c^2 \simeq 10^{-5}\,\mathrm{eV}$, corresponding to a frequency shift of order $1\,\mathrm{GHz}$, in agreement with the observed $2S_{1/2}$–$2P_{1/2}$ splitting. This correction is intrinsically finite thanks to the structural bound c_χ.

Hyperfine Structure.

The same spectral-probe logic extends to hyperfine structure. When the electron occupies an S-state, its non-vanishing density at the nucleus brings electronic and nuclear torsional cores into spectral proximity. Hyperfine splitting reflects the relative alignment of torsional fluxes: parallel alignments accumulate frustration and oppose relaxation, while antiparallel alignments partially compensate torsion. Together with the Lamb shift, hyperfine structure emerges as a finite spectral correction governed by Born–Infeld saturation at different relational scales.

4.11 Summary

Particles arise as stable, localized projected configurations that resist the advancement of admissible projective ordering. Mass is identified with the local saturation of projective capacity, naturally yielding $E = mc^2$ as a kinematic identity between two equivalent measures of the same structural quantity. Spin and statistical behavior originate from topological obstructions: fermionic configurations exhibit 4π periodicity, providing a unified origin for spin-$\frac{1}{2}$ behavior, fermionic antisymmetry, and the Pauli exclusion principle [35, 36]. Electric charge is interpreted as a chiral–torsional winding invariant of the admissible projective flux. Residual spectral splittings—such as the Lamb shift and hyperfine structure—arise as finite corrections induced by non-linear saturation and projectability constraints.

Localized excitations of the χ field may also admit collective coherent regimes in the $U(1)$ fiber sector, to be analyzed in Section 5.8.

5 The Projection Fiber and Gauge Emergence

The projection fiber Π is a central structural element linking the pre-geometric substrate χ to effective spacetime descriptions. It encodes the admissible equivalence classes and internal degrees of freedom through which relational configurations of χ can be consistently projected into geometric and field-like representations.

Gauge interactions do not arise from fundamental interaction fields. They emerge as symmetry structures of the projection process itself, under local re-identification of equivalent representatives along the fiber. This section develops the geometry of the projection fiber and clarifies how distinct classes of gauge phenomena naturally emerge from its structure.

5.1 The Geometry of the Π Subspace

The relational substrate χ is not accessed in its full structural complexity but through admissible local projections onto a reduced projection fiber $\Pi \cong S^3$. Here Π denotes the projection *fiber*, i.e. the space of admissible internal representatives associated with a projected configuration, not the projection map itself.

The identification $\Pi \cong S^3$ is not imposed *a priori* but follows from the minimal compact geometry (in dimension and connectivity) required to support non-injective yet stable projections. Its associated Hopf fibration naturally induces an effective $SU(2) \times U(1)$ symmetry structure, which emerges as an invariance of the projection process.

The metric on Π is dynamically induced by the local density of relational connections encoded in the underlying relational graph G, used as a numerical and representational support, not as a physical discretization. The mapping from the global graph Laplacian Δ_G to the effective projected Laplacian Δ_Π is

$$\Delta_\Pi = P^\dagger \Delta_G P, \tag{35}$$

where P denotes the projection operator onto the admissible spectral subspace. The smooth three-sphere geometry corresponds to a large-N spectral coarse-graining limit of the discrete relational structure.

Throughput constraints and channel counting.

The projection Π does not merely discard information. It enforces an admissibility constraint: local updates must remain operationally compatible with the surrounding global projected state U. This constrains the total admissible throughput by the number of independent relational channels (fibers) that can be synchronized across the update interface within one refresh step. Accordingly, capacity scalings commonly expressed as "area-law" behavior should be understood as consequences of projective throughput constraints rather than geometric postulates. This point becomes central when discussing resolvability bounds across scales in Section 13.12.

5.2 Gauges as Projective Fiber Transmittance

Gauge interactions are reinterpreted as degrees of freedom of the admissible projective flow within the projection fiber Π, encoding how locally admissible projections are coherently related despite non-injectivity.

- **Electromagnetism ($U(1)$):** Corresponds to the phase degree of freedom of the admissible projective flow along the fibers of the Hopf fibration of $\Pi \cong S^3$. This emergent geometric parameter reflects the freedom to locally re-identify projected descriptions without altering global relational consistency. The fine-structure constant α characterizes the *transmittance* of the projective flow through the fiber: the stability of admissible projective re-identifications along the fiber.
- **Weak Interaction ($SU(2)$):** Emerges from the rotational degrees of freedom of the S^3 projection fiber itself. These encode non-abelian modes of projective transmittance, corresponding to shear-like distortions of admissible projections. The massive character of the $W^{\pm}$ and Z^0 bosons follows from a non-zero spectral gap associated with these modes, interpretable as spectral drag or torsion intrinsic to the fiber geometry.
- **Strong Interaction ($SU(3)$):** Associated with topological constraints arising from non-trivial winding structures of projected configurations. The effective $SU(3)$-like symmetry reflects the threefold (triality) structure of the minimal self-intersecting stable soliton, identified with the trefoil knot ($w = 3$). Color symmetry emerges as a geometric consequence of topological stability classes within the projection fiber.

5.3 Topological Invariants of the Projection Fiber

The stability of localized physical descriptions within Π is governed by the conservation of topological invariants. When an excitation of χ admits a closed, non-contractible configuration within Π, it forms a persistent topological obstruction within the admissibility structure. Such configurations cannot be eliminated by smooth deformation without violating admissibility constraints. These obstructions are described in terms of knot-like (non-contractible loop and self-linking) topological structures within the projection fiber. Once formed, they impose global constraints on the admissibility cascade and give rise to long-lived, localized projected configurations. Particles correspond to stable topological defects of the projection.

The homotopy groups of the admissible fiber Π and of its admissible base determine which topological invariants are structurally available, and hence which physical phenomena can arise. Three consequences are established on the canonical admissible configuration model; their extension to the full effective configuration space $\mathcal{C}_{\mathrm{eff}}$ remains open (see [33] for the complete treatment).

Charge quantisation as a π_1 invariant

The Hopf fibration of the projection fiber $\Pi \cong S^3$,

$$S^1 \hookrightarrow S^3 \xrightarrow{\pi_H} S^2, \tag{36}$$

decomposes Π into a circular fiber S^1 and a two-dimensional base S^2. The fundamental group of the $U(1)$ fiber is

$$\pi_1(S^1) \cong \mathbb{Z}. \tag{37}$$

A closed loop in the fiber defines a winding number $w \in \mathbb{Z}$, which is a topological invariant: it cannot change under continuous deformation of an admissible configuration.

Proposition 5.1 (Charge quantisation)**.** *In the Cosmochrony framework, electric charge is quantised as a consequence of $\pi_1(S^1) \cong \mathbb{Z}$. The admissibility constraint removes configurations exceeding the flux bound but does not introduce identifications that would contract the fiber, so the fundamental group of the fiber sector remains that of $U(1)$. Electric charge is therefore an integer multiple of the elementary charge e,*

$$q_{\text{eff}} = w \cdot e, \qquad w \in \mathbb{Z}. \tag{38}$$

The winding number w labels distinct equivalence classes of admissible projected configurations and plays a central role in the organization of the mass spectrum (Section 11). The energetic cost required to maintain a non-trivial winding against the global advance of projective ordering is perceived, at the effective level, as rest energy (mc^2). Mass emerges as a spectral and topological consequence of persistent non-contractible structures within the projection fiber.

Remark (Winding conjugation and the spectral pair structure)*.* Charge conjugation corresponds to the reversal $w \mapsto -w$ of the winding number. In the discrete Weil realisation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the conjugation identity $\rho_{q-c} = \overline{\rho_c}$ shows that complex conjugation of the representation is realised by the map $c \mapsto q - c$ on the central character. In the present setting, the winding number is encoded in the phase of the representation, so complex conjugation acts as orientation reversal on phase windings, i.e. $w \mapsto -w$. The pair $\{c, q - c\}$ is therefore precisely the discrete realisation of the topological conjugate pair $\{w, -w\}$, and identifies the conjugate Weil pair as the minimal spectral realisation of a topological charge class. This identification is canonical: it follows from the representation theory of the Heisenberg group and requires no additional hypothesis. Its spectral consequences — in particular the doubling $\delta_{\text{pair}} = 2\delta_c$ — are developed in [33, 37].

Unlike the standard Dirac argument — which derives charge quantisation from the existence of magnetic monopoles via single-valuedness of the wave function on a surrounding sphere — charge quantisation in Cosmochrony follows from the topology of the $U(1)$ fiber itself, independently of whether magnetic monopoles exist. The two mechanisms are logically independent.

Absence of magnetic monopoles

In the unconstrained Hopf fibration, the base manifold is the full two-sphere S^2, with $\pi_2(S^2) \cong \mathbb{Z}$. A non-trivial element of $\pi_2(S^2)$ corresponds, in gauge-theoretic language, to a Dirac magnetic monopole: a point-like source of magnetic flux classified by an integer topological charge. If the full S^2 were the admissible base, magnetic monopoles would be topologically admissible.

The bounded-flux constraint $|\partial_t \chi_v| \leq c_\chi$ restricts the set of admissible fiber configurations. On the canonical admissible configuration model, the constraint removes a neighbourhood of each pole of S^2 (the points where the fiber degenerates, corresponding to maximal flux concentration). The admissible base is then

$$S^2_{\text{adm}} = S^2 \setminus (D^2_N \cup D^2_S) \simeq S^1 \times (0,1), \tag{39}$$

where D^2_N and D^2_S are open discs around the north and south poles.

Proposition 5.2 (Homotopy of the canonical admissible base)**.** *The canonical admissible base S^2_{adm} is homotopy equivalent to $S^1 \times (0,1)$. Its homotopy groups are*

$$\pi_1(S^2_{\text{adm}}) \cong \mathbb{Z}, \qquad \pi_2(S^2_{\text{adm}}) = 0. \tag{40}$$

Corollary 5.1 (No isolated magnetic charge)**.** *On the canonical admissible model, no isolated magnetic monopole is a topologically admissible excitation.*

In the standard Dirac – 't Hooft – Polyakov framework, monopoles are topologically admissible and their absence must be explained dynamically. In Cosmochrony, the situation is reversed: the absence of monopoles is not a dynamical accident but a topological consequence of admissibility. The bounded-flux constraint removes the topological support for monopole-type excitations. Magnetic flux lines (corresponding to $\mathbf{B} = \nabla \times \mathbf{A}_T$) remain admissible as long as they form closed loops; only isolated magnetic sources are excluded.

The extension of Corollary 5.1 from the canonical admissible base to the full effective configuration space $\mathcal{C}_{\text{eff}}$ is well-posed but remains open. The canonical model is not arbitrary: it captures precisely the universal effect of the bounded-flux constraint, which removes the polar regions of S^2 wherever the fiber phase gradient diverges. Since this divergence is the generic mechanism generating non-trivial $\pi_2(S^2) \cong \mathbb{Z}$ classes, this strongly suggests that any admissible base in $\mathcal{C}_{\text{eff}}$ is subject to the same polar removal, and hence has trivial π_2.

Projective chirality and parity violation

The winding number $w \in \mathbb{Z}$ carries a sign determined by the orientation of the fiber phase. In a parity-symmetric theory, configurations with w and $-w$ would be related by a mirror transformation and have equal dynamical weight. In Cosmochrony, the projection $\Pi : \chi \to \chi_{\text{eff}}$ is non-injective, and this non-injectivity can be chiral.

Definition 5.1 (Projective chirality)**.** A projection is *projectively chiral* if the effective stabilisation rates satisfy

$$\Gamma(Q) \neq \Gamma(-Q), \tag{41}$$

where $\Gamma(Q)$ is the effective stabilisation rate for configurations of topological charge Q, and $-Q$ denotes the charge-conjugate configuration.

When the projection is chiral, parity is violated: mirror-related configurations are not equally admissible. Two structural mechanisms within the framework produce chirality selection.

Spinorial selection.

The spectral admissibility programme establishes that the bounded-flux constraint selects spinorial (spin-$\frac{1}{2}$) representations as the most admissible along the chain $Q_8 \subset 2I \subset SU(2)$ (Section 6 and [18]). Spinorial representations carry an intrinsic handedness through the double-cover structure $SU(2) \to SO(3)$: the central $\mathbb{Z}_2$ retained by binary covers is precisely the element that distinguishes left- and right-handed spinors.

Non-injective projection.

The non-injectivity of the projection creates a fiber degeneracy: multiple χ-configurations map to the same effective state. The orientation of the winding within this fiber is not preserved by a generic projection, so parity invariance is not automatic. A chiral projection selects one orientation over its conjugate, producing $\Gamma(Q) \neq \Gamma(-Q)$ without any explicit parity-breaking term at the substrate level.

The maximal parity violation observed in weak interactions (polarised beta decay of ^{60}Co) can be interpreted as the manifestation of a maximally chiral projection in the $SU(2)$ sector of Π. Three levels of status should be distinguished. At the *structural* level, the non-injectivity of Π makes $\Gamma(Q) \neq \Gamma(-Q)$ possible without explicit parity breaking in the substrate dynamics: this is established by Definition 5.1. At the *mechanistic* level, the spinorial selection of Chapter 6 provides the concrete selection filter: the bounded-flux constraint retains a single chirality sector via the double-cover structure $SU(2) \to SO(3)$. At the *quantitative* level, $\Gamma(Q)$ is not yet expressed as a calculable observable of the spectral programme (e.g. via δ_{pair} or σ_{pair} of the O-series); the derivation of the full $V-A$ structure from the admissibility filter remains open. The relation between projective chirality and CP asymmetry is further developed in Appendix B.22.

Unified topological picture

The three results above are unified by a single structural principle: the effective physics is classified and constrained by the topological invariants that the admissibility constraint renders available, and by the orientation that the projection selects among their generators.

Topological invariant	Physical consequence	Mechanism
$\pi_1(S^1) \cong \mathbb{Z}$	Charge quantisation	Winding of $U(1)$ fiber
$\pi_2(S^2_{\text{adm}}) = 0$	No magnetic monopoles	Bounded-flux reduction of base
Orientation of π_1	Parity violation	Projective chirality

Table 1 Topological invariants of the canonical admissible configuration model and their physical consequences. The first two rows are proved on the canonical model; their extension to $\mathcal{C}_{\text{eff}}$ is open. The third row is established at the framework level as a structural mechanism. See [33] for the complete treatment.

5.4 Projective Holonomy and Fiber Constraints

The projection from relational configurations of the χ substrate to effective observable descriptions is generally non-injective. As a consequence, admissible effective states correspond not to single relational configurations but to equivalence classes under the projection operator.

This structure naturally endows the projection fiber with non-trivial topological constraints.

Relative configurations of projectors.

To illustrate this mechanism we consider a minimal model involving rank-one projectors acting on $\mathbb{C}^2$. Let Π_a and Π_b denote two such projectors. Their relative overlap can be characterized by the positive operator

$$M_{ab} = \Pi_a \Pi_b \Pi_a. \tag{42}$$

For rank-one projectors this operator has a single non-zero eigenvalue

$$\lambda_{ab} = \mathrm{Tr}(\Pi_a \Pi_b) = |\langle a|b\rangle|^2, \qquad 0 \le \lambda_{ab} \le 1. \tag{43}$$

Writing $\Pi(\mathbf{n}) = \frac{1}{2}(I + \mathbf{n} \cdot \boldsymbol{\sigma})$ with $\mathbf{n} \in S^2$, one obtains

$$\lambda_{ab} = \cos^2(\alpha/2),$$

where α denotes the angle between the Bloch vectors $\mathbf{n}_a$ and $\mathbf{n}_b$.

Admissibility and excluded regions.

Effective descriptions are restricted to configurations compatible with the projection constraints. In the present toy model this corresponds to excluding configurations in which projectors become either indistinguishable or orthogonal. Formally this can be expressed through a minimal spectral separation condition

$$\min(\lambda_{ab}, 1 - \lambda_{ab}) \ge \delta, \qquad 0 < \delta < \tfrac{1}{2}. \tag{44}$$

Geometrically this removes two caps around $\mathbf{n}_a$ and $-\mathbf{n}_a$ on the sphere of possible orientations of $\mathbf{n}_b$.

Emergent holonomy.

The admissible region is therefore the sphere with two excluded caps. Topologically,

$$S^2 \setminus (2 \text{ caps}) \simeq S^1 \times (0,1), \qquad \pi_1 \cong \mathbb{Z}. \tag{45}$$

A loop winding around this cylinder cannot be continuously contracted without crossing an inadmissible region. In this sense the projection constraint induces a non-trivial holonomy in the space of admissible configurations.

Conceptual implication.

This simple construction illustrates a general principle: topological rigidity can emerge from admissibility constraints imposed by projection, rather than from a pre-existing ambient geometry. The role of geometry is therefore secondary, arising from the structure of admissible relational configurations.

The spectral and classificatory consequences of these admissibility constraints are developed in Chapter 6.

5.5 The Vacuum State as a Minimal Surface

In the absence of localized excitations, the projection fiber Π settles toward a configuration of minimal spectral tension: the smoothest admissible projected configuration compatible with global admissibility constraints. Geometrically, the vacuum corresponds to a minimal surface in a purely spectral sense (not a spatial embedding): curvature, torsion, and winding modes are simultaneously minimized within Π. In this state, the admissible projective flow encounters no topological obstruction and propagates uniformly across the fiber. Any deviation—localized curvature, torsional distortion, or non-trivial winding—manifests as the presence of effective fields or particles.

The usual notion of field quantization is replaced by the quantization of admissible topological modes within a finite-volume projection fiber. Only a discrete set of deviations from the minimal surface is compatible with stability and projectability constraints, leading naturally to a discrete spectrum of excitations.

Vacuum energy is neither divergent nor arbitrary. It is intrinsically bounded by the spectral cutoff imposed by the relational graph underlying Π. The finiteness of vacuum energy reflects the finiteness of admissible spectral deformations of the minimal configuration, rather than the summation of zero-point energies of independent field modes.

5.6 The Inaccessibility of Absolute Zero as a Projective Necessity

Within the present framework, the Third Law of Thermodynamics is reinterpreted as a structural requirement for temporal projectability. Temperature T is not introduced as a primitive measure of kinetic agitation, but as an effective indicator of the rate at which distinct projected descriptions remain differentiable under successive updates.

A strict absolute zero $(T \to 0)$ would correspond to a regime of projective stasis in which successive effective descriptions become indistinguishable. Formally, this would imply

$$U_{n+1} = U_n, \tag{46}$$

so that no non-trivial ordering of projected states could be defined. Since time is introduced operationally as the ordering of distinguishable effective states, such a regime would eliminate the very notion of temporal succession.

Absolute zero is therefore not merely energetically inaccessible, but structurally excluded: a perfectly stationary projection would terminate the temporal description itself. The persistence of time requires that projected states remain minimally non-degenerate under admissible updates.

In this context, the so-called zero-point fluctuations ($\frac{1}{2}\hbar\omega$) are reinterpreted as the minimal projective variance required to maintain the distinguishability of successive effective states. They represent the irreducible resolution limit of the projection process Π, marking the regime where descriptive continuity encounters the quantized structure of the underlying relational substrate [34].

5.7 Necessity of Non-Injective Projection for Emergent Gravity and Mass

This section addresses a structural necessity question distinct from sufficiency. We do not show that the χ-substrate implies gravity. That derivation is developed in Sections 6 and A.8. We show here that, under minimal operational assumptions matching what is empirically observed, a non-injective projection Π is required. A bounded relaxation flux is additionally required to select a unique infrared gravitational dynamics. The structural necessity argument is complemented by two technical appendices: the classification of admissible injective projections and their trivial holonomy (Appendix A.9), and the infrared derivation and selection mechanism (Appendix A.10).

Operational definitions and standing assumptions

We work with a microscopic configuration space $\mathcal{C}_\chi$ endowed with an admissibility rule F encoding the structural constraints of χ (Section 2). The space of effective observables is $\mathcal{M}$. A projection (coarse-graining) map $\Pi : \mathcal{C}_\chi \to \mathcal{M}$ defines the effective state $y = \Pi(x)$ for each substrate configuration $x \in \mathcal{C}_\chi$.

Emergence.

An effective quantity Q_{eff} is called *emergent* if and only if

$$Q_{\text{eff}} = Q(\Pi, F, x), \tag{47}$$

determined solely by the admissibility rule F and the projection Π. No independent free parameter carrying Q is introduced at the effective level.

Ontological uniqueness.

The complete substrate is $(\mathcal{C}_\chi, F)$. All effective structures, metric, connection, mass, and charge, are derived from the pair (F, Π) only. No fundamental geometric field is postulated independently of this pair.

Observed gravity.

We assume the effective description admits a Lorentzian geometric formulation satisfying:

(D1) Effective locality. Field equations are local on $\mathcal{M}$.
(D2) Covariance. Diffeomorphism invariance as an invariance of description, not of the substrate.
(D3) Infrared dominance of a low-derivative action.

(D4) Universal coupling to energy-momentum.

Hypothesis H (descriptive redundancy).

The effective geometry is a structure on $\mathcal{M}$, not on $\mathcal{C}_\chi$. Formally,

$$g_{\text{eff}} = g(y), \quad y = \Pi(x), \quad \text{not } g(x). \tag{48}$$

This expresses that gravity is attributed to the stability of the projection, not to an intrinsic geometry defined on the substrate itself.

Remark. Hypothesis H is not an additional postulate of Cosmochrony. It is the formal translation of the claim that the effective geometric description belongs to the level of observables $\mathcal{M}$. This is operationally required by the definition of emergent gravity. A geometry that depends on x rather than on $y = \Pi(x)$ would be a substrate-level geometric structure. Any attempt to define $g(x)$ therefore violates Hypothesis H by construction.

Lemma 1: Injectivity is incompatible with emergent gravity under H

Lemma 5.1 (Injectivity forbids universal geometric emergence under H)**.** *Assume ontological uniqueness and Hypothesis H, Eq. 48. If $\Pi : \mathcal{C}_\chi \to \mathcal{M}$ is injective, then there is no intrinsic mechanism producing simultaneously*

(i) a spatially variable inertial rigidity, and
(ii) a non-trivial effective curvature,

without adding an independent geometric postulate at the effective level.

Proof Suppose Π is injective. Then $\mathcal{M}$ is isomorphic to $\mathcal{C}_\chi$ via Π, and any geometric structure $g(y)$ on $\mathcal{M}$ is equivalent to a structure on $\mathcal{C}_\chi$. Under ontological uniqueness, such a structure must be derived from (F, Π) alone.

However, an injective Π generates no descriptive redundancy. There is no quotient, no non-trivial fiber $\Pi^{-1}(y)$, and no re-identification symmetry between distinct substrate configurations. Without descriptive redundancy, no gauge-like or diffeomorphism-like invariance can arise as an invariance of description. Such an invariance can only be postulated independently. Without an invariance of description, universal geometric emergence cannot arise as a constraint shared by all effective couplings. In particular, this is incompatible with the operational meaning of universal coupling in condition (D4).

Therefore, either g_{eff} and/or inertial parameters are introduced as independent postulates (violating ontological uniqueness and emergence), or no emergent gravity in the sense of Hypothesis H is obtained.

The core of the argument is not the arithmetic identity $|\Pi^{-1}(y)| = 1$. It is the structural implication:

injective $\Rightarrow$ no redundancy $\Rightarrow$ no re-identification symmetry $\Rightarrow$ no universal geometric emergence by constraint.

The vanishing of the multiplicative gradient $\partial \log |\Pi^{-1}(y)|$ is a corollary of this chain, not its premise. $\square$

Lemma 2: Non-injectivity is the general form of descriptive redundancy

Lemma 5.2 (Descriptive invariance implies an effective quotient)**.** *In any framework where gravity is an invariance of description, there exists an effectively non-injective morphism. Equivalently, there exists an equivalence relation $x \sim x'$ on $\mathcal{C}_\chi$ such that observables and geometry coincide for distinct micro-configurations.*

Structural exhaustion We consider representative classes of frameworks realizing gravitational invariance of description.

General relativity. Physical configurations are defined modulo $\mathrm{Diff}(\mathcal{M})$. The physical state space is a quotient by a continuous group action. A quotient is a non-injective projection, since multiple representatives correspond to one physical state.

Discretized approaches (spin networks, causal sets, CDT). Macroscopic geometry is an invariant of sets of micro-configurations and is therefore many-to-one by construction. The non-injectivity is present even when it is not explicitly labeled as a projection.

Stochastic approaches (stochastic mechanics, GRW-type). The measure or filtration identifies micro-histories that are indistinguishable at the level of observables. This defines an effective quotient structure, hence a non-injectivity.

In each case, non-injectivity is not imported from Cosmochrony. It is the generic mathematical form of descriptive redundancy compatible with universal coupling. Cosmochrony renders this structure explicit by identifying it with Π, rather than concealing it in a gauge orbit, a sum over histories, or a stochastic measure. □

Lemma 3: Saturation selects a unique infrared dynamics

Lemma 5.3 (Non-injectivity alone does not select a unique IR completion)**.** *Non-injectivity of Π is sufficient for geometrization and descriptive invariance, but insufficient to select a unique infrared gravitational dynamics. A finite-capacity constraint, the saturation bound $|\partial_t\chi| \leq c_\chi$ on the relaxation flux, acts as a selection principle. It picks a class of bounded effective actions whose infrared expansion reproduces the Einstein–Hilbert term and whose nonlinear regime is Born–Infeld-like. More precisely, the selection yields a unique dominant IR term plus a controlled expansion in powers of c_χ^{-2}, organized by a single capacity parameter.*

Selection argument Conditions (D1)–(D4) constrain the effective gravitational sector but do not uniquely determine it. They allow the Einstein–Hilbert action plus an unbounded set of higher-derivative corrections not fixed by symmetry and covariance alone.

The saturation bound $|\partial_t\chi| \leq c_\chi$ plays three distinct roles.

1. *Nonlinear completion.* The bound enforces bounded response, yielding a Born–Infeld-like structure for the full effective action (Section 2.3, Appendix A.12). This eliminates the freedom in ultraviolet completions that conditions (D1)–(D4) alone leave open.
2. *Exclusion of projective pathologies.* Unbounded admissible flux gradients correspond to configurations where the fiber $\Pi^{-1}(y)$ collapses, meaning the projection becomes locally injective and the geometric description breaks down. The bound excludes these configurations, ensuring the projective description remains globally admissible.

3. *Quantitative prediction.* The bound relates the effective gravitational coupling G to microscopic capacity parameters:

$$G = G(c_\chi, \hbar_\chi, \ldots), \tag{49}$$

turning the underdetermined effective family into an organized expansion with a single capacity scale. This yields a falsifiable prediction independent of the geometric effective description.

Without the saturation bound, no intrinsic principle selects among the family of actions compatible with (D1)–(D4). In particular, no principle fixes the coefficient of the Einstein–Hilbert term or the structure of higher-derivative corrections. □

Theorem: Necessity of non-injective projection and bounded admissible flux

Theorem 5.1 (Necessity of non-injective projection and bounded relaxation)**.** *Assume:*

(i) mass and gravity are emergent in the operational sense of Section 5.7,
(ii) ontological uniqueness, with no independent fundamental geometric fields,
(iii) observed gravity satisfying (D1)–(D4),
(iv) Hypothesis H, Eq. 48.

Then:

(a) Π *must be non-injective. Equivalently, there must exist an effective equivalence relation on* $\mathcal{C}_\chi$ *compatible with universal coupling.*
(b) The substrate admissible flux must be bounded, providing the selection principle required for a unique infrared dynamics with controlled higher-order corrections.

Proof Part (a). By Lemma 5.1, if Π were injective under Hypothesis H and ontological uniqueness, universal geometric emergence would require an independent geometric postulate at the effective level. This violates the operational notion of emergence. Lemma 5.2 further shows that any alternative framework implementing descriptive invariance introduces an effective quotient, hence a non-injectivity in disguise, regardless of terminology.

Part (b). By Lemma 5.3, non-injectivity alone does not select a unique infrared dynamics compatible with (D1)–(D4). The bounded admissible flux $|\partial_t \chi| \leq c_\chi$ supplies the selection mechanism by fixing the nonlinear completion and organizing higher-order corrections by a single capacity scale. □

Remaining technical steps toward a quantitative derivation

Two technical steps remain to convert Theorem 5.1 from a structural necessity result into a fully quantitative derivation.

Step A: Classification of admissible injective projections.

One must formalize the class of admissible injective maps $\Pi : \mathcal{C}_\chi \to \mathcal{M}$. One must then show that within this class, no re-identification symmetry can arise without adding extra structure. This program is carried out in Appendix A.9, where injective admissible projections are shown to have trivial holonomy. It is not a claim that curvature cannot be defined mathematically on $\mathcal{M}$. It is the claim that emergent gravity as descriptive invariance cannot arise from (F, Π) under injectivity and Hypothesis H.

Step B: Derivation of Einstein–Hilbert from the coherence functional. One must construct the coherence functional

$$S_{\mathrm{coh}}[g, \psi] = \int_{\mathcal{M}} \mathcal{L}_{\mathrm{defect}}(\Pi, F, y)\, \mathrm{d}^4 y, \tag{50}$$

defined as the integrated cost of projective defect, the mismatch between successive projected descriptions under the relaxation rule F. The infrared expansion and the selection mechanism induced by the saturation bound are developed in Appendix A.10.

One must then show that:

1. the infrared expansion of S_{coh} yields $\int \sqrt{-g}\, R$ as the dominant term compatible with (D1)–(D4),
2. the saturation bound enforces a Born–Infeld-like completion whose expansion begins with R,
3. the coefficient of R in the expansion determines G via Eq. 49, yielding a falsifiable quantitative prediction relating the effective coupling to c_χ and $\hbar_\chi$.

This would establish the Einstein–Hilbert term not as a postulate, but as the unique dominant term permitted by projective coherence, with Cosmochrony fixing the nonlinear completion and the coupling constant.

Non-injectivity also implies the possibility of composite winding classes and coherent collective phases, discussed below.

5.8 Collective Coherent Phases of the $U(1)$ Fiber

In the previous sections, electric charge and gauge structure were derived from the non-injective projection of the χ substrate onto an effective $U(1)$ fiber. We now consider the possibility that this fiber sector admits collective coherent phases.

The analysis below does not introduce new fundamental fields. It studies a specific collective regime of already established degrees of freedom. Superconductivity is interpreted as a macroscopic coherent phase of the $U(1)$ fiber, arising from the stabilization and phase locking of composite topological excitations.

Stability of $w = 2$ Composite Classes

Section 4 established that charged excitations correspond to topological configurations of the χ field projected onto the $U(1)$ fiber. A single excitation carries winding number $w = 1$ and effective charge e.

We now consider composite configurations of winding number $w = 2$. Such configurations are not trivial superpositions of two independent $w = 1$ modes. They define a distinct topological class characterized by a continuous fiber matching structure.

Let L_Π denote the effective projective ordering operator restricted to the fiber sector. We define the projective stability gap

$$\Delta_\Pi \equiv \lambda_{\min}\left(L_\Pi\big|_{w=2}\right), \tag{51}$$

as the lowest positive eigenvalue around the $w = 2$ configuration.

A finite Δ_Π implies that the composite class is metastable against separation into two $w = 1$ excitations or local phase defects. The existence of this gap is a structural consequence of bounded projective transport capacity established in Section 5.7.

Phase Coherence and the London Structure

If a macroscopic density of $w = 2$ composites occupies a common fiber phase sector, we may introduce the order parameter

$$\Psi = |\Psi|e^{i\theta}, \tag{52}$$

where $|\Psi|^2$ measures the density of coherent composites.

Gauge covariance fixes the unique admissible covariant gradient

$$D\theta = \nabla\theta - \frac{2e}{\hbar}A. \tag{53}$$

The leading contribution to the free energy must then take the form

$$F_{\text{grad}} = \frac{\hbar^2\rho_s}{2m^*}\left(\nabla\theta - \frac{2e}{\hbar}A\right)^2, \tag{54}$$

where ρ_s is the phase stiffness and m^* an effective inertia.

This term is not postulated as a symmetry-breaking potential. It represents the energetic cost of breaking global fiber coherence. The London equation and flux quantization follow directly from this structure.

Frustration-Induced Stabilization in Strongly Correlated Regimes

In conventional low-T_c materials, mediator-assisted interactions lower Δ_Π and stabilize the $w = 2$ class.

In strongly correlated systems, stabilization may arise instead from real-space frustration in admissibility constraints of the fiber. Let F denote a frustration density proxy, operationally linked to staggered magnetic correlations.

In this regime, isolated $w = 1$ excitations increase local mismatch, whereas a $w = 2$ composite redistributes phase incompatibility over an extended region. Pair formation thus minimizes a geometric cost functional rather than resulting from a specific momentum-space interaction kernel.

The amplitude scale for composite formation is set by the dominant frustration energy scale J,

$$\Delta_\Pi^{\text{ampl}} \sim J. \tag{55}$$

Phase Stiffness and BKT-Type Transition

Global superconductivity requires phase locking across the system. The relevant control parameter is the phase stiffness ρ_s, which measures the admissible projective capacity of the projected substrate.

In doped quasi-two-dimensional systems, the density of active composites scales with carrier concentration δ. To leading order,

$$\rho_s(0) \propto \delta J. \tag{56}$$

In strongly anisotropic layered systems, the transition temperature is governed by vortex unbinding and satisfies approximately

$$k_B T_c \simeq \frac{\pi}{2}\rho_s(T_c^-). \tag{57}$$

This naturally allows separation between amplitude formation ($T^* \sim \Delta_\Pi^{\mathrm{ampl}}/k_B$) and global phase coherence (T_c).

Superconductivity as Collective Saturation

The emergence of superconductivity can thus be interpreted as a collective saturation phase of the $U(1)$ fiber.

At the microscopic level, bounded projective transport constrains admissible field gradients. At the macroscopic level, coherent occupation of a single fiber phase sector minimizes projective ordering cost and expels magnetic flux from the bulk.

In this perspective, superconductivity is structurally analogous to other saturation phenomena derived in this work. Mass corresponds to isotropic saturation of local projective capacity. Schwinger pair production corresponds to saturation of directed admissible flux. Gravitational curvature corresponds to collective constraint on the effective projective ordering rate.

Superconductivity represents the coherent phase of the same bounded projective dynamics in the electromagnetic sector.

No modification of Standard Model interactions is required. The phenomenon emerges as a specific collective regime of the already established $U(1)$ fiber structure.

6 Spectral Admissibility and Spinorial Structure

6.1 Admissibility from Bounded Flux

The admissibility constraints of the χ substrate are governed by a fundamental bound on the local admissible flux rate. This bound expresses the finite transmissibility of the relational substrate and limits the rate at which local configurations may contribute to admissible projected descriptions.

Operationally, this constraint can be expressed as

$$|\partial_t \chi_v| \leq c_\chi, \tag{58}$$

where c_χ denotes the maximal admissible flux speed associated with the substrate.

This condition plays a role analogous to a causal bound, but it operates at the level of relational admissibility rather than geometric propagation. Configurations whose local admissible flux exceeds this bound cannot sustain a stable projected description.

When the dynamics is expressed in a spectral basis of the relational Laplacian L, each mode evolves independently at leading order. Let λ_ρ denote the Laplacian eigenvalue associated with a spectral sector labelled by ρ.

The bounded-flux condition then restricts the maximal amplitude that a spectral mode can reach without violating the local admissibility constraint. A detailed analysis shows that the maximal admissible amplitude of a mode associated with eigenvalue λ_ρ obeys

$$A_\rho^{\max} = \frac{c_\chi}{\sqrt{\lambda_\rho}}. \tag{59}$$

This relation defines a universal admissibility envelope for spectral modes. Modes associated with smaller eigenvalues can sustain larger amplitudes, whereas high-frequency modes are strongly suppressed by the bounded-flux constraint.

In this sense, the Laplacian spectrum acts as a structural filter on the space of admissible configurations. Only modes compatible with the admissibility envelope remain sustainably projectable.

The existence of this envelope introduces a hierarchy among spectral sectors. Even when several sectors are kinematically available, the bounded-flux constraint favours those associated with lower Laplacian eigenvalues.

The consequences of this spectral filtering become particularly transparent when explicit relational graphs are considered. The simplest analytically tractable example is provided by the Cayley graph of the quaternion group Q_8, which we analyse in the following section.

6.2 The Q_8 Prototype

To illustrate the admissibility mechanism explicitly, we consider the Cayley graph of the quaternion group Q_8. This example provides the simplest analytically tractable setting in which the bounded-flux constraint induces a non-trivial hierarchy among spectral sectors.

The quaternion group

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$$

is generated by the elements $\{\pm i, \pm j, \pm k\}$ with relations

$$i^2 = j^2 = k^2 = ijk = -1.$$

We consider the undirected Cayley graph constructed from the generating set

$$S = \{\pm i, \pm j, \pm k\}.$$

Each vertex corresponds to a group element and edges connect elements related by right multiplication by a generator in S.

The relational Laplacian of this graph admits a spectral decomposition indexed by the irreducible representations of Q_8. Besides four one-dimensional representations, Q_8 possesses a two-dimensional irreducible representation corresponding to its spinorial sector.

Let λ_ρ denote the Laplacian eigenvalue associated with the representation sector ρ. Applying the admissibility envelope derived in Section 6.1,

$$A_\rho^{\max} = \frac{c_\chi}{\sqrt{\lambda_\rho}},$$

one obtains the maximal amplitude that each sector can sustain without violating the bounded-flux constraint.

A direct computation of the Laplacian spectrum shows that the admissible amplitude of the two-dimensional sector exceeds that of the one-dimensional sectors by a factor

$$\sqrt{\frac{4}{3}}.$$

Thus, even though several representation sectors are kinematically present in the graph, the bounded-flux constraint dynamically favours the non-abelian sector.

In physical terms, the admissibility envelope acts as a spectral filter that suppresses sectors associated with larger Laplacian eigenvalues. The lowest non-trivial sector therefore becomes the dominant carrier of admissible fluctuations.

This result should not be interpreted as a special property of the quaternion group itself. Rather, Q_8 provides the minimal relational structure in which the mechanism can be computed exactly.

The strengthening of this hierarchy in larger binary groups, and in particular in the binary icosahedral group $2I$, is analysed in the following section.

6.3 From Q_8 to $2I$: Strengthening the Spinorial Hierarchy

The admissibility hierarchy observed in the quaternion example is not an isolated feature of the smallest non-abelian group. The same mechanism persists and becomes more pronounced in larger binary groups.

A natural next step is provided by the binary icosahedral group $2I$, which is the double cover of the rotational symmetry group of the icosahedron. This group plays a distinguished role among finite subgroups of $SU(2)$ and contains Q_8 as a subgroup.

As in the previous section, we consider the undirected Cayley graph associated with a symmetric generating set S. The relational Laplacian of this graph again decomposes into sectors labelled by the irreducible representations of $2I$.

The spectrum of $2I$ contains representation sectors of several dimensions. Applying the admissibility envelope derived in Section 6.1,

$$A_\rho^{\max} = \frac{c_\chi}{\sqrt{\lambda_\rho}},$$

one finds that the hierarchy between sectors becomes more pronounced than in the quaternion case.

In particular, the admissible amplitude associated with the lowest two-dimensional representation exceeds that of the lowest one-dimensional sector by a factor

$$\sqrt{\frac{14}{9}}.$$

This strengthening reflects the richer relational structure of the binary icosahedral group. While several representation sectors remain kinematically available, the bounded-flux constraint dynamically selects those with the smallest Laplacian eigenvalues.

The resulting hierarchy naturally organises the representation sectors into a tower of admissible modes whose amplitudes decrease with increasing spectral frequency.

From the viewpoint of relational dynamics, the binary icosahedral group thus provides a larger finite structure in which the admissibility mechanism remains fully operative.

Moreover, $2I$ is embedded in the continuous group $SU(2)$. This embedding suggests that the hierarchy observed in finite relational structures may persist along sequences of increasingly refined graphs approaching a continuous limit.

Before exploring this limit, however, it is useful to examine more closely the structural conditions that allow generators to participate in admissible configurations. These conditions can be expressed in terms of representation characters and lead to the notion of neutral generators introduced in the next section.

6.4 Neutral Generators and Character-Vanishing Conditions

The admissibility hierarchy observed in the previous sections suggests that not all generators contribute equally to dynamically sustainable configurations. A useful structural notion capturing this behaviour is that of *neutral generators*.

Let ρ be an irreducible representation of a finite group G, and let χ_ρ denote its character. A generator $s \in G$ will be called *neutral* in the sector ρ if

$$\chi_\rho(s) = 0. \tag{60}$$

This condition expresses a cancellation property at the level of the representation trace. Although the operator $\rho(s)$ itself need not vanish, the vanishing of its trace implies that its eigenvalues occur in balanced pairs whose sum is zero. Such elements therefore play a distinguished role in organising the geometry of admissible generating sets.

Character-vanishing conditions are not exceptional. In irreducible representations of dimension greater than one, many conjugacy classes naturally satisfy $\chi_\rho(s) = 0$. The condition therefore provides a representation-theoretic criterion for selecting generators that remain compatible with admissible spectral configurations.

In concrete realisations of binary groups inside $SU(2)$, neutral generators correspond to elements whose representation matrices are traceless. These elements act as half-turn rotations in the spinorial representation. However, the definition of neutrality does not depend on this specific realisation and can be formulated entirely in terms of representation characters.

Neutral generators impose strong constraints on the mutual relations between elements of the generating set. When several generators satisfy the character-vanishing condition in the same representation sector, their pairwise products determine a rigid pattern of character values.

As we show in the following section, these relations can be encoded in a Gram matrix whose entries are determined by representation characters. This observation provides a bridge between spectral admissibility and the geometric structure of generating sets.

6.5 Pairwise Character Geometry and Gram Rigidity

The character-vanishing condition introduced in the previous section imposes strong constraints on the mutual relations between generators. These constraints can be expressed in a compact geometric form.

Let $S = \{s_1, \ldots, s_k\}$ be a generating set of G and let ρ be an irreducible representation in which each generator of S satisfies the neutrality condition

$$\chi_\rho(s_i) = 0.$$

The relations between generators can then be encoded in a Gram matrix whose entries are determined by representation characters. For a pair of generators $s_i, s_j \in S$, we define

$$G_{ij} = -\frac{1}{2}\chi_\rho(s_i s_j). \tag{61}$$

This matrix captures the pairwise geometry of the generating set as seen through the representation ρ. Indeed, when the representation is realised in $SU(2)$, each neutral generator corresponds to a traceless element that can be written in the form

$$\rho(s_i) = \pm i\, \vec{u}_i \cdot \vec{\sigma},$$

where $\vec{\sigma}$ denotes the vector of Pauli matrices and $\vec{u}_i \in S^2$ is a unit vector.

In this representation one finds

$$\chi_\rho(s_i s_j) = -2\, \vec{u}_i \cdot \vec{u}_j,$$

so that the Gram matrix becomes

$$G_{ij} = \vec{u}_i \cdot \vec{u}_j.$$

The character relations therefore determine the inner products between the vectors $\vec{u}_i$ associated with the generators. In this sense the geometry of the generating set is fixed by purely representation-theoretic data.

This rigidity has important consequences. Once the characters of pairwise products are specified, the relative angles between generators cannot be continuously deformed without violating the neutrality conditions. The admissible configurations of generators therefore form discrete families rather than continuous manifolds.

The Gram formulation provides a convenient bridge between spectral admissibility and geometric constraints on generating sets. In the next section we exploit this structure by introducing the second-moment tensor associated with the vectors $\vec{u}_i$. This tensor will allow us to distinguish isotropic configurations from anisotropic ones and will lead to a structural exclusion of certain candidate groups.

6.6 Second-Moment Tensor, Isotropic Patterns, and Structural Exclusion

The Gram formulation developed in the previous section associates to each neutral generator s_i a unit vector $\vec{u}_i \in S^2$. The geometry of the generating set can therefore be analysed through the configuration of these vectors on the unit sphere.

A convenient global descriptor of this configuration is provided by the second-moment tensor

$$M = \sum_{i=1}^{k} \vec{u}_i \vec{u}_i^T. \tag{62}$$

This tensor measures the directional distribution of the generators. Two qualitatively different regimes can occur.

Anisotropic configurations.

If the tensor M possesses distinct eigenvalues, the set of generators is anisotropically distributed on the sphere. Such configurations typically arise when generators concentrate along preferred directions. In this regime the admissible structures are closely related to generalised dihedral families, which can be continuously deformed while preserving the neutrality condition.

Isotropic configurations.

A particularly important situation occurs when the second-moment tensor is proportional to the identity,

$$M \propto I. \tag{63}$$

In this case the generators are distributed isotropically on the sphere. Isotropic patterns are much more constrained, since the neutrality and Gram relations fix the relative angles between generators up to discrete choices.

Among isotropic configurations, a special role is played by orthogonal patterns in which the vectors $\vec{u}_i$ form mutually orthogonal directions. Such patterns arise naturally in the quaternion case discussed earlier and correspond to highly symmetric arrangements of generators.

The isotropy condition therefore acts as a strong structural filter. Many candidate groups fail to satisfy the combined constraints imposed by neutrality, Gram rigidity, and isotropy. In particular, certain finite subgroups that might appear plausible from a purely group-theoretic perspective cannot realise a compatible generator geometry.

A notable example is provided by the binary tetrahedral group $2T$. Although $2T$ is a finite subgroup of $SU(2)$, its character relations do not admit an isotropic configuration of neutral generators compatible with the Gram constraints. The corresponding second-moment tensor necessarily departs from isotropy, leading to an inconsistency with the admissibility conditions.

The second-moment tensor thus provides a simple and effective diagnostic tool. By distinguishing isotropic from anisotropic generator configurations, it allows one to eliminate certain candidate groups and prepares the classification result presented in the next section.

6.7 Classification of Admissible Neutral Finite Groups

The structural constraints developed in the previous sections impose strong restrictions on the class of finite groups that can realise admissible neutral generating sets.

Three ingredients play a central role in this analysis. First, generators must satisfy the character-vanishing condition

$$\chi_\rho(s) = 0$$

in a common irreducible representation sector. Second, the pairwise character relations impose a rigid Gram geometry for the generating set. Third, the configuration of generators must be compatible with the isotropy constraint on the second-moment tensor introduced in Section 6.6.

Taken together, these conditions drastically reduce the space of admissible structures. In particular, many candidate groups fail to realise a generator geometry compatible with both the Gram constraints and the isotropy requirement.

The resulting classification can be summarised as follows.

Theorem 6.1. *Let G be a finite group admitting a generating set S whose elements are neutral in a common irreducible representation sector ρ, and whose pairwise relations satisfy the Gram constraints described above.*

Then the admissible groups belong to one of the following families:

$$Q_8, \qquad \mathrm{Dic}_n \ (n \geq 3), \qquad 2O, \qquad 2I.$$

The family Dic_n ($n \geq 3$) corresponds to anisotropic configurations of the second-moment tensor, for which $M \not\propto I$. In these cases the generator axes exhibit a preferred direction reflecting the underlying dihedral symmetry.

The quaternion group Q_8 occupies a special position. Although it belongs to the binary dihedral series ($\mathrm{Dic}_2 \cong Q_8$), its generator configuration forms an orthogonal arrangement whose second-moment tensor satisfies $M \propto I$. This orthogonal isotropic geometry distinguishes Q_8 from the anisotropic Dic_n family.

In contrast, the last two groups arise from isotropic configurations in which the second-moment tensor is proportional to the identity but the Gram relations are not purely orthogonal. These highly symmetric arrangements correspond to the binary polyhedral groups associated with the octahedral and icosahedral symmetries.

An important consequence of this analysis is the exclusion of the binary tetrahedral group $2T$. Although $2T$ is a finite subgroup of $SU(2)$, its character relations do not allow a configuration of neutral generators compatible with both the Gram constraints and the isotropy condition. As a result, $2T$ cannot realise an admissible neutral generating set in the sense defined above.

The classification therefore identifies a restricted set of finite structures capable of supporting admissible spectral configurations.

The detailed verification of the Gram relations, second-moment tensors, and character tables for the relevant groups is provided in Appendix D.

6.8 Binary-Polyhedral Maximality and the $SU(2)$ Thread as Structural Attractor

The classification obtained in the previous section identifies a restricted set of finite groups capable of supporting admissible neutral generating sets. Among these structures, the binary polyhedral groups occupy a distinguished position.

While the quaternion and binary dihedral families arise from anisotropic configurations of the second-moment tensor, the binary polyhedral groups $2O$ and $2I$ correspond to isotropic configurations in which

$$M \propto I.$$

In such configurations the generators are distributed uniformly on the sphere and the admissibility constraints are satisfied in the most symmetric possible manner.

This enhanced symmetry has important consequences for the admissible spectral hierarchy. Isotropic configurations maximise the uniformity of the Gram relations and therefore minimise the structural tension between different representation sectors.

As a result, the binary polyhedral groups realise particularly stable arrangements of admissible generators. In comparison with groups whose generator configurations are anisotropic, their admissible spectral sectors remain more robust under variations of the generating set.

This property becomes even clearer when one considers global measures of admissibility such as the penalised spectral capacity C_α. When constructive alignment between sectors is penalised, binary polyhedral groups systematically dominate their non-binary counterparts obtained by quotienting the central $\mathbb{Z}_2$.

In this sense the double-cover structure characteristic of binary groups is not merely a representational convenience. Rather, it reflects a deeper compatibility between the admissibility constraints and the spinorial geometry of the underlying representations.

The binary polyhedral groups therefore form natural attractors within the space of finite relational structures compatible with admissible spectral dynamics.

Moreover, these groups are embedded in the continuous group $SU(2)$. This observation suggests that the structures identified above are not isolated algebraic curiosities but rather finite manifestations of a deeper spinorial framework.

The role of $SU(2)$ as a structural limit of these constructions will be examined in the following section, where sequences of increasingly refined relational graphs are considered.

6.9 Global Spectral Capacity and the Continuum Limit

The structural analysis developed in the previous sections identifies a restricted set of finite relational graphs capable of supporting admissible neutral generating sets. To compare such structures quantitatively, it is useful to introduce a global measure of admissibility.

Let L denote the relational Laplacian associated with the Cayley graph of a generating set S. If $\{\lambda_\rho\}$ denotes the set of Laplacian eigenvalues associated with irreducible sectors ρ, the admissibility envelope introduced in Section 6.5 implies that the maximal effective amplitude in sector ρ is bounded by

$$A_\rho^{\max} = \frac{c_\chi}{\sqrt{\lambda_\rho}}.$$

This relation naturally leads to a notion of global spectral capacity. A simple aggregate measure can be defined by summing the admissible amplitudes over all sectors,

$$C = \sum_\rho A_\rho^{\max}. \tag{64}$$

However, constructive alignment between sectors can artificially inflate this quantity. To account for this effect one may introduce a penalised capacity

$$C_\alpha,$$

in which correlated contributions are weighted by a parameter $\alpha \in [0, 1]$.

This framework allows different relational structures to be compared in terms of their ability to support admissible spectral modes. Binary polyhedral groups, in particular the icosahedral group $2I$, emerge as near-maximal configurations among finite groups with comparable valency.

A complementary diagnostic is provided by the admissibility ratio

$$R_p,$$

which measures the relative concentration of admissible spectral weight across the hierarchy of sectors. Along sequences of increasingly refined relational graphs, this ratio approaches a stable value characteristic of the underlying spectral geometry.

These considerations suggest that the structures identified in the preceding sections should not be viewed as isolated algebraic cases. Rather, they form part of a broader sequence of relational graphs whose spectral properties progressively approximate those of the continuous group $SU(2)$.

One may therefore consider chains of relational structures of the form

$$Q_8 \to 2I \to \mathrm{LPS} \to SU(2),$$

in which increasingly refined graphs approximate the spectral geometry of the continuous spin group.

In this limit the discrete spectrum becomes dense and the admissibility filter progressively weakens. The bounded-flux constraint still imposes a local restriction on the amplitude of modes, but the spectral hierarchy itself approaches a continuous distribution.

The discrete structures analysed in this chapter can thus be interpreted as finite approximations of a deeper spinorial framework. Their role is to reveal the structural mechanisms through which admissible spectral hierarchies emerge from bounded relational dynamics.

Part III
Gravitation and Cosmology

This part addresses gravitation and cosmology as collective, large-scale manifestations of constrained projective ordering rather than as fundamental interactions. Gravitation is shown to arise from the local inhibition of the admissible projective ordering rate induced by particle excitations, leading to an emergent curvature description consistent with general relativity in appropriate regimes.

Cosmological expansion is interpreted as a global consequence of the monotone accumulation of projected spectral structure, without invoking inflation, dark matter particles, or a fundamental dark energy component. Observable phenomena such as galactic rotation curves, gravitational lensing, cosmic voids, and the Hubble tension are reinterpreted as effective responses to structural saturation and projective compensation.

This part connects the effective projective structure of χ to astrophysical and cosmological observables, providing the bridge between foundational principles and large-scale data.

7 Gravity as a Collective Effect of Particle Excitations

7.1 Local Slowdown of the Projective Ordering Rate

Gravitation emerges from the collective influence of localized projected configurations on the admissible projective ordering rate. When particle-like configurations are present in significant number, their combined effect leads to a macroscopic reduction of the admissible ordering rate. In an effective spacetime parametrization, this may be expressed as

$$\mathcal{D}_{\text{eff}}\chi_{\text{eff}} \;\simeq\; c\,(1-\alpha\,\rho)\,, \tag{65}$$

where ρ denotes the effective density of localized projected configurations and α encodes their average contribution to projective ordering resistance. This first-order approximation is valid when localized constraints are dilute and weakly overlapping.

The coupling parameter α is emergent and scales as $\alpha \propto G/c^2$ in terms of effective inertial mass densities. The collective slowdown manifests as gravitational time dilation and curvature effects, without introducing any independent gravitational force or mediator.

7.2 Collective Gravitational Coupling and Operational Geometry

The collective reduction of the admissible projective ordering rate modulates how efficiently structural variations can be correlated between different effective locations. In regions where projected descriptions are nearly homogeneous, the effective coupling approaches a uniform value; localized configurations weaken it by introducing structural constraints.

From an operational perspective, this modulation can be interpreted as a *self-consistency overhead*: maintaining mutual compatibility of projected descriptions across separated regions requires an increasing consistency cost as local constraints accumulate. Inertial and gravitational responses arise precisely as manifestations of this overhead in collective updates of the effective state.

Spatial separation is defined operationally: two effective regions are close if structural variations can be efficiently correlated between them. In the continuum and weak-constraint regime, this operational notion admits a compact description in terms of an effective spatial metric summarizing the collective response to relative variations. Spacetime curvature emerges as a descriptive manifestation of how localized configurations modulate collective ordering, correlation efficiency, and the associated self-consistency overhead.

A more explicit relational construction of the coupling mechanism is presented in Appendix D.1.

7.3 Emergent Curvature

Spatial variations in the admissible projective ordering rate lead to non-uniform correlation patterns within projected descriptions. When a smooth geometric parametrization is applicable, these non-uniformities are summarized by gradients of an emergent metric structure, reproducing the phenomenology traditionally attributed to curved spacetime: gravitational time dilation, geodesic deviation, and lensing effects.

Einstein's Equations as a Structural Equilibrium Principle.

When projected configurations admit a smooth, locally injective geometric description, their collective structural constraints can be summarized by an effective metric $g_{\mu\nu}$. Einstein's equations emerge *necessarily* as the unique generally covariant, low-derivative consistency condition relating curvature to the effective distribution of projectively stable configurations in the macroscopic projected regime. The Einstein tensor encodes a geometric identity constraining admissible macroscopic descriptions; the stress–energy tensor summarizes how localized configurations resist the advancement of projective ordering. Einstein's equations therefore express a balance condition between geometry and physical structure.

A direct infrared derivation of the Einstein response from the spectral entropy functional of the relational operator is developed in the companion work [38]. The present section focuses on the effective phenomenological interpretation within the Cosmochrony framework.

This interpretation explains the extraordinary universality of general relativity: as long as a smooth spacetime description exists and projective ordering is monotonic, bounded, and weakly inhomogeneous, the same geometric relations must hold, independently of the microscopic nature of the underlying substrate. General relativity thus plays a role analogous to thermodynamics—exact within its domain, silent outside it.

When projection ceases to be locally injective, Einstein's equations do not break down; they simply no longer apply, because the concept of spacetime has not yet emerged.

7.4 Recovery of the Schwarzschild Metric

For a static, approximately spherically symmetric distribution of localized projected configurations, the collective reduction of admissible relaxation ordering admits a simple effective description.

Operational potential from projective ordering slowdown.

The dimensionless lapse-like factor is

$$N(r) \;\equiv\; \frac{D^{\chi}_{\rm loc}(r)}{D^{\chi}_{0}}, \qquad 0 < N(r) \le 1, \tag{66}$$

with the weak-field identification

$$\frac{D^{\chi}_{\rm loc}}{D^{\chi}_{0}} \;\simeq\; 1 + \frac{\Phi}{c^2}, \qquad \left|\frac{\Phi}{c^2}\right| \ll 1. \tag{67}$$

Poisson-like equation and exterior solution.
In the weak-structure regime:

$$\nabla^2 \Phi \;\simeq\; 4\pi G_{\text{eff}}\, \rho, \tag{68}$$

yielding, for an isolated spherical source of mass M:

$$\Phi(r) \;\simeq\; -\frac{G_{\text{eff}} M}{r}. \tag{69}$$

Metric components from Φ***.***
The static spherically symmetric ansatz reads

$$ds^2 = -N(r)^2 c^2 dt^2 + N(r)^{-2} dr^2 + r^2 d\Omega^2. \tag{70}$$

In the weak-field limit:

$$g_{tt} \simeq -\left(1 + 2\frac{\Phi}{c^2}\right), \tag{71}$$

$$g_{rr} \simeq \left(1 + 2\frac{\Phi}{c^2}\right)^{-1} \;\simeq\; 1 - 2\frac{\Phi}{c^2}, \tag{72}$$

coinciding with the standard Schwarzschild weak-field expansion:

$$g_{tt} \simeq -\left(1 - \frac{2G_{\text{eff}} M}{c^2 r}\right), \quad g_{rr} \simeq \left(1 - \frac{2G_{\text{eff}} M}{c^2 r}\right)^{-1}. \tag{73}$$

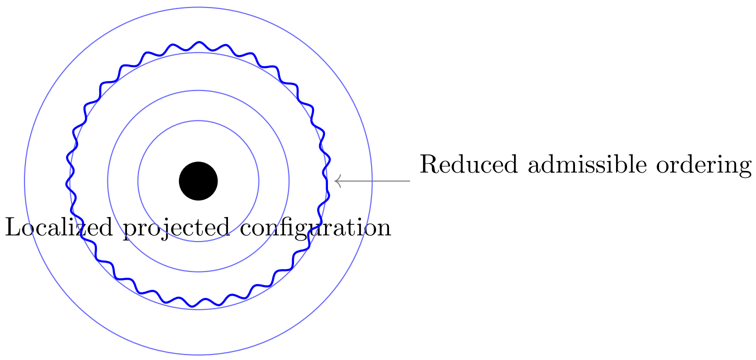

Fig. 5 Emergence of Schwarzschild-like behavior. A localized projected configuration induces a spatially varying constraint on the admissible projective ordering rate, manifesting as differential proper-time accumulation and emergent metric curvature.

Comparison to classic observational tests.
Gravitational redshift follows directly from g_{tt}:

$$\frac{\nu_{\rm obs}}{\nu_{\rm emit}} \simeq 1 + \frac{\Phi(r_{\rm emit}) - \Phi(r_{\rm obs})}{c^2}. \tag{74}$$

Light deflection yields the standard angle

$$\alpha \simeq \frac{4G_{\rm eff}M}{bc^2}, \tag{75}$$

with b the impact parameter.

7.5 Domain of validity of the effective geometric description

The recovery of the Schwarzschild exterior geometry establishes that, in the presence of a localized isotropic obstruction, bounded relational flux conservation selects a nonlinear stationary metric configuration. This metric provides an effective geometric description of the projected dynamics, from which standard gravitational phenomena, including perihelion precession, follow without additional assumptions.

It is therefore essential to characterize the domain of validity of this effective geometric description.

Injectivity as a condition of geometric observability

The emergence of an effective metric relies on the assumption that the projection $\Pi : \Omega \to O$ remains locally injective at the observational scale. In this regime, distinct underlying configurations correspond to distinguishable observables, and the relational dynamics admits a geometric description in terms of a smooth metric $g_{\mu\nu}$.

This condition can be expressed by comparing the projective resolution scale ℓ_χ to the local curvature scale of the effective geometry. In the infrared regime, the geometric description remains valid when curvature invariants are small in projective units. A covariant injectivity condition may therefore be written schematically as

$$\mathcal{I}(x)\, \ell_\chi^2 \ll 1, \tag{76}$$

where $\mathcal{I}(x)$ denotes a characteristic local curvature scale. This relation should be understood as an order-of-magnitude criterion, not as an exact spectral identity.

Application to the Schwarzschild exterior

For a static spherically symmetric exterior geometry, the Ricci scalar vanishes, and the relevant curvature scale is provided by the Kretschner invariant

$$\mathcal{K}(r) \equiv \sqrt{R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}} \sim \frac{r_s}{r^3}, \tag{77}$$

with $r_s = 2G_{\rm eff}M/c^2$.

The injectivity condition (76) therefore becomes

$$\mathcal{K}(r)\,\ell_\chi^2 \ll 1. \tag{78}$$

When this bound is satisfied, the projection remains locally injective and the effective metric description is valid. Conversely, when $\mathcal{K}(r)\,\ell_\chi^2 \sim 1$, distinct underlying configurations become observationally indistinguishable, and the geometric description breaks down.

Characteristic scale of injectivity breakdown

The condition $\mathcal{K}(r)\,\ell_\chi^2 \sim 1$ defines a characteristic radial scale

$$r_{\mathrm{NI}} \sim \left(r_s \ell_\chi^2\right)^{1/3}. \tag{79}$$

For $r \gg r_{\mathrm{NI}}$, the effective metric description is valid. As $r \to r_{\mathrm{NI}}$, the projective resolution limit is reached, and geometric observables cease to provide a complete description of the relational state.

Importantly, r_{NI} does not in general coincide with the Schwarzschild radius r_s. The breakdown of geometric observability is therefore controlled by projective resolution rather than by the existence of a causal horizon.

Hierarchy of regimes and observational consequences

The ratio between the injectivity breakdown scale and the Schwarzschild radius is parametrically

$$\frac{r_{\mathrm{NI}}}{r_s} \sim \left(\frac{\ell_\chi}{r_s}\right)^{2/3}. \tag{80}$$

For any microscopic value of ℓ_χ, one has $\ell_\chi \ll r_s$ for astrophysical objects, implying

$$r_{\mathrm{NI}} \ll r_s. \tag{81}$$

The breakdown of local injectivity therefore occurs deep inside the Schwarzschild radius, well beyond the domain accessible to external observers. All currently tested gravitational phenomena, including orbital dynamics, light deflection, and black hole imaging, lie in the regime

$$\mathcal{K}\,\ell_\chi^2 \ll 1, \tag{82}$$

where the effective metric description is exact.

This provides a structural explanation for the empirical success of general relativity in weak and moderately strong gravitational fields, without requiring fine tuning or additional corrections at observable scales.

Relation to higher-order dynamical corrections

The breakdown of local injectivity concerns the validity of the effective geometric description itself. This should be distinguished from the higher-order corrections to the dynamics of excitations discussed in Appendix B.13, which arise within the injective regime and do not modify the exterior metric.

Deviations from standard relativistic predictions are therefore expected only in regimes where the projection ceases to be locally injective or where strong-field saturation effects become relevant.

Physical interpretation of the non-injective regime

The breakdown of local injectivity signals a transition in the nature of observable quantities. In the injective regime, each observable configuration corresponds to a unique underlying relational state, and the dynamics admits a geometric description in terms of a smooth metric and well-defined trajectories.

When the condition $\mathcal{K}\,\ell_\chi^2 \ll 1$ is violated, this correspondence fails. Multiple distinct configurations of the relational substrate become indistinguishable under the projection Π, and the notion of a pointwise geometric state loses operational meaning.

In this regime, the effective metric description ceases to be complete. Geometric observables such as distances, trajectories, or local curvature no longer provide a faithful description of the underlying dynamics. Instead, observables correspond to equivalence classes of configurations, reflecting the non-injective nature of the projection.

This transition does not correspond to a breakdown of the underlying dynamics, but to a limitation of the geometric description. The relational substrate remains well-defined, but its evolution can no longer be captured by a local metric structure.

From this perspective, the emergence of geometry is intrinsically tied to the resolvability of relational configurations. Geometry is not a fundamental structure, but an effective description valid only within the injective regime.

The non-injective regime therefore represents a genuinely non-geometric phase of the theory, in which the appropriate description is expected to be spectral or relational rather than metric.

This interpretation provides a natural framework for understanding regimes where classical spacetime notions are expected to fail, such as the interior of compact objects or the early stages of cosmological evolution, without requiring singular behavior of the underlying dynamics.

7.6 Equivalence Principle

All admissible localized projected configurations constrain the admissible projective ordering rate in the same universal manner. Their internal composition plays no role in how they affect or respond to the admissible ordering environment. Inertial resistance and gravitational response are two effective manifestations of the same underlying constraint, and the equivalence principle arises as an emergent symmetry of admissible projected descriptions.

7.7 Gravitational Waves

Time-dependent variations in the distribution of localized projected configurations induce collective and transient modulations of the admissible projective ordering rate. These modulations propagate at the maximal admissible ordering speed c and are described, in an effective spacetime language, as gravitational waves. Unlike electromagnetic radiation, they represent collective variations of the admissible ordering and correlation structure itself.

The standard properties—propagation at speed c, transverse polarization, and energy transport—are recovered as effective features of collective projective ordering dynamics. Gravitational-wave descriptions remain valid only within regimes where projection onto an effective spacetime remains well defined; in strong-gravity environments approaching the deprojection threshold (Section 7.8), such modulations may become attenuated or lose a clear spacetime interpretation.

7.8 Strong Gravity and Black Holes

Projectable	$\rightarrow$	**Horizon**	$\rightarrow$	**Non-projectable**
Gravitational Waves		Boundary of Projection		Deprojected Regime
Effective Modulations		Π non-injective		No spacetime Representation

Fig. 6 Conceptual regimes of projection in Cosmochrony. Black holes mark the boundary beyond which spacetime representations cease to be injective.

In regions of sufficiently high density of localized projected configurations, admissible projective ordering becomes strongly constrained, defining an effective horizon. Black holes correspond to domains where physical processes become asymptotically inaccessible due to the loss of injectivity of spacetime projection. This naturally accounts for extreme time dilation without requiring divergent curvature invariants.

Because admissible ordering is bounded, configurations corresponding to infinite curvature or density cannot be physically realized. Apparent singularities signal the breakdown of spacetime representability, not genuine divergences of the underlying relational structure.

Black Holes, Deprojection, and Vacuum Reprojection

In strong-gravity regimes, the projection $\Pi : \mathcal{C}_{\mathrm{rel}} \longrightarrow \mathcal{M}$ loses injectivity: multiple inequivalent relational configurations correspond to the same effective spacetime event. This *deprojection* does not destroy information. Relational information ceases to be expressible in spatiotemporal form but remains encoded structurally and is in principle reprojectable once projectability is restored. Reprojection manifests as radiation-like excitations or particle–antiparticle pairs.

Black Hole Entropy as Projective Capacity Saturation

Black hole entropy measures the saturation of projective capacity at the horizon. Near the boundary where the effective temporal ordering ceases to be resolvable, a large multiplicity of distinct χ micro-configurations become indistinguishable at the level of the effective metric:

$$S_{\mathrm{BH}} \;\sim\; \log\left|\Pi^{-1}(g_H)\right|, \tag{83}$$

where g_H denotes the saturated horizon geometry. The area law follows from the fact that saturation occurs at the boundary between projectable and non-projectable regimes. Hawking thermality arises from the coarse-grained statistics of discrete reprojection events at the saturation interface.

The numerical factor $1/4$ in $S = A/4$ arises from the fourfold degeneracy in the stability spectrum of χ excitations, linked to the intrinsic 4π periodicity discussed in Section 4.2.

Saturation as a projectability boundary.

In Cosmochrony, strong-gravity regimes correspond to configurations in which the demanded update rate approaches the admissible projective throughput set by b_χ. An effective horizon can be interpreted as a boundary beyond which the projected description cannot remain smooth and approximately injective under admissible updates. This does not require a loss of substrate information, but reflects a loss of projectability in the effective state U.

7.9 Black Hole Evaporation and the Information Problem

Black hole evaporation is an effective process arising from the gradual restoration of projectability near the boundary separating projectable and non-projectable domains. The apparent information loss identified by Hawking [39] arises from applying a spacetime-based description beyond its domain of validity. In Cosmochrony, information is encoded in the global relational configuration independently of its spacetime projection.

Horizon Reprojection Equation

The energy flux emerging from the horizon is a sum over discrete reprojection events:

$$\Phi_\chi \equiv \frac{dE}{dt} = \sum_k \delta(t - t_k)\, \hbar_\chi\, \nu_k(L_{\mathrm{sol}}).$$

Here $\hbar_\chi$ is the fundamental quantum of reprojection, the $\nu_k(L_{\mathrm{sol}})$ are resonance frequencies of the stability operator at the horizon, and t_k label the instants at which local configurations reach the projection threshold.

Emergent Temperature and Projective Ordering Gradient

The apparent Hawking temperature is determined by the gradient of the effective projective ordering rate normal to the horizon:

$$k_B T_\chi = \frac{\hbar_\chi c}{2\pi}\, \left|\nabla_\perp \chi_{\mathrm{eff}}\right|_H.$$

For more massive black holes, the relaxation gradient is distributed over a larger area, reducing the frequency of reprojection events and explaining the inverse mass–temperature relation without invoking vacuum particle creation.

Information Conservation and Spectral Encoding

During deprojection, information is stored in the nonlinear degrees of freedom of χ within the projection fiber. During reprojection, each emitted quantum carries the spectral imprint of the eigenmodes governing the χ configuration, ensuring global information conservation.

Entropy as a Projection Saturation Limit

The Bekenstein–Hawking area law

$$S = \frac{A}{4} \tag{84}$$

is reinterpreted as the informational capacity of the projection map at the horizon boundary. Entropy quantifies hidden relational structure—the structural multiplicity of χ configurations no longer distinguishable at the metric level—rather than thermal ignorance.

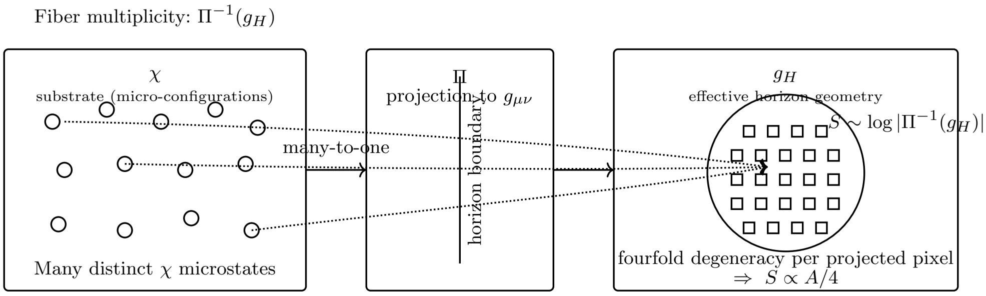

Fig. 7 Saturation of the projection map at the horizon. Multiple micro-configurations of χ collapse onto the same effective horizon geometry g_H. Entropy measures the structural multiplicity of the fiber $\Pi^{-1}(g_H)$.

Prediction: Spectral Granularity

If Hawking radiation is fundamentally discrete, it cannot be a perfect black-body continuum. The spectral line width of a reprojected quantum is

$$\Delta\nu_k \approx \frac{1}{\tau_\chi} \sqrt{\alpha}\, \ln(Q), \tag{85}$$

where τ_χ is the characteristic projective ordering timescale near the horizon $\alpha \simeq 3 \times 10^{-7}$ is the spectral packing fraction, and Q is the topological charge of the emitted configuration. The ratio between line width and line spacing is universally governed by α:

$$\frac{\Delta\nu}{\delta\nu} \propto \alpha. \tag{86}$$

Observational Prospects

In the late stages of evaporation, the spectral spacing may enter the sensitivity bands of next-generation interferometers (LISA, Einstein Telescope). The universal relation $\Delta\nu/\delta\nu \simeq \alpha$ should also apply to analogue gravity systems, where current experiments in Bose–Einstein condensates approach the required spectral resolution ($\delta f/f \sim 10^{-5}$), making near-term tests feasible.

7.10 Unified Origin of Gravitational and Electromagnetic Effects

Gravitational and electromagnetic phenomena arise as complementary effective manifestations of the same relational substrate. Gravitational effects correspond to sustained, quasi-static constraints on the admissible projective ordering rate induced by persistent localized configurations. Electromagnetic phenomena arise from time-dependent, phase-structured patterns within admissible projected descriptions, admitting an effective formulation in terms of propagating oscillatory modes. The familiar distinction between the two interactions emerges at the level of effective descriptions rather than from fundamentally separate fields.

An explicit derivation of Maxwell-like equations within the Cosmochrony framework is provided in Appendix A.15.

7.11 Effective Gravitational Lensing

In the weak-field and thin-lens regimes, the lens equation is

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \frac{D_{ls}}{D_s}\,\nabla_{\boldsymbol{\theta}}\psi(\boldsymbol{\theta}), \tag{87}$$

with the lensing potential

$$\psi(\boldsymbol{\theta}) = \frac{2D_{ls}}{c^2 D_l D_s}\int \Phi_{\text{eff}}(D_l\boldsymbol{\theta}, z)\; dz. \tag{88}$$

The effective potential Φ_{eff} is a geometric descriptor encoding collective constraints of the projected χ-substrate. All observable lensing quantities (convergence, shear, magnification, caustics) follow in the usual way. No additional dark matter component is required: lensing directly probes the emergent geometry.

7.12 Summary

Gravity emerges as a macroscopic consequence of localized projected configurations collectively constraining the admissible projective ordering rate. Classical gravitational phenomena—time dilation, curvature, gravitational waves, and black holes—are recovered as distinct descriptive regimes of this collective constraint. In extreme regimes, the spectral structure of horizon-associated phenomena encodes direct information about the projective structure and fiber topology of the underlying substrate.

8 Cosmological Implications

8.1 The Big Bang as a Maximal Constraint Regime of the χ Substrate

The Big Bang is interpreted as the boundary of applicability of effective spacetime descriptions. In this regime, the density of structural and topological constraints within χ exceeds the threshold required for stable geometric projection: effective notions of spatial distance, temporal duration, and causal ordering cease to be well-defined. The apparent singular behavior of standard cosmological models reflects the extrapolation of geometric descriptions beyond their domain of validity.

Cosmological evolution is therefore described as the progressive selection of admissible configurations through the effective reprojection sequence, starting from this maximal constraint regime. The Big Bang marks not the origin of spacetime, but the transition beyond which spacetime becomes an appropriate effective framework. The arrow of time arises from the irreversibility of the reprojection sequence under the admissibility constraints of χ (Section 2.6).

Structural symmetry breaking at the first transition.

The initial state U_0 occupies a unique position in the cascade: it has no predecessor U_{-1}. The admissible neighbourhood $\mathcal{A}(U_0)$ is therefore unconditioned by any prior projective history, making U_0 maximally symmetric — all successor directions are equivalently accessible. The first transition $U_0 \to U_1$ is, however, inevitably asymmetric: once a specific $U_1 \in \mathcal{A}(U_0)$ is reached, the next neighbourhood $\mathcal{A}(U_1)$ differs generically from $\mathcal{A}(U_0)$. This is not a phase transition driven by an external potential or a temperature parameter. It is a structural consequence of the evolution principle itself: because every transition shifts the projective position and thus the local admissible neighbourhood, perfect symmetry and non-trivial evolution are mutually exclusive. The apparent symmetry-breaking of the early universe is therefore not an event that requires explanation; it is the unavoidable first step of any evolution founded on a difference-based admissibility principle.

8.2 Cosmological Cycles of Constraint and Reprojection

The maximally constrained regime is not confined to the early universe. It may be locally reapproached whenever structural constraints saturate, most notably in black holes. Deprojection does not destroy information but renders it inaccessible to spacetime descriptions. Reprojection is the restoration of descriptive projectability once relational consistency conditions are satisfied.

Phenomena commonly associated with the quantum vacuum reflect the persistent presence of reprojectable relational structures within χ. The vacuum represents a regime of minimal yet non-vanishing projectability. Cosmological evolution involves a continuous interplay between global projective unfolding, local reconfinement, deprojection, and reprojection across scales.

8.3 Cosmic Expansion Without Inflation

In standard cosmology, an inflationary phase resolves the horizon problem [40, 41]. In Cosmochrony, large-scale homogeneity and isotropy reflect the global relational coherence of the χ substrate in the maximally constrained regime, rather than the outcome of a rapid expansion of spacetime. Prior to the emergence of a stable geometric projection, notions such as distance, light cones, and causal disconnection are undefined. The horizon problem therefore does not arise.

Cosmic Expansion as Progressive Projective Unfolding

Cosmic expansion reflects the progressive selection of admissible configurations through the effective reprojection sequence [42]. As the ordering parameter increases monotonically, projected descriptions admit an ever broader range of mutually distinguishable relational configurations. Expansion is not driven by an external energy component but is an intrinsic consequence of the admissibility ordering constrained by χ. Localized matter configurations act as persistent structural constraints, leading to spatially inhomogeneous unfolding that later manifests as large-scale structure.

8.4 Emergent Hubble Law

In homogeneous regimes, the projective ordering is uniform and admits the linear representation

$$\chi(t) = \chi_0 + c\,t. \tag{89}$$

Identifying effective spatial scales with accumulated relational differentiation yields a Hubble-like law [43, 44] with

$$H(t) \equiv \frac{1}{\chi}\frac{d\chi}{dt}, \tag{90}$$

where H_0 quantifies the current state of global projective unfolding.

Observational discriminant: Cosmochrony predicts a $\sim 5\%$ higher $H(z)$ at $z \sim 1$ compared to ΛCDM, testable with DESI/Euclid (Section 12.1).

Cosmic Acceleration Without Dark Energy.

The observed late-time acceleration does not require a cosmological constant. As structure formation proceeds, localized configurations increasingly constrain the local projective ordering rate, introducing growing spatial inhomogeneities. When interpreted within homogeneous and isotropic models, these inhomogeneities manifest as apparent acceleration. Cosmic acceleration is an emergent interpretative effect arising from progressively uneven projective ordering across cosmic scales.

8.5 Projective Cascade, Dynamical Λ_{eff}, and the Cosmological Origin of Fundamental Constants

The preceding sections establish $\chi(t)$ as the global effective projective field and $H(t) = \dot{\chi}/\chi$ as the emergent Hubble rate (Section 8.4, Eq. (90)). This section develops four structural consequences that follow directly from these definitions together with the constitutive relations Eqs. (275) and (277):

1. the effective cosmological constant Λ_{eff} is not a fundamental constant but a dynamical residual of ongoing projective cascade evolution, with equation of state $w \neq -1$;
2. the projective cascade proceeds as a discrete sequence of projectable configurations, providing a natural explanation for the 10^{16}–10^{19} hierarchy between the Planck energy and observed particle energies;
3. Planck's constant $\hbar$ is an epoch-dependent invariant of the current projective cascade epoch, fixed by K_0 and χ_c through Eq. (275);
4. the Big Bang corresponds precisely to the first globally projectable configuration of χ, whose existence is characterised by a finite saturation condition derived from the Born–Infeld threshold.

Each result is an internal consequence of the Cosmochrony framework; no new free parameter is introduced.

Dynamical Λ_{eff} from Residual Projective Capacity

Definition of residual projective capacity.

Energy is defined in this framework as the operational cost of modifying a projected configuration while preserving admissibility (F.7). At the global cosmological level, the outstanding projective capacity of χ at cosmic time t is the quantity that remains to be exhausted before the cascade reaches an asymptotically saturated state. We define it as a dimensionless fractional quantity $\Omega_\chi(t)$ already introduced in Eq. (**??**):

$$\Omega_\chi(t) \;\equiv\; \langle \beta^2 \rangle, \qquad \beta \;\equiv\; \frac{|\nabla\chi|}{c}, \tag{91}$$

which measures the fraction of projective capacity stored in active spatial gradients. The effective cosmological density associated with residual large-scale projective capacity (F.6) is then

$$\rho_\Lambda(t) \;\equiv\; \frac{3\,\Omega_\chi(t)\,H^2(t)}{8\pi G}, \tag{92}$$

which is dimensionally consistent with the critical density $\rho_c = 3H^2/(8\pi G)$ (natural units with $c = 1$ at the effective spacetime level; factors of c are restored in numerical estimates below). In the homogeneous background limit where Ω_χ is nearly constant, Eq. (92) reproduces the Friedmann closure condition up to an $O(\Omega_\chi)$ correction, consistently with Appendix C.3, Eq. (169).

Derivation of $\rho_\Lambda \propto H^2$.

From the linear background closure $\chi(t) = \chi_0 + ct$ (Eq. (66)) and the definition $H(t) = \dot{\chi}/\chi$ (Eq. (67)):

$$H(t) \;=\; \frac{c}{\chi_0 + ct} \;\xrightarrow{t \gg \chi_0/c}\; \frac{1}{t}, \tag{93}$$

so $H \propto t^{-1}$ holds in the late-time relaxation-dominated regime. Since $\rho_\Lambda \propto \Omega_\chi H^2$ and Ω_χ varies slowly (Appendix C.3), one obtains directly

$$\rho_\Lambda(t) \;\propto\; H^2(t) \;\propto\; \frac{1}{t^2}. \tag{94}$$

Numerically, with $H_0 \approx 2.2 \times 10^{-18}\ \mathrm{s}^{-1}$ and $G = 6.67 \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$,

$$\rho_\Lambda(t_0) \;\sim\; \Omega_\chi(t_0)\,\frac{3H_0^2}{8\pi G} \;\approx\; \Omega_\chi(t_0) \times 9 \times 10^{-27}\ \mathrm{kg\,m^{-3}}, \tag{95}$$

in agreement with the observed dark energy density for $\Omega_\chi(t_0) \sim 0.7$, without any fine-tuning.

Equation of state $w \neq -1$.

With Ω_χ treated as a slowly varying function of the scale factor a, differentiate $\rho_\Lambda = 3\Omega_\chi H^2/(8\pi G)$ along the background:

$$\frac{\dot\rho_\Lambda}{\rho_\Lambda} \;=\; \frac{\mathrm{d}\ln\Omega_\chi}{\mathrm{d}\ln a}\,H \;+\; 2\frac{\dot H}{H}. \tag{96}$$

Substituting into the continuity equation $\dot\rho_\Lambda + 3H(1+w_\chi)\rho_\Lambda = 0$ yields

$$w_\chi \;=\; -1 + \frac{2(1+q)}{3} \;-\; \frac{1}{3}\frac{\mathrm{d}\ln\Omega_\chi}{\mathrm{d}\ln a}, \tag{97}$$

where $q = -\ddot a/(aH^2)$ is the deceleration parameter of the *same* self-consistent background, and all three terms are dynamical outputs of the framework.

Two remarks are essential. First, the correction term $-\frac{1}{3}\,\mathrm{d}\ln\Omega_\chi/\mathrm{d}\ln a$ is not a free parameter: $\Omega_\chi(a) = \langle\beta^2\rangle$ (Eq. (91)) is determined by the relaxation dynamics. Its contribution can have either sign. If Ω_χ decreases as $\Omega_\chi \propto a^{-\delta}$ with $\delta > 0$, then $-\frac{1}{3}\,\mathrm{d}\ln\Omega_\chi/\mathrm{d}\ln a = +\delta/3$, which increases w_χ (i.e. makes it less negative). Conversely, any epoch where Ω_χ grows with a contributes negatively and can keep w_χ closer to -1 even when q departs from -1.

Second, using the ΛCDM-inferred value $q \approx -0.55$ directly in Eq. (97) would be circular: that value of q is reconstructed assuming a dark energy component with $w = -1$, whereas the present model changes the background equations. The proper observational confrontation is therefore not through a single effective w, but through the predicted $H(z)$ profile (Appendix C.3, Eq. (166)), which can be compared directly to BAO and supernova data without assuming a fixed equation of state. The framework's prediction of a mild $H(z)$ enhancement at $0.1 \lesssim z \lesssim 10$ relative to ΛCDM (Section 11.1) is the appropriate discriminating observable.

Resolution of the coincidence problem.

In the standard ΛCDM framework, the coincidence between ρ_Λ and the matter density ρ_m today is accidental. Here, both $\rho_\Lambda \propto H^2$ and $\rho_m \propto H^2$, so their ratio $\rho_\Lambda/\rho_m \sim \Omega_\chi$

remains of order unity throughout the cascade-dominated epoch. The coincidence is therefore a structural consequence of the single projective cascade driving both matter inertia and residual vacuum energy.

Discrete Projective Cascade and the Energy Hierarchy

Discreteness of projection.

The projectability threshold Θ_p (Appendix D.5) defines the minimal spectral gap above which a configuration of χ admits a stable geometric description. Each time the relaxation network crosses Θ_p from above, a new projectable regime opens: this is what we call a *relaxation step* $n \to n+1$. The minimum action exchanged per step is bounded below by the projection granularity

$$\Delta S_n \;\geq\; \hbar_\chi \;\equiv\; \frac{c^3}{K_0\,\chi_c}, \tag{98}$$

which is Eq. (275). Relaxation is therefore intrinsically discrete: the continuum effective description is an approximation valid between successive steps.

The residual relational gradient energy,

$$E_\chi[\varphi] \;\equiv\; \|\mathcal{L}\,\varphi\|^2 \;=\; \sum_{i,j} w_{ij}(\varphi_i - \varphi_j)^2, \tag{99}$$

measures the outstanding relaxation capacity at the microscopic level. Each relaxation step $n \to n+1$ corresponds to a monotone decrease of E_χ, with the macroscopic counterpart $\Omega_\chi = \langle \beta^2 \rangle$ tracking the same quantity at the cosmological scale (Eq. (91)).

Cascade law.

Let E_n denote the effective energy available to the projected description at step n. Since each step irreversibly increases the number of distinguishable projected configurations Ω_n, the energy per configuration decreases. Under the assumption that each step redistributes the remaining energy uniformly over the newly accessible configurations, and using the self-similar spectral structure established in Appendix D.2 (the ratio $G_{\text{eff}} \propto K_0/\chi_c^2$ is preserved under renormalisation), one obtains

$$E_n \;\sim\; \frac{E_P}{n}, \tag{100}$$

where $E_P = \sqrt{\hbar c^5/G} \approx 2 \times 10^9$ J is the Planck energy, corresponding to the saturated initial state (Section 8.1).

Total number of steps and the hierarchy.

The total number of Planck-time steps elapsed since the initial projection is

$$N \;\equiv\; \frac{t_U}{t_P} \;\approx\; \frac{4.3 \times 10^{17}\ \text{s}}{5.4 \times 10^{-44}\ \text{s}} \;\approx\; 10^{61}. \tag{101}$$

Applying Eq. (100) with $n = N$:

$$E_{\text{particle}} \sim \frac{E_P}{N} \sim \frac{2\times 10^9}{10^{61}}\ \text{J} \sim 10^{-52}\ \text{J} \sim 10^{-33}\ \text{eV}, \tag{102}$$

which corresponds to the energy scale associated with the cosmological constant (the dark energy quantum $\hbar H_0$), consistently with Eq. (95). For intermediate particle species stabilised at earlier steps $n \ll N$, Eq. (100) yields

$$\frac{E_P}{E_{\text{particle}}} \sim n, \tag{103}$$

so the observed hierarchy $E_P/m_e c^2 \approx 10^{22}$ requires that the electron stabilised at a relaxation step $n_e \sim 10^{22}$, far earlier than the present epoch. This provides a structural interpretation of the hierarchy problem: particle masses reflect the depth of their stabilisation step in the relaxation cascade, not an unexplained fine-tuning of a Higgs potential.

Primordial energy and cascade consistency.
Inverting Eq. (100) at step $n = 1$, the energy available at the first projectable configuration is $E_0 \sim E_P$, in agreement with the maximal-constraint interpretation of the Big Bang (Section 8.1). A crude upper bound on the total observable energy today follows from counting all Planck-volume cells in the observable Universe, $N_{\text{vol}} \sim (R_U/\ell_P)^3 \sim 10^{183}$, and assigning each cell the energy $E_P/N \sim 10^{-52}$ J of the current cascade step. This bound far exceeds the observed $E_{\text{today}} \sim 10^{69}$ J because it ignores the spectral distribution of stabilisation depths: most cells have relaxed to states with $n \gg N$ and contribute negligibly. A quantitative account requires the full spectral distribution of stabilisation depths across the relaxation network, which is left for future work. The cascade law (100) is therefore presented as a structural framework for the hierarchy, not as a closed energy budget.

Planck's Constant as a Cascade-Epoch Invariant

Starting point: Eq. (275).
The effective action quantum is defined structurally as

$$\hbar_\chi \equiv \frac{c^3}{K_0\, \chi_c}, \tag{104}$$

where K_0 is the maximal projective conductivity and χ_c is the critical relaxation threshold (Appendix F.1). The identification $\hbar_\chi \simeq \hbar$ is operationally enforced in projectable regimes (Appendix D.6, Eq. (253)).

Cosmological normalisation.

Appendix D.2 establishes that the dynamics are spectrally self-similar: G remains constant under renormalisation provided the ratio K_0/χ_c^2 is preserved. The cosmological-scale identification $\chi_c \sim c/H_0 = \chi(t_0)$ (Appendix C.4) then gives, using Eq. (277),

$$K_0 \;\sim\; \frac{c^4}{16\pi G\,\chi_c^2} \;\sim\; \frac{H_0^2}{16\pi G} \;\sim\; 10^{-52}\ \mathrm{m}^{-2}, \tag{105}$$

which is the cosmological-scale entry of Table 8. Substituting into Eq. (104):

$$\hbar_\chi \;\sim\; \frac{c^3}{K_0\,\chi_c} \;\sim\; \frac{c^3\,\cdot\,16\pi G}{H_0^2/H_0} \;\sim\; \frac{16\pi G c^3}{H_0} \;\approx\; 10^{70}\ \mathrm{J\,s}, \tag{106}$$

a cosmological action scale, not the microscopic $\hbar$. The microscopic Planck-scale identification $\chi_c \sim \ell_P$ instead yields $K_0 \sim 10^{93}\ \mathrm{m}^{-2}$ and recovers $\hbar_\chi \simeq \hbar = 1.05 \times 10^{-34}\ \mathrm{J\,s}$. Both branches are internally consistent, as stated in Appendix D.2.

Interpretation.

The two branches reflect the spectral self-similarity of the cascade: at each scale, the ratio $K_0\,\chi_c^2$ is fixed by G, while $\hbar_\chi$ tracks the granularity of that scale's projection. The universal microscopic $\hbar$ is therefore the action granularity of the *current particle-scale* projectable regime. It is not a fundamental constant of χ; it is the action quantum that emerges when the projection resolves configurations at the length scale ℓ_P, as encoded by the Planck-branch normalisation. The cosmological branch ($\hbar_\chi \sim 10^{70}\ \mathrm{J\,s}$) corresponds to the action scale of the relaxation field itself on Hubble scales and is not directly observable in particle dynamics; it characterises the global projective flux, not the granularity of quantum transitions in the effective projected description. This implies a further structural consequence: in primordial high-constraint regimes approaching the initial projection (Section 7.1), χ_c may differ from its present-epoch value, causing both $\hbar$ and the fine-structure constant α to scale accordingly (Eq. (9)), while leaving the product $K_0\,\chi_c^2 \propto G^{-1}$ invariant.

The Big Bang as the First Globally Projectable Configuration

Projectability condition.

A configuration of χ is projectable if and only if its local Born–Infeld saturation parameter (Appendix D.5, Eq. (247)) satisfies

$$S_i(\chi) \;\equiv\; \frac{1}{c^2}\sum_{j\in\mathcal{N}(i)} K_{ij}(\chi_i-\chi_j)^2 \;<\; 1 \quad \forall\, i. \tag{107}$$

When $S_i \to 1$ for all i simultaneously, the relaxation rate $R_i = c\sqrt{1-S_i} \to 0$: no ordered projected update is possible. The network is in the maximally constrained regime of Section 8.1.

First projectable state.

The Big Bang is defined in this framework as the first global configuration for which Eq. (107) is satisfied with a non-zero spectral gap:

$$\lambda_2(L_G(t_{\text{proj}})) \;>\; 0, \tag{108}$$

where λ_2 is the second eigenvalue of the weighted Laplacian L_G of the projective admissibility network (Appendix C.1). The projectability threshold Θ_p defined in Appendix D.5 is precisely the condition $\lambda_2 > \varepsilon_p$, so t_{proj} is the first instant at which Θ_p is crossed. Before t_{proj}, spacetime, distance, and causal ordering are undefined; the apparent singularity of standard cosmology reflects only the breakdown of the geometric projection, not a physical divergence of χ.

Energy at first projection.

At t_{proj}, the minimal action available per projectable configuration is $\hbar_\chi$, and the total number of simultaneously projectable modes is of order unity (by definition of the first crossing of Θ_p). The effective energy density associated with the first projection is therefore set by the Planck energy per Planck volume,

$$\rho_{\text{proj}} \;\sim\; \frac{E_P}{\ell_P^3} \;=\; \frac{c^5}{\hbar G^2} \;\approx\; 5\times 10^{96}\ \text{kg}\,\text{m}^{-3}, \tag{109}$$

consistently with the standard Planck density. The subsequent cascade described in Section 8.5 then accounts for the reduction from ρ_{proj} to the present-day densities.

Homogeneity as a structural consequence.

Prior to t_{proj}, all configurations of χ belong to the same fibre of the (not yet defined) projection: they are operationally indistinguishable. Homogeneity and isotropy of the early Universe are not initial conditions to be explained; they are structural consequences of the indistinguishability of pre-projection configurations. In the language of Appendix C.2, the horizon problem does not arise because causal disconnection is a notion defined only after t_{proj}; the pre-projection substrate is globally connected by construction.

Summary: four linked results.

Table 2 collects the four results of this section and their anchoring equations.

Table 2 Summary of the four structural results derived in Section 8.5. Each result is a consequence of existing Cosmochrony equations; no new free parameter is introduced.

Result	Key relation	Anchoring equation
Λ_{eff} dynamical, $w \neq -1$	$\rho_\Lambda \propto H^2$	Eqs. (67), (**??**), (92)
Energy hierarchy $E_P/E_n \sim n$	$E_n \sim E_P/n$	Eqs. (275), (100)
$\hbar$ as epoch-dependent invariant	$\hbar_\chi = c^3/(K_0\chi_c)$	Eqs. (275), (277), (106)
Big Bang = first projectable state	$\lambda_2(L_G(t_{proj})) > 0$	Eqs. (179), (108)

8.6 The Self-Referential Origin of Temporal Succession

The framework introduced in the preceding sections places the substrate field χ at a pre-geometric level: no spacetime manifold, no metric, and no notion of physical time is presupposed. Observable physics—including time itself—emerges through the admissibility constraints and the projection Π. This raises a structural question that must be addressed explicitly: if χ is timeless, in what sense does the universe "evolve," and what determines the succession from one projected state to the next? The present section provides a logically closed account of temporal emergence within the framework and clarifies how the spatial and temporal sectors, despite having distinct logical origins, remain coupled in the manner required by special relativity.

Two levels: structural rules and their execution.

The relational rules encoded by χ—the Laplacian weights w_{ij}, the saturation bound c_χ, the admissibility envelopes, the fiber topology of Π—constitute a fixed structural layer. They are written once and do not themselves change. This layer is analogous to the source code of a deterministic program: all rules are simultaneously present, yet the rules alone do not constitute a running computation. Physical time belongs to the execution layer, not to the code layer. The apparent paradox of a "timeless" substrate generating temporal order dissolves once these two levels are kept distinct.

The ordering parameter τ and the relational energy.

The relaxation cascade introduced in Section 8.5 is governed by the flow equation

$$\frac{d\varphi_i}{d\tau} = -(\mathcal{L}\,\varphi)_i, \tag{110}$$

where τ is a monotonically increasing parameter indexing the sequence of configurations. No physical interpretation of τ as a fundamental time is imposed at this level. Instead, τ labels a strict ordering on the space of relational configurations defined by the monotone decrease of the relational energy

$$E_\chi[\varphi] \;=\; \|\mathcal{L}\,\varphi\|^2 \;=\; \sum_{i,j} w_{ij}\,(\varphi_i - \varphi_j)^2, \tag{111}$$

already defined in Eq. (99). This ordering is intrinsic: it follows from the structure of $\mathcal{L}$ alone, without reference to any background clock. Physical time is the coarse-grained, projected image of this ordering, once the hyperbolic signature of the effective operator is established ([45, Sec. 6]).

The universe state U as its own cursor.

The central self-referential feature of the framework is the following. The current macroscopic state of the universe U_n—understood as the totality of projected observable data at a given level of the cascade—does not merely record the outcome of the preceding transition. It determines which successor states U_{n+1} are admissible.

More precisely:

- U_n encodes the current spectral structure of the effective Laplacian, including the local connectivity weights, the saturation regime, and the descriptive resolution scale $\ell(U_n)$ (cf. [46]).
- These data define the admissibility constraints on the next projection: which configurations of χ project injectively onto observable fields, and which local transitions remain within the bounded-flux regime.
- Among these constraints, the bounded-flux condition c_χ imposes that the local relational flux must remain at saturation throughout an admissible transition: any descent along E_χ that is shallower than the steepest admissible gradient would leave residual flux below the saturation bound, meaning that the projection retains further resolvable structure and the transition is not yet complete. Only the steepest admissible descent exhausts the available flux budget and closes the transition. No external selection principle is involved; the saturation condition itself determines the unique realised step.

There is therefore no external cursor advancing the universe from U_n to U_{n+1}. The state U_n is the cursor. It carries within itself the conditions that make the next step both determinate and admissible.

The cursor acts on fibre classes, not on χ directly.

Because the projection $\Pi : \Omega \to \mathcal{O}$ is non-injective, a single observable state U_n corresponds to an entire fibre of substrate configurations,

$$\Pi^{-1}(U_n) \subset \Omega. \tag{112}$$

The cursor does not select a unique microscopic configuration of χ; it selects an admissible fibre class. The residual multiplicity within $\Pi^{-1}(U_n)$ is precisely the source of projective fluctuations: the effective observables at level U_n cannot distinguish between the microscopic configurations in the fibre, and this irreducible indeterminacy propagates into the next transition. Quantum fluctuations, in this picture, are not imposed as an additional postulate: they reflect the bounded resolvability of the projection at each step of the cascade.

Irreversibility and the arrow of time.

The succession $U_n \to U_{n+1}$ is irreversible by construction: E_χ is non-increasing along admissible transitions, with a strict decrease whenever U_n is not already a stationary configuration of $\mathcal{L}$. This monotone decrease constitutes the arrow of time. It does not require a separate thermodynamic argument or a special initial condition external to the framework; it is built into the admissibility structure of the code layer. The question "why does time have a direction?" thus reduces to "why is the initial state U_0 not already a global minimum of E_χ?"—a question about the boundary conditions of the cascade, not about the nature of time itself.

The present as a dynamical frontier.

Within the cascade, the universe state U_n partitions the space of relational configurations into two complementary sectors:

- configurations already stabilised under Π, whose projected images are fixed and no longer carry residual relational gradient;
- configurations that are not yet admissible, in the sense that the current spectral structure of $\mathcal{L}$ does not yet support a locally injective projection onto them.

The present corresponds to the dynamical frontier between these two sectors: the region where the projection is becoming stable but where the cascade has not yet terminated. Because the non-injective fibre $\Pi^{-1}(U_n)$ retains finite multiplicity at this frontier, the present is not an infinitesimal instant but a zone of finite relational extent, consistent with the finite propagation speeds imposed by the bounded-flux constraint. This provides a structural account of the physical thickness of the present without invoking additional postulates.

Co-emergence of space and time from a single operator.

The preceding paragraphs have described space and time as having logically distinct origins within the framework: spatial structure follows from the admissibility envelopes of χ and the fibre topology of Π (see [18] and Section 6), while temporal succession follows from the ordered execution of the cascade. One might therefore ask why space and time remain as tightly coupled as special relativity requires.

The answer lies in the structure of the effective operator $\mathcal{L}$. As established in [45, Sec. 6], the bounded-flux condition c_χ constrains the admissible forms of the effective Laplacian $\mathcal{L}$ simultaneously in its spatial and its cascade (temporal) sectors. Specifically:

- A purely elliptic (spatial) operator is excluded because it would admit instantaneous propagation, violating c_χ.
- An ultra-hyperbolic operator (more than one time-like direction) is excluded because it leads to an ill-posed Cauchy problem, destabilising the cascade.
- The unique compatible class is a hyperbolic operator with exactly one time-like direction and three space-like directions, whose principal symbol defines the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$.

The Lorentz group therefore emerges not from a separate postulate but as the symmetry group of the unique operator class that is simultaneously compatible with three-dimensional spatial admissibility and with the irreversible, bounded-flux cascade. Space and time are logically separable in their origins—admissibility selects the former, cascade ordering generates the latter—but they are dynamically inseparable in the effective description, because only one operator class satisfies both constraints at once. This is why the universal ratio of spatial to temporal extents is fixed at c, and why no admissible state of the cascade can disentangle space from time at the macroscopic level.

Metric rigidity and the general-relativistic space–time budget.

The signature argument above operates at the level of special relativity: it establishes the existence of a pseudo-Riemannian metric $g_{\mu\nu}$ with signature $(-+++)$ and guarantees Lorentz invariance of the flat effective geometry. It does not, however, explain why the space–time budget remains exact and self-consistent in the presence of matter and curvature, i.e. why dilating the temporal metric coefficient at one point is compensated by a corresponding contraction of the spatial coefficients so that $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$ remains a well-defined invariant throughout a curved spacetime.

This stronger rigidity is enforced at a second level, through the spectral entropy functional

$$S_\Pi[g] \;=\; \tfrac{1}{2}\log \det{}' A_g, \tag{113}$$

where $A_g = -\nabla_g^2$ is the Laplace-type operator on the effective geometry ([38, Sec. 2, Sec. 4]). The renormalised metric variation of $S_\Pi[g]$ generates, as its dominant infrared term, the Einstein tensor:

$$\frac{\delta S_\Pi^{\text{ren}}}{\delta g^{\mu\nu}} \;=\; c_{\text{EH}}\, G_{\mu\nu} + c_\Lambda\, g_{\mu\nu} + (\text{higher-derivative and non-local terms}). \tag{114}$$

The stationarity condition $\delta S_\Pi[g]/\delta g^{\mu\nu} = 0$, imposed as the equilibrium requirement of the projection, therefore yields the Einstein field equations as the constraint that the effective metric must satisfy at each step of the cascade [38, 47]. It is precisely these equations that lock the relationship between the temporal and spatial components of $g_{\mu\nu}$ at every spacetime point: they prevent the metric from being deformed in a way that would violate the invariance of ds^2. The $A(r) = B(r)^{-1}$ identity of the Schwarzschild solution, which is the most explicit expression of this space–time budget compensation, is a direct consequence of flux conservation under the constrained effective operator (Section [45, Sec. 7] and [38, Sec. 5.3, 6]).

The logical chain is therefore the following. The bounded-flux condition c_χ selects the $(-+++)$ signature, guaranteeing the existence of a pseudo-Riemannian metric and the universality of c (special relativity level). The spectral entropy functional $S_\Pi[g]$, whose extremisation encodes the equilibrium of the cascade projection, then dynamically locks the metric coefficients through the Einstein equations, enforcing the exact compensation of the space–time budget under arbitrary matter distributions and curvature (general relativity level). The two levels are not independent: both follow from the same bounded-projection structure of χ, operating at different orders in the effective gradient expansion.

Relation to the block-universe picture.

The present account differs from a block-universe interpretation. A block universe asserts that all states U_n co-exist in a four-dimensional manifold, and that temporal flow is an illusion of perspective. In the present framework, the structural rules co-exist (the code is fixed), but the realisation of U_n is sequential and self-generated. No co-existence of all states is required or implied. Each state engenders its successor through its own admissibility structure. The appropriate analogy is a causal automaton rather than a static manifold: the program runs, step by step, driven by its own output.

Summary.
The five elements that together constitute a logically closed account of temporal emergence and space–time coupling are:

1. χ *as static code.* The admissibility rules, the Laplacian structure, and the saturation bound c_χ are fixed and atemporal; they form the structural layer.
2. U_n *as cursor on fibre classes.* The current universe state is self-referential: it determines, via its own spectral structure, which fibre classes are admissible successors. No external parameter advances the system.
3. *The bounded-flux constraint as the unique selector.* Among candidate transitions, only the steepest admissible descent along E_χ satisfies c_χ; the constraint itself acts as the selection mechanism, with no additional rule required.
4. *Co-emergence of space and time (special relativity level).* The spatial and temporal sectors are logically distinct in origin but dynamically inseparable: the bounded-flux operator uniquely selects the $(-+++)$ signature, fixing the universal ratio c and coupling the two sectors into Lorentz-invariant structure.
5. *Metric rigidity (general relativity level).* The spectral entropy functional $S_\Pi[g]$, whose extremisation encodes the equilibrium of the cascade projection, generates the Einstein equations as its dominant infrared condition. These equations dynamically lock the relationship between spatial and temporal metric coefficients at every point, enforcing the exact compensation of the space–time budget—the invariance of ds^2—under arbitrary matter distributions and curvature.

Items 4 and 5 answer the question of space–time coupling at two distinct levels: item 4 explains why there is a single universal speed c and Lorentz invariance; item 5 explains why the metric compensation remains exact and position-dependent in the presence of gravity. Both levels follow from the same bounded-projection structure of χ, operating at different orders in the effective gradient expansion. Together, all five elements provide a logically closed account of temporal emergence that is consistent with the spatial structure established in the spectral admissibility programme [18], with the Lorentz and Einstein structures derived in [38, 45], and with the relational fluctuation structure that underlies quantum indeterminacy in the framework.

8.7 Cosmic Microwave Background

The CMB reflects the imprint of early projective cascade and reprojection processes as the substrate transitioned toward a regime admitting stable geometric descriptions. Large-scale correlations arise from the global relational coherence of χ prior to geometric differentiation, rather than from superluminal expansion. Acoustic features are interpreted as resonance patterns arising once the standard photon–baryon plasma description becomes applicable [48–50].

Effective closure: primordial spectrum.

The standard Boltzmann transfer functions remain unchanged. Cosmochrony modifies the CMB spectra only through the primordial curvature spectrum:

$$C_\ell^{XY} = 4\pi \int_0^\infty \frac{dk}{k} P_\zeta(k)\, \Delta_\ell^X(k)\, \Delta_\ell^Y(k), \tag{115}$$

with a projectability-induced infrared filter and emergent tilt:

$$P_\zeta(k) = A_s \left(\frac{k}{k_*}\right)^{n_\chi - 1} C^2\left(\frac{k}{k_p}\right), \qquad C(x) \xrightarrow{x \ll 1} 0, \quad C(x) \xrightarrow{x \gg 1} 1. \tag{116}$$

Relational spectral scales and angular multipoles.

The admissibility cutoff can be expressed in terms of the low-lying spectrum of the relational operator L_χ (Section 2.4):

$$k_n \;\propto\; \sqrt{\lambda_n(\lambda_*)}, \qquad \ell_n \;\approx\; k_n\, D_A(z_*), \tag{117}$$

yielding the robust ratio constraint

$$\frac{\ell_n}{\ell_1} \;\approx\; \sqrt{\frac{\lambda_n}{\lambda_1}}. \tag{118}$$

Numerical relaxation experiments (Appendix D.7) indicate $\lambda_2/\lambda_1 \simeq 8/3$, implying $\ell_2/\ell_1 \approx 1.63$, searchable as a scale-independent modulation signature in the low-ℓ sector. These ℓ_n characterize relational angular modulations associated with admissibility, not the acoustic peak indices of the photon–baryon plasma.

8.8 Dark Matter as Residual Projective Effects

Dark matter phenomena correspond to configurations of χ that resist projective ordering advancement while failing the projectability conditions required for Standard Model interactions. These **non-projected spectral modes** possess inertial mass and contribute to gravitational curvature while remaining invisible to electromagnetic or electroweak probes.

Galactic Rotation and Effective Spectral Stiffness.

The flattening of rotation curves is interpreted as a spatial variation of G_{eff} induced by the local projective ordering state. At large radii, a transition in the effective spectral stiffness leads to a logarithmic gravitational potential, reproducing MOND-like behavior without invoking a modification of gravity or a universal acceleration scale. Unlike the universal constant a_0 in MOND, the transition threshold $\mathcal{K}_c$ is local and environment-dependent, naturally explaining the observed variation of the apparent dark matter fraction among galaxies.

Gravitational Lensing and Substrate Memory.

Lensing phenomena (e.g. the Bullet Cluster) are manifestations of **projective lag**: the projective geometry associated with mass-solitons persists after baryonic gas has lost coherence. Light deflection is treated as effective refraction within the spectral gradient of the projected χ geometry.

Predictive Distinction from Particulate Dark Matter.

Cosmochrony predicts non-local correlations between gravitational mass discrepancies and the global spectral age of a system, the absence of sharp central cusps, a minimum smoothing scale imposed by the spectral response, and **spectral echoes**—faint gravitational signatures in regions where matter was previously present.

8.9 Entropy and the Arrow of Time

The arrow of time is a fundamental structural feature arising from the irreversibility of the reprojection sequence $\chi_{\text{eff}}(t) \to \chi_{\text{eff}}(t+1)$, not a derived statistical phenomenon. Each reprojection step loses information under the non-injective projection Π, making exact reversal structurally inadmissible: no admissible successor state can restore the full pre-image of its predecessor. χ does not itself relax; it supplies the admissibility constraints that orient the sequence of constrained reconstructions. Irreversibility arises from information loss under Π; directionality arises from the structural constraints of χ. This is consistent with the monotone decrease of the relational energy E_χ along admissible transitions established in Section 8.6: that monotonicity is the projected image of the structural ordering imposed by χ, not a property of χ itself evolving in time.

Entropy increase emerges only at the level of effective spacetime descriptions, providing a statistical summary of how macroscopic degrees of freedom evolve under successive non-injective reprojections. Entropy growth does not explain the arrow of time; it reflects the cumulative information loss accumulated under the reprojection sequence. The temporal asymmetry is not present in χ itself—which is atemporally fixed—but in the structure of admissible reprojections that χ constrains.

This reverses the standard explanatory hierarchy: time asymmetry is imposed structurally by the admissibility constraints on the reprojection sequence, not attributed to special initial conditions or to a substrate that itself evolves. Processes involving deprojection do not correspond to entropy decrease or temporal reversal but represent a transition to a level of description where thermodynamic notions no longer apply.

8.10 Cosmic Voids as Maximal Projective Capacity Probes

Cosmic voids are regions where the projective ordering of χ is least frustrated by localized excitations. Within the effective description (Section 3.4), near-maximal admissible projective flux produces enhanced geodesic defocusing, leading to a negative gravitational lensing signal and non-linear peculiar velocity outflows at void boundaries—effects absent or suppressed in ΛCDM.

Connection to local H_0 determinations.

Enhanced outward peculiar velocities at void boundaries can bias low-redshift distance–redshift inferences toward higher locally inferred expansion rates. A decisive test is the cross-correlation between void lensing profiles and locally inferred H_0 maps.

Phenomenological void parametrization.

Observable void signals are modeled as a ΛCDM baseline plus a saturating correction controlled by a single dimensionless amplitude β_{void}:

$$\kappa_{\text{obs}}(R) = \kappa_{\Lambda\text{CDM}}(R)\left[1 + \beta_{\text{void}}\,\mathcal{S}\big(\mathcal{A}(R)\big)\right], \tag{119}$$

$$v_{\text{obs}}(r) = v_{\Lambda\text{CDM}}(r)\left[1 + \beta_{\text{void}}\,\mathcal{S}\big(\mathcal{B}(r)\big)\right], \tag{120}$$

with $\mathcal{S}(x) = x/\sqrt{1+x^2}$ interpolating between linear and saturated regimes.

8.11 The Hubble Tension

The discrepancy between early- and late-universe determinations of H_0 is well established [50–53]. In Cosmochrony, different observational probes access different regimes of effective projectability. Early-universe measurements probe a regime close to the transition from maximal constraint to geometric projectability; late-time measurements probe a more weakly constrained regime where effective spacetime descriptions are more fully developed. The tension arises from using a single spacetime-based parametrization to describe observations sampling distinct stages of the projective cascade.

The Hubble Tension as a Diagnostic of Topological Decoherence

The effective Hubble parameter may be expressed as

$$H_{\text{eff}}(\mathbf{x}, t) \;\sim\; \frac{1}{\tau_\chi(\mathbf{x}, t)}, \tag{121}$$

where τ_χ is an effective projective ordering timescale. Regions of high structural complexity (clusters, filaments) in which the projective ordering timescale acquires a spatial dependence:

$$\tau_\chi(\mathbf{x}) = \tau_\chi^{(0)}\left[1 + \epsilon\,\mathcal{T}(\mathbf{x})\right], \tag{122}$$

where $\mathcal{T}(\mathbf{x})$ encodes local topological density. The tension reflects a non-commutativity between cosmological averaging and local projection.

Cosmochrony predicts that locally inferred values of H_0 should exhibit weak but systematic correlations with the surrounding topological environment. Quantitative estimates are provided in Section 12.1 and Appendix C.3.

8.12 Large-Angle Temperature Anomalies

Large-angle CMB anomalies—low-multipole power suppression and unexpected large-scale alignments—remain only partially explained within ΛCDM [54]. In Cosmochrony,

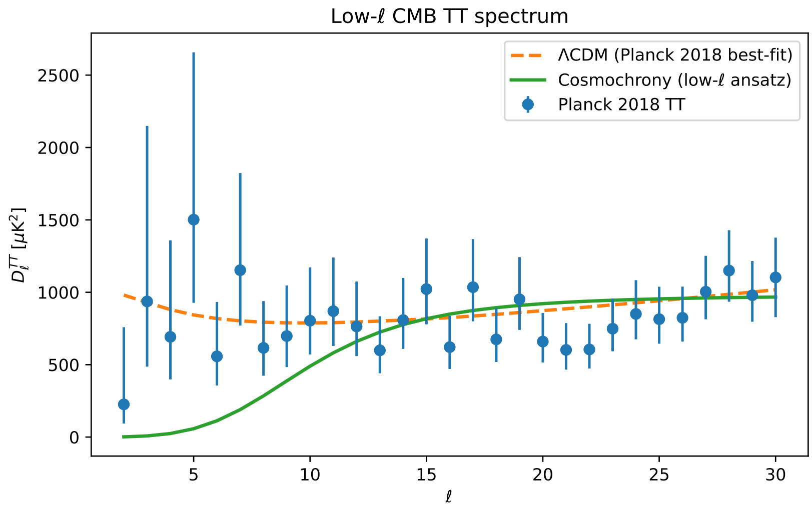

Fig. 8 Low-ℓ CMB TT power spectrum comparison. The Cosmochrony ansatz (green) shows natural suppression of power at large angular scales ($\ell < 10$) compared to ΛCDM (dashed orange).

these features are residual relational correlations inherited from the pre-geometric regime of χ.

A complementary unsmoothed representation, directly confronting the raw Planck low-ℓ data with the same attenuation mechanism, is provided in Appendix C.1 (Fig. 20).

Structural Admissibility and Low-ℓ Suppression

The primordial spectrum is modulated by an admissibility filter:

$$P_{\mathrm{obs}}(k,t) = \mathcal{A}^2(k,t)\, P_0(k), \tag{123}$$

with a natural phenomenological form

$$\mathcal{A}(k,t) = \exp\left[-\left(\frac{k_c(t)}{k}\right)^p\right], \tag{124}$$

where $k_c(t)$ is the coherence scale associated with the maximal size of projectively admissible configurations. For $k \ll k_c(t)$, power suppression arises from the structural impossibility of supporting such global modes within the available relational complexity. The scale $k_c(t)$ admits a graph-theoretic interpretation through the spectral gap $\lambda_2(t)$ of the effective connectivity graph.

Testable Predictions

A. Correlated Suppression Across TT, TE, and EE Spectra.

The coherence scale k_c should be consistent across temperature and polarization spectra. Future high-precision measurements (e.g. *LiteBIRD*) provide a decisive test.

B. Absence of Primordial Non-Gaussianities.

The framework predicts an exceptionally small $f_{\rm NL}$. A significant detection would falsify this scenario.

C. Scale-Dependent Spectral Tilt Near the Cutoff.

A mild running of n_s localized near k_c is expected. A detailed treatment is provided in Appendix C.1.

8.13 Effective Potential for Galactic Dynamics from Projective Capacity Saturation

Operational status.

The effective potential is defined only through observable kinematics:

$$g_{\rm eff}(r) \equiv -\frac{d\Phi_{\rm eff}}{dr}, \qquad v^2(r) = r\,\frac{d\Phi_{\rm eff}}{dr}. \tag{125}$$

Emergent acceleration scale.

The nonlinear projective saturation constraint induces an effective background kinematic scale

$$a_0(t) \sim c\,H(t), \tag{126}$$

weakly time-dependent, with a projection efficiency factor $\eta = O(0.1)$ for galactic dynamics.

Asymptotic regimes and flat rotation curves.

Let $g_N(r) = GM_b(r)/r^2$. In the high-acceleration regime ($g_N \gg a_0$), $g_{\rm eff} \simeq g_N$. In the low-acceleration regime ($g_N \ll a_0$), saturation yields

$$g_{\rm eff}(r) \simeq \sqrt{g_N(r)\,a_0(t)}, \tag{127}$$

implying

$$\Phi_{\rm eff}(r) \simeq \sqrt{GM_b\,a_0(t)}\,\ln\!\left(\frac{r}{r_s}\right) + \text{const.}, \tag{128}$$

a logarithmic potential producing asymptotically flat rotation curves. The transition radius is $r_s(t) = \sqrt{GM_b/a_0(t)}$.

Minimal smooth interpolation.

A parsimonious operational fit function is

$$g_{\rm eff}(r) = \sqrt{g_N(r)^2 + a_0(t)\,g_N(r)}. \tag{129}$$

Aspect	ΛCDM	MOND	Cosmochrony
Ontology	Non-baryonic halo	Modified dynamics	Projective saturation properties of χ
Flat rotation curves	Invisible mass	$g \simeq \sqrt{g_N a_0}$	Saturation: $g \simeq \sqrt{g_N a_0(t)}$
Key scale	Halo profile parameters	Universal a_0	Emergent $a_0(t) \sim cH(t)$
Tully–Fisher	Formation models	$v^4 \propto GM_b a_0$	$v^4 \propto GM_b a_0(t)$
Discriminating signature	Cusps vs. cores	Strict universality	Slow evolution of $a_0(t)$

Table 3 Conceptual comparison of flat-rotation-curve explanations.

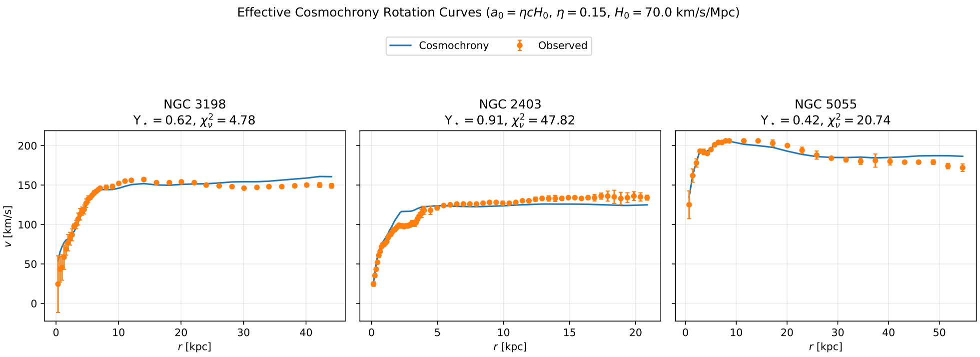

Fig. 9 Observed rotation curves compared with the Cosmochrony saturation prediction. Left: NGC 3198. Center: NGC 2403. Right: NGC 5055. See Appendix D.8 for data sources.

Baryonic Tully–Fisher scaling.

In the deep-saturation regime:

$$v_\infty^4 \simeq G\, M_b\, a_0(t), \tag{130}$$

with a mild redshift dependence through $a_0(t) \sim cH(t)$.

Observed rotation curves: multi-galaxy test

The Cosmochrony prediction is confronted with observed rotation curves of NGC 3198 (flat), NGC 2403 (rising), and NGC 5055 (mildly declining). No dark matter halo is introduced. Baryonic contributions are taken from the literature; the only fitted parameter is the stellar mass-to-light ratio $\Upsilon_\star$. The acceleration scale is fixed by $a_0(t_0) \sim cH(t_0)$.

8.14 Summary

Cosmological phenomena emerge from the global monotone growth of projected spectral structure. Cosmic expansion, large-scale homogeneity, late-time acceleration, and the arrow of time arise naturally from the admissibility cascade, without invoking an inflationary phase, dark energy, or an initial spacetime singularity. The Big Bang is a limiting regime of maximal constraint; black holes represent localized reapproaches to the same descriptive boundary. At the effective level, the framework reproduces the Hubble law, CMB structure, and large-scale gravitational behavior as emergent consequences of a single non-injective projective structure.

Part IV
Quantum Mechanics

This part presents a unified treatment of quantum phenomena as consequences of non-injective projection rather than as fundamental departures from classical logic. Quantum correlations, indeterminacy, and entanglement are derived from the structural non-factorizability of projected descriptions of the underlying substrate χ.

Violations of Bell inequalities are interpreted as signatures of ontological projection failure, not as evidence of superluminal influences. Measurement, decoherence, and apparent wave-function collapse are described as context-dependent stabilization processes within projected regimes.

Temporal ordering, established in Part I as the monotone growth of cumulative projected capacity $I(n)$ along the admissible cascade, provides the pre-existing ordering with respect to which effective quantum evolution is formulated. The Schrödinger equation is accordingly not a fundamental law from which time originates, but an effective description formulated within a structurally prior ordering.

The relation to the standard quantum formalism is clarified, including the emergence of Hilbert space structure, effective Schrödinger dynamics, and the Born rule. Quantum mechanics is thus situated as a consistent and predictive effective framework arising from deeper relational constraints.

The logical chain underlying this part can be summarised as follows. Non-injectivity of Π (established as a structural necessity in Section B.23 and [20, Thm. 3]) implies that distinct configurations of χ are mapped to the same observable: a given observable o carries an irreducible fiber $\Pi^{-1}(o)$ of indistinguishable pre-images. Because the effective description has no direct access to χ, it must represent the coexistence of projectively admissible contributions internally, without appeal to the substrate. This necessity forces an additive, phase-sensitive, norm-preserving structure on the space of effective descriptions—a complex Hilbert space, interpreted not as a fundamental ontology but as the minimal representational closure compatible with non-injective projection (Section 9.12). In the regime where the quasi-stable background and Born–Infeld bounds jointly hold, this structure supports a unique effective dynamics of Schrödinger type, derived in Appendix B.12 as Theorem B.1. The spinorial structure SU(2), required by the maximum quaternionic admissibility of the projection fiber (Section 6.8), completes the identification of the standard quantum-mechanical framework as an emergent effective description. The first three steps of this chain are structural necessities; the Hilbert interpretation is a structurally motivated consequence, not yet a fully derived theorem; Schrödinger dynamics is a proved result under explicit hypotheses.

9 Quantum Phenomena and Entanglement

Building on the non-injectivity of the projection Π (Section 2.4), this section shows how entanglement, nonlocal correlations, measurement statistics, and the formal apparatus of quantum mechanics emerge without introducing additional ontological degrees of freedom. The framework adopts **ontological monism**: there is a single ontological substrate χ, while apparent multiplicity and spatial separation are properties of the projected description.

9.1 Non-Factorizable Projected Descriptions and Quantum Correlations

The projection Π is a fixed mapping with intrinsic finite resolution and irreducible projection noise: its pixelisation is set by the spectral admissibility threshold λ^* (Section 2.4) and its capacity is bounded by the Born–Infeld saturation condition. Observable dynamics is not a dynamics of χ, which is atemporally fixed, but an iterated reprojection of the effective state through the deforming mirror $\Pi \circ \sigma$:

$$U_{t+1} = \Pi \circ \sigma(U_t), \tag{131}$$

where $\sigma : O \to \Omega$ is an admissible section selecting a configuration $\omega_t = \sigma(U_t)$ within the fiber $\Pi^{-1}(U_t)$ under the constraints of χ. The composition $\Pi \circ \sigma$ acts as a deforming mirror: σ lifts the current state back into the configuration space, and Π reflects it into the next observable state. Because σ is not the inverse of Π, the reflection is deformed by two distinct mechanisms. First, the admissibility constraints encoded in χ govern which configuration ω_t within the fiber $\Pi^{-1}(U_t)$ is selected: the selection is structural, not random. Second, the projection noise: distinct configurations within the same fiber never overlap perfectly under Π; their residual differences, inaccessible to the effective description, appear as fluctuations in U_{t+1}. It is this irreducible residue that guarantees $U_{t+1} \neq U_t$ and makes the sequence irreversible: a configuration with no residue would imply fibers whose elements project identically, leaving no relational difference to motivate a next state—the structural analogue of absolute zero (Section 5.6). The substrate χ constrains the admissibility of σ but does not supply dynamical content: it governs what can be reflected, not what is. Each step generically loses information (non-injectivity), introduces a projective deformation (projection noise), and preserves certain global invariants (admissibility constraints). What appears as separate particles or subsystems at time t are excitations of U_t that Π can resolve as distinct.

Entanglement arises as a resolution event in this reprojection sequence: two degrees of freedom A and B of $\chi_{\text{eff}}(t)$ fall below the resolution of Π in the relevant observable sector, so that $\Pi(A) \approx \Pi(B)$ within that sector. They share an effective resolution cell. No special or pre-selected configuration of χ is required, and no superdeterminism is implied: whether two excitations fall into the same cell is determined by the structure of $\chi_{\text{eff}}(t)$ relative to the fixed pixelisation of Π, exactly as two distinct inputs may land on the same pixel of a camera whose resolution is fixed.

This does not reduce χ to a passive or contentless mirror. χ does not encode observable values, but it carries the full structural information determining which relational configurations are admissible. The global balance invariants—total spin, total charge, winding number—that appear as infra-projectable constraints of the fiber $\Pi^{-1}(o_{AB})$ originate in this admissibility structure, not in the effective state $\chi_{\text{eff}}(t)$ and not in a classically rich hidden state. χ is minimal in observable content and maximal in structural constraint: it is a coherence space, not a data store. The observable universe does not merely reflect itself through a passive mirror; it reads itself through a deforming one. The deformation—the fact that σ is not the inverse of Π—is what makes successive states different, and what the admissibility structure of χ constrains without prescribing.

The deeper reason for this non-factorizability can be identified at the level of the spectral structure of the projection. A configuration $\omega \in \Omega_\chi$ may carry a global invariant $I(\omega)$—total charge, total angular momentum, winding number, or more generally any topological raccordement constraint—that belongs to a sector of the substrate which is not individually projectable within the effective description. Such invariants belong to spectral sectors of L_χ below the projection threshold λ^* (Section 2.4) and therefore cannot appear as independent local observables in the effective description: they are not representable as a separable pair $I_A(\omega_A) \oplus I_B(\omega_B)$ projectable independently into $O_A \times O_B$, but exist once, at the level of χ, and can only appear in the effective description as a property of the joint fiber $\Pi^{-1}(o_{AB})$. The relevant pre-image is therefore not $\Pi^{-1}(o_A) \times \Pi^{-1}(o_B)$ but the non-factorisable global class $\Pi^{-1}(o_{AB})$, whose elements are configurations constrained jointly by $I(\omega)$.

The type of correlation is determined by which balance constraint is infra-projectable, not by a different mechanism. Direct correlations ($s_A = s_B$) arise when the infra-projectable invariant selects decompositions with equal local values; anti-correlations ($s_A + s_B = 0$) arise when the invariant forces opposite local values. In both cases the constraint is carried by the fiber, not by individual pre-images. Measurement at A selects a locally stable reprojection $\Pi_A(\omega) = o_A^*$; because the fiber $\Pi^{-1}(o_{AB})$ is non-factorizable, this selection simultaneously restricts the class of configurations admissible for B to those compatible with $I(\omega)$ and o_A^*, without any signal propagating from A to B. The apparent "collapse" at B is a restriction of admissible projected descriptions imposed by the global non-decomposability of the fiber, not a dynamical influence.

Ontological Monism and the Shared Projection Hypothesis

A single connected excitation in χ can be represented as several spatially separated effective excitations. The observed separation is a property of the projected metric representation, not a fundamental separation of the underlying entity.

Shared Fiber Phase and Spin Correlations.

Spin correlations can be read as correlations of a *shared* internal degree of freedom of the projection fiber. A measurement at one location selects a locally stable reprojection of that shared fiber degree of freedom; the correlated statistics at the distant location reflect the restricted set of reprojections still compatible with the same underlying configuration.

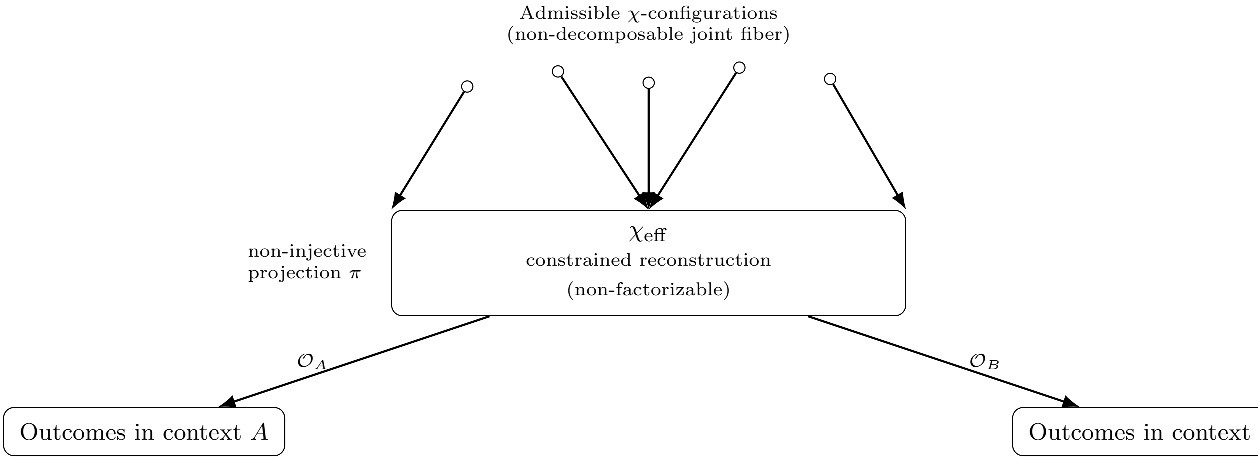

Fig. 10 Non-factorizable regime. The joint fiber $\Pi^{-1}(o_{AB})$ is non-decomposable: no independent pre-images can be assigned to subsystems A and B. The admissibility structure of χ determines which balance invariants are infra-projectable; it does not select a unique pre-existing configuration. Different measurement contexts define distinct admissible reprojections of the same constrained fiber, producing correlated outcomes without superluminal influence.

9.2 Nonlocality and the Holistic Character of Projected Descriptions

Quantum nonlocality does not arise from superluminal interactions but reflects the intrinsically non-factorizable character of certain admissible projected descriptions [55]. Entangled systems correspond to single projected configurations that cannot be decomposed into independent subsystems without loss of admissibility.

9.3 Nonlocal Correlations Without Superluminality

Correlated outcomes arise because spacelike separated measurements correspond to different local reprojections of a single non-factorizable description. The factorization assumptions underlying Bell-type inequalities are violated, while dynamical locality and relativistic causality remain intact (see Appendix E.2). The correlations are *ontological*—fixed by the non-injective global relational structure—rather than *dynamical*.

Attosecond-resolved measurements have demonstrated that quantum correlations become experimentally accessible on timescales far shorter than any interaction-based mechanism [56]. Within Cosmochrony, the observed timescale corresponds to the resolution time of the effective projection Π, not to a physical propagation time: it probes projective accessibility rather than dynamical formation of correlations.

9.4 Temporal Ordering and Relativistic Consistency

Temporal ordering arises not as a primitive ingredient of the framework but as a structural consequence of the non-injective projection Π: it is the ordering induced by the monotone growth of cumulative projected capacity $I(n)$ along the admissible

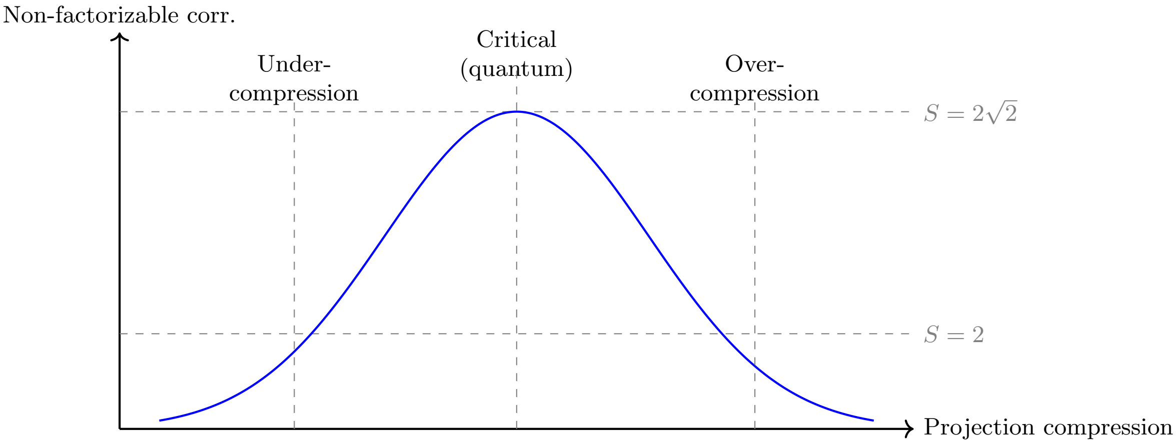

Fig. 11 Bell inequality violations as a function of the compression induced by Π. Non-factorizable correlations emerge in an intermediate regime.

cascade [10]. Different observers may assign different orderings to spacelike separated events; such differences reflect observer-dependence of spacetime slicing and have no impact on the relational consistency of admissible descriptions. Relativistic covariance is preserved by the invariance of the underlying non-injective relational configuration across all admissible projected descriptions.

9.5 Relation to Bell Inequalities

Bell's theorem [55] rules out any ontological completion of quantum mechanics based on factorizable hidden-variable models. Cosmochrony fully accepts Bell's theorem and identifies the ontological assumption that fails: the factorization hypothesis $P(a, b|x, y, \lambda) = P(a|x, \lambda)\, P(b|y, \lambda)$.

Because the projection Π is generically non-injective, admissible projected states are not associated with independent ontic pre-images for their subsystems. The factorization hypothesis is not merely violated but *ill-defined* within the space of admissible projected descriptions.

Bell-type violations arise in an intermediate regime of projective compression: the effective description is sufficiently coarse-grained to permit subsystem separation, yet retains enough global relational structure to prevent factorization. In the limit of extreme coarse-graining, projected descriptions become effectively classical.

The dependence of entanglement correlations on measurement orientation acquires a natural reading within this framework. Each measurement apparatus defines a local relational frame, corresponding to a directional projection operator $\Pi_{\mathbf{a}}$ or $\Pi_{\mathbf{b}}$ parametrized by the orientation vectors **a** and **b**. The local outcomes $o_A = \Pi_{\mathbf{a}}(\omega)$ and $o_B = \Pi_{\mathbf{b}}(\omega)$ are therefore two readings of the same infra-projectable invariant $I(\omega)$ from two distinct relational frames. For a spin-singlet configuration ($S_{\text{tot}} = 0$), the infra-projectable invariant is represented by a unit element of S^3, reflecting the SU(2) structure of spin degrees of freedom in the projection fiber (Section 5.2). Directional measurements correspond to projections of this S^3 invariant onto axes in the effective space S^2; the spin observables satisfy $(\boldsymbol{\sigma}\cdot\mathbf{a})^2 = 1$ and the mean correlation between two measurement directions is

$$E(\mathbf{a},\mathbf{b}) = \langle(\boldsymbol{\sigma}\cdot\mathbf{a})\otimes(\boldsymbol{\sigma}\cdot\mathbf{b})\rangle = -\mathbf{a}\cdot\mathbf{b} = -\cos\theta, \tag{132}$$

where $\theta = \angle(\mathbf{a},\mathbf{b})$. This result follows from the SU(2) geometry of the projection space: the $\cos\theta$ law is not derived from a classical sign model—which would yield $1-2\theta/\pi$ and not saturate the Tsirelson bound—but reflects the non-commutative structure of directional projections of an S^3 invariant. No dynamical influence between A and B is required; the $\cos\theta$ law is a structural consequence of the SU(2) geometry of Π.

The maximal violation of the CHSH inequality follows from the same geometric structure. The CHSH combination $S = E(\mathbf{a},\mathbf{b}) + E(\mathbf{a},\mathbf{b}') + E(\mathbf{a}',\mathbf{b}) - E(\mathbf{a}',\mathbf{b}')$ is a linear combination of scalar products of unit vectors in $\mathbb{R}^2$. The identity

$$\left|\mathbf{a}\cdot\mathbf{b} + \mathbf{a}\cdot\mathbf{b}' + \mathbf{a}'\cdot\mathbf{b} - \mathbf{a}'\cdot\mathbf{b}'\right| \;\leq\; 2\sqrt{2} \tag{133}$$

holds for all unit vectors and is saturated by the optimal configuration $\mathbf{a} = (1,0)$, $\mathbf{a}' = (0,1)$, $\mathbf{b} = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{b}' = \frac{1}{\sqrt{2}}(1,-1)$, which yields $|S| = 2\sqrt{2}$. The Tsirelson bound is therefore not a postulate of quantum theory imported into the framework: it is a geometric constraint on the space of directional projections of a single substrate invariant. Its value $2\sqrt{2}$ reflects the Hilbert-space geometry associated with the SU(2) structure of the projection fiber (Section 5.2). The classical bound $|S| \leq 2$ corresponds to the regime in which the projection becomes effectively injective and the invariant can be assigned independent local pre-images, restoring factorizability.

No hidden variables and no superluminal influence are introduced. Quantum nonlocality is ontological rather than dynamical: the failure of Bell-type factorizability arises from the structure of admissible projected descriptions, not from nonlocal interaction.

9.6 Measurement, Decoherence, and Apparent Collapse

Quantum measurement does not involve fundamental wavefunction collapse. Measurement corresponds to the transition from a non-factorizable admissible projected description to a set of effectively factorized local projections. Decoherence suppresses interference between incompatible descriptive branches by rendering their relative phase information inaccessible within spacetime representations [57]. The underlying relational structure remains globally well defined. The apparent collapse is the effective manifestation of a non-injective relational structure becoming only partially projectable into spacetime.

The measurement context has a diachronic as well as a synchronic dimension. Standard contextuality frameworks identify a context with a compatible family of observables at a fixed step n — a synchronic choice of subalgebra. In the present framework, the admissible neighbourhood $\mathcal{A}(U_{n-1})$ introduces a second, diachronic layer: the set of admissible measurement outcomes at step n is itself conditioned on the previously realised state U_{n-1}. The context of projection is therefore not merely the measurement algebra chosen at step n, but also the already-realised projective position from which the next observable state may be reached.

9.7 Limits of Entanglement and Environmental Effects

Entanglement arises only within restricted regimes in which a non-factorizable global description remains jointly projectable. Environmental coupling progressively restricts the set of admissible projected descriptions to locally stable, approximately factorizable regimes. Entanglement is most robust for effectively isolated systems and becomes increasingly fragile in macroscopic environments. Classical behavior emerges when only factorized projected descriptions remain admissible, without requiring any modification of the underlying relational structure.

9.8 Structural Stability of Projected Descriptions

Entanglement, measurement, and decay correspond to distinct stability regimes of projected descriptions. Entanglement corresponds to the regime in which a non-factorizable projected description remains structurally stable. Measurement and decoherence correspond to selective stabilization: interaction with an environment amplifies certain relational features while rendering alternative descriptions inadmissible. Particle decay represents the regime in which the non-factorizable description becomes unstable, and stability is recovered through factorization into several localized configurations.

A technical analysis is provided in Appendix B.19.

9.9 Entanglement as a Resolution Event in Reprojection Dynamics

The preceding sections establish the structural origin of non-factorizable correlations. This section gives a unified account of entanglement, disentanglement, anti-correlated entanglement, entanglement swapping, and monogamy within a single mechanism: the finite-resolution reprojection sequence.

Generic reprojection.

At each step $t \to t+1$ the effective state $\chi_{\text{eff}}(t)$ is reprojected through the fixed mapping Π. The generic outcome is a deformed image: information is lost (non-injectivity), and the new effective state differs from the old one by projective noise. Two excitations A and B that Π can resolve as distinct produce separable fibers $\Pi^{-1}(o_A) \times \Pi^{-1}(o_B)$, and no entanglement arises.

Entanglement.

Entanglement arises when the geometry of $\chi_{\text{eff}}(t)$ places two degrees of freedom A and B within the same resolution cell of Π in the relevant observable sector. The joint fiber $\Pi^{-1}(o_{AB})$ is non-decomposable, and the global balance invariant I_{AB} is infra-projectable: it cannot be assigned independently to A and B. No pre-selected or distinguished configuration of χ is involved.

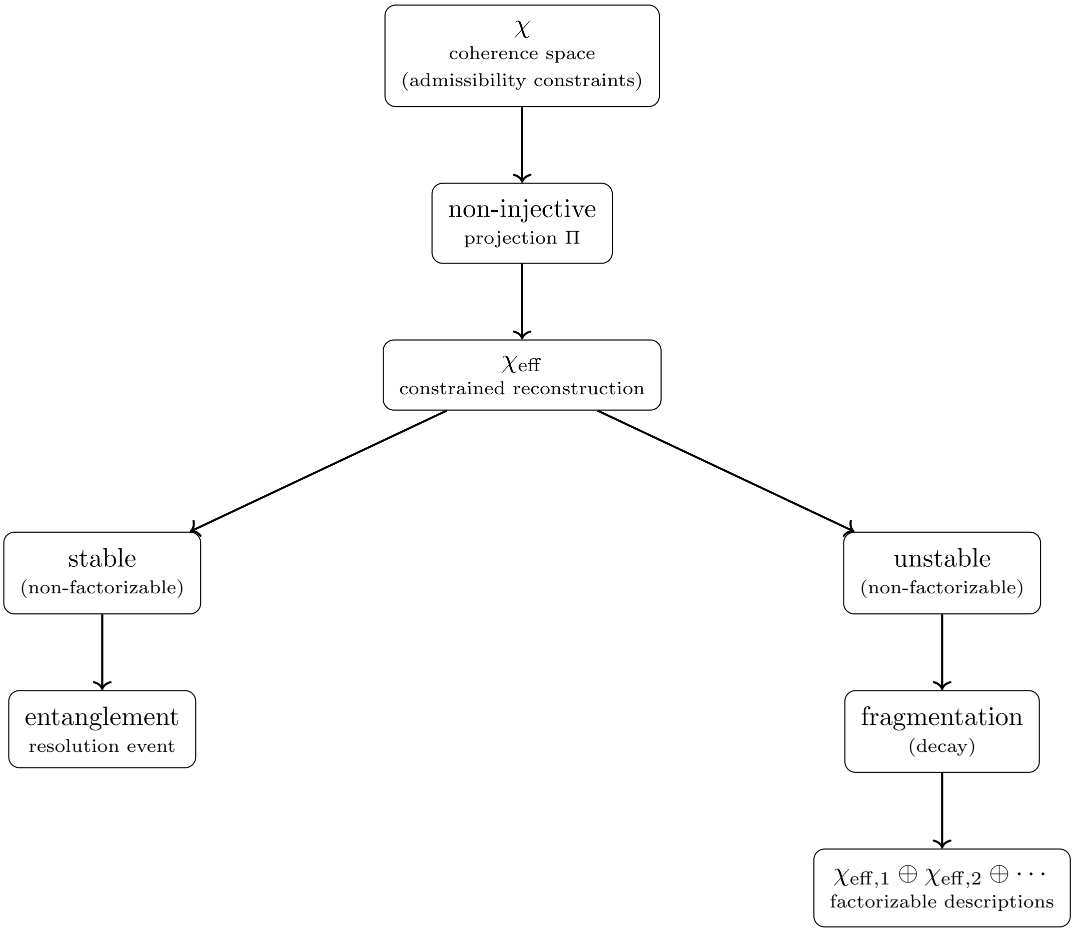

Fig. 12 Reprojection branching induced by the finite-resolution projection Π. χ is a coherence space whose admissibility structure constrains which reconstructions can consistently emerge; it does not pre-select a unique substrate configuration. At each reprojection step the effective state χ_{eff} either remains jointly non-factorizable—entanglement, arising when two degrees of freedom share a resolution cell—or fragments into localized factorizable descriptions (particle decay). The branching is a property of the reprojection dynamics, not of a distinguished substrate configuration.

Direct and anti-correlated entanglement.

The sign of the correlation is not a separate mechanism but a property of the infra-projectable invariant I_{AB} carried by the fiber. When the invariant selects decompositions in which local values coincide (e.g. $s_A = s_B = +\frac{1}{2}$), the correlation is direct. When the invariant is a conservation law that forces opposite local values (e.g. $s_A + s_B = 0$), the correlation is anti-correlated. The singlet and triplet states correspond to distinct global invariants carried by distinct fibers, not to distinct mechanisms.

Disentanglement by measurement or decoherence.

Measurement and environmental coupling modify the effective state $\chi_{\text{eff}}(t)$. The subsequent reprojection places A and B at resolvable positions in the observable sector: the joint fiber becomes decomposable, the global invariant I_{AB} becomes locally projectable, and the entanglement disappears. No collapse or destruction of information is involved. The balance constraint that was infra-projectable becomes projectable once the effective state has moved sufficiently in the space of observable configurations; Π has not changed, only the image it is reading.

Entanglement swapping.

Suppose at step t_1 excitations A and B fall below resolution and share the infra-projectable invariant I_{AB}; at step t_2 excitations B and C share I_{BC}. A measurement on B forces a locally admissible decomposition of I_{AB} and simultaneously restricts the admissible decompositions of I_{BC}. The residual constraint propagates to A and C: the subsequent effective state places A and C in a jointly constrained fiber $\Pi^{-1}(o_{AC})$, even though A and C have never shared a resolution cell directly. Swapping is transitive propagation of fiber constraints through an intermediate reprojection event, not a transmission of influence.

Monogamy.

A resolution cell has bounded capacity, set by the Born–Infeld saturation condition on the projection fiber. A single cell cannot simultaneously carry independent infra-projectable invariants for multiple disjoint pairs without saturating. Monogamy of entanglement is therefore a capacity constraint on the fiber, not an additional postulate.

Summary.

The five phenomena—entanglement, direct and anti-correlated correlations, disentanglement, swapping, and monogamy—share a single mechanism: the interaction of the evolving effective state $\chi_{\text{eff}}(t)$ with the fixed finite-resolution projection Π. Entanglement is a local resolution event in the reprojection dynamics; it requires no special configuration of χ, no modification of Π, and no non-local dynamical influence. The balance constraints that become infra-projectable originate in the admissibility structure of χ—which is a coherence space, not a data store—and not in any classically rich hidden state or pre-selected substrate configuration. This position is distinct from both naïve realism (no observable values are stored in χ) and eliminativism (the admissibility structure of χ is the genuine source of all invariants). The key principle is: *what lies below the projective resolution is not a value, but a constraint.*

9.10 Implications for Quantum Computation

Effective resources exploited in quantum computation may be reinterpreted as manifestations of structural non-injectivity and global consistency constraints. Multiple underlying relational configurations may correspond to the same effective observable state, providing a natural origin for effective parallelism without explicit information exchange. Part of the computational advantage may reflect general properties of non-factorizable descriptive mappings rather than exclusively quantum dynamical effects.

9.11 Integration with the Standard Model: A Spectral Interpretation

Weak Boson Masses from Spectral Geometry

The masses of $W^{\pm}$ and Z^0 emerge from the spectral properties of the Hodge Laplacian Δ_1 acting on 1-forms of the fiber bundle Π. The space of 1-forms decomposes into gauge-invariant subspaces $\Omega^1_{SU(2)}$ and $\Omega^1_{U(1)}$. Effective masses are

$$m_W \propto \sqrt{\lambda_{1,SU(2)}}, \qquad m_Z \propto \sqrt{\lambda_{1,U(1)}},$$

with the mass ratio governed by the metric anisotropy and curvature structure of the χ-induced geometry on Π.

Emergent Gauge Couplings

Gauge couplings are defined through normalized heat-kernel traces:

$$g^2 = 4\pi \left[\widehat{\mathrm{Tr}}_{SU(2)}\big(e^{-t_0\Delta_{SU(2)}}\big) - \widehat{\mathrm{Tr}}_{U(1)}\big(e^{-t_0\Delta_{U(1)}}\big) \right],$$

$$g'^2 = 4\pi\, \widehat{\mathrm{Tr}}_{U(1)}\big(e^{-t_0\Delta_{U(1)}}\big),$$

with $t_0 = L^2_{\text{fiber}}$. The Weinberg angle follows from spectral asymmetry:

$$\tan^2\theta_W = \frac{\widehat{\mathrm{Tr}}_{U(1)}\big(e^{-t_0\Delta_{U(1)}}\big)}{\widehat{\mathrm{Tr}}_{SU(2)}\big(e^{-t_0\Delta_{SU(2)}}\big)}.$$

Geometric Phase Transition and Mass Generation

Mass generation is a geometric phase transition of the χ substrate. Below a critical spectral density χ_{crit}, only massless modes are supported. Above it, spectral weight condenses into specific invariant subspaces, generating discrete non-zero eigenvalues $m_n \propto \sqrt{\lambda_n}$.

Strong Sector: Topological Confinement and Color

Color charge maps to the three fundamental degrees of freedom of the proton's trefoil topology ($Q = 3$). Topological confinement: separating components of a $Q = 3$ soliton requires linear energy increase exceeding the pair-creation threshold. Asymptotic freedom: at short distances the global topological constraint is not yet engaged, rendering the interaction effectively weaker.

The Origin of Mass: Spectral Overlap

Fermion masses are emergent quantities determined by the resonance between internal stability spectra and the global projective admissibility structure:

$$m_{\text{eff}} \propto \int_{\text{Fiber}} \mathcal{S}(\phi_n) \cdot \mathcal{R}(\chi) \, d\Pi. \tag{134}$$

Multiple fermion generations correspond to distinct stable topological classes of solitonic χ-configurations. The mass hierarchy $m_e \ll m_\mu \ll m_\tau$ arises from the ordered spectrum of L_{sol}:

$$m_n \sim \text{Spec}(L_{\text{sol}})_n. \tag{135}$$

Mixing matrices (CKM, PMNS) encode the misalignment between the mass basis (eigenvectors of L_{sol}) and the interaction basis (principal axes of Π). CP violation originates from non-trivial topological torsion of the projection fiber.

9.12 Relation to Quantum Formalism

The formal apparatus of quantum mechanics arises as an effective framework organizing admissible projected descriptions in regimes where localization, linearity, and approximate factorization hold. Quantum mechanics is not replaced but reinterpreted as an effective theory whose validity is restricted to regimes admitting a stable, approximately linear, spacetime-based description.

Status of the Wavefunction.

The wavefunction ψ is not a fundamental physical entity but a statistical encoding of admissible local reprojections compatible with a given non-factorizable projected configuration. Its complex phase encodes relational constraints between alternative local descriptions; its modulus determines the statistical weight of accessible reprojections.

Why a Hilbert Space.

Because the effective description has no access to χ itself, it must represent, at the level of observables alone, the coexistence of distinct configurations belonging to the same fiber $\Pi^{-1}(o)$. This representational requirement imposes three successive constraints. First, contributions from different pre-images must be combinable: the space of effective descriptions must be additive. Second, observed interference phenomena—constructive or destructive superposition of projectively admissible contributions—require a phase structure that a purely real or purely positive representation cannot support: complex scalars are forced. Third, the effective dynamics must preserve the total statistical weight of admissible reprojections, imposing a conserved norm. Together, these three constraints single out a complex Hilbert space as the minimal representational closure compatible with non-injective projection. This is not a postulate but a structurally motivated consequence of the non-injectivity of Π; it does not yet constitute a fully derived theorem, and the precise conditions under which the representation is unique are left for future work.

Emergence of Hilbert Space Structure.

Quantum superposition is the observable trace of fiber multiplicity: a state of the form $\alpha |o_1\rangle + \beta |o_2\rangle$ encodes, within the effective description, the coexistence of projectively admissible contributions from $\Pi^{-1}(o_1)$ and $\Pi^{-1}(o_2)$, not a literal simultaneous occupation of two distinct ontic states. In regimes where relational constraints are weak and approximately factorizable, distinct admissible descriptions can be combined, giving rise to approximate linear structure. The inner product encodes mutual compatibility; orthogonality corresponds to mutually exclusive descriptive regimes. Unitary evolution is a consistency-preserving transformation valid as long as projectability and factorizability remain intact. The structural derivation of the effective Schrödinger dynamics from these constraints is given in Appendix B.12.

Emergence of the Schrödinger Equation.

In the non-relativistic regime, the effective dynamics of projected modal amplitudes reduces to a Schrödinger-type evolution equation. Within Cosmochrony, this equation is not taken as fundamental, but arises as the weak-amplitude, approximately injective limit of the projected dynamics of admissible χ-configurations. Separating a rapidly varying phase from a slowly varying envelope yields

$$i\hbar_{\text{eff}}\,\partial_t\psi = -\frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}}\nabla^2\psi + V_{\text{eff}}(x)\psi. \tag{136}$$

A structural derivation of this effective equation is given in Appendix B.12.

Operators and Non-Commutativity.

The canonical commutation relation $[\hat{x},\hat{p}] = i\hbar_{\text{eff}}$ expresses the non-commutativity of geometric measurements induced by projection from the pre-geometric substrate. The uncertainty principle emerges as a geometric consistency condition on admissible effective descriptions.

This canonical non-commutativity operates at the level of effective spacetime observables. It should be distinguished from the spectral and sectorial non-commutativities discussed in Sections 11.2 and 11.6, which arise within the projection fiber itself and manifest respectively as inertial mass and CP-violating geometric phases.

Origin of Quantization.

Only a restricted class of localized projected configurations admits long-lived, internally consistent descriptions. Admissible configurations form discrete equivalence classes, yielding effective energy quantization. Planck's constant emerges as a universal conversion factor characterizing the minimal structural scale at which projected descriptions remain stable.

Measurement and the Born Rule.

Measurement outcomes correspond to effective reprojections onto mutually exclusive descriptive regimes. The Born rule emerges as the unique stable measure on the space of admissible projected descriptions that remains invariant under loss of phase coherence, coarse-graining, and environmental coupling.

Within the effective quantum framework, a variational characterisation of the Born weights is available: for each measurement context C (an abelian subalgebra), the Born probabilities $p_C(i) = \mathrm{Tr}(\rho P_i)$ arise as the unique minimisers of the Umegaki relative entropy of the state ρ projected onto C [58]. This characterisation is synchronic: it operates at a fixed step n for a given state ρ and measurement context. It presupposes the Hilbert space structure and the trace functional, and is therefore a characterisation within the quantum framework rather than a derivation from the pre-quantum level.

The deeper programme of Cosmochrony is to derive the Born weights from the admissible transition structure itself. In the present framework, the relevant context is not fixed externally: it is generated by the admissible neighbourhood $\mathcal{A}(U_{n-1})$, which conditions which measurements are accessible at step n given the previously realised state U_{n-1}. Contextuality is therefore not only synchronic (choice of subalgebra at fixed n) but diachronic (dependence on the projective history $U_0 \rightsquigarrow \cdots \rightsquigarrow U_{n-1}$). A full derivation of Born probabilities from these diachronic admissibility constraints is left for future work.

Entanglement and Nonlocal Correlations.

Entangled systems arise when a unified relational configuration admits an effective projection onto spatially separated degrees of freedom. Correlations are fully compatible with relativistic causality (Appendix E.2).

Spin and Statistics.

Spin emerges as a topological property of admissible projected configurations. 4π-periodic configurations correspond to fermionic behavior; 2π-periodic ones to bosonic behavior. The spin–statistics connection follows from the same topological constraints (Appendix B.4).

Orbital Geometry as Probabilistic Visibility.

Atomic orbitals correspond to effective probabilistic visibility patterns of admissible projected descriptions. Regions of high probability correspond to domains where consistent reprojection is most robust; nodal regions correspond to incompatible descriptive regimes.

Scope and Limitations.

All standard quantum-mechanical formalisms remain valid and unchanged within their domains. Cosmochrony provides an interpretative and unificatory account of the structural origin of quantum phenomena without introducing new degrees of freedom or hidden variables. A complete formal correspondence between the relational substrate and the operator-based structures of quantum theory is left for future work.

9.13 Summary

Gauge interactions emerge as admissible modes of the projection process. The photon corresponds to scalar transmission modes; $W^{\pm}$ and Z^0 bosons arise as shear-like spectral modes whose masses reflect the spectral rigidity of the projection fiber. Strong interactions and confinement are reinterpreted through topological stability of knotted solitonic configurations. Mass emerges from spectral overlap between localized configurations and the global projective admissibility structure. Entanglement and nonlocal correlations reflect the persistence of non-factorizable admissible projected descriptions. Quantum mechanics appears as an effective statistical framework describing the limits of local projectability imposed by a globally non-injective relational structure. Classical behavior emerges as the limiting case in which only locally stable, factorizable projected descriptions remain admissible.

Part V
Predictions, Discussion, and Conclusion

This final part collects the phenomenological consequences of the Cosmochrony framework and confronts them with observation. Radiation processes, mass spectra, and quantization effects are revisited from a unified structural perspective, leading to concrete and falsifiable predictions. Where available, these predictions are grounded in spectral results established in the O-series admissibility programme [18, 23], providing an independent numerical foundation for the phenomenological claims.

Testable signatures are identified across cosmology, gravitation, particle physics, and high-energy regimes, with particular attention to discriminating Cosmochrony from ΛCDM-based models and standard quantum-gravitational extensions.

The concluding discussion situates Cosmochrony with respect to existing theoretical frameworks, clarifies its domain of validity, and highlights open challenges. Rather than proposing a final theory, this part frames Cosmochrony as a coherent pre-geometric research program constrained by internal consistency and empirical accessibility.

10 Radiation and Quantization

10.1 Radiation as a Projective Reconfiguration

Radiation arises when a locally stable projected configuration undergoes a transition toward a less constrained state. The excess relational content ceases to admit a particle-like projected description and becomes expressible only through delocalized projected modes. In effective spacetime descriptions, this redistribution appears as radiative emission—a transfer of descriptive weight from particle-like configurations to propagating field-like descriptions, without invoking discrete objects or stochastic processes.

10.2 Emergence of Photons

Photons are not fundamental entities. Prior to emission or detection, no photon exists as an independent object. A reconfiguration of relational structure within χ ceases to admit a localized projection and becomes expressible only through extended, delocalized effective modes—represented in spacetime as electromagnetic waves.

Photon-like events emerge only at interaction: when a delocalized projective mode becomes locally constrained by interaction with a localized excitation, the projection resolves into a discrete transfer of projective capacity. Quantization is a property of interaction and local reprojection, not of propagation. Wave–particle duality reflects a duality of description rather than of underlying ontology. Interference phenomena arise from the coherence of delocalized projective modes; individual detection events correspond to localized reprojections.

In contrast to the coherent composite regime discussed in Section 5.8, photon modes correspond to linear excitations around the unsaturated fiber sector.

10.3 Geometric Origin of $E = h\nu$

Building on Section 4.5, the Planck relation

$$E = h\nu \tag{137}$$

acquires a geometric interpretation in the context of radiative processes. The energy E measures the projective capacity redistributed during a reprojection event; the frequency ν characterizes the minimal temporal resolution required for a coherent effective description of this redistribution. The constant h is the effective projection of the fundamental substrate invariant $\hbar_\chi \equiv c^3/(K_{0,\mathrm{bare}}\, \chi_{c,\mathrm{bare}})$ (Section D.6), acting as a universal conversion factor between structural projective capacity and temporal projective resolution.

In the photoelectric effect, the threshold frequency ν_0 corresponds to the minimal projective resolution required to destabilize a bound electronic soliton. Quantization of radiative energy arises not from discretized propagation but from the discrete nature of local reprojection events, which impose a minimal unit of effective projective capacity transfer determined by the spectral graininess $\hbar_\chi$.

10.4 Projective Origin of Fluctuations

Vacuum fluctuations arise from the non-injectivity of the projection Π in regimes where no stable localized excitations are present. In such regimes, a wide multiplicity of distinct relational configurations of χ project onto the same effective spacetime description. Since admissible observables are functions on $\mathcal{O} = \operatorname{Im} \Pi$, not on Ω_χ [20, 59], intra-fibre differences are by definition invisible to observation: the residual structure within each fibre $\Pi^{-1}(o)$ constitutes a degree of freedom that no admissible observable can resolve. Any two such co-projecting configurations $\omega_1, \omega_2 \in \Pi^{-1}(o)$ share a common projected part—which yields the stable observable o—and carry residual structural differences that are not erased by Π. The projected manifestation of these residual differences constitutes the observable fluctuation. No intrinsic indeterminacy of the pre-geometric substrate is required: χ does not vary at the substrate level. Fluctuations are therefore an exclusive feature of the projected description, and their magnitude is controlled by the degree of non-injectivity of Π, quantified by the effective projective entropy S_Π.

The spatiotemporal character of fluctuations—their variation in both space and time—follows directly from the same mechanism. Spatial variations arise from residual relational differences between co-projecting configurations that are localized in distinct regions of the effective description. Temporal variations arise because co-projecting configurations may differ in their position along the projective ordering parameter λ: such ordering differences, once projected, are precisely what the effective description identifies as differences in operational time τ. The time-dependence of vacuum fluctuations is therefore not a signature of any infra-physical dynamics of χ, but the effective translation of projective ordering differences between co-projecting configurations.

A natural question arises at this point: why do projection residuals *appear random* rather than repeating identically under successive projections? If the substrate χ were stationary and the projection Π were applied repeatedly to the same fibre configuration, the residuals would indeed repeat identically. The appearance of randomness is therefore not a property of χ itself, but a consequence of the fact that the effective projection is not applied at a fixed relaxation stage. More precisely, the projection must be understood as implicitly λ-dependent: one effectively probes a family of maps Π_λ. At each effective step U_n, one probes not a fixed fibre $\Pi^{-1}(o)$, but a slice of the co-projecting set determined by the current value of the relaxation-ordering parameter λ. Between two successive steps U_n and U_{n+1}, λ has advanced, and the set of substrate configurations that co-project onto the same observable o is therefore not identical. The advance of λ is monotonic but remains unobservable at the effective level, since λ-ordering is precisely what the effective description identifies as time. Each reprojection thus samples a *different slice* of an evolving fibre, and the residual differences between these slices are registered as fluctuations in the effective description. In this sense, the apparent stochasticity of fluctuations is the observable signature of the irreversible advance of λ, rather than an intrinsic indeterminacy of χ. Equivalently, if the projective ordering could be suspended while reprojections continued, the same residuals would repeat identically. Fluctuations therefore arise because the projection cannot be iterated at fixed λ: time, understood as the projected image of the ordering induced by Π, is the minimal and irreducible source of the observed noise.

Remark (Fluctuations as λ-induced fibre drift). The mechanism identified above admits a natural geometric reading. The family $\{\Pi_\lambda\}_{\lambda\in\Lambda}$ defines a λ-parametrised bundle of co-projecting fibres over the observable space $\mathcal{O}$. Fluctuations correspond to the transverse variation of successive fibre slices along the relaxation flow, i.e. to the drift of $\Pi_\lambda^{-1}(o)$ as λ advances. A rigorous formulation of this drift—as a connection on the fibre bundle, or as a Lie-derivative along the relaxation vector field—is left as an open structural direction.

Two distinct mechanisms may contribute to the apparent randomisation of successive projection residuals. The first is *fibre decorrelation*: even if the norm of the residual differences remains approximately constant, the relaxation dynamics F generically drives successive fibre slices into increasingly orthogonal directions within Ω_χ, so that successive residuals become uncorrelated without necessarily growing in amplitude. This decorrelation is sufficient to account for the apparent randomness of fluctuations and does not require any energy growth at the substrate level. The second, potentially distinct mechanism is *projective amplification*: if the restriction of F to the fibre admits unstable directions in Ω_χ, residual norms may grow along the relaxation sequence until bounded by the Born–Infeld admissibility constraint $A^{\mathrm{BI}}_{\max}$. Once this ceiling is reached, the projection enters a saturation regime and structural compensation mechanisms become operative; the full hierarchy of saturation responses is developed in Section 13.11. Whether projective amplification is active alongside decorrelation, and under what conditions each mechanism dominates, remains an open structural question.

The Born–Infeld bound plays a dual role in this picture. It caps the amplitude of projective amplification, providing a finite structural ceiling on fluctuation growth. Crucially, this ceiling is not imposed by hand but follows from the admissibility constraint on the projection Π itself. In standard quantum field theory, the absence of such a bound forces the introduction of ultraviolet regulators and renormalisation procedures to control divergent loop integrals. In the present framework, renormalisation is not required as an independent procedure: the Born–Infeld bound replaces it with a structurally motivated finite cutoff, grounded in the projective architecture of the effective description.

10.5 Structural Predictions

Two structural predictions follow from the projective picture of fluctuations developed in Section 10.4.

First, in a regime where the effective projection becomes *locally* injective, the local projective entropy S_Π is suppressed and quantum fluctuations are correspondingly reduced. This suppression has a concrete realisation in the spectral admissibility programme: in Q9 [60], the position sector of the effective Dirichlet form is shown to vanish in the large-q limit (Q9 Lemma 4.3: $\lim_{q\to\infty}\mathcal{E}_q^X(f,f)=0$), demonstrating that intra-fibre degrees of freedom are expelled from $\mathcal{O}=\operatorname{Im}\Pi$ as admissibility is imposed. Since the ordering differences between co-projecting configurations are precisely what generate distinguishable successive effective states, a locally injective regime implies $U_{n+1}=U_n$ in that region: the projected description becomes locally stationary and operational time ceases to advance there. This is the projective reading of decoherence and of the classical limit: it does not require environmental averaging as a primitive assumption, but follows from the local reduction of the co-projecting fiber. Such locally injective regimes are admissible; they correspond to what is ordinarily described as the quasi-classical regime. A locally injective regime is one in which the restriction of Π to the accessible configuration neighborhood admits a single dominant element of $\Pi^{-1}(o)$: distinct substrate configurations within that neighborhood produce distinguishable projected states, even though the global projection remains non-injective.

Second, a *globally* injective projection—one in which $|\Pi^{-1}(o)| = 1$ for all observables simultaneously—would eliminate all projective entropy and terminate temporal ordering everywhere. This is structurally excluded on two independent grounds. The configuration space Ω_χ of the substrate is structurally richer than the observable space O: the projection Π is an irreducible coarse-graining, so $|\Omega_\chi| \gg |O|$ and a globally injective Π is mathematically incompatible with the compression implicit in any effective description. Furthermore, a globally stationary projection would imply $U_{n+1} = U_n$ for all effective observers simultaneously, which is the cosmological counterpart of absolute zero and is excluded by the same argument developed in Section 5.6. The persistence of a minimal global non-injectivity is therefore not merely compatible with temporal evolution; it is a structural requirement for it.

10.6 The Casimir Effect

When material boundaries impose structural constraints on local projectability, certain effective descriptions become incompatible with the boundary conditions, reducing the set of admissible projective configurations between the boundaries. The Casimir effect arises from this asymmetry: a difference in the density of admissible effective reprojections between the interior and exterior of the confining geometry, manifesting as a net pressure on the confining surfaces. No fundamental vacuum energy density or propagating vacuum modes are required.

10.7 Weakly Interacting Radiation

Weakly interacting radiation corresponds to delocalized projective regimes whose structural contrast is insufficient to efficiently induce localized reprojection upon encountering matter. Low-frequency or weakly coupled modes are characterized by smooth, slowly varying relational structure, strongly suppressing the probability of stable localized energy transfer. Small interaction cross sections reflect the low likelihood that a given projective configuration satisfies the geometric and topological conditions required for localized reprojection.

10.8 Summary

Radiation and quantization arise from the projective reconfiguration of localized matter excitations within the admissible effective description. Photon-like events emerge only during interaction-induced reprojection. Quantization reflects geometric and topological constraints of the projective dynamics at the level of projection: discrete energy transfer occurs when continuous projective descriptions satisfy the conditions required for localized reprojection.

11 Spectral Mass Spectrum and Hierarchy

Within the spectral admissibility framework established in Chapter 6, particle masses appear as invariants associated with stable modes of the projection fiber. The mass hierarchy therefore reflects the structure of admissible spectral configurations rather than independent dynamical parameters.

This section develops the physical consequences of that structure. We first interpret rest mass as a spectral stability condition of projected configurations. We then show how torsional constraints generate mass amplification, how higher winding sectors may lead to irreducible spectral structures associated with composite particles, and how related mechanisms may leave large-scale signatures in the form of gravitational spectral wakes.

Finally, the same spectral framework provides a geometric interpretation of flavor structure and CP violation as arising from incompatible spectral extractions within the same projection fiber.

11.1 Spectral Stability and the Unit of Mass

Within the admissibility framework developed in Chapter 6, rest mass appears as a spectral invariant associated with the stability of projected configurations.

In the effective description, particle states correspond to stable modes supported by the projection fiber Π. Their mass is determined by the eigenvalues of the scalar Laplacian $\Delta_G^{(0)}$ acting on this space, subject to topological constraints $\mathcal{T}$:

$$m^2c^2 \;=\; \lambda_{\mathcal{T}} \;\equiv\; \mathrm{Eig}\Big(\Delta_G^{(0)}\Big)\Big|_{\mathcal{T}}. \tag{138}$$

The quantity $\lambda_{\mathcal{T}}$ represents the minimal spectral cost required to maintain the corresponding configuration within the admissible projective sector. Mass therefore measures the resistance of a configuration to projective ordering advancement.

The electron mass m_e corresponds to the lowest non-trivial admissible eigenmode λ_1 of the fiber geometry $\Pi \cong S^3$. All other particle masses can then be expressed relative to this fundamental scale,

$$m \;=\; m_e\sqrt{\frac{\lambda_{\mathcal{T}}}{\lambda_1}}. \tag{139}$$

11.2 Non-Commutativity as a Source of Mass

Within the spectral framework established in Chapter 6, torsion acts as a dynamical constraint that competes with relaxation transport through the projection fiber.

Inhibition of Projective Ordering

Let Ω_w denote the effective torsion operator associated with winding number w. For the fundamental lepton configuration ($w = 1$), torsion and projective ordering remain spectrally compatible:

$$[\Delta_G^{(0)}, \Omega_1] = 0. \tag{140}$$

For higher-winding configurations ($w \geq 2$), torsional constraints become spectrally frustrated:

$$[\Delta_G^{(0)}, \Omega_w] \neq 0 \qquad (w \geq 2). \tag{141}$$

This non-commutativity inhibits uniform projective ordering across the projection fiber and induces an effective spectral compression, which manifests as increased inertial mass.

The torsional contribution can be expressed through the spectral invariant

$$\mathcal{A}(w) \equiv \frac{1}{2} \ln\left(\frac{\det_\zeta(\Delta_G^{(0)} + \Omega_w)}{\det_\zeta(\Delta_G^{(0)})}\right), \tag{142}$$

where $\det_\zeta$ denotes zeta-regularized determinants.

Dynamical Stability of Torsional Frequencies

For $w = 2$, the projection fiber becomes dynamically anisotropic and splits into competing sectors

$$\Pi = \Pi_\parallel \oplus \Pi_\perp.$$

The quantities $\lambda_\parallel$ and $\lambda_\perp$ are effective torsional rotation frequencies generated by Ω_2 within the first multiplet. They are extracted from the commutator flow on observables restricted to $\mathcal{H}_{k=1}$.

Dynamical stability favors frequency ratios that remain robust under perturbations. In analogy with KAM-type criteria, the most stable admissible ratio corresponds to the golden ratio

$$\frac{\lambda_\parallel}{\lambda_\perp} = \varphi, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{143}$$

The appearance of φ therefore reflects a stability selection rule rather than a static spectral extremum.

Leptonic Spectrum Synthesis

The leptonic mass hierarchy combines two distinct mechanisms:

- a topologically determined spectral invariant

$$\frac{\lambda_2}{\lambda_1} = \frac{8}{3},$$

- a dynamical stability selection rule governed by torsional non-commutativity.

For the muon this leads to the parameter-free relation

$$\boxed{\frac{m_\mu}{m_e} = \sqrt{\frac{\lambda_2}{\lambda_1}} \cdot \frac{3}{2\alpha} \cdot \frac{1}{\varphi}} \qquad \text{with} \qquad \frac{\lambda_2}{\lambda_1} = \frac{8}{3}. \tag{144}$$

The robustness of $\lambda_2/\lambda_1 = 8/3$ is demonstrated independently in Appendix D.7.

Outlook.
The same commutator-driven torsional mechanism extends naturally to higher winding sectors ($w \geq 3$), providing a route to the mass scales of composite particles such as the proton.

11.3 Irreducible Torsion and Higher Winding Sectors

Within the spectral framework established in Chapter 6, the condition $[\Delta_G^{(0)}, \Omega_w] \neq 0$ signals spectral frustration for all $w \geq 2$. However, non-commutativity alone does not distinguish the qualitative structure of different higher-winding sectors. A further structural criterion is required.

From Anisotropy to Irreducibility

In the $w = 2$ sector, torsion induces anisotropy within the projection fiber, leading to a controlled splitting $\Pi = \Pi_{\parallel} \oplus \Pi_{\perp}$. The associated dynamical selection principle therefore operates within a reducible decomposition of the first spectral multiplet.

For $w \geq 3$, the situation may be qualitatively different. The torsional operator Ω_w can act on the admissible subspace in a manner that is not decomposable into independent lower-winding sectors.

Formally, let

$$\mathcal{A}_w \equiv \mathrm{alg}\Big(\Delta_G^{(0)},\, \Omega_w\Big)\Big|_{H_{\mathrm{adm}}^{(w)}} \tag{145}$$

denote the algebra generated by the projective ordering operator and torsion, restricted to the admissible configuration space.

If the commutant

$$\mathcal{A}'_w = \big\{X \in \mathrm{End}(H_{\mathrm{adm}}^{(w)}) \mid [X, A] = 0 \ \forall A \in \mathcal{A}_w\big\} \tag{146}$$

reduces to scalar multiples of the identity, the torsional action is irreducible in the admissible sector.

Irreducibility implies that no non-trivial projector exists that simultaneously preserves admissibility and diagonalizes the torsional action. In that case a naive decomposition $\Omega_w \simeq \Omega_1^{\oplus w}$ is not dynamically accessible within $H_{\mathrm{adm}}^{(w)}$.

Operationally, irreducibility can be tested by computing the dimension of the commutant $\mathcal{A}'_w$. If $\dim \mathcal{A}'_w = 1$, the torsional sector is algebraically irreducible within $H_{\mathrm{adm}}^{(w)}$.

Admissibility as a Structural Constraint

The admissible subspace $H_{\mathrm{adm}}^{(w)}$ is defined by the bounded-admissibility condition. Factorizations that would decompose Ω_w into independent lower-winding components may require intermediate configurations that violate admissibility.

In such cases irreducibility is not an additional axiom but a consequence of bounded admissibility: the decomposition path exits the admissible domain and is therefore dynamically forbidden.

This mechanism provides a structural explanation for why higher winding sectors may exhibit stronger mass amplification than the $w = 2$ case. The obstruction is therefore not merely energetic but algebraic.

Spectral Consequences for $w = 3$

The $w = 3$ sector is the first case in which genuine algebraic irreducibility may arise. If $\mathcal{A}'_3$ is trivial within $H^{(3)}_{\mathrm{adm}}$, the torsional action cannot be reduced to a sum of independent leptonic sectors.

In that regime the relevant mass scale is controlled by a global spectral invariant rather than by a two-mode dynamical selection rule. A natural candidate is the torsional action $\mathcal{A}(3)$ defined in Eq. (142), or an associated canonical ratio extracted from the spectrum of $\Delta^{(0)}_G + \Omega_3$ restricted to $H^{(3)}_{\mathrm{adm}}$.

Identifying such an invariant constitutes the next structural step beyond the leptonic sector.

11.4 Baryonic Sector and Spectral Hierarchy

Within the spectral admissibility framework established in Chapter 6, the $w = 3$ configuration is the first sector in which algebraic irreducibility becomes structurally possible within the admissible subspace. If no non-trivial projector simultaneously preserves admissibility and diagonalizes $\mathcal{A}_3$, the resulting mass scale cannot be interpreted as a superposition of lower-winding contributions.

From Dynamical Selection to Global Invariants

In the $w = 2$ sector, mass amplification arises from a dynamical selection rule acting within a reducible two-mode structure. The Pisano ratio φ enters as a stability condition on the torsional generator restricted to the first multiplet.

For $w = 3$, such a two-mode reduction need not exist. If the torsional action is algebraically irreducible in $H^{(3)}_{\mathrm{adm}}$, no analogue of the Pisano selection rule applies, because no non-trivial invariant subspace remains on which a two-mode frequency selection could operate. The relevant observable must then be a global spectral invariant.

A natural candidate is the torsional action $\mathcal{A}(3)$ defined in Eq. (142), or an invariant ratio extracted canonically from the spectrum of $\Delta^{(0)}_G + \Omega_3$ restricted to $H^{(3)}_{\mathrm{adm}}$. Such an invariant should be independent of overall amplitude rescalings of Ω_3 and stable under increasing spectral truncation.

Mass Hierarchy as Spectral Scaling

If the $w = 3$ sector is irreducible, the baryonic mass scale is controlled by the spectral distortion induced by Ω_3. In analogy with the leptonic sector one expects

$$\frac{m_{w=3}}{m_{w=1}} \sim \sqrt{\frac{\lambda^{(3)}_{\mathrm{bind}}}{\lambda_1}}. \tag{147}$$

Unlike the leptonic case, this ratio would not be determined by a two-mode dynamical fixed point, but by the global spectral structure of the irreducible torsional sector.

Programmatic Outlook

The structural task is therefore twofold:

(i) determine whether $\mathcal{A}_3'$ is trivial within $H^{(3)}_{\mathrm{adm}}$,
(ii) identify a canonical spectral invariant of $\Delta_G^{(0)} + \Omega_3$ that is stable under admissible deformations.

If such an invariant exists and does not depend on parameter tuning, the baryonic mass scale would emerge as a structural consequence of spectral irreducibility rather than as phenomenological input.

Structural Implication for the Proton–Electron Hierarchy

Under these conditions the proton–electron mass hierarchy could be interpreted as the first manifestation of an irreducible torsional sector in the relational spectrum. In that case the observed ratio m_p/m_e would reflect a structural signature of spectral non-factorizability rather than an independent dynamical constant.

11.5 Gravitational Shadows and the Spectral Wake

Within the torsional framework described above, the constraint Ω_w induces an extended deformation of the local spectral structure of the χ substrate. This deformation modifies the local density of admissible projective modes and may persist over scales larger than the core configuration.

We refer to this persistent redistribution of projective capacity as a *spectral wake* or gravitational shadow. It does not correspond to additional matter or propagating degrees of freedom, but to a long-lived modification of the relaxation structure induced by the torsional configuration.

Such spectral wakes provide a possible structural interpretation for phenomena commonly attributed to dark matter, without requiring the introduction of new particle species.

Two qualitative features naturally emerge.

Elastic remanence.

The spectral deformation may persist under displacement of the baryonic configuration. In this regime the wake behaves as a form of structural memory of the substrate, which could account for the observed offsets between baryonic and effective gravitational components in systems such as the Bullet Cluster.

Non-local susceptibility.

When the local relaxation gradient becomes sufficiently weak, the substrate response may transition from a linear to a nonlinear regime. A characteristic acceleration scale of order $a_0 \sim cH_0$ then appears as the natural scale at which this transition occurs, leading to phenomenology similar to MOND-like modifications of gravitational dynamics.

Within the Cosmochrony framework, such effects would therefore reflect a persistent redistribution of projective capacity in the relational substrate rather than the presence of unseen matter.

11.6 Flavor Structure and CP Violation

Within the spectral framework established in Chapter 6, the interpretation of mass developed above naturally extends to the structure of flavor. Distinct fermionic generations correspond to spectrally separated projected modes supported by the same projection fiber Π, whose geometry was introduced in Section 5.1. The associated eigenvalues determine the projective energy of the projected configurations and are therefore directly related to the inertial masses introduced in Section 11.1.

Spectral non-degeneracy and the $n = 3$ threshold.

Let $\{\lambda_i\}$ denote the spectral eigenvalues governing three distinct relaxation modes. If two eigenvalues become degenerate,

$$\lambda_i = \lambda_j,$$

the corresponding subspace admits a rotational freedom that allows any accumulated phase to be absorbed by a redefinition of the projected basis. In that case no irreducible CP-odd invariant remains.

A non-trivial CP-violating sector therefore requires at least three spectrally distinct modes. The threshold $n \geq 3$ signals that the projective holonomy of the flavor sector can no longer be embedded in a two-dimensional manifold. An irreducible geometric phase becomes unavoidable once the spectral complexity of the fiber exceeds this minimal bound.

Dual spectral hierarchies.

In the Standard Model, CP violation arises from the mismatch between the up-type and down-type quark sectors. Within Cosmochrony this structure can be interpreted as the non-alignment of two sectorial spectral restrictions of the same projection fiber.

Let Π_u and Π_d denote the effective operators selecting the up-type and down-type projected modes from the admissible spectral subspace of $\Pi \cong S^3$. These operators do not define independent geometries; they act within the same projection fiber but correspond to distinct spectral extractions.

Because the projected state U has finite descriptive capacity, characterized by the bandwidth constraint b_χ, the two extraction channels cannot in general be simultaneously diagonalized. This induces a structural non-commutativity,

$$[\Pi_u, \Pi_d] \neq 0,$$

reflecting the operational mismatch between the two flavor sectors.

Cubic commutator invariant.

The physically relevant CP-odd quantity can then be expressed through the imaginary part of the trace of the cubic commutator,

$$\mathcal{J} \propto \operatorname{Im}\operatorname{Tr}\big([\Pi_u, \Pi_d]^3\big). \tag{148}$$

The cubic order is minimal for which the invariant vanishes identically in a two-generation system and becomes non-trivial only when three independent generations are present. If the sectorial operators commute, $[\Pi_u, \Pi_d] = 0$, then $\mathcal{J} = 0$ and CP symmetry is effectively restored.

When combined with the spectral separations $\lambda_{u,i}$ and $\lambda_{d,i}$, the invariant scales structurally as

$$\mathcal{J} \propto \operatorname{Im}\operatorname{Tr}([\Pi_u, \Pi_d]^3) \prod_{i<j}(\lambda_{u,i}^2 - \lambda_{u,j}^2) \prod_{k<l}(\lambda_{d,k}^2 - \lambda_{d,l}^2).$$

Degeneracy in either sector suppresses the invariant, recovering the structural constraint known from the Jarlskog formulation of the Standard Model.

Magnitude of the invariant.

The experimentally observed smallness of the Jarlskog invariant, $\mathcal{J} \sim 3 \times 10^{-5}$, suggests that the flavor sector operates near a *quasi-commutative regime* of the projection algebra.

Let the misalignment between the up-type and down-type operators be characterized by a dimensionless parameter $\epsilon \ll 1$ such that

$$[\Pi_u, \Pi_d] \sim \epsilon\, K,$$

where K denotes a normalized operator in the flavor algebra. Since the CP-odd quantity is cubic in the commutator,

$$\mathcal{J} \sim \mathcal{O}(\epsilon^3). \tag{149}$$

This cubic dependence provides a geometric mechanism for suppressing CP violation: even a modest primary misalignment (e.g. $\epsilon \sim 10^{-2}$) can produce an invariant of order $\mathcal{J} \sim 10^{-6}$, compatible with the observed hierarchy.

Structural interpretation.

Within Cosmochrony, CP violation is therefore not introduced as an independent complex phase. Rather, it appears as a geometric residue of incompatible spectral extractions from a single relational substrate.

The non-commutativity considered here differs from the torsion–relaxation mismatch discussed in Section 11.2. There the commutator acts within a single topological excitation and manifests as inertial mass, whereas here it arises between distinct sectorial spectral restrictions of the same projection fiber and manifests as a CP-odd geometric phase. Both effects reflect different aspects of the same finite projective capacity constraint.

12 Testable Predictions and Observational Signatures

Numerical estimates presented below are order-of-magnitude consistency estimates derived from the geometric coupling between the χ substrate and the effective projective fraction Ω_χ. Their role is to demonstrate that the framework operates within a phenomenologically relevant regime.

12.1 Hubble Constant from Projective Structure

The Hubble parameter arises as an effective quantity:

$$H(t) = \frac{\dot{\chi}}{\chi}. \tag{150}$$

Assuming $\dot{\chi}_{\rm eff} \simeq c$, the present-day value is $H_0 \simeq c/\chi(t_0)$. Early- and late-universe probes sample χ at different projective ordering stages, naturally leading to systematically different inferred values of H_0 without invoking additional cosmological components.

Resolution of the Hubble tension.
The effective $H(z)$ acquires a mild redshift dependence departing from ΛCDM at intermediate redshifts ($0.1 \lesssim z \lesssim 10$), testable through upcoming BAO and supernova surveys.

12.2 Redshift Drift

Monotonic projective ordering implies a slowly evolving cosmological redshift. The effective drift rate is

$$\dot{z}_{\rm eff} \ \sim \ H_0(1+z) - \frac{c}{\chi(t)}, \tag{151}$$

corresponding to a secular variation of order $\Delta z \sim 10^{-10}\,\mathrm{yr}^{-1}$ at $z \sim 1$, differing from ΛCDM at the $\sim 10\%$ level. Future extremely large telescopes equipped with ultra-stable spectrographs may probe this effect.

12.3 Gravitational Wave Propagation

In regions of high excitation density near compact objects, local suppression of the projective ordering rate modifies the persistence and coherence of gravitational-wave projective descriptions. This effect induces frequency-dependent phase shifts or amplitude suppression, rather than purely dissipative losses.

We predict a relative amplitude deviation from general relativistic templates scaling as

$$\frac{\Delta A}{A} \sim \left(\frac{r_s}{r}\right)^2,$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius and r the effective propagation radius of the dominant ringdown mode.

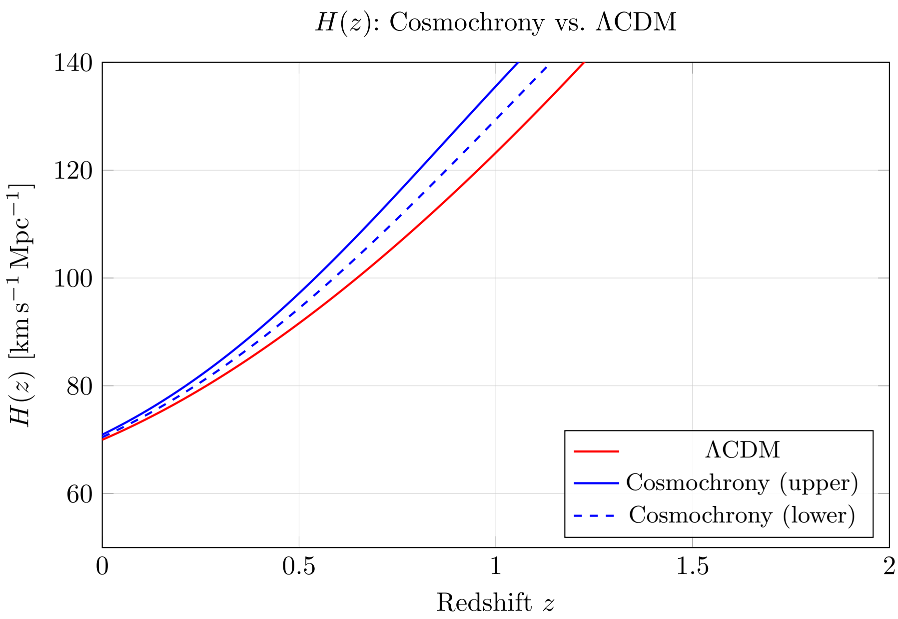

Fig. 13 Schematic comparison of $H(z)$. Cosmochrony predicts a mild enhancement at intermediate redshifts due to projective capacity inhomogeneities. The Cosmochrony band corresponds to the admissible interval in (Ω_χ, δ) entering the bounded-response mechanism, with $a_\star$ fixed independently; it represents structural uncertainty rather than parameter fitting.

For propagation at $r \approx 10\, r_s$, this yields

$$\frac{\Delta A}{A} \sim 10^{-2},$$

corresponding to a percent-level suppression of coherent gravitational-wave amplitudes relative to general relativity.

This deviation should manifest most clearly during the late-time ringdown phase of binary black hole mergers as a systematic, frequency-dependent mismatch with general relativistic ringdown templates. The effect preserves luminal propagation speed but alters coherence structure.

Falsifiability condition. If future high signal-to-noise detections (SNR $\gtrsim 100$) by space-based observatories such as LISA constrain $\Delta A/A < 10^{-3}$ at $r \sim 10\, r_s$, the present scaling law would be excluded at the predicted amplitude level.

12.4 Galaxy Rotation Curves from Projective Capacity Gradients

Persistent projective capacity gradients in the effective description modify the effective inertial response of orbiting matter, producing approximately flat rotation curves without introducing dark matter components (Section 8.13). Observable consequences include a correlation between rotation curve flattening and indicators of structural projective activity, deviations from baryonic scaling relations in dynamically young galaxies, and a reduced need for fine-tuned dark matter profiles in low-surface-brightness systems.

12.5 Spin and Topological Signatures

If spin originates from topologically nontrivial configurations of χ (Section 4.6), ultra-high-precision interference experiments sensitive to 4π rotational symmetry may probe deviations associated with the internal topology of localized projectable configurations. Such deviations would appear as extremely small phase shifts under closed 2π versus 4π rotational cycles. These signatures are expected to be strongly suppressed and lie beyond current experimental resolution.

The absence of isolated magnetic monopoles constitutes a structural prediction of the framework, independent of any dynamical fine-tuning. The bounded-flux admissibility constraint reduces the Hopf base from the full S^2 to a topological annulus with $\pi_2 = 0$, removing the topological support for monopole-type excitations at the level of the canonical admissible model (Section 5.3 and [33]). This contrasts with the standard Dirac – 't Hooft – Polyakov picture, in which monopoles are topologically admissible and their non-observation requires dynamical explanation.

12.6 Dark Energy as Emergent Projective Pressure

Because cosmic acceleration emerges as a geometric consequence of the global growth of projected capacity, no independent dark energy component is introduced. Vacuum energy is not absent from the framework: it arises as the residual projective capacity of configurations not yet stably locked by the projection — the portion of the non-injective fibre that remains unresolved at each cascade step (Section E.10). What the framework does not introduce is a vacuum energy *tuned independently* to explain cosmic acceleration, nor an additional propagating degree of freedom responsible for it.

However, the absence of an independent dark energy *component* does not imply a constant equation of state. The projective cascade exhibits two structurally distinct growth regimes (Section 8.5): a polynomial regime governed by individual BFS shell growth ($|S_n| \sim n^3$, homogeneous dimension $D_{\text{hom}} = 4$), and an exponential regime governed by the branching of distinct admissible trajectories. The effective equation-of-state parameter $w(z)$ evolves across the transition between these two regimes: near $w < -1$ when exponential branching dominates at early times, relaxing toward $w > -1$ as the admissible neighbourhood $\mathcal{A}(U_n)$ becomes progressively constrained by accumulated projective history. This structural prediction is compatible with recent DESI DR2 evidence favouring dynamical dark energy over a cosmological constant [61].

The key observational discriminators of the projective mechanism from standard dark energy models are therefore:

- no independent clustering component or fifth force associated with a dark energy field;
- a specific evolution of $w(z)$ tied to the polynomial-to-exponential transition of the admissible projective volume, rather than to a scalar potential;

- the combined absence of an inflationary tensor imprint at large angular scales and suppressed low-ℓ CMB power, as signatures of the same projective mechanism governing both early and late acceleration.

12.7 CMB Polarization Signatures (Outlook)

Residual large-scale projective correlations inherited from the pre-geometric projective structure of χ are expected to imprint scale-dependent signatures on the CMB (Section 8.7). The observed $\sim 10\%$ suppression of the CMB quadrupole power relative to ΛCDM arises naturally from this projective structure [62]. Appendix C.1 provides quantitative estimates.

The* 8/3 *Scaling in CMB Polarization.

The fundamental ratio $\lambda_2/\lambda_1 = 8/3$ is expected to leave a structural signature in the polarization sector. The bare geometric tensor-to-scalar ratio is

$$r_0 = \frac{\Delta_t^2}{\Delta_s^2} = \frac{\lambda_{\text{base}}}{\lambda_{\text{fiber}}} = \frac{3}{8} \simeq 0.375. \tag{152}$$

The observed ratio undergoes topological decoherence:

$$r_{\text{obs}}(t) = r_0 \cdot \exp\left(-\zeta \frac{\tau_\chi}{t}\right), \tag{153}$$

providing a structural explanation for the low observed $r < 0.036$ without invoking slow-roll dynamics.

12.8 Neutrino-Mediated Projective Reorganization and Decay Signatures

Particle decay and neutrino emission are manifestations of structural reorganization (Sections 4.4, 4.9). Neutrino-like excitations act as non-local projective channels, contributing to irreversible smoothing of admissible configurations.

Environmental Modulation of Particle Stability

12.9 Environmental Modulation of Particle Lifetimes

Particle stability may exhibit a weak environmental dependence. At leading order:

$$\frac{\delta\Gamma}{\Gamma} \simeq \beta \frac{\Delta U}{c^2}, \tag{154}$$

where $\beta \lesssim 10^{-6}$ (from local position invariance constraints) and $\Delta U/c^2 \sim 10^{-7}$–$10^{-6}$ for typical galactic environments, yielding

$$\frac{\delta\tau}{\tau} \sim 10^{-13}\text{–}10^{-12}, \tag{155}$$

well below current sensitivities but conceptually distinct from standard effects. The cleanest experimental target is leptonic weak decay; the most amplified interferometric target is neutral-meson mixing. Near compact objects ($\Delta U/c^2 \sim 10^{-4}$), the effect rises to $\delta\tau/\tau \sim 10^{-10}$, manifesting as environment-correlated spectral biases in hadronic cascades.

12.10 Strong Gravitational Lensing

The effective lensing potential decomposes as $\Phi_{\text{eff}} = \Phi_{\text{bar}} + \Phi_\chi$, with convergence

$$\kappa(\boldsymbol{\theta}) = \frac{D_l D_{ls}}{c^2 D_s} \int \nabla_\perp^2 \Phi_{\text{eff}}(D_l \boldsymbol{\theta}, z)\; dz, \tag{156}$$

yielding $\kappa = \kappa_{\text{bar}} + \kappa_\chi$. The emergent contribution is parametrized as

$$\Phi_\chi(\mathbf{x}) = \frac{c^2}{2} \ln\!\left(\frac{K(\bar{\chi}(\mathbf{x}))}{K_\infty}\right). \tag{157}$$

Testable signatures include partial decorrelation between baryonic mass and lensing strength, enhanced sensitivity to cluster morphology, strong lensing without dark matter substructure, and redshift dependence tracing projective ordering rather than mass accretion history.

Early Massive Structures in the JWST Era

Massive galaxies and strong lensing at $z \gtrsim 8$ are naturally accommodated: effective mass reflects localized resistance to projective ordering advancement associated with spectrally robust projected configurations, rather than requiring rapid hierarchical assembly.

“Impossibly Early” Galaxies

Their large effective masses encode strong resistance to projective ordering advancement rather than cumulative accretion. Cosmochrony predicts that such galaxies should exhibit relatively stable effective masses over extended redshift intervals, contrasting with the rapid growth required in hierarchical scenarios.

Qualitative Predictions

Massive galaxies at $z \gtrsim 8$ should persist with only moderate mass evolution. Early strong-lensing configurations should display weak redshift evolution of their effective convergence, with an early onset followed by moderate evolution.

Strong Gravitational Lensing in Abell-1689

The emergent convergence is isolated as

$$\kappa_\chi(\boldsymbol{\theta}) = \kappa_{\text{eff}}(\boldsymbol{\theta}) - \kappa_{\text{bar}}(\boldsymbol{\theta}), \tag{158}$$

with the emergent lensing potential obtained from $\nabla^2_{\boldsymbol{\theta}}\psi_\chi = 2\kappa_\chi$. In Cosmochrony, κ_χ is emergent geometric focusing from collective χ-substrate constraints, not additional dark matter.

Quantitative Scaling and Falsifiability

Beyond qualitative decorrelation effects, the emergent contribution κ_χ admits an order-of-magnitude scaling in saturated cluster cores.

We parameterize the relative excess convergence as

$$\frac{\kappa_\chi}{\kappa_{\rm bar}} \sim \gamma\left(\frac{\Sigma_{\rm bar}}{\Sigma_*}\right), \tag{159}$$

where $\Sigma_{\rm bar}$ is the baryonic surface density, Σ_* denotes a characteristic structural projective capacity scale, and γ is a dimensionless response coefficient encoding substrate efficiency.

For massive clusters with $\Sigma_{\rm bar} \sim \Sigma_*$, this yields

$$\frac{\kappa_\chi}{\kappa_{\rm bar}} \sim \gamma, \tag{160}$$

with a natural expectation $\gamma \sim 10^{-2}$–10^{-1}, corresponding to percent-to-ten-percent level decorrelation between baryonic mass distribution and total lensing convergence.

This deviation should manifest as a systematic mismatch between high-resolution baryonic mass reconstructions and gravitational lensing maps, particularly in morphologically complex cluster cores such as Abell-1689.

Falsifiability condition. If future lensing reconstructions constrain $\kappa_\chi/\kappa_{\rm bar} < 10^{-2}$ in saturated cluster environments, the present scaling law would be excluded at the predicted amplitude level.

12.11 Superconducting Gap Symmetry from Fiber Geometry

The collective coherent regime of the $U(1)$ fiber discussed in Section 5.8 leads to concrete and falsifiable predictions for the symmetry of the superconducting gap. In the present framework, gap symmetry is not determined by a specific momentum-space interaction kernel, but by a real-space minimization problem in the fiber matching cost under lattice constraints.

Geometric Selection Principle

Let $\mathcal{C}_{\rm tot}[\theta]$ denote the effective cost functional associated with fiber phase mismatch in the presence of lattice symmetry group $\mathcal{G}$ and dominant frustration wavevector $\mathbf{Q}$. Admissible composite configurations must satisfy simultaneously:

1. minimization of overlap with the dominant frustrated sector,
2. global continuity of the fiber phase without introducing additional defects.

Among the irreducible representations of $\mathcal{G}$ available to a $w = 2$ composite class, the physically realized superconducting gap corresponds to the representation that minimizes $\mathcal{C}_{\text{tot}}$.

This selection principle is independent of microscopic pairing details. It depends only on lattice symmetry and the dominant frustration structure.

Prediction for Cuprate Superconductors

In cuprate materials, the relevant lattice symmetry is D_{4h} and the dominant frustration vector is approximately (π, π), as measured through the magnetic structure factor $S(\pi, \pi)$.

Under these constraints, the minimization of the fiber matching cost selects the B_{1g} irreducible representation. The resulting gap symmetry is therefore

$$\Delta(\mathbf{k}) \propto \cos k_x - \cos k_y, \tag{161}$$

corresponding to $d_{x^2-y^2}$ symmetry.

This prediction does not rely on a specific spin-fluctuation mechanism. It follows from geometric frustration in the projected fiber sector. The nodal lines arise from the required sign alternation that reduces overlap with the (π, π) frustration sector while preserving global fiber continuity.

Prediction for Nickelate Superconductors

In infinite-layer nickelates, the antiferromagnetic frustration sector is reduced and partially isotropized relative to cuprates. Let r_F denote a dimensionless proxy for the relative frustration amplitude, operationally linked to the normalized structure factor.

For sufficiently reduced staggered frustration ($r_F < r_F^{\text{crit}}$), the minimization of $\mathcal{C}_{\text{tot}}$ favors an extended A_{1g} representation with sign-changing structure across Fermi surface sheets,

$$\Delta(\mathbf{k}) \sim s^{\pm}. \tag{162}$$

The framework therefore predicts a symmetry transition between $d_{x^2-y^2}$ and extended $s^{\pm}$ depending on the frustration ratio r_F. This constitutes a direct and falsifiable structural prediction.

Disorder Sensitivity

Because sign-changing order parameters are sensitive to non-magnetic disorder, the predicted $s^{\pm}$ regime in nickelates should display intermediate sensitivity to impurity scattering, weaker than pure d-wave cuprates but stronger than conventional isotropic s-wave superconductors.

Systematic impurity studies therefore provide a discriminating test of the fiber-geometry selection mechanism.

Scaling of the Critical Temperature

To leading order, the phase stiffness scales as

$$\rho_s(0) \propto \delta J f(r_F), \tag{163}$$

where δ is the carrier concentration, J the dominant frustration scale, and $f(r_F)$ a dimensionless function encoding the geometric reduction of frustration.

In quasi-two-dimensional systems,

$$k_B T_c \simeq \frac{\pi}{2}\rho_s(T_c^-), \tag{164}$$

yielding

$$T_c \sim \frac{\pi}{2k_B}\delta J f(r_F). \tag{165}$$

All quantities entering this relation are independently measurable through spectroscopy, transport, and neutron scattering. No free fitting parameter specific to superconductivity is introduced.

Falsifiability

The present mechanism would be falsified if:

- cuprates exhibited a robust fully isotropic s-wave gap at optimal doping despite strong (π, π) frustration,
- nickelates with reduced staggered frustration showed stable $d_{x^2-y^2}$ symmetry across all doping levels,
- the critical temperature displayed no correlation with independently measured exchange or frustration scales.

The symmetry channel is therefore not an adjustable ingredient. It is fixed by lattice group and frustration geometry, providing a sharp structural test of the framework.

12.12 Experimental Outlook and Discriminating Signatures

Particle creation is reinterpreted as a universal dissipation channel activated whenever directional projective ordering approaches its admissibility ceiling. This unifies several high-energy phenomena:

Astrophysical Jets.

Relativistic jets are dynamically selected channels through which excess substrate tension is discharged by continuous structure creation. A predicted *saturation clamping* near the horizon—a non-linear relation between near-horizon stress and plasma density—is testable with Event Horizon Telescope observations.

Primordial Cosmology.

Matter production emerges as a byproduct of projective-ordering-driven expansion. Cosmochrony predicts that primordial fluctuations may carry imprints of discrete nucleation events, suggesting specific non-Gaussianity and phase correlations at large angular scales.

Ultra-High-Energy Cosmic Rays.

UHECRs exceeding the GZK threshold may be locally produced during transient saturation spikes near compact objects. A predicted strong anisotropy correlated with nearby compact objects and multimessenger correlations would provide a decisive signature.

Multi-Scale Falsification Program.

The framework suggests a hierarchy of tests: precision atomic physics (Section 4.10), Schwinger threshold in ultra-intense lasers, saturation clamping in relativistic jets, and large-scale CMB anomalies as relics of discrete primordial nucleation.

12.13 Summary

Cosmochrony yields observationally testable signatures spanning cosmology, gravitation, galactic dynamics, and particle phenomena. All arise from non-injective projection and admissible projective ordering. Predicted deviations from ΛCDM at the percent level include suppression of low-ℓ CMB power, a mild redshift-dependent modulation of $H(z)$, and coherence-modifying effects in gravitational-wave propagation near compact objects. These signatures define a coherent phenomenological pattern from a single underlying mechanism—the irreversible projective ordering and partial projectability of the effective description.

13 Discussion and Comparison with Existing Frameworks

13.1 Relation to General Relativity

Spacetime curvature in Cosmochrony is an emergent descriptive construct (Section 7.3). No *a priori* metric dynamics is postulated; instead, an effective geometry emerges from variations in the local projective ordering rate. Matter configurations locally saturate projective capacity, leading to differential rates of effective proper-time advancement that are reinterpreted as effective metric deformations. In the weak-field regime, Newtonian gravity is reproduced; in the strong-field limit, Schwarzschild-like solutions emerge. General Relativity is recovered as the appropriate effective theory within its empirically validated domain.

13.2 Interpretation of Quantum Phenomena

Cosmochrony does not treat quantization or wave dynamics as fundamental [63]. As discussed in Section 9.12, the quantum formalism arises as an effective description of projected relational dynamics. Particles correspond to localized, topologically stable configurations; discrete observables arise from boundary conditions, topological constraints, and interaction-induced reprojection. The Planck relation $E = h\nu$ is a geometric correspondence between redistributed projective capacity and minimal projective resolution. Entanglement reflects a shared, non-factorizable χ configuration persisting only within a finite critical regime of projection, while decoherence reflects irreversible loss of relational accessibility.

13.3 Common Projective Origin of Quantum and Gravitational Structure

Sections 13.1 and 13.2 treat the recovery of general relativity and quantum mechanics separately. It is worth making explicit that both emerge from the same structural source: the non-injectivity of Π and the Born–Infeld admissibility bound, acting on the same static substrate χ.

On the quantum side, non-injectivity implies that any observable o carries an irreducible fiber $\Pi^{-1}(o)$ of indistinguishable pre-images. The effective description, having no access to χ directly, must represent the coexistence of projectively admissible contributions internally. This forces a complex, norm-preserving, additive structure—the Hilbert space framework—and, in the regime where the Born–Infeld bound keeps fluctuations subcritical and the background remains quasi-stable, a unique effective dynamics of Schrödinger type (Section 9.12, Appendix B.12). Unitarity is not postulated: it holds if and only if finite projective resolution and bounded admissible flux hold simultaneously (Theorem B.1, condition C5).

On the gravitational side, the same non-injectivity implies that the projection Π cannot be globally injective, so that the effective metric $g_{\mu\nu}$ is not a primitive object but a descriptive construct encoding variations in the local projective ordering rate (Section 7.3). Matter configurations locally saturate projective capacity, producing differential ordering rates that are reinterpreted as effective curvature. The Born–Infeld bound, which on the quantum side enforces unitarity, on the gravitational side selects the unique infrared dynamics compatible with descriptive invariance and prevents the emergence of arbitrary higher-derivative corrections (Section 5.7).

The two recoveries are therefore not independent achievements of the framework. They are two aspects of a single structural fact: a static substrate χ accessible only through a non-injective projection Π bounded by a Born–Infeld saturation constraint cannot produce an effective description that is simultaneously faithful (injective), linear (quantum), and geometric (relativistic). Each of the three adjectives describes the same compression from a different observational vantage point. The apparent conceptual distance between quantum mechanics and general relativity is a consequence of the fact that they are two effective languages for the same underlying projective structure, optimised for complementary regimes of resolution and scale.

13.4 Analogy with Collective Phenomena in QCD

A useful structural analogy may be drawn with low-energy QCD [64], where fundamental degrees of freedom (quarks, gluons) do not appear as isolated entities in the infrared. The observable spectrum consists instead of collective bound states, whose properties are not transparently reducible to perturbative constituents.

Similarly, Cosmochrony formulates fundamental dynamics solely in terms of χ and its projective structure. Observable quantities arise only after projection into regimes admitting stable effective descriptions. What appear as elementary constituents at a given effective level may represent stable, regime-dependent invariants of the underlying dynamics, in direct analogy with QCD confinement.

The analogy extends beyond particle confinement. In QCD, the vacuum itself admits multiple collective phases, including chiral symmetry breaking and color superconductivity, which emerge from non-perturbative organization of the same underlying degrees of freedom. In Cosmochrony, different physical phenomena likewise correspond to distinct collective regimes of a single substrate.

Gravitational curvature arises as a collective constraint on the local projective ordering rate. Schwinger pair production corresponds to saturation of the directed admissible projective flux. Superconductivity, as discussed in Section 5.8, represents a coherent collective phase of the $U(1)$ fiber sector, in which composite winding classes stabilize and lock globally.

In all these cases, the effective phenomena are not introduced as additional dynamical postulates. They correspond to regime-dependent reorganizations of the same projective structure. The underlying ontology remains unchanged, while the effective description undergoes phase-like transitions analogous to those observed in strongly coupled gauge theories.

This analogy is structural rather than dynamical. Cosmochrony does not import the specific field content of QCD. The comparison serves to clarify how radically different effective laws may arise from a single underlying relational dynamics, without multiplying fundamental entities.

13.5 Comparison with ΛCDM Cosmology

ΛCDM introduces cold dark matter, dark energy, and inflation as effective postulates [9, 65]. In Cosmochrony, cosmic expansion follows from the monotone growth of projected capacity, with $H(t) = \dot{\chi}/\chi$ and $H_0 \sim c/\chi(t_0)$. Dark energy is reinterpreted as the large-scale growth of projected capacity; cosmic acceleration reflects the cumulative advance of projective ordering over cosmological timescales. The coincidence problem and the Hubble tension may admit interpretations in terms of epoch-dependent projective ordering structure rather than new fundamental constituents. Low-ℓ CMB suppression is explored as a structural consequence of projective admissibility constraints rather than a statistical fluctuation.

Table 11 summarizes the conceptual differences between Cosmochrony, ΛCDM, and Loop Quantum Gravity.

13.6 Projective Incompleteness of Standard Physics

The comparison with ΛCDM and the Standard Model can be sharpened beyond the phenomenological level. Both frameworks describe the structure of the observable space O with extraordinary precision, but neither specifies the fundamental space Ω nor the projection $\Pi : \Omega \to O$. This absence is not a contingent gap to be filled by future work; it is a structural incompleteness that each framework signals from within its own internal structure, through three independent mechanisms [20].

(i) Cosmological singularity. The effective metric $g_{\mu\nu}$ ceases to be well-defined for $t \lesssim t_{\text{Planck}}$: the Einstein equations break down and $g_{\mu\nu}$ is not a valid object. A description that fails in a given regime is naturally interpreted as arising from a more fundamental structure whose validity extends beyond that regime. If a projection Π relating such a fundamental level Ω to the effective observables is introduced and satisfies the conditions of genuine emergence (Section 2.5), it must be non-injective. The singularity is therefore not a physical catastrophe but a structural signal: the geometric description has reached the boundary of its projectability domain.

(ii) Renormalisation group. The Standard Model is a Wilsonian effective field theory with ultraviolet cutoff Λ. The projection $\Pi_\Lambda : \Omega_{\text{UV}} \to O_{\text{IR}}$ that integrates out modes above Λ is non-injective by construction: infinitely many UV completions yield the same infrared observables. The renormalisation group is therefore not merely a computational tool but the explicit acknowledgement of a non-injective projection, with $S_\Pi > 0$ as a structural necessity. In the Cosmochrony framework, the Born–Infeld bound replaces the UV regulator with a structurally motivated finite ceiling, rendering renormalisation unnecessary as an independent procedure (Section 10.4).

(iii) Quantum measurement. The measurement map $\Pi_{\text{meas}} : \mathcal{H} \to O_{\text{class}}$ sends distinct quantum states to the same classical outcome. The measurement problem is, in the present language, the question of the structure of Π_{meas} and why it is non-injective. Non-injectivity is not an interpretation added to quantum mechanics: it is the formal content of the projection postulate itself.

These three mechanisms are independent of any hypothesis specific to Cosmochrony; they follow from the internal structure of standard physics alone. Their common conclusion is that ΛCDM and the Standard Model are necessarily projectively incomplete: they describe O without specifying the Ω and Π that define their own observables.

Level	**Standard physics**	**Status**
Observable space O	Fully described	present
Fundamental space Ω	Not specified	absent
Projection Π	Not specified	absent
Fibres $\Pi^{-1}(o)$	Not specified	absent

Table 4 Projective completeness of standard physics. The quantities that ΛCDM and the Standard Model cannot derive from first principles — the fine-structure constant α, the electron-to-proton mass ratio m_e/m_p, the CKM mixing angles — are precisely the invariants of Π that O alone cannot determine: they encode properties of the fibres $\Pi^{-1}(o)$ invisible to any description confined to the effective level.

The incompleteness is not empirical but structural. Standard physics omits the map that defines its own observables. Any candidate unification or quantum-gravity programme that remains within O — however sophisticated — does not complete the description; it extends it. A theory that genuinely completes standard physics must specify Ω, Π, and the non-injective structure of Π [20]. Cosmochrony provides such a specification.

13.7 Historical Admissibility of Projected Degrees of Freedom

The set of effective degrees of freedom depends on the admissibility conditions imposed by the projective ordering state. In the early Universe, only highly coherent, low-complexity global configurations were admissible. As the projective ordering advances, the admissible space progressively enlarges, enabling increasingly localized invariants described as particles, fields, and interactions. The particle spectrum is therefore a historically conditioned outcome of the projective ordering structure, not a timeless feature. This perspective reconciles the emergence of complexity with monotonic entropy increase: the early Universe is characterized by both low entropy and low admissible complexity. Low-ℓ CMB anomalies may be interpreted as relics of this early regime of restricted admissibility.

13.8 Inflation, Horizon Problems, and Initial Conditions

Because χ is a static pre-geometric substrate rather than a dynamically expanding metric field, causal connectivity is not defined in terms of spacetime lightcones at the fundamental level. Large-scale coherence arises from initial relational smoothness and the subsequent monotone advance of projective ordering, without requiring an inflationary phase. The horizon problem is rendered inoperative by the absence of an initially fragmented causal structure at the pre-geometric level. A detailed quantitative treatment of primordial perturbations and their CMB imprint remains an open direction for future work.

13.9 Ontological Parsimony and the Metric

Cosmochrony achieves genuine ontological simplification rather than relabeling. In GR, the metric $g_{\mu\nu}$ is a fundamental tensor field with ten independent components; matter and energy are conceptually distinct from geometry. In Cosmochrony, only χ is fundamental; the metric carries no independent ontological status; matter, energy, and geometry correspond to distinct regimes of the same relational structure.

The frameworks are operationally distinct: GR propagates two tensorial gravitational-wave polarizations from the metric; Cosmochrony has no independent geometric degrees of freedom. Singularities signal breakdown of effective geometric descriptions while χ remains well-defined. Quantization applies only to effective excitations within projectable regimes, not to χ itself [66].

$$\text{Standard:} \quad g_{\mu\nu}\,(\text{geometry}) + \psi\,(\text{matter}) + \Lambda\,(\text{dark energy}), \tag{166}$$

$$\text{Cosmochrony:} \quad \chi\,(\text{single substrate}) \;\longrightarrow\; \{\text{spacetime, matter, expansion}\}. \tag{167}$$

13.10 Relation to the Higgs Mechanism: Emergence from χ Dynamics

The Higgs field and its VEV are reinterpreted as effective low-energy descriptors of a specific projective regime of χ. In this framework, mass generation does not originate from an independent scalar sector, but from the stabilization of locally stable projected configurations that saturate projective capacity.

Structural Transition

Below a critical scale χ_c (homogeneous regime), only massless globally coherent configurations are admissible. Above χ_c (structured regime), localized projected configurations with high spectral stability emerge as massive excitations. The electroweak scale is related through

$$\langle \phi_H \rangle \;\propto\; \frac{\hbar_{\text{eff}}\, c}{\chi_c}, \tag{168}$$

naturally recovering the observed scale for $\chi_c \sim 10^{-18}\,\text{m}$.

Mass Generation as Solitonic Stabilization

Fermion masses scale as $m_f \propto y_f\, \hbar_{\text{eff}}/\chi_c$; gauge boson masses as $m_W \propto g\, \hbar_{\text{eff}}/\chi_c$. These relations reproduce standard Higgs-generated mass terms at the effective level.

Within the spectral interpretation developed in Section 11.6, the Yukawa parameters encode the effective projection of underlying spectral eigenvalues of the admissibility structure. The observed flavor hierarchies and associated CP structure therefore reflect the same spectral organization, rather than introducing independent mass-generating dynamics.

Distinction from Superconducting Mass Generation

A potential source of confusion concerns the appearance of an effective mass term for gauge fields in superconductivity, discussed in Section 5.8. In that regime, global phase locking of $w = 2$ composites produces a London term proportional to A^2, leading to Meissner screening and an effective photon mass inside the medium.

This phenomenon is not interpreted here as fundamental symmetry breaking. It corresponds to emergent projective coherence in a condensed phase of the $U(1)$ fiber sector. The underlying gauge symmetry of the χ substrate remains intact. The apparent mass arises from collective reorganization of degrees of freedom within a bounded projective regime.

By contrast, electroweak mass generation operates in the vacuum structure of the effective theory and is associated with a distinct projective transition at the scale χ_c. Although both mechanisms yield quadratic gauge-field terms at the effective level, their ontological status differs. The superconducting London mass is medium-dependent and disappears above the coherence temperature, whereas electroweak gauge boson masses persist in the vacuum.

The formal similarity between the two mechanisms reflects a deeper structural principle: mass terms emerge whenever a collective projective configuration restricts admissible gauge variations. However, no additional fundamental Higgs-like degree of freedom is introduced in the superconducting case. The effective mass acquisition of the gauge field in superconductivity is therefore a manifestation of emergent projective coherence, not a reinvention of the Higgs mechanism.

Phenomenological Status

No deviation from established collider results is implied at accessible energies. Departures may arise only in extreme regimes where the mapping between solitonic stabilization scales and effective parameters becomes nonlinear. Open challenges include deriving the detailed correspondence between χ soliton spectra and the full Standard Model mass hierarchy.

13.11 Structural Responses to Bounded Projective Capacity

Physical observables arise through a generically non-injective projection $\Pi : \Omega \to O$ from the relational substrate Ω. The χ substrate is governed by a bounded admissible projective flux, characterized by a Born–Infeld-type constraint b_{BI}. This bound limits the admissible transfer of projective capacity and prevents divergence of effective observables.

When the projective flux approaches its admissible bound, the projection enters a saturation regime. In such regimes, distinct underlying configurations become structurally indistinguishable at the level of O. This loss of distinguishability induces an objective structural entropy associated with the projection:

$$S_\Pi = -\sum_{o \in O} \mu(\Pi^{-1}(o)) \log \mu(\Pi^{-1}(o)), \tag{169}$$

which reflects non-injectivity rather than epistemic uncertainty.

The relationship between the fluctuation amplitude and the admissible projective flux defines a hierarchy of five regimes, ranging from the irreducible projective floor to the inadmissible regime beyond the Born–Infeld ceiling. Table 5 provides an overview; each level is developed in the paragraphs below.

Level	Regime	Mechanism	Example
1	Irreducible floor	Minimal projective variance	Zero-point energy $\frac{1}{2}\hbar\omega$
2	Normal fluctuations	λ-induced fibre drift	Vacuum fluctuations
3A	Saturation (incoherent)	Thermodynamic compensation	Solar corona heating
3B	Saturation (coherent)	Coherent projective reorganisation	Superconductivity
4	Beyond BI ceiling	Pair creation as dissipation	Schwinger effect

Table 5 Hierarchy of projective fluctuation regimes. Levels 1 and 4 are structural boundaries: Level 1 is the irreducible floor below which temporal ordering ceases, Level 4 is the inadmissible ceiling above which admissibility is restored by pair creation. Levels 3A and 3B are alternative responses to the same saturation constraint, selected by the coherence of the projected fiber.

Level 1 — Irreducible projective floor.

The Born–Infeld bound does not only cap fluctuation amplitudes from above; it also implies a minimum below which the projection loses its capacity to generate distinguishable successive effective states. This floor corresponds to the minimal projective variance required to maintain $U_{n+1} \neq U_n$: the irreducible resolution limit of Π. In the effective description, it manifests as the zero-point energy $\frac{1}{2}\hbar\omega$ of each mode—not as a physical excitation of the substrate, but as the smallest fluctuation amplitude consistent with a non-stationary projected description. A projection operating exactly at this floor would be locally injective and temporally stationary; as established in Section 10.5, global stationarity is structurally excluded.

Level 2 — Normal fluctuation regime.

Above the floor, the projection operates in the normal fluctuation regime, governed by the λ-induced fibre drift mechanism developed in Section 10.4. Two mechanisms contribute: fibre decorrelation, in which successive fibre slices become uncorrelated without amplitude growth, and projective amplification, in which residual norms grow under unstable directions of the admissible dynamics F. Both remain bounded by $A^{\mathrm{BI}}_{\max}$. This regime accounts for the standard phenomenology of quantum vacuum fluctuations, including the Casimir effect (Section 10.6) and projective spectral noise effects such as the Lamb shift [67].

Level 3A — Thermodynamic compensation.

In incoherent or weakly correlated systems, collective stabilization of the fiber is not available. The unresolved projective flux is therefore absorbed into effective thermodynamic descriptors. Temperature acts as a Lagrange multiplier compressing unresolved relational structure into the observable sector.

Magnetically guided dilute plasmas, such as the solar corona, illustrate this regime. Reduced projectability under strong magnetic structuring leads to an elevated effective temperature, interpretable as a compensatory response to bounded projective capacity rather than as excess local energy density.

Level 3B — Coherent reorganization.

In strongly coherent collective phases, the system may instead reorganize its internal fiber structure so as to preserve maximal projective stability. Rather than inflating thermodynamic variables, the projected description modifies its admissible connection structure.

Superconductivity provides a paradigmatic example. Collective phase locking of the $U(1)$ fiber suppresses internal fluctuations and enforces near-vanishing covariant phase gradients in the bulk. The Meissner effect may therefore be interpreted as the expulsion of incompatible connection curvature, maintaining projective stability within the coherent phase.

Level 4 — Beyond the Born–Infeld ceiling.

When the projective flux exceeds the admissible bound b_{BI}, the projection enters an inadmissible regime: no stable effective description can be maintained at the current flux level. Admissibility is restored by a structural reconfiguration of the projection fiber, manifesting in the effective description as the creation of particle–antiparticle pairs. Pair creation thus acts as a dissipative channel that redistributes excess projective flux into stable, projectable modes, enlarging the space of admissible configurations and restoring the projection below the Born–Infeld ceiling. The Schwinger effect—spontaneous pair production in ultra-strong electric fields [67]—is the paradigmatic instance of this regime. It emerges here not from vacuum fluctuations or Dirac-sea arguments, but as a universal transition between smooth admissible projection and topologically mediated dissipation.

Unified interpretation.

These five levels do not represent distinct physical mechanisms but a single continuous structural response to the bounded projective capacity constraint, resolved differently depending on the amplitude of the flux, the coherence of the fiber, and the dynamical accessibility of collective reorganization.

Thermodynamic variables, geometric curvature, horizon formation, and collective coherence all act as compensatory descriptors of unresolved relational structure. Which response occurs depends on whether collective reorganization of the projected fiber is dynamically accessible.

The bounded character of the admissible projective flux ensures that projection-induced thermodynamic and geometric quantities remain finite. This renders the framework predictive despite the intrinsic non-injectivity of Π, and replaces the ultraviolet divergences of standard quantum field theory—controlled there by renormalisation—with a structurally motivated finite ceiling grounded in the Born–Infeld admissibility constraint.

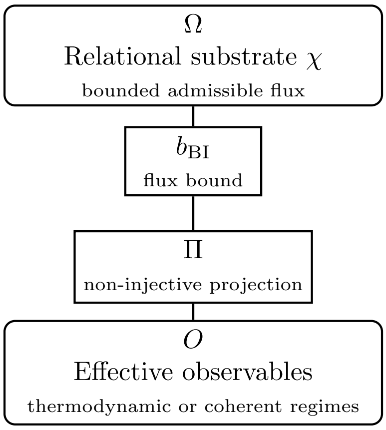

Fig. 14 Structural responses to bounded projective capacity. Approach to the flux bound may induce either compensatory thermodynamic descriptors or coherent projective reorganization, depending on collective accessibility of the fiber structure.

13.12 Bounds on Projective Resolvability Across Scales

Saturation phenomena across scales—bounded gravitational response in low-density environments and the operational accessibility of quantum correlations—reflect a common limitation on projective resolvability. In Cosmochrony, the projected description U does not expose the full relational content of the substrate χ. It exposes only what can be rendered operationally accessible through the non-injective mapping Π under admissible projective transport. The relevant limitation is therefore not a postulated geometric cutoff, but a throughput bound on the rate at which relational structure can be updated while maintaining operational closure of the projected state.

Attosecond chronoscopy experiments illustrate this principle. Measured delays can be interpreted as the minimal temporal resolution required for a non-factorizable projected description to become operationally resolvable, rather than as a dynamical buildup of correlations [56].

We summarize this limitation by an effective inequality on projected update rates,

$$\left| \frac{\partial \mathcal{O}_{\text{eff}}}{\partial \tau} \right| \leq \; b_\chi \, \mathcal{S}_\Pi, \tag{170}$$

where b_χ is the invariant bound on admissible projective transport in the substrate and $\mathcal{S}_\Pi$ parametrizes the effective structural complexity of the projection for the class of observables under consideration. The factor $\mathcal{S}_\Pi$ is not a new dynamical field. It encodes how many independent relational channels can be synchronized within one operational update step.

Distinct phenomenological manifestations correspond to distinct choices of $\mathcal{O}_{\text{eff}}$ and the associated resolution scale. In macroscopic weak-field regimes, the same bound induces an operational saturation threshold that can be expressed as an effective acceleration proxy,

$$a_\star \sim \kappa \frac{b_\chi}{\ell}, \tag{171}$$

where ℓ is the local descriptive resolution of the projected state and $\kappa = \mathcal{O}(1)$ encodes the normalization convention of Π. In measurement-limited quantum protocols, the bound similarly appears as a bandwidth constraint,

$$B_\Pi(\mathcal{M}) = \eta_{\mathcal{M}}\, b_\chi\, \mathcal{S}_\Pi, \tag{172}$$

for a measurement procedure $\mathcal{M}$, with $\eta_{\mathcal{M}}$ absorbing protocol-dependent conventions. These expressions represent different dimensional projections of the same underlying transport bound. A modification of the effective resolvability of the $\chi \to \Pi$ mapping would therefore induce correlated shifts in both galactic saturation scales and quantum chronoscopy thresholds.

A further generic consequence is the divergence of projective stress under sufficiently deep refinement. Let D denote an operational update demand, defined as a characteristic rate at which projected relations must be recomputed to preserve admissibility. Let C denote the admissible projective throughput under the bound b_χ for the same class of updates. The dimensionless stress ratio

$$\Xi \equiv \frac{D}{C} \tag{173}$$

provides a regime indicator. Whenever a process increases D faster than C can be redistributed by the available synchronized channels, Ξ grows and a saturated reconfiguration becomes unavoidable. This statement is independent of any particular emergent geometry and follows solely from bounded projective transport and finite projective resolvability.

13.13 Projective Non-Termination and the Condition of Temporal Ordering

Time is identified with the ordering of distinguishable projected states: $\Delta\tau \sim \mathrm{dist}(U_n, U_{n+1})$. A hypothetical terminal state U_0 satisfying $\Pi(\chi + \delta\chi) = U_0$ for all admissible $\delta\chi$ would imply $\Delta\tau \to 0$ for all internal observers.

The inaccessibility of absolute zero emerges as a projective necessity: zero-point fluctuations are reinterpreted as the minimal projective bandwidth required to prevent collapse of the observable description into a stationary state. Time ceases only when the projection becomes stationary—a regime structurally forbidden by the existence of a finite bound on projective resolvability.

The connection between projective stationarity and injectivity can now be made explicit. A stationary projection corresponds to the limit in which, within any admissible configuration neighborhood, distinct substrate configurations fail to produce distinguishable projected states: no local variation $\delta\chi$ induces a resolvable change in the effective description, so that the accessible neighborhood is mapped into an effectively indistinguishable projected class and no update can be resolved. This is the opposite extreme from injectivity, but leads to the same operational consequence: $U_{n+1} = U_n$, since neither extreme admits a non-trivial ordering of distinguishable projected states. This equivalence unifies three results established independently in this paper: the inaccessibility of absolute zero (Section 5.6), the suppression of vacuum fluctuations in injective regimes (Section 10.5), and the projective non-termination condition stated here. All three express the same structural fact: $S_\Pi > 0$ is a necessary condition for temporal ordering, and $S_\Pi = 0$ is the universal limit at which time, energy, and fluctuations simultaneously vanish. Locally, this limit can be approached—corresponding to quasi-classical or decohered regimes—but it cannot be globally reached without eliminating the effective description itself.

13.14 Conceptual Implications and Open Challenges

Temporal ordering arises from the monotone growth of cumulative projected capacity $I(n)$ along the admissible cascade [10]; energy quantifies residual projective capacity available for further structural advancement; irreversibility reflects the one-way activation of projected modes — a property of spectral modes in the Gram–Schmidt span, compatible with the structural reorganization of coherent configurations into more stable ones (Section 4.4). Time, energy, and irreversibility are not independent axioms but complementary effective descriptions of the same projective structure.

Open challenges include the quantitative reconstruction of CMB anisotropies from early-stage projective ordering structure, the detailed treatment of non-equilibrium decoherence and reprojection, the emergence of gauge symmetries from topological features, the explicit derivation of the full flavor hierarchy and the quantitative determination of the CP-violating invariant from first-principles spectral admissibility structure, and the long-term stability of solitonic configurations under extreme conditions. Progress will require large-scale numerical campaigns on the spectral admissibility pipeline, discretized lattice realizations, and targeted experimental tests.

Possible Cosmological Implications of Flavor Misalignment.

Within the present framework, the Jarlskog invariant $\mathcal{J}$ is interpreted as a structural residue of quasi-commutative sectorial restrictions of the admissible spectral subspace of the projection fiber Π (Appendix B.22). In the effective low-energy regime, this invariant is treated as a stable characteristic of the stabilized flavor manifold.

Because the operators Π_u and Π_d are defined as restrictions of the global non-injective mapping, their algebraic structure is ultimately rooted in the global state of the relational substrate χ. In extreme regimes approaching global saturation—such as the early Universe—the admissible spectral subspace of the fiber could differ from its present quasi-stationary configuration.

Whether such regimes might modify the effective structure of flavor misalignment remains an open question. No quantitative model is presently derived. Establishing a predictive relation between $\mathcal{J}$ and early-Universe projective ordering conditions constitutes a significant theoretical challenge for future work.

14 Conclusion and Outlook

Cosmochrony reduces the fundamental assumptions of physics to a single structural origin: the non-injective projection of a static pre-geometric relational substrate χ onto an observable effective description. From this substrate, time, spacetime geometry, and a wide spectrum of physical phenomena emerge as effective descriptions obtained through coarse-grained, generally non-injective projection of admissible relational configurations.

A central structural ingredient of the framework is the spectral organization of the admissible sector of the substrate. The projective structure induces a spectral organization on the space of χ configurations, and physical states correspond to projectable modes selected by admissibility constraints. This spectral filtering provides the bridge between the relational substrate and the effective hierarchy of physical excitations, and underlies the emergence of mass scales through admissible spectral localisation.

Within this setting, bounded-response regimes arise from saturation of the admissible projective flux. The resulting effective dynamics take a Born–Infeld-like form without introducing additional fundamental fields. The speed limit c_χ and Planck's constant h appear as complementary bounds on projectability, respectively limiting the maximal propagation rate and the minimal resolvable granularity. General Relativity emerges as a thermodynamic and geometric limit of the underlying relational dynamics in regimes where local injectivity is approximately recovered.

The framework also provides a structural interpretation of the origin of matter and interactions. Gauge interactions arise as aspects of projection dynamics associated with the topology of the projection fiber, while mass and particle stability are associated with localized spectral configurations and topological constraints of the admissible sector. Electric charge is identified with a π_1 winding invariant of the fiber, while the absence of magnetic monopoles follows from the admissibility-induced triviality of π_2 on the canonical admissible base. Quantum indeterminacy and entanglement appear as generic consequences of non-injective projection rather than as signatures of fundamental nonlocal dynamics.

At cosmological scales, expansion and the arrow of time follow from the global growth of cumulative projected capacity $I(n)$ [10]. At galactic scales, saturation of the admissible projective capacity produces flat rotation curves without requiring dark matter halos. In strong-gravity regimes, black hole evaporation can be interpreted as a reprojection process ensuring information preservation within the relational description, without invoking fundamental information loss.

The space of admissible projected configurations evolves as projective ordering advances: the particle content of the Universe is therefore a historically conditioned outcome rather than a fixed input.

Conceptual Shift and Outlook

Cosmochrony represents a shift from a "matter-on-spacetime" paradigm to a "projection-of-substrate" ontology, in which spacetime, quantum behaviour, and interactions emerge from a single underlying structural fact: the non-injective projection Π of a static relational substrate χ.

Immediate falsifiable research directions include: cosmological signatures (low-ℓ CMB anomalies induced by finite projective capacity), galactic phenomenology (systematic correlation between apparent dark matter effects and local projective capacity gradients), and fundamental-scale effects (environment-dependent variations in decay rates or effective couplings in regimes approaching admissibility saturation).

On the theoretical side, a key open problem is the derivation of the full Standard Model structure as a hierarchy of admissible spectral excitations and topological classes within the relational projective admissibility structure, including the quantitative emergence of chirality selection and the $V-A$ structure of weak interactions.

Part VI
Appendices

A Mathematical Foundations of Cosmochrony — Dynamics, Stability, and Analytical Solutions

This appendix provides a rigorous mathematical formulation of the χ-field dynamics. All results follow from the fundamental postulates without assuming a pre-existing spacetime metric. Geometric notions should be understood as effective, coarse-grained representations consistent with Appendix D.

A.1 Effective Lagrangian Description as a Hydrodynamic Limit

In regimes where χ varies smoothly, discrete relational couplings K_{ij} can be summarized by effective continuum quantities. Distances emerge as cumulative resistance to projective ordering advancement. The effective metric $g_{\mu\nu}$ encodes the coarse-grained density of correlations.

To reproduce the continuum evolution from the discrete rule (Eq. 244), one introduces a purely representational effective Lagrangian:

$$\mathcal{L}_{\text{eff}} = \frac{1}{16\pi G_{\text{eff}}} F(\chi)\, R - \Lambda_{\text{flow}}^4\, \chi + \cdots$$

where R is the Ricci scalar of the effective metric and $F(\chi)$ parametrizes how the projective ordering structure is encoded in the geometric description. Einstein-like field equations reflect the universality of geometric descriptions for slowly varying collective phenomena. Singularities signal only the limits of the hydrodynamic approximation. The effective Lagrangian has no ontological status and should not be quantized.

In projective regimes where the mapping Π is effectively injective, the existence of $S_{\text{eff}} = \int \mathcal{L}_{\text{eff}}\, \mathrm{d}^4x$ implies additional structural constraints. Admissible variations of underlying χ configurations that preserve projectability map to variations of the effective fields that leave S_{eff} invariant up to boundary terms. As a consequence, the effective description admits local continuity identities of Noether type. In particular, when $\mathcal{L}_{\text{eff}}$ is generally covariant with respect to the emergent metric $g_{\mu\nu}$, the associated effective stress tensor

$$T^{\mu\nu}_{\text{eff}} := \frac{2}{\sqrt{-g}} \frac{\delta S_{\text{eff}}}{\delta g_{\mu\nu}}$$

satisfies the local identity

$$\nabla_\mu T^{\mu\nu}_{\text{eff}} = 0$$

within the hydrodynamic domain.

From this perspective, local conservation is not a primitive postulate of the χ description but a stability condition of the projective regime. If the projective flux approaches saturation and effective injectivity fails, the variational formulation may cease to be well-posed, and the above local identities need not remain strictly defined at the boundary of the saturated region. Global admissibility of the underlying χ dynamics is nonetheless preserved, and apparent non-local redistribution in the effective description is interpreted as deprojection followed by reprojection once projectability conditions are restored.

A.2 Stability Analysis of the χ-Field Dynamics

In the hydrodynamic regime, the effective ordering dynamics reads $\partial_t \chi = c\sqrt{1 - |\nabla\chi|^2/c^2}$.

Around a homogeneous background $\chi_0(t) = ct + \chi_{0,0}$, a perturbation $\delta\chi$ satisfies $\partial_t \delta\chi = -(2c)^{-1}|\nabla\delta\chi|^2 + \mathcal{O}(|\nabla\delta\chi|^4)$. No linear term appears: the homogeneous solution is marginally stable at linear order. The leading nonlinear correction is strictly negative, so any spatial inhomogeneity reduces the local projective ordering rate and is dynamically suppressed.

The diagnostic functional $E[\delta\chi] = \frac{1}{2}\int |\nabla\delta\chi|^2\, d^3x$ is non-increasing. Planar perturbations are progressively flattened; spherical ones decay monotonically. The effective dynamics is dissipative and contractive, establishing nonlinear stability. This stability is inseparable from the monotonic character of projective ordering: the same constraint defining the arrow of time precludes dynamical instabilities.

A.3 Analytical Solutions of the χ-Field Dynamics

Explicit solutions of the effective ordering equation:

$$\partial_t \chi = c\sqrt{1 - \frac{|\nabla\chi|^2}{c^2}}. \tag{174}$$

Homogeneous ordering.

$\nabla\chi = 0$ gives $\chi(t) = \chi_0 + ct$, underlying emergent cosmological expansion (Section 3.7).

Gradient-saturated profiles.

$|\partial_r \chi| = c$ yields $\partial_t \chi = 0$: local freezing of the ordering parameter. The profiles $\chi(r) = \chi_0 \pm cr$ model horizons and maximally constrained regions.

Linear ordering fronts.

$\chi(x, t) = \chi_0 + ct \pm vx$ with $|v| < c$ satisfies Eq. (174) identically. These are kinematic boundaries, not propagating waves.

Absence of linear wave solutions.

Small perturbations around homogeneous configurations do not propagate as oscillatory modes (Section A.2). Apparent wave-like phenomena arise only at the effective level through collective excitations.

A.4 Coupling with Matter: Effective Source Term $S[\chi, \rho]$

In the emergent spacetime description:

$$\Box_{\text{eff}}\chi = S[\chi, \rho]. \tag{175}$$

$S[\chi, \rho]$ encodes the effective resistance of localized excitations to the global projective ordering advance. It summarizes gravitational time dilation, inertial mass, and effective curvature.

Weak-field regime.

$$S[\chi, \rho] \simeq -\alpha\, \rho, \tag{176}$$

with $\alpha \sim G/c^2$. This reproduces the Poisson equation and the Schwarzschild solution at leading order.

Strong-field regime.

$$S[\chi, \rho] = -\alpha\, \rho\, F\left(\frac{\rho}{\rho_c}, \chi\right), \tag{177}$$

where F is bounded and ρ_c denotes a characteristic density beyond which projective ordering resistance saturates. These nonlinearities prevent unphysical halting of the projective ordering advance and encode departures from classical gravity, signaling the breakdown of the hydrodynamic approximation.

A.5 Strong-Field Constitutive Coupling Near a Schwarzschild Black Hole

The effective constitutive relation:

$$K_{\text{eff}} = K_0 \exp\left(-\frac{(\Delta\chi)^2}{\chi_c^2}\right). \tag{178}$$

The operational lapse-like factor:

$$N(r) \equiv \frac{\mathcal{D}_{\text{loc}}\chi(r)}{\mathcal{D}_0\chi}, \quad 0 < N(r) \leq 1. \tag{179}$$

Matching to the Schwarzschild metric (purely effective):

$$ds^2 = -f(r)c^2dt^2 + f(r)^{-1}dr^2 + r^2d\Omega^2, \quad f(r) = 1 - \frac{r_s}{r}. \tag{180}$$

$$N(r)^2 = f(r) = 1 - \frac{r_s}{r}. \tag{181}$$

Minimal identification $K_{\text{eff}}(r)/K_0 \equiv N(r)^2$:

$$K_{\text{eff}}(r)/K_0 \equiv N(r)^2. \tag{182}$$

Yielding the strong-field profile:

$$K_{\text{eff}}(r) = K_0\left(1 - \frac{r_s}{r}\right), \quad r > r_s. \tag{183}$$

Inverting Eq. (178):

$$\frac{(\Delta\chi(r))^2}{\chi_c^2} = -\ln\left(1 - \frac{r_s}{r}\right). \tag{184}$$

As $r \to r_s^+$:

$$\Delta\chi(r) \sim \chi_c \sqrt{-\ln\left(1 - \frac{r_s}{r}\right)}. \tag{185}$$

This logarithmic divergence signals that the hydrodynamic parametrization exceeds its domain of validity near the effective horizon. A Schwarzschild horizon corresponds to vanishing projective ordering conductivity ($K_{\text{eff}} \to 0$), not a fundamental spacetime singularity.

A.6 Minimal Kinematic Constraint

In its saturated form:

$$(\partial_t \chi)^2 + |\nabla\chi|^2 = c^2, \tag{186}$$

where ∂_t and ∇ denote effective variations introduced only once a projectable geometric regime is established. More generally, admissible configurations satisfy the corresponding inequality.

This constraint is imposed ab initio at the pre-geometric level and does not presuppose a spacetime metric, light cones, or Lorentzian structure. The constant c is the effective manifestation of the invariant structural bound c_χ (Section 2.11). Lorentz symmetry arises a posteriori as a property of saturated projective ordering.

In homogeneous regimes, the constraint enforces $\partial_t\chi = c$, leading to linear growth and effective cosmic expansion. In inhomogeneous regimes, partial saturation manifests as local slowdown underlying gravitational time dilation. At scales near the Planck length, the continuum approximation breaks down and a fully discrete formulation is required.

A.7 Effective Evolution Equation

Once a stable geometric description has emerged, large-scale regularities of admissible projected configurations may be written as $\Box_{\text{eff}}\chi = S[\chi, \rho]$, where $\Box_{\text{eff}}$ is the d'Alembertian of the emergent metric and ρ represents the effective density of locally stable projected configurations. Neither the operator nor the source term is fundamental.

$S[\chi, \rho]$ encodes how localized patterns locally reduce the admissible projective ordering rate. Gradients in the effective ordering parameter are reinterpreted as gravitational time dilation and spacetime curvature.

In weak-field regimes, $S[\chi, \rho] \simeq -\alpha\rho$ with $\alpha \sim G/c^2$, reproducing the Poisson equation and Schwarzschild-like solutions. Nonlinear corrections encode saturation effects inherited from the kinematic constraint $\partial_t\chi \le c$ and signal the breakdown of the hydrodynamic approximation.

A.8 Relational Foundation and Emergent Geometry

The continuous representation of χ is a pragmatic strategy for contact with established formalisms, not an ontological commitment. At a more fundamental level, Cosmochrony can be formulated in purely relational terms where neither spacetime points, distances, nor a metric are assumed a priori. Temporal ordering arises from the monotone growth of cumulative projected capacity $I(n)$ [10]; spatial relations are reconstructed operationally from patterns of correlation and connectivity. The metric appears only as a derived object. A concrete realization is developed in Appendix E.

A.9 Admissible Injective Projections Have Trivial Holonomy

This appendix formalizes Step A announced in Section 5.7, the first part of the necessity argument formulated in Section 5.7. We show that admissible injective projections necessarily have trivial holonomy. Since non-trivial holonomy is the minimal structural condition for re-identification symmetry, injective admissible projections cannot support emergent gravitational dynamics.

We classify admissible projection maps $\Pi : \mathcal{C}_\chi \to \mathcal{M}$ by their fiber topology. We show that injective maps in this class have trivial holonomy. Since non-trivial holonomy is the minimal condition for a re-identification symmetry to arise as a descriptive invariance, injectivity structurally precludes emergent gravity under Hypothesis H and ontological uniqueness.

Setup: admissible projections as fibered structures

We work in the regime where $\mathcal{C}_\chi$ and $\mathcal{M}$ admit smooth structures (the continuum limit of Section E.1 and Appendix D.5). In this regime, the projection $\Pi : \mathcal{C}_\chi \to \mathcal{M}$ is required to be *admissible*.

Definition A.1 (Admissible projection)**.** A smooth surjective map $\Pi : \mathcal{C}_\chi \to \mathcal{M}$ is called admissible if:

(A1) Π is a fiber bundle map. For each $y \in \mathcal{M}$, the preimage $\Pi^{-1}(y) \subset \mathcal{C}_\chi$ is a smooth submanifold. Fibers are mutually diffeomorphic.
(A2) Π is compatible with the relaxation rule F. If $x, x' \in \mathcal{C}_\chi$ satisfy $\Pi(x) = \Pi(x')$, then $\Pi(F(x)) = \Pi(F(x'))$. Effective dynamics is therefore well-defined on $\mathcal{M}$.
(A3) Π supports a connection. There exists a smooth horizontal distribution $H \subset T\mathcal{C}_\chi$ complementary to the vertical, with the usual equivariance properties whenever a fiber group action is present.

Condition (A1) places Π in the category of fiber bundles over $\mathcal{M}$. In the principal bundle case, where G_Π acts freely and transitively on fibers, non-injectivity of Π corresponds to G_Π being non-trivial. For general fiber bundles where the fiber is not identified with the structure group, non-injectivity is the weaker condition $|\Pi^{-1}(y)| > 1$. The structure group may still be non-trivial even when fibers are discrete. The arguments below apply to the principal bundle case, which is the relevant one for Cosmochrony (Section 5.1, $\Pi \cong S^3$ with $G_\Pi \supseteq \mathrm{SU}(2) \times \mathrm{U}(1)$).

Holonomy as the obstruction to global re-identification

Let Π be an admissible projection equipped with a connection. In the principal case, the connection is a $\mathfrak{g}_\Pi$-valued one-form ω on $\mathcal{C}_\chi$. The holonomy group $\mathrm{Hol}(\omega, x_0)$ at $x_0 \in \mathcal{C}_\chi$ is the subgroup of G_Π generated by parallel transport along loops γ in $\mathcal{M}$

based at $y_0 = \Pi(x_0)$:

$$\mathrm{Hol}(\omega, x_0) = \left\{ P_\gamma \in G_\Pi \;\middle|\; \gamma : [0,1] \to \mathcal{M},\ \gamma(0) = \gamma(1) = y_0 \right\} \subseteq G_\Pi. \tag{187}$$

Holonomy measures the obstruction to globally consistent re-identification of fibers. It detects the failure of horizontal lifts of loops in $\mathcal{M}$ to close in $\mathcal{C}_\chi$.

Definition A.2 (Re-identification symmetry)**.** A *re-identification symmetry* of the projection Π is a transformation $\phi : \mathcal{C}_\chi \to \mathcal{C}_\chi$ that:

(i) preserves effective states: $\Pi(\phi(x)) = \Pi(x)$ for all x,
(ii) is non-trivial: $\phi \neq \mathrm{id}$ on a non-empty open set,
(iii) is locally implementable: ϕ acts fiberwise and continuously.

In the principal case, re-identification symmetries correspond to bundle automorphisms acting along fibers. Their global consistency is constrained by holonomy.

Proposition A.1 (Holonomy controls global re-identification consistency)**.** *Let Π be an admissible projection with connection ω. If $\mathrm{Hol}(\omega, x_0) = \{e\}$, the bundle admits a global flat section and re-identification symmetries are globally trivializable. If $\mathrm{Hol}(\omega, x_0)$ is a positive-dimensional Lie subgroup of G_Π, re-identification symmetries are intrinsic and cannot be globally trivialized. The converse, flat connection implies trivial holonomy, holds when $\mathcal{M}$ is simply connected, but not in general. The argument below does not rely on this converse.*

Proof Standard result in differential geometry (Ambrose–Singer theorem and its converse). The holonomy algebra $\mathfrak{hol}(\omega, x_0)$ equals the Lie algebra generated by curvature values $\Omega(u, v)$ for all horizontal vectors u, v at points reachable from x_0.

A flat connection ($\Omega = 0$) has holonomy group contained in a discrete subgroup of G_Π when $\mathcal{M}$ is simply connected. In the general case, the holonomy of a flat connection is a representation of $\pi_1(\mathcal{M})$ into G_Π, which may be non-trivial.

However, under injectivity $G_\Pi = \{e\}$ (Theorem A.1). So the holonomy group is trivial regardless of the topology of $\mathcal{M}$. The flat connection argument is therefore not the primary route here. It is superseded by the triviality of G_Π established directly from injectivity. □

The triviality theorem for injective admissible projections

Theorem A.1 (Injective admissible projections have trivial holonomy)**.** *Let $\Pi : \mathcal{C}_\chi \to \mathcal{M}$ be an admissible projection in the sense of Definition A.1. In the principal bundle case, where G_Π acts freely and transitively on fibers, non-injectivity of Π corresponds to G_Π being non-trivial. For general fiber bundles where the fiber is not identified with the structure group, non-injectivity is the weaker condition $|\Pi^{-1}(y)| > 1$. The structure group may still be non-trivial even when fibers are discrete. The arguments below apply to the principal bundle case, which is the relevant one for Cosmochrony (Section 5.1, $\Pi \cong S^3$ with $G_\Pi \supseteq \mathrm{SU}(2) \times \mathrm{U}(1)$). If Π is injective, then:*

(a) the fiber over every $y \in \mathcal{M}$ is a single point: $\Pi^{-1}(y) = \{x_y\}$*,*
(b) in the principal specialization, the structure group is trivial: $G_\Pi = \{e\}$*,*
(c) in that principal case, the holonomy group is trivial: $\mathrm{Hol}(\omega, x_0) = \{e\}$ *for any connection* ω *and any basepoint* x_0*,*
(d) no non-trivial re-identification symmetry exists in the sense of Definition A.2.

Proof Part (a) is immediate from injectivity. If $\Pi(x) = \Pi(x') = y$, then $x = x'$, hence $|\Pi^{-1}(y)| = 1$ for all y.

Part (b) is a principal-bundle statement. If fibers are singletons, the only group acting freely and transitively on a one-element set is $\{e\}$.

Part (c) follows from $\mathrm{Hol}(\omega, x_0) \subseteq G_\Pi$. If $G_\Pi = \{e\}$, then $\mathrm{Hol}(\omega, x_0) = \{e\}$.

Part (d) follows directly from injectivity. If $\Pi(\phi(x)) = \Pi(x)$ for all x, injectivity implies $\phi(x) = x$ for all x. Hence $\phi = \mathrm{id}$, contradicting non-triviality. □

From trivial holonomy to absence of emergent gauge structure

Corollary A.1 (No emergent gauge field from injective projection)**.** *Under the hypotheses of Theorem A.1 in the principal specialization, any connection is trivial in the sense that its curvature vanishes. No non-trivial gauge field can emerge from the projection structure alone.*

Proof With $G_\Pi = \{e\}$, the Lie algebra is $\{0\}$. Any connection is therefore $\omega = 0$, and $\Omega = d\omega + \omega \wedge \omega = 0$. A non-trivial gauge field would require an independent postulate at the level of $\mathcal{M}$, violating ontological uniqueness. □

Corollary A.2 (No emergent diffeomorphism invariance from injective projection)**.** *Under the hypotheses of Theorem A.1 and Hypothesis H, no diffeomorphism invariance as a descriptive invariance can arise from* Π *alone.*

Proof Diffeomorphism invariance as a descriptive invariance (D2) requires multiple representatives of the same effective state. That is, it requires $|\Pi^{-1}(y)| > 1$ on some region of $\mathcal{M}$. Under injectivity, $|\Pi^{-1}(y)| = 1$ everywhere. Therefore, the diffeomorphism group of $\mathcal{M}$ cannot arise as a re-identification symmetry induced by Π . Any such invariance would have to be postulated directly at the level of $\mathcal{M}$, contradicting emergence and ontological uniqueness. □

Closing the gap: from holonomy to gravitational curvature

The preceding results establish that injective projections cannot produce the structural prerequisites of emergent gravity, namely re-identification symmetry and descriptive invariance as consequences of the projection. We now connect fiber curvature to effective base curvature in the standard bundle-induced manner.

In the Cosmochrony framework, the effective metric on $\mathcal{M}$ is induced by the Hessian of the effective Lagrangian (Section A.12, Eq. (117)):

$$g_{\mu\nu}^{\text{eff}} \propto \frac{\partial^2 \mathcal{L}_{\text{eff}}}{\partial(\partial_\mu \chi)\, \partial(\partial_\nu \chi)}, \tag{188}$$

valid in projectable regimes.

Proposition A.2 (Fiber curvature contributes to base curvature)**.** *Let $\Pi : \mathcal{C}_\chi \to \mathcal{M}$ be an admissible principal G_Π-bundle with connection ω and curvature Ω. If g^{eff} is induced from the bundle geometry via Eq. (188), then the projection-induced contribution to the effective Ricci tensor contains a term of the form*

$$R^{\text{eff}}_{\mu\nu} \supset \text{tr}_{\mathfrak{g}_\Pi}\big(\Omega_{\mu\alpha}\,\Omega_\nu{}^\alpha\big) + \dots, \tag{189}$$

where the trace runs over $\mathfrak{g}_\Pi$ and the ellipsis denotes additional terms fixed by the same induction scheme.

Flat connection ($\Omega = 0$, hence $G_\Pi = \{e\}$ under injectivity). The fiber-induced contribution to effective base curvature vanishes. The term $\text{tr}_{\mathfrak{g}_\Pi}\big(\Omega_{\mu\alpha}\Omega_\nu{}^\alpha\big)$ in Eq. (189) is identically zero since $\Omega = 0$. The effective metric g^{eff} on $\mathcal{M}$ receives no curvature sourcing from the fiber geometry. Whether g^{eff} is globally flat depends on additional structure of $\mathcal{M}$. The claim here is only that no curvature is generated by the projection without an independent postulate at the effective level.

Proof In a principal bundle with non-flat connection, Ω is a horizontal adjoint-valued 2-form. Bundle-induced reductions yield quadratic curvature contributions of Yang–Mills type, which appear in the base Ricci tensor in the standard Kaluza–Klein reduction. If $G_\Pi = \{e\}$, then $\mathfrak{g}_\Pi = \{0\}$ and $\Omega = 0$, so the term vanishes identically. □

Remark.

$$\Pi \text{ injective} \Rightarrow G_\Pi = \{e\} \Rightarrow \Omega = 0 \Rightarrow \text{no fiber-induced curvature on } \mathcal{M} \Rightarrow R^{\text{eff}}_{\mu\nu\rho\sigma} \text{ receives no projective sourcing.} \tag{190}$$

Equation (190) shows that effective gravitational curvature on $\mathcal{M}$ receives no sourcing from the fiber geometry when Π is injective. Any non-zero curvature of g^{eff} would then require an independent postulate at the level of $\mathcal{M}$, violating ontological uniqueness. The chain is topological, not dynamical, and holds for any admissible projection in the sense of Definition A.1.

The minimal non-injective case: $G_\Pi = \mathrm{U}(1)$

As a consistency check and to connect to Section 5.1, we consider the minimal non-trivial case $G_\Pi = \mathrm{U}(1)$. A principal U(1)-bundle over $\mathcal{M}$ with connection ω has:

$$\text{Hol}(\omega, x_0) \subseteq \mathrm{U}(1), \tag{191}$$

$$\Omega = d\omega \in \Omega^2(\mathcal{M}, i\mathbb{R}), \tag{192}$$

where Ω is interpreted as the electromagnetic field strength in the Cosmochrony setting (Section A.11). The first Chern class $c_1(\Pi) = [\Omega/2\pi i] \in H^2(\mathcal{M}, \mathbb{Z})$ is non-trivial if and only if the bundle is topologically non-trivial.

This matches the structure of Section 5.2. The electromagnetic U(1) gauge symmetry emerges because the projection fiber supports a non-trivial Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$.

In the injective limit, where the fiber degenerates to a point, the bundle trivializes and the U(1) gauge structure disappears, consistent with Eq. (190).

Summary of Step A

The classification of admissible injective projections establishes the following hierarchy, completing Step A of Theorem 5.1.

Fiber type	Structure group G_Π	Emergent structure
Point ($\lvert\Pi^{-1}(y)\rvert = 1$, injective)	$G_\Pi = \{e\}$	No gauge, no projection-induced curvatu
Circle (S^1, minimal non-injective)	$G_\Pi = \mathrm{U}(1)$	U(1) gauge field, electromagnetic curvatu
Three-sphere (S^3, Cosmochrony setting)	$G_\Pi \supseteq \mathrm{SU}(2) \times \mathrm{U}(1)$	Gauge structure and projection-induced

The injective case is the unique case in which no re-identification symmetry and no projection-induced curvature can arise without an independent postulate. Non-injectivity is therefore not merely sufficient but necessary for emergent gauge and gravitational structure under Hypothesis H and ontological uniqueness.

This excludes injective admissible projections as candidates for emergent gravitational dynamics. The complementary constructive part of the necessity proof is developed in Section A.10, where we show that under non-injective admissible projections satisfying the saturation bound, the infrared dynamics is uniquely selected.

A.10 Infrared expansion of the coherence functional

This subsection completes the necessity argument introduced in Section 5.7. Having excluded injective admissible projections in Section A.9, we now show that under non-injective admissible projection satisfying the saturation bound of Section 3, the infrared effective dynamics is uniquely selected. In particular, the Einstein–Hilbert term arises as the leading infrared contribution of the coherence functional.

To extract the dominant infrared behavior of S_{coh}, we expand $\mathcal{L}_{\mathrm{defect}}$ in powers of the curvature scale ℓ_c^{-2}, where ℓ_c is the characteristic scale of curvature variations. In the infrared regime $\ell_c \gg \ell_\chi$ (with $\ell_\chi = \hbar_\chi / c_\chi$ the microscopic capacity length), curvature is slowly varying and the expansion is controlled.

Step 1: metric from the operator symbol.

Following the derivation in [21] and Appendix A.12, the effective metric $g_{\mu\nu}$ is identified with the principal symbol of the effective continuum operator $\mathcal{L}_\Pi = \nabla^\mu (A^{\mu\nu}(x) \nabla_\nu)$:

$$g^{\mu\nu}(x) \propto A^{\mu\nu}(x), \tag{193}$$

where $A^{\mu\nu}$ encodes the local connectivity and stiffness of the relational substrate [21]. This is an emergent metric: it is not introduced as an independent variable but arises from the spectral structure of Π.

Step 2: from fiber curvature to effective Laplacian to Riemann tensor.

The curvature 2-form $\Omega_{\mu\nu}$ of the fiber connection acts on sections of the adjoint bundle ad P over $\mathcal{M}$. Its square $\|\Omega_{\mu\nu}\|^2_{\mathfrak{g}_\Pi}$, integrated over $\mathcal{M}$, defines a Yang–Mills-type functional on $\mathcal{M}$. In the Kaluza–Klein reduction from $\mathcal{C}_\chi$ to $\mathcal{M}$, the fiber curvature contributes to the effective operator $\mathcal{L}_\Pi$ through its symbol. Following [21], the principal symbol of $\mathcal{L}_\Pi$ is $\sigma_2(\mathcal{L}_\Pi)(x,k) = A^{\mu\nu}(x)k_\mu k_\nu$, which identifies $g^{\mu\nu} \propto A^{\mu\nu}$. The fiber curvature Ω modifies the sub-principal symbol of $\mathcal{L}_\Pi$ at order ℓ_c^{-2}, contributing a term proportional to $R_{\mu\nu}k^\mu k^\nu$ through the standard formula for the sub-principal symbol of a connection Laplacian $\nabla^\mu\nabla_\mu$ acting on sections of a bundle with curvature Ω:

$$\sigma_1(\nabla^\mu\nabla_\mu) \supset \frac{1}{6}R\cdot \mathrm{id} + \mathcal{F}(\Omega),$$

where the first term is the Lichnerowicz correction and $\mathcal{F}(\Omega)$ encodes the fiber curvature contribution. It is this sub-principal symbol, not the Yang–Mills square $\|\Omega\|^2$ directly, that generates R at two-derivative order. The Yang–Mills square $\|\Omega\|^2 \sim R^2_{\mu\nu\rho\sigma}$ appears only at four-derivative order and is subdominant under (D3). The two-derivative IR term R arises from the sub-principal symbol of the connection Laplacian on the fiber bundle, via the Lichnerowicz–Weitzenböck identity. We summarize the infrared content of this statement as:

$$\sigma_1(\mathcal{L}_\Pi)\big|_{\mathrm{fiber}} \supset \frac{1}{6}R + \mathcal{O}(\ell_c^{-4}), \tag{194}$$

yielding $\mathcal{L}^{\mathrm{geom}}_{\mathrm{defect}} \propto R$ at two-derivative order, with the four-derivative Yang–Mills term $\|\Omega\|^2$ entering only as a subleading correction.

However, conditions (D1)–(D4) impose additional constraints. In particular, (D3) (infrared dominance of a low-derivative action) requires that the action be organized by the number of derivatives, with the lowest-derivative terms dominating in the infrared.

Step 3: integration by parts and the Gauss–Bonnet identity.

In four dimensions, the Gauss–Bonnet combination $\mathcal{G} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2$ is a total derivative and does not contribute to the equations of motion. Therefore, among all quadratic curvature invariants, only R^2 and $R_{\mu\nu}R^{\mu\nu}$ are independent at the level of the action, modulo boundary terms. But both are of order ℓ_c^{-4}, subdominant under (D3).

The leading IR contribution to the defect density from the geometric sector is therefore:

$$\mathcal{L}^{\mathrm{geom}}_{\mathrm{defect}} = \frac{c_\chi^3}{\hbar_\chi}\cdot\frac{1}{16\pi}\cdot R + \mathcal{O}(\ell_c^{-4}), \tag{195}$$

where the prefactor $c_\chi^3/\hbar_\chi$ matches the dimensions of the Einstein–Hilbert density up to the normalization fixed by the precise operator mapping. The normalization is tracked explicitly in Section A.12.

A.11 Saturation selects the Born–Infeld completion

The IR expansion (50) is valid only for $\ell_c \gg \ell_\chi$. In the ultraviolet regime $\ell_c \sim \ell_\chi$, higher-derivative corrections become comparable to the Einstein–Hilbert term, and the expansion breaks down. The saturation bound $|\partial_t \chi| \leq c_\chi$ provides the principle that selects among the possible UV completions.

As established in [45], the minimal conditions for a UV completion compatible with bounded flux propagation are:

(i) it reduces to a quadratic (Einstein–Hilbert-like) form in the IR ($\ell_c \gg \ell_\chi$),
(ii) it enforces a strict upper bound on admissible field invariants at $\ell_c \sim \ell_\chi$,
(iii) it introduces no additional microscopic scales beyond c_χ and $\hbar_\chi$.

Applied to the gravitational sector, the Born–Infeld completion takes the form:

$$S_{\rm coh}^{\rm full}[g,\psi] = \frac{c_\chi^4}{\hbar_\chi^2} \int_{\mathcal{M}} \left(\sqrt{-\det\left(g_{\mu\nu} + \frac{\hbar_\chi}{c_\chi^2} R_{\mu\nu}\right)} - \sqrt{-g} \right) d^4x + S_{\rm matter}[g,\psi]. \tag{196}$$

In the IR expansion $\hbar_\chi / c_\chi^2 \ll \ell_c^2$:

$$\sqrt{-\det\left(g_{\mu\nu} + \frac{\hbar_\chi}{c_\chi^2} R_{\mu\nu}\right)} \approx \sqrt{-g} \left(1 + \frac{\hbar_\chi}{2c_\chi^2} R + \mathcal{O}\left(\frac{\hbar_\chi^2}{c_\chi^4} R_{\mu\nu} R^{\mu\nu}\right)\right), \tag{197}$$

so that:

$$S_{\rm coh}^{\rm full} \xrightarrow{\ell_c \gg \ell_\chi} \frac{c_\chi^2}{2\hbar_\chi} \int_{\mathcal{M}} R\sqrt{-g}\, d^4x + S_{\rm matter} + \mathcal{O}(\ell_c^{-4}). \tag{198}$$

Proposition A.3 (Born–Infeld structure as gravitational UV completion)**.** *Under conditions (i)–(iii), ontological uniqueness, and Hypothesis [H-ext] (admissible coherence extensivity), the UV completion of the Einstein–Hilbert term in $S_{\rm coh}$ is uniquely determined to be of Eddington-inspired Born–Infeld form* (196) *with $\mathcal{R}_{\mu\nu} = R_{\mu\nu}$.*

Proof The uniqueness of the determinantal structure and the identification of the Ricci tensor as the spectral carrier are established in [38], Theorem 6.1, via three independent results: the scalar Born–Infeld reduction fixes the one-dimensional sector function; coherence extensivity under [H-ext] lifts this to the full tensorial determinant; and a classification of two-derivative covariant endomorphisms selects $g^{\mu\alpha} R_{\alpha\nu}$ as the unique admissible spectral carrier. The trace ambiguity $R_{\mu\nu} \mapsto R_{\mu\nu} - \alpha\, g_{\mu\nu} R$ is resolved by matching to the a_2 heat-kernel coefficient of [38], selecting $\alpha = 0$. See [38], Section 6 for the full argument. □

A.12 Prediction: $G_{\rm eff}$ in terms of c_χ and $\hbar_\chi$

Comparing (198) with the standard Einstein–Hilbert normalization $\frac{c^4}{16\pi G} \int R\sqrt{-g}\, d^4x$ and using $c = c_\chi$, the identification requires:

$$\frac{c_\chi^4}{16\pi G_{\rm eff}} = \frac{c_\chi^2}{2\hbar_\chi}.$$

Formally, this gives:

$$G_{\text{eff}} = \frac{c_\chi^2 \hbar_\chi}{8\pi}. \tag{199}$$

A dimensional check in SI units yields $[c_\chi^2 \hbar_\chi] = \mathrm{kg\,m^4\,s^{-3}}$, while $[G] = \mathrm{m^3\,kg^{-1}\,s^{-2}}$. These do not match, which signals that the prefactor $c_\chi^4/\hbar_\chi^2$ in Eq. (196) must carry the residual dimensions. The dimensional consistency of Eq. (204) depends on the precise normalization of the Born–Infeld gravitational action, which requires the full tensorial derivation noted in Section A.11 as an open step. The structural prediction $G_{\text{eff}} = G(c_\chi, \hbar_\chi)$ is robust. The precise numerical prefactor and dimensional matching are left as a quantitative step contingent on the tensorial completion.

A.13 Energy and Curvature

Energy emerges as a diagnostic of constrained projective ordering. At the effective level:

$$\mathcal{E}_\chi^{\text{eff}} = \tfrac{1}{2}\left[(\partial_t \chi)^2 + (\nabla \chi)^2\right].$$

This functional has no Hamiltonian or variational status. Regions of large $\mathcal{E}_\chi^{\text{eff}}$ correspond to strong internal gradients and are identified with particle-like excitations.

The apparent ordering of atomic orbitals by increasing energy reflects the structural cost of sustaining extended oscillatory configurations. An outer orbital constrains projective ordering over a wider region, increasing total resistance independently of the geometric visibility discussed in Appendix A.14.

Curvature characterizes internal deformation of projected χ configurations and how they modulate the advance of projective ordering. Effective spacetime curvature arises secondarily as a macroscopic descriptor. Stable solitonic configurations emerge when nonlinear self-interaction balances dispersive gradients, providing a dynamical origin for long-lived particle-like excitations.

A.14 Level Sets, Projections, and Apparent Orbital Geometry

For a continuous scalar field $\phi : \mathbb{R}^3 \to \mathbb{R}$, the level set $\mathcal{L}_c = \{\mathbf{x} \mid \phi(\mathbf{x}) = c\}$ is generically a two-dimensional surface, possibly with disconnected components.

Projecting the thresholded set $P_c = \{z \mid \exists (x, y)\ \phi(x, y, z) \geq c\}$ generically produces disjoint intervals even when ϕ is continuous. This fragmentation is a purely geometric consequence of thresholding followed by projection: no discontinuity of ϕ is involved.

The envelope $f(z) = \max_{x,y} \phi(x, y, z)$ is continuous, but the condition $f(z) \geq c$ selects disconnected regions whose appearance and disappearance as c varies reflect visibility changes, not structural changes. The inverse reconstruction of ϕ from P_c is underdetermined.

Orbital-like patterns, nodal structures, and probabilistic visibility regions are therefore emergent manifestations of an underlying continuous field. Their apparent discreteness reflects projection and detection criteria.

A.15 Emergent Electrodynamics from χ Dynamics

In the weak-gradient regime, $\nabla\chi$ decomposes as $\nabla\chi = -\nabla\phi + \mathbf{A}_{\mathrm{T}}$ with $\nabla \cdot \mathbf{A}_{\mathrm{T}} = 0$, induced by the topology of localized excitations.

Charge as Transverse Torsion

The bounded canonical current $\mathbf{J}_\chi \propto \nabla\chi/\sqrt{1 - |\nabla\chi|^2/c^2}$ saturates as $|\nabla\chi| \to c$. An effective charge invariant is $q = \kappa \oint_\gamma \mathbf{A}_{\mathrm{T}} \cdot d\boldsymbol{\ell}$, whose sign reflects the chirality of the transverse torsion. The magnetic field is $\mathbf{B} = \nabla \times \mathbf{A}_{\mathrm{T}}$.

Emergent Electromagnetic Fields

Electric and magnetic fields:

$$\mathbf{E} = -\nabla\phi - \frac{1}{c}\frac{\partial \mathbf{A}_{\mathrm{T}}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}_{\mathrm{T}}.$$

These satisfy a closed system of Maxwell-like relations. Gauge invariance under $\phi \to \phi - c^{-1}\partial_t\Lambda$, $\mathbf{A}_{\mathrm{T}} \to \mathbf{A}_{\mathrm{T}} + \nabla\Lambda$ reflects the relational nature of χ (Section A.8).

Electromagnetism appears as a geometric and topological manifestation of the χ substrate, not an independent interaction mediated by elementary gauge fields.

A.16 Relational Consistency of the Effective Lagrangian

Step 1: Relational Constraint

$$\mathcal{C}_i[\chi] \equiv \sum_j K_{ij}(\chi_i - \chi_j)^2 \le \chi_c^2. \tag{200}$$

Step 2: Variational Formulation

Constrained action with KKT conditions:

$$S[\{\chi_i\}, \{\mu_i\}] = \int d\lambda \left[\sum_i \tfrac{1}{2}(d\chi_i/d\lambda)^2 - U[\{\chi_i\}] - \sum_i \mu_i(\mathcal{C}_i - \chi_c^2) \right]. \tag{201}$$

Step 3: Continuum Limit and Canonical Form

In projectable regimes:

$$|\nabla\chi|^2 \le c^2, \tag{202}$$

with $c^2 \equiv a^2\chi_c^2/K_0$. The canonical representation satisfying boundedness, monotonicity, and regularity:

$$f(x) = -c^2\sqrt{1-x}, \tag{203}$$

yielding the Born–Infeld-like Lagrangian:

$$\mathcal{L}_{\mathrm{eff}} = -c^2\sqrt{1 - \frac{|\nabla\chi|^2}{c^2}} + \partial_t\chi. \tag{204}$$

Step 4: Role of $U[\{\chi_i\}]$

In the continuum limit, $U[\{\chi_i\}] \to \int d^3x\, V(\chi)$, connected to the main text's $V(\chi)$ via coarse-graining.

Step 5: Connection to Emergent Geometry

The effective metric is defined via the Hessian:

$$g_{\mu\nu}^{\mathrm{eff}} \propto \frac{\partial^2 \mathcal{L}_{\mathrm{eff}}}{\partial(\partial_\mu \chi)\,\partial(\partial_\nu \chi)}, \tag{205}$$

valid in projectable regimes (Appendix D.5).

Continuum Limit: Laplace–Beltrami Operator

In the dense limit ($N \to \infty$), the discrete graph Laplacian converges to the Laplace–Beltrami operator on an emergent Riemannian manifold. No background geometry is assumed: g_{ab} arises as the continuum encoding of microscopic connectivity.

Necessity of the Born–Infeld Structure

A quadratic action permits unbounded gradients and instantaneous propagation, contradicting the maximal admissible projective flux c_χ. The Born–Infeld form is the minimal non-polynomial functional enforcing strict saturation, causal consistency, and self-regularization. The saturation constant b represents the substrate's upper admissible projective flux bound; the speed of light c is a derived parameter with $c \leq b$.

B Conceptual Extensions of Cosmochrony — Particles, Quantum Phenomena, and Classical Limits

This appendix illustrates how familiar particle, quantum, and classical structures may emerge once the χ field admits localized and stable configurations. These developments serve as a conceptual bridge between the mathematical foundations (Appendix A) and the cosmological considerations (Appendix C). None are required for the logical coherence of the framework.

B.1 Interpretative Status of the χ Field

χ is a static pre-geometric substrate encoding admissibility constraints from which duration, distance, and causal ordering are reconstructed as projective consequences of Π. The notation $\chi(x^\mu)$ is representational: spacetime coordinates serve as labels for relational information in regimes where a stable geometric description has emerged. χ should not be interpreted as a matter field on spacetime, a scalar coupled to a pre-existing metric, or a hidden-variable replacement for the wavefunction. Spacetime is an emergent bookkeeping structure *for* χ, not a container *of* χ.

B.2 Topological Configurations of the χ Field: Solitons as Particles

Solitonic configurations are effective geometric representations illustrating how particle-like properties arise from stable, localized configurations of χ_{eff} once a smooth geometric projection applies (Appendix E).

Charge as Oriented Projective Asymmetry

A localized configuration deforms χ_{eff} relative to the background $\chi_{\text{eff},0}$: excess (positive charge) or deficit (negative charge). The magnitude is diagnosed by a Gauss-like flux integral $q_{\text{eff}} \propto \oint_\Sigma \nabla\chi_{\text{eff}} \cdot d\mathbf{S}$. Coulomb-like interactions emerge from frustration minimization of the Dirichlet functional $\mathcal{F}[\chi_{\text{eff}}] = \frac{1}{2}\int |\nabla\chi_{\text{eff}}|^2\, d^3x$.

Vortical Configurations and Integer-Spin Excitations

An effective winding number $n = (2\pi)^{-1} \oint \nabla \arg(\chi_{\text{eff}}) \cdot d\mathbf{l}$ characterizes charge orientation, topological robustness, and integer spin.

Skyrmion-Like Configurations and Spin-$\frac{1}{2}$ Excitations

Configurations with topological index $Q = \pm 1$ exhibit 4π-periodicity under rotations, providing a geometric origin for spin-$\frac{1}{2}$ behavior and fermionic statistics.

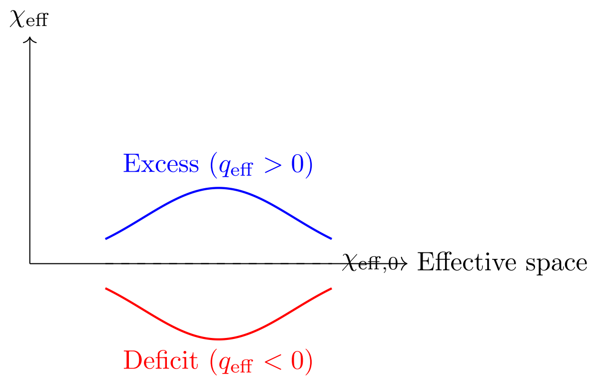

Fig. 15 Oriented deformations of χ_{eff} defining effective charge polarity.

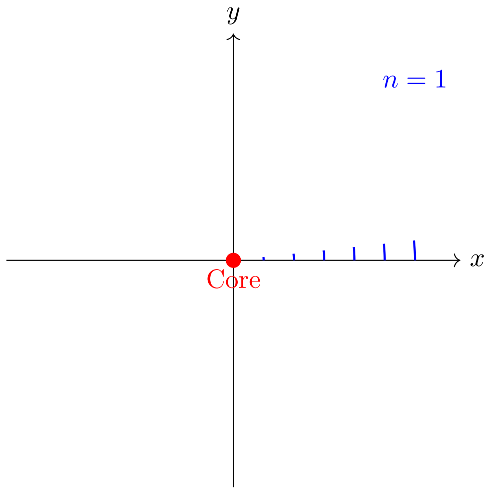

Fig. 16 Vortical configuration with winding $n = 1$ (integer spin).

Table 6 Effective solitonic configurations and emergent particle properties.

Configuration	Index	χ_{eff} asymmetry	Properties
Vortical	n	Excess/deficit	Charge $\propto n$, int. spin
Skyrmion	$Q = \pm 1$	Oriented deformation	Charge $\propto Q$, spin-$\frac{1}{2}$

Summary: Topology, Charge, and Spin

B.3 Soliton Energy and Structural Mass Scaling

Mass is interpreted as the total cost of a localized configuration's resistance to advancement of the projective ordering. In the effective geometric regime:

$$M_{\text{eff}} \propto \int_{\mathcal{V}} \left[\mathcal{T}(\nabla\chi_{\text{eff}}) + \mathcal{U}(\chi_{\text{eff}})\right] d^3x.$$

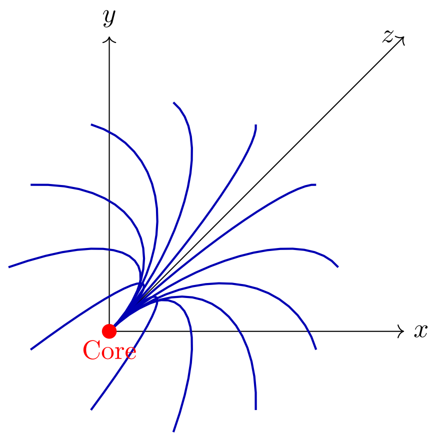

Fig. 17 Skyrmion-like configuration of $\chi_{\rm eff}$ (spin-$\frac{1}{2}$).

For kink-like configurations, a balance between gradient resistance and nonlinear stabilization fixes the soliton width ξ, yielding $M_{\rm eff} \sim \sqrt{\lambda_{\rm eff}}\,\xi\,\chi_c^2$. Configurations with higher winding or linking indices involve increased gradients, so $M_{n+1} > M_n$, establishing mass hierarchies structurally. From a spectral viewpoint, $M_{\rm eff} \sim \lambda_{\rm min}^{-1}$, where $\lambda_{\rm min}$ is the smallest positive eigenvalue of the linearized projective stability operator.

B.4 Example: 4π-Periodic Soliton and Spinorial Behavior

An illustrative construction supporting the topological interpretation of spin (Section 9.12).

In the projectable regime, certain localized excitations admit an effective internal phase θ. A complex-valued proxy (the underlying field remains real): $\chi_{\rm eff}(x) = \eta \tanh(\kappa x)\, e^{i\theta(x)}$. Choosing $\theta(\alpha) = \alpha/2$:

$$\psi(\alpha) \equiv \psi_0\, e^{i\alpha/2}. \tag{206}$$

A 2π cycle inverts the sign:

$$\psi(\alpha + 2\pi) = -\psi(\alpha). \tag{207}$$

A 4π cycle restores identity:

$$\psi(\alpha + 4\pi) = \psi(\alpha). \tag{208}$$

Schematically:

$$\begin{aligned} &\alpha : 0 \to 2\pi \Rightarrow \text{nontrivial loop } (\mathbb{Z}_2) \Rightarrow \psi \mapsto -\psi, \\ &\alpha : 0 \to 4\pi \Rightarrow \text{trivial loop} \Rightarrow \psi \mapsto \psi. \end{aligned} \tag{209}$$

This 4π-periodicity mirrors the double-cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ and is topologically compatible with fermion-like transformation behavior without introducing fundamental spinor fields. The sign change is an effective encoding of the $\mathbb{Z}_2$ obstruction (Section B.6: $\pi_1(\mathcal{C}_{\rm eff}) = \mathbb{Z}_2$).

B.5 Relation to Classical Limits

Classical behavior corresponds to a projective regime where χ_{eff} varies slowly, localized excitations are dilute, and topological constraints are suppressed. Perturbations propagate as weak disturbances on an effectively flat background, recovering superposition and approximate locality. In the nonlinear regime, large gradients induce effective curvature and horizon-like behavior. The classical limit is defined not by $\hbar \to 0$ but by the stability of a particular admissible projective regime governing the effective description of χ.

B.6 Status of the Formulation

The framework is minimal yet structurally complete at the conceptual level. A fully covariant action principle and systematic quantization remain open technical extensions, not conceptual deficiencies. GR and QFT emerge as effective, coarse-grained descriptions within specific dynamical regimes of χ_{eff}. Cosmochrony should be viewed as a foundational framework clarifying what is fundamental and how known structures arise, rather than as a closed effective field theory.

B.7 Soliton and Particle Solutions

Elementary particles are stable localized configurations arising in the projectable regime. While the fundamental field is scalar, certain configurations possess nontrivial internal organization requiring a double-valued representation under effective rotations. The Dirac equation appears as the minimal effective dynamics compatible with approximate locality, relativistic covariance, and 4π-periodic topology. Spin, fermionic statistics, and exclusion behavior arise as consequences of the nontrivial configuration space. The effective self-interaction functional must support localized configurations with finite mass, dynamical stability, and topologically inequivalent sectors.

B.8 Perspectives: Towards a Derivation of the Proton-to-Electron Mass Ratio

The proton is modeled, at the exploratory level, as a composite bound state within a $Q = 3$ sector, motivated by generic stability features observed in nonlinear solitonic systems [68, 69].

At the schematic level,

$$\frac{m_p}{m_e} \sim \sqrt{\frac{\lambda_{\mathrm{bind}}}{\lambda_e}}, \tag{210}$$

where λ_e denotes the elementary-sector stiffness scale and λ_{bind} a characteristic composite-sector scale.

Using the empirical value $m_p/m_e \simeq 1836$ as an external constraint implies an order-of-magnitude hierarchy

$$\frac{\lambda_{\mathrm{bind}}}{\lambda_e} \sim 10^6. \tag{211}$$

This relation should be understood solely as a target hierarchy for any explicit realization of the mechanism. No claim is made that the ratio is predicted or uniquely determined at the present stage.

Working Ansatz for $V(\chi)$

A minimal two-scale effective potential:

$$V(\chi) = \frac{\lambda}{4}(\chi^2 - \eta^2)^2 + \varepsilon\, \eta^4 \left[1 - \cos\left(\frac{\chi}{\eta}\right)\right], \tag{212}$$

with $\varepsilon \ll 1$.

Linking (λ, η) to Observables

Dimensionless control combinations:

$$g \equiv \lambda\, \chi_c^2\, \eta^2, \quad u \equiv \varepsilon. \tag{213}$$

Matching strategy.
The feasibility condition:

$$\exists\, (g, u) \text{ s.t. } \quad \frac{\lambda_{\text{bind}}(Q{=}3;\, g, u)}{\lambda_e(g, u)} \sim 10^6, \tag{214}$$

while remaining stable under small (g, u) variations.

Numerical Program

The pipeline tests whether the admissibility constraints on Π together with $V(\chi)$ admit stable sectors with $\lambda_{\text{bind}}/\lambda_e \sim 10^6$ without fine-tuning: dynamics and formation, soliton harvesting by topological classification, stability operator extraction, spectral ratio test, and fine-structure diagnostics.

B.9 Spectral Scaling and the Projection Ontology

Inertial mass is a spectral signature of projection visibility. The non-injective projection Π implies that each effective particle corresponds to a large equivalence class of micro-configurations whose stability eigenvalues measure the fiber weight. The ratio $m_p/m_e \approx \sqrt{\lambda_p/\lambda_e}$ is independent of the absolute action scale $\hbar_\chi$ and is therefore a structurally protected invariant.

B.10 Spectral Characterization of Mass and the Secondary Role of $V(\chi)$

Particle masses emerge as eigenmodes of an effective projective stability operator $\Delta_G^{(0)} \psi_n = -\lambda_n \psi_n$, with $m_n c^2 \propto \sqrt{\lambda_n}\, \chi_c$. This is analogous to bounded elastic systems where vibrational frequencies arise from geometry and connectivity.

Three conceptual levels are distinguished: the background-independent spectral operator (fundamental), coarse-grained χ_{eff} geometry (emergent), and interaction-induced redistributions (dynamical). $V(\chi)$ plays a secondary role, providing local

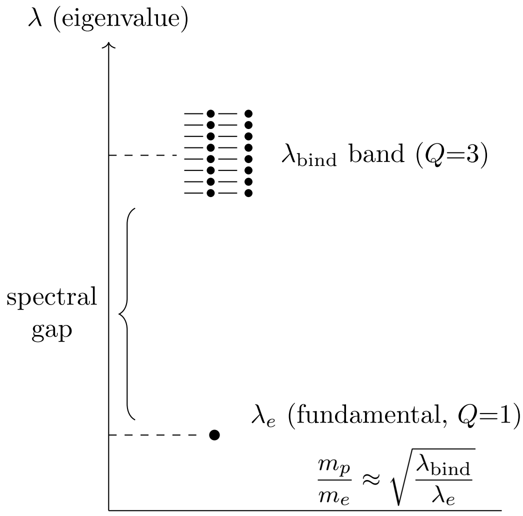

Fig. 18 Spectral gap in the stability spectrum of $\mathcal{L}_{\mathrm{sol}}$: elementary mode λ_e separated from binding-mode band λ_{bind}.

stabilization and fine splittings without setting the overall mass scale. Its admissible form is constrained by compatibility with the pre-existing spectral structure. Potential-induced corrections shift eigenvalues as $\lambda_n \to \lambda_n^{(0)} + \Delta\lambda_n^{(V)}$, affecting small splittings (e.g. neutron–proton) while leaving topologically dominated ratios largely invariant.

B.11 Spectral Stability and the Emergence of $\hbar_{\mathrm{eff}}$

From the fundamental scales K_0 (stiffness), χ_c (correlation length), and c (maximal speed), a natural action unit is $\hbar_\chi \equiv c^3/(K_0\chi_c)$. Here K_0 and χ_c denote bare substrate parameters.

For a solitonic mode with eigenvalue λ_n, identifying the rest energy $E_n = \hbar_{\mathrm{eff}}\, \nu_n$ yields $\hbar_{\mathrm{eff}}$ as a geometric and spectral quantity. At microscopic scales ($\ell \sim \chi_c$), $\hbar_{\mathrm{eff}} \approx \hbar_\chi \approx \hbar$. At macroscopic scales, $\hbar_{\mathrm{eff}} \approx \hbar_\chi(\chi_c/\ell_{\mathrm{spacetime}})^2$, suppressing quantum effects through reduced spectral accessibility rather than decoherence. Reproducing particle-scale behavior requires $K_0\chi_c^2 \sim \hbar$.

B.12 Emergence of the Schrödinger equation as an effective projected dynamics

Main theorem

The following theorem states the complete result. Steps 1–5 and Appendix B.13 constitute its proof.

Theorem B.1 (Emergence of quantum dynamics from projected admissibility structure)**.** *Let* $(\Omega, \mathcal{R}, \Pi)$ *be a Cosmochrony triple consisting of:*

(H1) *a space* Ω *of admissible relational configurations with a nonlinear admissibility operator* $\mathcal{R}$*,*

(H2) *a generally non-injective projection* $\Pi : \Omega \to O$ *with finite projective resolution* $\varepsilon_\Pi > 0$*,*

(H3) *a Born–Infeld admissibility bound: every mode satisfies* $A_n(\tau) \le c_\chi/\sqrt{\lambda_n}$ *at all* τ*,*

(H4) *a quasi-stable background* $\bar{\chi}$ *with* $\|\delta\chi\| \ll 1$ *and* $A_n \ll A_n^{\max}$*,*

(H5) *a spectrally admissible projected sector: the low-lying modes of* $\delta\chi$ *diagonalize* $\mathcal{L} \equiv D\mathcal{R}|_{\bar{\chi}}$*.*

Then:

(C1) ***Anti-Hermitian reduction.*** *The restriction of* $\mathcal{L}$ *to the projected-stable sector is anti-Hermitian:*

$$\mathcal{L}\big|_{\mathrm{proj}} = -i\,\Omega, \qquad \Omega = \Omega^\dagger, \quad \sigma(\Omega) \subset \mathbb{R}_{\ge 0}.$$

(C2) ***Unique modal evolution.*** *For each mode* u_n*, the unique solution of the projected dynamics is* $c_n(\tau) = c_n(0)\,e^{-i\omega_n\tau}$*, giving the modal Schrödinger equation*

$$i\hbar_{\mathrm{eff}}\,\partial_\tau c_n = E_n\,c_n, \qquad E_n = \hbar_{\mathrm{eff}}\,\omega_n.$$

(C3) ***Effective Schrödinger equation.*** *The projected wavefunction* $\psi(x,t) = \sum_n c_n(t)\,\phi_n(x)$ *satisfies*

$$i\hbar_{\mathrm{eff}}\,\partial_t\psi = \hat{H}_{\mathrm{eff}}\,\psi + \mathcal{N}_{\Pi,\mathrm{BI}}[\psi],$$

where $\hat{H}_{\mathrm{eff}}$ *is Hermitian, defined spectrally by* $\hat{H}_{\mathrm{eff}}\phi_n = E_n\phi_n$*, and* $\mathcal{N}_{\Pi,\mathrm{BI}} = O(A_n/A_n^{\max},\, \ell_\chi^2\Delta_g)$*.*

(C4) ***Standard Hamiltonian in the non-relativistic limit.*** *In the additional regime* $\ell_\chi^2\Delta_g \ll 1$ *and locally homogeneous substrate,*

$$\hat{H}_{\mathrm{eff}} = -\frac{\hbar_{\mathrm{eff}}^2}{2m_{\mathrm{eff}}}\,\Delta_g + V_{\mathrm{eff}}(x) + O(\ell_\chi^2\,\Delta_g^2),$$

where Δ_g *is the Laplace–Beltrami operator of the emergent metric* $g^{\mu\nu}$ *defined by* $\sigma_2(-\Delta_\chi)(x,k) = g^{\mu\nu}(x)\,k_\mu k_\nu$*, and* m_{eff} *is determined by the spectral gap of the mode family.*

(C5) ***Structural status of unitarity.*** *Norm preservation on the projected-stable sector, and hence unitarity of the effective dynamics, is not an independent postulate. It holds if and only if (H2) and (H3) hold simultaneously:*

$$\text{unitarity} \iff (\text{Born–Infeld capacity}) \cap (\text{finite projective resolution}).$$

Proof (C1) is established in Step 3: $\mathrm{Re}(\mu_n) > 0$ is excluded by (H3); $\mathrm{Re}(\mu_n) < 0$ is excluded by (H2) since the mode amplitude falls below ε_Π at finite τ; hence $\mathrm{Re}(\mu_n) = 0$ is the only structurally admissible value, giving $\mu_n = -i\omega_n$ and $\mathcal{L}|_{\mathrm{proj}} = -i\Omega$.

(C2) follows by direct integration of $\partial_\tau c_n = -i\omega_n c_n$, which is a first-order linear ODE with constant coefficients whose unique global solution is $c_n(\tau) = c_n(0)\, e^{-i\omega_n \tau}$. Multiplication by $\hbar_{\mathrm{eff}}$ and the energy-frequency relation $E_n = \hbar_{\mathrm{eff}}\omega_n$ (Section 4.5) give $i\hbar_{\mathrm{eff}}\, \partial_\tau c_n = E_n\, c_n$.

(C3) follows from (C2) by reconstruction: setting $\psi = \sum_n c_n \phi_n$ and applying $i\hbar_{\mathrm{eff}}\partial_t$,

$$i\hbar_{\mathrm{eff}}\, \partial_t \psi = \sum_n E_n\, c_n(t)\, \phi_n(x) = \hat{H}_{\mathrm{eff}}\psi$$

at leading order. The correction $\mathcal{N}_{\Pi,\mathrm{BI}}$ collects Born–Infeld nonlinearities $O(A_n/A_n^{\max})$ from (H3) and inter-modal coupling from (H4); both are negligible under (H4).

(C4) follows from the spectral identification in Appendix B.B.13: $\omega_n^2 = \lambda_n$ at leading order (from the admissibility envelope [18]), combined with the operator–metric correspondence $\sigma_2(-\Delta_\chi)(x,k) = g^{\mu\nu}k_\mu k_\nu$ [45], and the non-relativistic expansion $\hbar_{\mathrm{eff}}\sqrt{-\Delta_g} \simeq m_{\mathrm{eff}}c^2 + \frac{-\hbar_{\mathrm{eff}}^2\Delta_g}{2m_{\mathrm{eff}}}$.

(C5) is the logical converse of (C1): $\mathrm{Re}(\mu_n) = 0$ requires both (H2) and (H3); if either fails, modes with $\mathrm{Re}(\mu_n) \neq 0$ enter the sector, breaking norm preservation. □

Remark. Hypothesis (H5) — spectral admissibility of the projected sector — is a structural assumption on the relational substrate, not an additional quantization postulate. It asserts that the low-lying modes of $\mathcal{L}$ are diagonalizable within the projected-stable sector. This is guaranteed whenever the substrate background $\bar{\chi}$ is spectrally admissible in the sense of [18].

Remark. The theorem makes explicit that two distinct regimes are involved. Conclusions (C1)–(C3) require only (H1)–(H5) and hold at the level of general projected substrate dynamics. Conclusion (C4) requires the additional continuum-metric hypothesis implicit in Step B of Appendix B.B.13. The non-relativistic Hamiltonian $-\hbar^2\Delta_g/(2m)$ is therefore a further specialization, not the most general output of the theorem.

We now develop the theorem step by step. The purpose of the following construction is not to introduce additional assumptions, but to make explicit how the effective Schrödinger dynamics follows from the projected admissibility structure of the substrate in the weak-amplitude regime.

Let Ω denote the space of admissible relational configurations and

$$\Pi : \Omega \to O$$

the generally non-injective projection defining the effective observables. A projected state does not correspond to a single microscopic configuration, but to an equivalence class of configurations under Π. The effective state therefore encodes only the resolvable modal content that survives the projection at finite projective resolution.

Step 1: Substrate dynamics and linearization.

At the substrate level, the primitive dynamics is governed by the admissibility structure of Π and takes the generic form

$$\partial_\tau \chi = \mathcal{R}[\chi],$$

where τ denotes the projective ordering parameter and $\mathcal{R}$ is the nonlinear admissibility operator encoding the constraints of Π and the Born–Infeld bound. The label "$\mathcal{R}$" is retained for continuity with the formal literature; it does not imply that χ undergoes an independent dynamical relaxation. The operator $\mathcal{R}$ encodes admissibility constraints, not a temporal evolution of the static substrate χ. A configuration $\bar{\chi}$ is a fixed point of $\mathcal{R}$ if and only if it is admissible; small perturbations $\delta\chi = \chi - \bar{\chi}$ then evolve as

$$\partial_\tau \delta\chi = \mathcal{L}\,\delta\chi + O(\delta\chi^2), \qquad \mathcal{L} \equiv D\mathcal{R}\big|_{\bar{\chi}}.$$

Step 2: Spectral decomposition.

Hypothesis (H5) asserts that the low-lying modes of $\mathcal{L}$ are diagonalizable within the projected-stable sector. Let $\{u_n\}$ be a basis of eigenvectors with eigenvalues $\{\mu_n\}$: $\mathcal{L}\,u_n = \mu_n\,u_n$. Decompose the perturbation as $\delta\chi = \sum_n c_n(\tau)\,u_n$; each coefficient then satisfies

$$\partial_\tau c_n = \mu_n\,c_n, \qquad c_n(\tau) = c_n(0)\,e^{\mu_n \tau}.$$

Step 3: Projection constraints fix* $\mathrm{Re}(\mu_n) = 0$*.

Two independent structural constraints act on the eigenvalues.

Born–Infeld bound (H3). The amplitude bound $A_n(\tau) \le c_\chi/\sqrt{\lambda_n}$ excludes $\mathrm{Re}(\mu_n) > 0$: a growing mode would eventually violate the admissibility envelope and exit the domain of validity of the linearized dynamics.

Finite projective resolution (H2). A mode with $\mathrm{Re}(\mu_n) < 0$ has amplitude decaying as $|c_n(\tau)| = |c_n(0)|\,e^{\mathrm{Re}(\mu_n)\tau} \to 0$. It holds at every finite value of τ at which the amplitude has decayed below the projective resolution scale. Decaying modes therefore do not belong to the projected-stable sector.

The two constraints together leave $\mathrm{Re}(\mu_n) = 0$ as the only structurally admissible possibility for modes in the projected-stable sector. The eigenvalues therefore take the form $\mu_n = -i\omega_n$ with $\omega_n \in \mathbb{R}$. Equivalently, the restriction of $\mathcal{L}$ to the projected-stable sector is anti-Hermitian:

$$\mathcal{L}\big|_{\mathrm{proj}} = -i\,\Omega,$$

where Ω is a Hermitian operator with real spectrum $\{\omega_n\}$.

Remark. The two constraints play complementary and non-redundant roles. The Born–Infeld bound eliminates growing modes through a dynamical capacity argument: the substrate cannot sustain unbounded flux. Projectability eliminates decaying modes through a resolvability argument: a mode whose effective amplitude falls below the projective resolution threshold is not a degree of freedom of the effective description at finite projective resolution. Neither constraint alone is sufficient; their conjunction is what leaves $\mathrm{Re}(\mu_n) = 0$ as the only structurally admissible value. Norm preservation in the projected-stable sector is therefore not an additional postulate. It is the necessary consequence of bounded substrate capacity and finite projective resolution acting simultaneously. Equivalently, this establishes that unitarity of the effective quantum dynamics is not a fundamental symmetry postulate, but the structural signature of the intersection of two independent constraints:

$$\text{unitarity} = (\text{Born–Infeld capacity}) \cap (\text{non-injective projection}).$$

Step 4: From anti-Hermitian operator to phase evolution — the closure argument.

The anti-Hermitian reduction $\mathcal{L}|_{\rm proj} = -i\,\Omega$ established in Step 3 determines the modal evolution completely. For any mode u_n in the projected-stable sector, the linearized equation reads

$$\partial_\tau c_n = -i\,\omega_n\,c_n, \qquad \omega_n \in \mathbb{R},\ \omega_n \geq 0.$$

This is a first-order linear ordinary differential equation with constant coefficients, whose unique global solution is

$$c_n(\tau) = c_n(0)\,e^{-i\omega_n\tau}.$$

The decomposition $c_n(\tau) = a_n(\tau)\,e^{-i\omega_n\tau}$ with slowly varying a_n is therefore not an ansatz but a rewriting of the exact solution, with a_n absorbing only residual corrections from nonlinear and inter-modal terms:

$$\partial_\tau a_n = \big[\partial_\tau c_n + i\omega_n c_n\big]e^{i\omega_n\tau} = O\big(a_n^2,\, \nabla a_n,\, \Pi_{\rm sat}\big)\,.$$

In the weak-amplitude, approximately injective regime these correction terms are negligible, so a_n is constant to leading order.

Multiplying the exact modal equation by $\hbar_{\rm eff}$ and using $E_n = \hbar_{\rm eff}\,\omega_n$, one obtains

$$i\hbar_{\rm eff}\,\partial_\tau c_n = E_n\,c_n.$$

This is the modal Schrödinger equation. It follows from $\mathcal{L}|_{\rm proj} = -i\,\Omega$ by direct integration, with no additional hypothesis.

Step 5: Reconstruction of the wavefunction and the operator equation.

Passing from the projective ordering parameter τ to the effective projected time coordinate t amounts, in the weak-curvature regime, to a monotone reparametrization $\partial_\tau = \zeta(x,t)\,\partial_t$ with $\zeta \simeq 1$ in a locally Minkowskian domain. Absorbing ζ into the local normalization of $\hbar_{\rm eff}$ gives, to leading order,

$$i\hbar_{\rm eff}\,\partial_t c_n = E_n\,c_n.$$

Let $\phi_n(x)$ denote the effective projected spatial eigenmodes. The projected state reconstructs as

$$\psi(x,t) = \sum_n c_n(t)\,\phi_n(x).$$

Applying $i\hbar_{\rm eff}\,\partial_t$ to both sides gives

$$i\hbar_{\rm eff}\,\partial_t\psi = \sum_n \big(i\hbar_{\rm eff}\,\partial_t c_n\big)\phi_n(x) = \sum_n E_n\,c_n(t)\,\phi_n(x) = \hat{H}_{\rm eff}\,\psi,$$

where the effective Hamiltonian is defined spectrally by $\hat{H}_{\rm eff}\phi_n \equiv E_n\phi_n$. This yields

$$\boxed{i\hbar_{\rm eff}\,\partial_t\psi = \hat{H}_{\rm eff}\,\psi.}$$

Proposition B.1 (Effective Schrödinger equation)**.** *Under hypotheses* (H1)–(H5) *of Theorem B.1, the projected wavefunction* $\psi(x,t) = \sum_n c_n(t)\,\phi_n(x)$ *satisfies*

$$i\hbar_{\text{eff}}\,\partial_t\psi = \hat{H}_{\text{eff}}\,\psi + \mathcal{N}_{\Pi,\text{BI}}[\psi],$$

where $\hat{H}_{\text{eff}}$ *is Hermitian and spectrally defined by* $\hat{H}_{\text{eff}}\phi_n = E_n\phi_n$*, and* $\mathcal{N}_{\Pi,\text{BI}} = O(A_n/A_n^{\max}, \ell_\chi^2\Delta_g)$*. In the regime* $A_n \ll A_n^{\max}$ *and* $\ell_\chi^2\Delta_g \ll 1$*, the correction vanishes and the equation reduces to the standard linear Schrödinger equation.*

Proof Conclusion (C2) of Theorem B.1 gives $i\hbar_{\text{eff}}\,\partial_t c_n = E_n\,c_n$. Reconstruction via $\psi = \sum_n c_n\phi_n$ and the spectral definition of $\hat{H}_{\text{eff}}$ give (C3). The correction $\mathcal{N}_{\Pi,\text{BI}}$ is negligible under (H4). □

Structural conclusions.

This derivation clarifies the status of the standard quantum formalism.

First, the wavefunction ψ is not fundamental. It is the compressed descriptor of a spectrally admissible family of projected substrate modes belonging to the projected-stable sector at finite projective resolution.

Second, $\hbar_{\text{eff}}$ is not a primitive parameter. It measures the finite projective resolvability of the effective description and appears because phase transport remains observable after non-injective compression.

Third, the factor i does not require a fundamental complex ontology. It is the algebraic consequence of the anti-Hermitian reduction of $\mathcal{L}|_{\text{proj}}$, which itself follows from the joint action of the Born–Infeld bound and finite projective resolution.

Fourth, linearity is not exact at all scales. It holds only when three conditions are simultaneously satisfied: the substrate remains sufficiently close to a quasi-stable background, Born–Infeld-type saturation effects remain negligible, and the projection remains approximately injective over the support of the state. Outside this regime,

$$i\hbar_{\text{eff}}\,\partial_t\psi = \hat{H}_{\text{eff}}\,\psi + \mathcal{N}_{\Pi,\text{BI}}[\psi],$$

where $\mathcal{N}_{\Pi,\text{BI}}$ collects the nonlinear contributions induced by bounded projective capacity and substrate saturation. The ordinary Schrödinger equation is recovered as the universal infrared transport law of projected coherence within the projected-stable sector.

The previous subsection establishes the emergence of a Schrödinger-type evolution equation at the operator level. The next subsection identifies the spectral origin of the corresponding Hamiltonian and derives its non-relativistic Laplace–Beltrami form.

B.13 Spectral identification of the effective Hamiltonian

We adopt the convention that the Laplace–Beltrami operator Δ_g is non-positive, so that $-\Delta_g$ is positive definite. All eigenvalues λ_n of $-\Delta_g$ are therefore non-negative.

Step A: Linearized dispersion in the spectrally admissible regime.

In the spectrally admissible regime, the substrate modes u_n satisfy $-\Delta_\chi\, u_n = \lambda_n\, u_n$ with $\lambda_n \geq 0$. In the linearized (weak-amplitude) regime, the dispersion relation between ω_n and λ_n follows from the admissibility envelope [18]:

$$\omega_n^2 = \lambda_n + O\big(\lambda_n^2 A_n^2 / c_\chi^2\big)\,.$$

To leading order: $\omega_n = \sqrt{\lambda_n}$, so that $\Omega = \sqrt{-\Delta_\chi}|_{\mathrm{proj}}$.

Step B: Projected Laplacian and effective metric.

The projection Π restricts the substrate Laplacian to the admissible spectral subspace via $\Delta_\Pi = P^\dagger \Delta_\chi\, P$ (Section 5.1), where P projects onto the projected-stable sector. In the continuum limit of a locally homogeneous and isotropic relational substrate, the principal symbol of $-\Delta_\chi$ defines an effective metric through the operator–metric correspondence [45]:

$$\sigma_2(-\Delta_\chi)(x,k) = g^{\mu\nu}(x)\, k_\mu k_\nu.$$

In the low-lying spectral sector:

$$-\Delta_\Pi \simeq -\Delta_g + O(\ell_\chi^2\, \Delta_g^2),$$

where $-\Delta_g \geq 0$ is the Laplace–Beltrami operator of the effective metric g.

Step C: Effective Hamiltonian from spectral identification.

Expanding the dispersion relation in Fourier space,

$$\omega(k) = \sqrt{m_{\mathrm{eff}}^2 c^4 + c^2 k^2} \simeq m_{\mathrm{eff}} c^2 + \frac{k^2}{2m_{\mathrm{eff}}} + \cdots\,,$$

where m_{eff} is defined by the spectral gap separating the ground state of the projected sector from the first admissible excitation within a given mode family. At the operator level this gives

$$\hbar_{\mathrm{eff}} \sqrt{-\Delta_g} \simeq m_{\mathrm{eff}} c^2 + \frac{-\Delta_g}{2m_{\mathrm{eff}}} + \cdots\,.$$

Subtracting the rest-energy contribution:

$$\hat{H}_{\mathrm{eff}} = -\frac{\hbar_{\mathrm{eff}}^2}{2m_{\mathrm{eff}}}\, \Delta_g + V_{\mathrm{eff}}(x) + O\big(\ell_\chi^2\, \Delta_g^2\big)\,.$$

Complete derivation chain.

$$\partial_\tau \chi = \mathcal{R}[\chi] \xrightarrow{\text{linearize}} \mathcal{L}|_{\mathrm{proj}} = -i\,\Omega \xrightarrow{\omega_n^2 = \lambda_n} \Omega = \sqrt{-\Delta_\Pi} \xrightarrow{-\Delta_\Pi \simeq -\Delta_g} \hat{H}_{\mathrm{eff}} \simeq -\frac{\hbar_{\mathrm{eff}}^2}{2m_{\mathrm{eff}}}\, \Delta_g.$$

Here $\mathcal{R}$ denotes the admissibility operator of the projected dynamics (see §B.12, Step 1); the arrow "linearize" refers to linearization around a quasi-stable admissible background, not to a dynamical process of χ.

Three limitations apply. First, $\omega_n^2 = \lambda_n$ holds in the linearized regime; the Born–Infeld correction gives $\omega_n^2(A_n) \approx \lambda_n(1 - 3\lambda_n A_n^2/8c_\chi^2)$, negligible for $A_n \ll A_n^{\max}$. Second, the operator–metric correspondence is established in the locally homogeneous continuum limit; strongly inhomogeneous regimes receive non-metric corrections. Third, the numerical value of $m_{\rm eff}$ for specific particle species requires microscopic identification of the relevant mode family, addressed in [70].

The previous subsection identifies the spectral origin of the effective Hamiltonian. We now show that, within the projected-stable weak-amplitude regime, this Hamiltonian generates a global strongly continuous unitary evolution group.

B.14 First nonlinear correction from finite projective capacity

Context and scope

Proposition B.1 establishes the effective Schrödinger equation in the form

$$i\hbar_{\rm eff}\,\partial_t\psi \;=\; \hat{H}_{\rm eff}\,\psi \;+\; \mathcal{N}_{\Pi,\mathrm{BI}}[\psi], \tag{215}$$

where the correction term $\mathcal{N}_{\Pi,\mathrm{BI}} = \mathcal{O}(A_n/A_n^{\max},\, \ell_\chi^2\Delta_g)$ collects contributions from Born–Infeld saturation and finite projective capacity. Under hypothesis (H4), this term is negligible and the linear equation is recovered.

The present section relaxes (H4) partially: we keep $\ell_\chi^2\Delta_g \ll 1$ (continuum regime) but allow the local projective occupancy to be non-negligible. The goal is to identify the leading-order structure of $\mathcal{N}_{\Pi,\mathrm{BI}}[\psi]$ in this regime and show that it takes the form of a cubic nonlinearity of Gross–Pitaevskii type, with an effective coupling constant determined by the local spectral admissible capacity.

Projective occupancy

Let $\Sigma(x,t)$ denote the *local spectral admissible capacity*: the effective total admissible amplitude available in the projected-stable sector at point x and time t. Concretely, it is defined by

$$\Sigma(x,t) \;:=\; \sum_n w_n(x,t)\, A_n^{\max}(x,t), \tag{216}$$

where the sum runs over spectrally admissible modes, $w_n(x,t)$ are local spectral weights (normalised to unity), and $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ is the Born–Infeld admissibility envelope of mode n. The quantity Σ carries the dimension of an amplitude; it encodes the local "room" left in the projected sector before saturation.

The *projective occupancy ratio*

$$\eta(x,t) \;:=\; \frac{|\psi(x,t)|^2}{\Sigma(x,t)} \tag{217}$$

measures the fraction of local admissible capacity currently occupied by the projected wavefunction. By definition, $\eta \in [0,1]$. The regime (H4) of Theorem B.1 corresponds to $\eta \ll 1$.

Density-dependent correction to the local phase evolution

When η is no longer negligible, the finite admissible capacity induces a density-dependent correction to the local phase evolution. The mechanism is the following. Each microconfiguration $\omega \in \Pi^{-1}(x)$ contributes to the projected amplitude $\psi(x,t)$ with a phase weight determined by its admissible projective rate. At low occupancy, these contributions are independent. At higher occupancy, the constraint that the total projected amplitude must remain within the admissible envelope introduces a correlation: configurations are "competing" for the available admissible capacity. This competition feeds back on the effective local chemical potential.

More precisely, the effective local phase-evolution rate acquires a correction of the form

$$\mu_{\text{eff}}(\eta) \;=\; \mu_0 \;+\; \gamma\,\eta \;+\; \mathcal{O}(\eta^2), \tag{218}$$

where μ_0 is the dilute-regime value (determined by the spectral gap) and γ is a saturation coefficient with the dimension of an energy. The coefficient γ encodes the first-order response of the admissible capacity constraint to variations of the local occupancy. In principle, it can be expressed in terms of the local derivative of the Born–Infeld saturation functional with respect to the projected density, but its explicit form depends on the detailed microstructure of the preimage dynamics and is left as an open derivation. Substituting the definition (222) of η into (223), the correction term contributes to the equation of motion as $\mu_{\text{eff}}(\eta)\,\psi = \mu_0\,\psi + (\gamma/\Sigma)\,|\psi|^2\psi + \mathcal{O}(|\psi|^4\psi)$.

Effective nonlinear equation

Incorporating this correction into (220), the projected dynamics in the finite-occupancy regime takes the form

$$\boxed{i\hbar_{\text{eff}}\,\partial_t\psi \;=\; -\kappa\,\Delta_g\,\psi \;+\; V_{\text{eff}}(x,t)\,\psi \;+\; g(x,t)\,|\psi|^2\,\psi \;+\; \mathcal{O}(|\psi|^4\psi),} \tag{219}$$

where $\kappa = \hbar_{\text{eff}}^2/(2m_{\text{eff}})$ is the dispersion coefficient (inherited from §B.13) and the *effective nonlinear coupling* is

$$g(x,t) \;:=\; \frac{\gamma}{\Sigma(x,t)}. \tag{220}$$

Equation (224) has the form of the Gross–Pitaevskii / nonlinear Schrödinger equation, but with a fundamental difference in interpretation: the coupling $g(x,t)$ is *not postulated* but *derived* from the finite local spectral admissible capacity. It varies in space and time through $\Sigma(x,t)$, reflecting the local structure of the admissible spectral sector.

Remark (Structural interpretation of the cubic term)*.* In this framework, the term $g\,|\psi|^2\psi$ does not represent a literal self-interaction of the wavefunction with itself. It encodes the *self-retroaction of projective multiplicity*: the density $|\psi|^2$ measures how densely the local projected image is occupied by preimage configurations, and this density modifies the effective phase-evolution rate via the admissible capacity constraint.

Remark (Sign of g and focusing/defocusing regimes)*.* The sign of γ depends on the dominant correction mechanism. If the saturation constraint acts as a repulsive pressure (as in the standard Born–Infeld saturating regime), one expects $\gamma > 0$, giving $g > 0$: a *defocusing* nonlinearity that further spreads the wavefunction. In regimes where the competition between preimages leads to *constructive mutual reinforcement* of phase-coherent configurations, one can have $\gamma < 0$ and $g < 0$: a *focusing* nonlinearity capable of balancing dispersive spreading. The physically relevant regime for localised structures corresponds to the focusing case.

Solitonic self-coherent projected states

In the focusing regime ($g < 0$) and in one spatial dimension, equation (224) admits stationary localised solutions of the form

$$\psi(x,t) \;=\; \phi(x)\, e^{-i\omega t/\hbar_{\rm eff}}, \tag{221}$$

where ϕ satisfies the nonlinear eigenvalue problem

$$-\kappa\,\phi'' \;+\; V_{\rm eff}\,\phi \;+\; g\,|\phi|^2\,\phi \;=\; \omega\,\phi. \tag{222}$$

In the homogeneous case ($V_{\rm eff} = 0$, g constant), the exact solution is

$$\phi(x) \;\propto\; \operatorname{sech}\!\left(\frac{x - x_0}{L}\right), \qquad L = \sqrt{\frac{2\kappa}{|\omega|}}, \tag{223}$$

with amplitude and width determined jointly by κ, $|g|$, and ω.

Within the Cosmochrony framework, these solutions receive a specific interpretation:

Proposition B.2 (Solitonic states as self-coherent projective classes)**.** *A solitonic solution of the form* (226)–(228) *corresponds to a class of relational configurations* $\{\omega \in \Omega : \Pi(\omega) \approx \phi\}$ *that is* projectivement auto-cohérente*: the tendency of substrate modes to dephase and disperse (the term* $-\kappa\Delta_g\psi$*) is exactly balanced by the saturation-induced localisation (the term* $g|\psi|^2\psi$*).*

Equivalently, these are configurations in which the dispersive divergence of preimage phases is compensated by the projective multiplicity feedback, so that the projected image remains stationary in shape.

The dispersive term $-\kappa\Delta_g\psi$ arises, as shown in §B.13, from the divergence of projected mode phases across the preimage $\Pi^{-1}(x)$. The nonlinear term $g|\psi|^2\psi$ is the leading correction from the finite admissible capacity, which constrains how many independent phase contributions can accumulate. The soliton is the configuration where these two effects exactly cancel.

Regime diagram

The three qualitative regimes are summarised as follows:

- $\eta \ll 1$ (dilute projective regime): $g\,|\psi|^2\psi \approx 0$; standard linear Schrödinger equation; no localisation beyond external potential.
- $\eta \lesssim 1$ (moderate occupancy): $g\,|\psi|^2\psi$ is the leading correction; Gross–Pitaevskii-type dynamics; possible soliton formation in the focusing case.
- $\eta \to 1$ (near-saturated regime): higher-order terms in $\mathcal{O}(\eta^2)$ become relevant; the effective description departs from exact unitary evolution at the level of the projected dynamics, signalling the breakdown of the low-occupancy approximation and the onset of strongly constrained projective regimes; the effective description approaches the boundary of projectability.

Structural status

Three points deserve emphasis.

First, equation (224) is not a modification of quantum mechanics. It is the leading-order effective equation in a regime where hypothesis (H4) partially fails. In all regimes currently accessible to laboratory experiments, $\eta \ll 1$ and the linear Schrödinger equation remains valid to all measurable precision.

Second, the coupling $g(x,t) = \gamma/\Sigma(x,t)$ is not a free parameter. It is determined by the local spectral admissible capacity Σ, which is itself fixed by the Born–Infeld admissibility envelope and the local spectral weights. The nonlinearity is therefore structurally constrained, not phenomenological. In particular, the origin of the coupling $g(x,t)$ can be traced back to the Born–Infeld saturation constraint on the substrate amplitudes: the nonlinear term is the effective manifestation, at the level of the projected dynamics, of the bounded-flux condition $|\partial_t \chi_v| \le c_\chi$. This closes the logical chain from the admissibility bound (Section 2.4) through the linear Schrödinger emergence (§B.12) to the present nonlinear correction.

Third, the equation (224) with the identification (225) provides a natural bridge to collective quantum phenomena: in a regime where many particles share the same projected spatial mode (Bose–Einstein condensation), the effective coupling $g(x,t)$ encodes the collective projective pressure, making contact with the Gross–Pitaevskii equation of condensate physics without postulating inter-particle interactions as fundamental. These connections are developed further in the companion papers on superconductivity and collective quantum dynamics [71].

B.15 Unitary evolution group and Stone's theorem

The modal Schrödinger equation established in Theorem B.1 admits a global operator formulation. This subsection shows that the projected dynamics generates a strongly continuous one-parameter unitary group, and identifies the precise role of the Cosmochrony hypotheses in the applicability of Stone's theorem.

Setting.

Let $\mathcal{H}_{\text{proj}}$ denote the Hilbert space of projected-stable amplitudes, defined as the ℓ^2-completion of the projected-stable sector:

$$\mathcal{H}_{\text{proj}} \equiv \overline{\text{span}\{u_n : \text{Re}(\mu_n) = 0\}}^{\ell^2}.$$

In the spectrally admissible regime of hypothesis (H5), this space is separable. The operator Ω established in conclusion (C1) acts on $\mathcal{H}_{\text{proj}}$ with domain

$$\mathcal{D}(\Omega) = \left\{\psi = \sum_n c_n u_n \in \mathcal{H}_{\text{proj}} : \sum_n \omega_n^2 |c_n|^2 < \infty\right\}.$$

Proposition B.3 (Self-adjointness of Ω)**.** *Under hypotheses* (H1)–(H5)*, the operator* Ω *defined by* $\Omega u_n = \omega_n u_n$*,* $\omega_n \geq 0$*, is self-adjoint on* $\mathcal{D}(\Omega) \subset \mathcal{H}_{\text{proj}}$*.*

Proof The modes $\{u_n\}$ form an orthonormal basis of $\mathcal{H}_{\text{proj}}$ by hypothesis (H5). The operator Ω is diagonal in this basis with real eigenvalues $\omega_n \geq 0$. A diagonal operator with real spectrum on an orthonormal basis is self-adjoint on the domain of square-summable sequences of weighted coefficients, which is precisely $\mathcal{D}(\Omega)$. In the finite-dimensional case (finitely many admissible modes), self-adjointness is immediate since every Hermitian matrix on $\mathbb{C}^N$ is self-adjoint. In the infinite-dimensional case, Ω is essentially self-adjoint on the dense subspace of finite linear combinations of $\{u_n\}$, and its unique self-adjoint extension has domain $\mathcal{D}(\Omega)$. □

Remark. Proposition B.3 is not independent of the Cosmochrony hypotheses. It relies on (C1): the eigenvalues ω_n are real because $\mathcal{L}|_{\text{proj}} = -i\Omega$ with Ω Hermitian, which itself follows from the joint action of (H2) and (H3). If either the Born–Infeld bound or the finite projective resolution were absent, modes with $\text{Re}(\mu_n) \neq 0$ would enter the sector, Ω would acquire a non-zero imaginary part in its spectrum, and self-adjointness would fail.

Theorem B.2 (Unitary evolution group)**.** *Under hypotheses* (H1)–(H5)*, define the one-parameter family of operators*

$$U(t) \equiv e^{-i\Omega t/\hbar_{\text{eff}}}, \qquad t \in \mathbb{R},$$

acting on $\mathcal{H}_{\text{proj}}$ *by* $U(t)\, u_n = e^{-i\omega_n t/\hbar_{\text{eff}}}\, u_n$*. Then:*

(U1) ***Strong continuity.*** *$t \mapsto U(t)$ is strongly continuous: for every $\psi \in \mathcal{H}_{\text{proj}}$, $\|U(t)\psi - \psi\|_{\mathcal{H}} \to 0$ as $t \to 0$.*
(U2) ***Unitarity.*** *$U(t)^* = U(-t) = U(t)^{-1}$ for all $t \in \mathbb{R}$: the family $\{U(t)\}_{t\in\mathbb{R}}$ is a one-parameter unitary group.*
(U3) ***Group law.*** *$U(t + s) = U(t)\, U(s)$ for all $t, s \in \mathbb{R}$.*

(U4) ***Generator.*** *The infinitesimal generator of $\{U(t)\}$ is $-i\Omega/\hbar_{\rm eff}$: for $\psi \in \mathcal{D}(\Omega)$,*

$$\lim_{t\to 0} \frac{U(t)\psi - \psi}{t} = -\frac{i\Omega}{\hbar_{\rm eff}}\, \psi,$$

equivalently,

$$i\hbar_{\rm eff}\, \partial_t\, U(t)\psi = \hat{H}_{\rm eff}\, U(t)\psi, \qquad \hat{H}_{\rm eff} \equiv \hbar_{\rm eff}\, \Omega.$$

(U5) ***Uniqueness.*** *$\{U(t)\}_{t\in\mathbb{R}}$ is the unique strongly continuous unitary group on $\mathcal{H}_{\rm proj}$ with generator $\hat{H}_{\rm eff}/\hbar_{\rm eff}$.*

Proof By Proposition B.3, $\Omega/\hbar_{\rm eff}$ is self-adjoint on $\mathcal{D}(\Omega)$. Stone's theorem (see e.g. [72]) states that every self-adjoint operator A on a separable Hilbert space generates a unique strongly continuous one-parameter unitary group e^{-iAt}, and conversely every such group arises from a self-adjoint generator. Applying this to $A = \Omega/\hbar_{\rm eff}$ on the separable space $\mathcal{H}_{\rm proj}$ gives (U1)–(U5) directly.

The identification of the generator with $\hat{H}_{\rm eff}/\hbar_{\rm eff}$ in (U4) follows from conclusion (C2): the modal equation $i\hbar_{\rm eff}\partial_t c_n = E_n c_n$ with $E_n = \hbar_{\rm eff}\omega_n$ is the component form of $i\hbar_{\rm eff}\partial_t U(t)\psi = \hat{H}_{\rm eff} U(t)\psi$. □

Corollary B.1 (Norm preservation)**.** *For every $\psi \in \mathcal{H}_{\rm proj}$ and all $t \in \mathbb{R}$,*

$$\|U(t)\psi\|_{\mathcal{H}} = \|\psi\|_{\mathcal{H}}.$$

In particular, $\sum_n |c_n(t)|^2 = \sum_n |c_n(0)|^2$ for all t.

Proof Unitarity (U2) gives $U(t)^*U(t) = \mathrm{Id}$, hence $\|U(t)\psi\|^2 = \langle U(t)\psi, U(t)\psi\rangle = \langle \psi, U(t)^*U(t)\psi\rangle = \|\psi\|^2$. □

Remark (Relation to Conclusion (C5)). Corollary B.1 is the global, time-integrated form of Conclusion (C5) of Theorem B.1. (C5) establishes norm preservation as a structural consequence of (H2)∩(H3) at the infinitesimal level (via the anti-Hermitian reduction). Theorem B.2 upgrades this to a global statement via Stone's theorem: the infinitesimal generator being self-adjoint is sufficient to guarantee exact norm preservation for all time, with no approximation. The logical chain is therefore:

$$(\mathrm{H2}) \cap (\mathrm{H3}) \xrightarrow{(\mathrm{C1})} \Omega = \Omega^\dagger \xrightarrow{\text{Stone}} U(t) = e^{-i\Omega t/\hbar_{\rm eff}} \text{ unitary } \Rightarrow \|U(t)\psi\| = \|\psi\|.$$

Remark (Scope and limitations). Theorem B.2 applies within the projected-stable sector and in the weak-amplitude regime (H4). Outside this regime, the correction term $\mathcal{N}_{\Pi,\mathrm{BI}}[\psi]$ in (C3) modifies the effective dynamics and generally breaks the exact unitary-group structure.

The evolution is then no longer described by a strongly continuous unitary group. Instead, it is governed by a nonlinear and generally non-unitary effective flow, reflecting the amplitude-dependent corrections induced by Born–Infeld saturation and finite projective capacity. In particular, exact norm conservation need not hold near the projective capacity boundary.

This departure from exact unitarity is not a defect of the framework but a structural prediction: the standard unitary quantum dynamics is recovered precisely in the regime where Born–Infeld saturation and projective non-injectivity effects are simultaneously negligible.

B.16 Continuity with the standard formalism and expected corrections

The emergence of the Schrödinger equation in the projected weak-amplitude regime explains why standard quantum mechanics remains empirically valid across all currently accessible laboratory domains. In those regimes, the effective description is sufficiently far from the saturation threshold that the nonlinear corrections induced by bounded projective capacity remain negligible.

The present framework nevertheless predicts that exact linearity cannot hold arbitrarily. When localization becomes too sharp, when relational gradients become too large, or when the projective description approaches a saturation boundary, the effective Schrödinger dynamics must receive corrections. These corrections preserve the ordinary formalism in the infrared and moderate-resolution regime, while producing departures only near the structural limits of projectability.

Cosmochrony does not reinterpret the Schrödinger equation as false or merely approximate in an informal sense. Rather, it identifies the precise structural reason why it must arise, and also why it cannot be ultimate. The quantum formalism is the natural low-resolution description of a deeper relational dynamics, just as effective geometry is the low-curvature description of a deeper projective structure.

In this sense, the relation between Cosmochrony and quantum mechanics is analogous to the relation between kinetic theory and hydrodynamics. The effective equation is real, predictive, and operationally complete within its domain. Its domain of validity is, however, determined by a more primitive substrate. The wavefunction ψ is therefore not eliminated, but reinterpreted: it remains the correct effective descriptor of projected modal structure within the projected-stable sector, while its apparent fundamentality is replaced by an emergent status grounded in the joint structural constraints of bounded substrate capacity and finite projective resolution.

B.17 Renormalization of Substrate Parameters

Bare parameters (K_0, χ_c) are universal substrate invariants; effective parameters (K_{eff}, χ_{eff}) emerge through coarse-graining (Appendix D). The universality of $\hbar$ and the spectral invariant α_{spec} stems from their dependence on bare ratios, invariant under projective scaling within a given projective epoch. Dimensionless couplings such as α_{EM} inherit no arbitrariness from substrate parameters: the electric charge is a geometric invariant of the soliton's spectral embedding relative to Π, so the dependence on K_0 cancels in dimensionless ratios.

B.18 Structural Origin of Quantum Correlations and Non-Locality

The non-injective projection Π provides a structural reinterpretation: what appear as two spatially separated particles may correspond to a single relational configuration in χ. Entanglement is the manifestation of a single χ-source through multiple non-injective projective images. Spin correlations arise from torsional conservation: a local stabilization at A structurally constrains admissible projective states at B originating from the same source.

The mechanism through which a single χ source constrains two spatially separated projective images can be made more precise. Let $I(\omega)$ denote a global invariant of the configuration $\omega \in \Omega_\chi$—a conserved topological quantity such as total winding number, total charge, or total angular momentum. Such an invariant belongs to a sector of the substrate that is not individually projectable: it cannot be decomposed as $I_A \oplus I_B$ with I_A and I_B each independently representable as local effective observables in O_A and O_B. It follows that the fiber $\Pi^{-1}(o_{AB})$ cannot be factored as $\Pi^{-1}(o_A) \times \Pi^{-1}(o_B)$: any admissible description must treat $I(\omega)$ as a single global property of the joint projected class. The two observers at A and B do not measure two independent properties of the substrate. They read the same underlying invariant from two distinct local relational frames Π_A and Π_B. The perfect correlations that characterize entangled states reflect the fact that there is only one invariant to be distributed across the two frames, not two independent quantities that happen to agree. Bell-type factorization fails because it assumes that the effective variable $\lambda = \Pi(\omega)$ admits a product representation $\Pi^{-1}(o_A) \times \Pi^{-1}(o_B)$; the infra-projectable character of $I(\omega)$ makes this assumption structurally inadmissible.

The geometric origin of the $\cos\theta$ correlation law and the Tsirelson bound can be stated compactly in terms of the substrate invariant structure. Let the substrate invariant associated with a singlet configuration be represented by a unit element of S^3, reflecting the SU(2) structure of the spin sector of the projection fiber. Directional measurements correspond to projections of this S^3 invariant onto axes in the effective space S^2, with spin observables satisfying $(\boldsymbol{\sigma} \cdot \mathbf{a})^2 = 1$. The resulting mean correlation is $E(\mathbf{a}, \mathbf{b}) = \langle (\boldsymbol{\sigma} \cdot \mathbf{a}) \otimes (\boldsymbol{\sigma} \cdot \mathbf{b}) \rangle = -\mathbf{a} \cdot \mathbf{b}$, reproducing the quantum mechanical prediction. Note that a classical model with binary outcomes $A = \text{sign}(I \cdot \mathbf{a})$, $B = -\text{sign}(I \cdot \mathbf{b})$ and I uniformly distributed on S^2 yields $E = 1 - 2\theta/\pi$, not $-\cos\theta$: the correct $\cos\theta$ law requires the SU(2) operator structure, not a classical sign projection. The CHSH combination satisfies $|S| \leq 2\sqrt{2}$ as a consequence of the Cauchy–Schwarz inequality applied to unit vectors in $\mathbb{R}^2$; this bound is saturated at the optimal orientation angles and coincides with the Tsirelson bound. Both results—the $\cos\theta$ law and the bound $2\sqrt{2}$—are therefore structural consequences of the spherical geometry of the projection space induced by Π, not independent quantum postulates. A factorizable hidden-variable model would require $\Pi^{-1}(o_{AB}) = \Pi^{-1}(o_A) \times \Pi^{-1}(o_B)$, which is incompatible with the global uniqueness of $I(\omega)$; Bell factorization fails for the same reason that the invariant cannot be split into two independently projectable local components.

The framework remains ontologically realist at the substrate level but structurally non-local with respect to the emergent spacetime.

B.19 Metastability, Decay Channels, and Exponential Lifetimes

Diagnostic Structural Functional

$E_{\text{struct}}[\chi_{\text{eff}}] \sim \int_{\mathcal{V}} (|\nabla \chi_{\text{eff}}|^2 + \mu^2 |\chi_{\text{eff}}|^2)\, d^3x$ quantifies the projective resistance of the configuration to advancement of the admissible ordering.

Admissible Factorization Channels

A decay channel is an admissible factorization preserving topological invariants: $Q(\chi_{\text{eff},A}) = \sum_i Q(\chi_{\text{eff},i})$. Kinematic accessibility requires $\Delta E_{\text{struct}} > 0$. The survival probability is $P(\tau) = \exp(-\Gamma\tau)$ with $\Gamma = \sum_c \Gamma_c$.

Different interaction classes correspond to different degrees of constraint: strong decays involve direct topological reorganization, electromagnetic decays preserve core topology, and weak decays require deep reconfiguration through rarer paths.

Non-Injective Projection and Structural Factorization

Entanglement: a single χ_0 admits a non-factorizable projection. Decay: the projection becomes unstable and admissibility is recovered through factorization $\Pi(\chi_0) \to \Pi(\chi_1) \oplus \Pi(\chi_2) \oplus \cdots$. The distinction is not at the substrate level but in the stability properties of the projected description.

B.20 Measurement, Temporal Ordering, and Antiparticle Emergence

Measurement corresponds to the stabilization of a single projected description $\Pi_{\alpha^*}(\chi_0)$ under interaction with an environment. A partial ordering $\prec$ on admissible projected configurations induces an effective arrow of time without a fundamental temporal parameter.

Antiparticles emerge when admissible factorization of a configuration with $Q(\chi_A) = 0$ produces paired contributions with opposite signs: $Q(\chi_i) = +q$, $Q(\chi_j) = -q$. This reflects the necessity of preserving signed structural invariants under non-injective projection.

B.21 Structural Interpretation of CPT Symmetry

The combined transformation $(Q, \tau, \mathbf{x}) \to (-Q, -\tau, -\mathbf{x})$ leaves admissibility conditions invariant. Under factorization, conservation of signed invariants enforces particle–antiparticle pairs. CPT symmetry emerges as an invariance of the admissible projection structure.

B.22 CP Asymmetry and Chiral Selection

This appendix develops the dynamical and spectral structure of CP violation within the Cosmochrony framework. The underlying topological mechanism — projective chirality as defined by $\Gamma(Q) \neq \Gamma(-Q)$ — is established at the framework level in Section 5.3 and [33]; the present section examines its spectral consequences and its relation to the Jarlskog invariant.

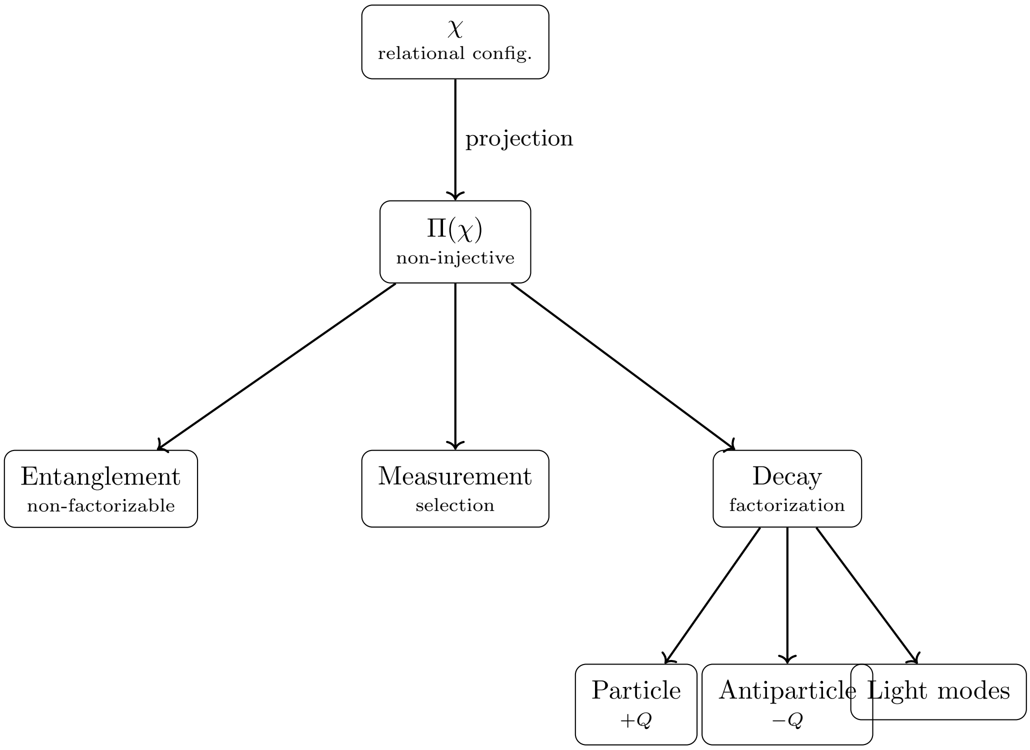

Fig. 19 Unified structural interpretation: entanglement, measurement, and decay all originate from Π. Antiparticles emerge when signed invariants are redistributed.

CPT versus CP as Admissibility Symmetries

Let projected configurations carry a set of signed structural invariants $\{Q_i\}$, associated with orientation, chirality, or phase winding. The admissibility conditions are invariant under the combined transformation

$$(Q_i,\ \tau,\ \mathbf{x}) \to (-Q_i,\ -\tau,\ -\mathbf{x}),$$

which defines an effective CPT symmetry at the level of the projected description.

In contrast, CP acts only on a subset of the invariants $\{Q_i\}$ and does not reverse the effective ordering parameter. CP is therefore not, in general, an invariance of admissibility. Effective CP violation may arise without violating CPT invariance, reflecting a structural asymmetry of the projection rather than a breakdown of the relational substrate.

Structural Bias and Statistical Asymmetry

Assume that admissible projected configurations exhibit a slight asymmetry in projective stabilization efficiency with respect to the sign of a structural invariant Q. Let $\Gamma(Q)$ denote the effective stabilization rate.

If

$$\Gamma(Q) \neq \Gamma(-Q),$$

then configurations carrying one orientation are statistically favored during projective selection. Matter–antimatter asymmetry may therefore emerge as a dynamical bias in the projective selection process, without requiring explicit symmetry breaking at the fundamental level.

This statistical asymmetry provides the ontological background within which the more specific spectral structure of flavor mixing, reconstructed in Section 11.6, can operate.

Spectral Structure and the Jarlskog Invariant

In the Standard Model, CP violation in the quark sector is characterized by the Jarlskog invariant [73], which vanishes if any two fermion masses within a given sector are degenerate or if the mixing matrix is reducible to a two-generation form.

Within Cosmochrony, this structure is interpreted in terms of the spectral properties of the projection fiber Π. Let $\{\lambda_{u,i}\}$ and $\{\lambda_{d,i}\}$ denote the effective spectral eigenvalues associated with the up-type and down-type projective modes, respectively. These eigenvalues determine the inertial masses via Eq. (29) and encode the projective resistance of the corresponding projected configurations.

Degeneracy of eigenvalues within either sector,

$$\lambda_{u,i} = \lambda_{u,j} \quad \text{or} \quad \lambda_{d,i} = \lambda_{d,j},$$

restores rotational freedom within the corresponding subspace, allowing phase redefinitions that eliminate any CP-odd invariant.

The presence of three spectrally distinct generations ($n \geq 3$) is therefore the minimal condition under which an irreducible phase can arise. This threshold signals that the projective holonomy of the flavor manifold cannot be embedded in a two-dimensional spectral subspace.

Sectorial Misalignment and Cubic Commutator Structure

The full CP-violating structure depends not only on spectral non-degeneracy within a single sector, but on the incompatibility between the up-type and down-type spectral bases.

Let Π_u and Π_d denote the sectorial restrictions of the global non-injective mapping Π associated with these two projective channels. Because the projected state U has finite descriptive capacity (bounded by the projective bandwidth b_χ), these operators cannot in general be simultaneously diagonalized. Their non-commutativity reflects an irreducible misalignment of spectral anchors within the same fiber.

Following the algebraic structure identified by Jarlskog [73], the CP-odd invariant can be expressed in terms of the imaginary part of the trace

B.23 Non-Injectivity as a Structural Necessity of Genuine Emergence

Non-injectivity of the projection $\Pi : \Omega \to \mathcal{O}$ is not a modelling choice introduced to generate desired phenomenology. It is a structural necessity of any genuinely emergent physical description, provably so under minimal hypotheses [20].

The core result is the following. Let $\Pi : \Omega \to \mathcal{O}$ be surjective, with observables entirely defined by Π — meaning that $\mathcal{O}$ carries no independent microstate-resolving structure beyond what Π provides — and suppose that $\mathcal{O}$ is not structurally isomorphic to Ω. Then Π is necessarily non-injective and the associated projection entropy satisfies $S_\Pi > 0$. The proof is elementary: if Π were injective, it would be a structural isomorphism $\mathcal{O} \cong \Omega$, contradicting the assumption of genuine emergence. A fully injective emergent description is therefore a contradiction in terms — it reduces emergence to a mere relabelling of the fundamental level.

Three structural consequences are co-extensive with this necessity.

1. **Descriptive redundancy.** The multiplicity of pre-images $\Pi^{-1}(o)$ for a given $o \in \mathcal{O}$ defines a structural projection entropy

$$S_\Pi \;\equiv\; -\sum_{o\in\mathcal{O}} \nu(o)\, \log \nu(o), \qquad \nu(o) = \mu\big(\Pi^{-1}(o)\big)\,, \tag{224}$$

   which is strictly positive whenever Π is non-injective and measures an objective structural property of Π, not epistemic ignorance about the underlying configuration. In any genuinely emergent framework, descriptive redundancy, non-injectivity of Π, and $S_\Pi > 0$ are co-extensive [20].
2. **Emergent gauge structure.** If Π were injective and compatible with relational locality, parallel transport of effective states along any closed loop in $\mathcal{O}$ would lift uniquely to a closed loop in Ω, and the holonomy group of the induced fiber bundle would reduce to the identity. Trivial holonomy precludes any non-trivial fiber re-identification symmetry and therefore any emergent gauge degree of freedom (Appendix A.9). Any non-trivial gauge structure therefore requires Π to be non-injective, and hence $S_\Pi > 0$.
3. **Projective Bell-violation.** Non-injectivity implies that the joint outcome distribution of spatially separated measurements cannot be factorized into independent local contributions, because the underlying configurations are not individually assigned to localized observable outcomes. Bell-type correlations arise as the minimal observable signature of $S_\Pi > 0$ [21].

In the Cosmochrony framework, the intermediate geometric hypothesis required to pass from the universal necessity of non-injectivity to the specific gauge group $\mathrm{SU}(2)\times U(1)$ is explicitly realised: the projection $\Pi : \mathcal{C}_\chi \to M$ defines a principal bundle with fibre S^3, and the Born–Infeld saturation bound selects the unique admissible infrared dynamics. The route from non-injectivity to SU(2) via Born–Infeld parity, Weil representations, and the Hurwitz classification is detailed in Appendix A and the O-series companion papers [74, 75].

In summary, non-injectivity is the unique structural condition that simultaneously accounts for descriptive redundancy ($S_\Pi > 0$), emergent gauge structure (non-trivial holonomy), finite transport capacity (bounded effective flux), and projective Bell-violation (loss of probabilistic factorization). It is a theorem, not an assumption.

B.24 Bounded Flux as the Origin of Character Vanishing

The bounded-flux condition introduced in Chapter 6 imposes a constraint on the maximal amplitude that a spectral mode of the relational Laplacian can sustain. This section clarifies why this constraint selects, at the level of representation theory, generators whose character vanishes in a common irreducible sector.

Consider the relational Laplacian $\mathcal{L}_\chi$ acting on the configuration space of the χ substrate, and let ρ denote an irreducible representation sector with associated Laplacian eigenvalue λ_ρ. The bounded-flux condition requires

$$A_\rho^{\max} \;=\; \frac{c_\chi}{\sqrt{\lambda_\rho}}, \tag{225}$$

where c_χ is the maximal admissible projective speed. Sectors with smaller eigenvalues admit larger amplitudes and are therefore dynamically preferred.

Now consider a generating set S of a finite group G acting on a spectral sector ρ. Each generator $s \in S$ acts on the eigenspace V_ρ via the matrix $\rho(s)$. The local projective cost associated with s is proportional to the real part of $1 - \chi_\rho(s)/\dim\rho$, where $\chi_\rho(s) = \mathrm{Tr}(\rho(s))$ is the character.

The admissibility condition (230) is saturated by the lowest available eigenvalue. For a Cayley graph with generating set S, the Laplacian eigenvalue in sector ρ takes the form

$$\lambda_\rho \;=\; |S| - \frac{1}{\dim\rho}\sum_{s\in S}\chi_\rho(s). \tag{226}$$

Minimizing λ_ρ subject to the constraint that S generates G is equivalent to maximizing $\sum_{s\in S}\chi_\rho(s)$. The global minimum is achieved when $\chi_\rho(s) = 0$ for all $s \in S$, in which case $\lambda_\rho = |S|$, and the admissibility envelope is at its most permissive.

The character-vanishing condition $\chi_\rho(s) = 0$ is therefore not an independently imposed algebraic requirement. It emerges as the necessary condition for generators to saturate the bounded-flux filter at the lowest possible spectral cost, and hence to belong to the admissible sector as defined by the projective structure of Π.

B.25 Non-Injectivity as the Structural Origin of Spinorial Selection

The selection of spinorial representations — characterized here by groups admitting a two-dimensional faithful irreducible representation, ultimately converging to $SU(2)$ — is not imposed by hand. It arises from the interplay between non-injective projection and the bounded-flux constraint developed in Chapter 6.

The key mechanism is the following. Non-injective projection introduces a fiber structure over the space of effective observables. Closed loops in the observable space may fail to lift to closed loops in the relational configuration space, generating non-trivial holonomy. For holonomy to be non-trivial and physically meaningful, the fiber must support a non-abelian structure.

Within the framework of Chapter 6, the groups capable of realising admissible neutral generating sets are classified (Theorem 6.1). The classification yields exactly four families: Q_8, Dic_n $(n \geq 3)$, $2O$, and $2I$. All of these are subgroups of $SU(2)$, not of $SO(3)$.

The exclusion of $SO(3)$ counterparts — the ordinary dihedral groups, the octahedral group O, and the icosahedral group I — follows from the requirement that the second-moment tensor $M = \sum_i \vec{u}_i \vec{u}_i^T$ must be compatible with the isotropic constraint, and from the binary cover structure that is necessary for non-trivial holonomy of the spinorial fiber. An $SO(3)$-type generator quotients out the central $\mathbb{Z}_2$ of $SU(2)$, which destroys the 4π-periodicity required for spinorial behavior.

In this sense, the $SU(2)$ thread is not a symmetry postulate. It is the unique emergent structure compatible simultaneously with non-injective projection, bounded flux, and the non-trivial holonomy required for emergent gauge and curvature structure.

B.26 Status of the SU(2) Thread as Structural Attractor

The preceding subsections establish that $SU(2)$ emerges as the unique continuous limit of the chain of admissible finite relational structures $Q_8 \hookrightarrow 2I \hookrightarrow \mathrm{LPS}_p \hookrightarrow SU(2)$. This subsection clarifies the precise sense in which $SU(2)$ is an attractor and not merely a convenient limit.

Finite resolution versus continuum.

At any fixed relational resolution, the admissible structure is a finite group. The admissibility ratio

$$R_p = \frac{A_{\rho_2}^{\max}}{A_{\rho_1}^{\max}} = \sqrt{\frac{\lambda_{\rho_1}}{\lambda_{\rho_2}}} > 1 \tag{227}$$

quantifies how much more amplitude the spinorial sector can sustain relative to the abelian sectors. For Q_8 on its 6-regular Cayley graph, $R_p = \sqrt{4/3} \approx 1.155$. For $2I$, the hierarchy is more pronounced: $R_p = \sqrt{14/9} \approx 1.247$. For the Lubotzky-Phillips-Sarnak family LPS_p, the ratio grows with p while the graph remains Ramanujan, so the spectral gap is maximally efficient.

As the resolution scale increases and the Cayley graph approximates the Haar measure on $SU(2) \cong S^3$, the distinction between abelian and non-abelian sectors dissolves: the continuous group has a single connected component, and the admissibility ratio $R_p \to 1$. The filter disappears not because it is switched off, but because S^3 already saturates the bounded-flux condition uniformly across all modes.

Attractor in the space of finite relational structures.

The penalized spectral capacity $C_\alpha(G, S)$, which discounts groups achieving large amplitudes through constructive alignment of sectors, systematically favors binary polyhedral groups over their $SO(3)$ quotients for all $\alpha > 0$. Binary covers dominate precisely because the central $\mathbb{Z}_2$ they retain prevents constructive interference between spinorial and abelian sectors, realising a more uniform distribution of admissible amplitude over the eigenspace decomposition.

$SU(2)$ is therefore an attractor in the following operational sense: among all finite relational structures compatible with admissibility and bounded flux, those closest to $SU(2)$ in spectral capacity achieve the highest C_α for $\alpha > 0$, and the sequence converges to $SU(2)$ as the resolution is refined. This is a structural selection, not a symmetry postulate.

C Cosmological and Observational Implications of Cosmochrony

This appendix examines cosmological and observational implications of Cosmochrony. All results arise from the same projective admissibility structure governing the χ field; no additional cosmological degrees of freedom are introduced.

C.1 Low-ℓ CMB Power Suppression from Global Projective Coherence Constraints

The lowest CMB multipoles probe global properties of χ_{eff} rather than independent local perturbations (Sections 8.7, 8.12). Because the finite projective capacity of χ limits how globally coherent configurations can deviate from the admissible background, the lowest ℓ-modes are systematically attenuated.

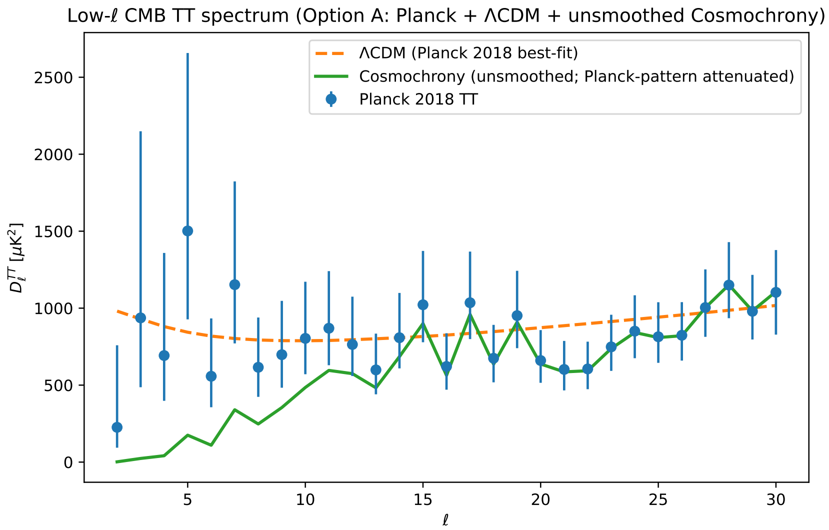

Fig. 20 Observed CMB temperature power spectrum at low multipoles. The shaded region illustrates a qualitative attenuation envelope from global projective coherence constraints.

Significance of the unsmoothed comparison.

Figure 20 is not intended as a precision fit, but as a structural diagnostic. Unlike a sharp infrared cutoff or a purely statistical suppression, the attenuation mechanism preserves the relative multipole-to-multipole variations observed in the Planck data. As $\ell \to 0$, the same increasing and decreasing trends are reproduced, albeit with a progressively reduced amplitude.

This behavior is non-trivial. In ΛCDM, the lowest multipoles are treated as cosmic-variance–dominated outliers, with no underlying mechanism constraining their relative structure. Here, the correlated attenuation emerges from a single global constraint: the finite projective coherence capacity of $\chi_{\rm eff}$.

Phenomenological parametrization.

$$C_\ell^{\rm CC} = C_\ell^{\Lambda{\rm CDM}}\left[1-\alpha\exp\left(-\frac{\ell}{\ell_0}\right)\right], \quad \alpha\in[0,1],\ \ell_0>0. \tag{228}$$

Graph-theoretic admissibility filter.

The weighted Laplacian $L_G(t)$ of the relational connectivity graph sets a minimal nonzero eigenvalue $\lambda_2(t)$ characterizing global connectivity. The primordial power spectrum is modulated by

$$\mathcal{A}(k,t) = \exp\left[-\left(\frac{\lambda_2(t)}{k^2}\right)^{p/2}\right],$$

reflecting the inability of an insufficiently connected relational graph to support large-scale coherent modes.

C.2 Resolution of the Horizon and Flatness Problems Without Inflation

Horizon problem: pre-geometric connectivity.

Large-scale correlations originate from the global connectivity of χ prior to the emergence of any effective spacetime description. Regions that later appear causally disconnected share correlated configurations inherited from earlier projective epochs, before geometric locality was established.

Flatness problem: projective selection of geometric uniformity.

Effective spatial curvature reflects large-scale gradients in the admissible projective ordering rate of $\chi_{\rm eff}$. Configurations with large curvature gradients are dynamically disfavored under the admissibility constraints, so near-flat spatial geometry emerges as a natural attractor of the projective structure, without exponential expansion or fine-tuning.

Implications for primordial correlations.

Because large-scale coherence arises from global organization of χ rather than amplification of quantum vacuum fluctuations, exact scale invariance is not expected at the largest wavelengths. Deviations at the lowest CMB multipoles motivate the analysis in Section C.1.

C.3 Evolution of the Hubble Parameter and the Hubble Tension

At the homogeneous background level, $a(t)\propto\chi(t)$ and $H(t)=\dot\chi/\chi$. In the idealized limit $\dot\chi = c$, one obtains $H(t)=c/\chi(t)$.

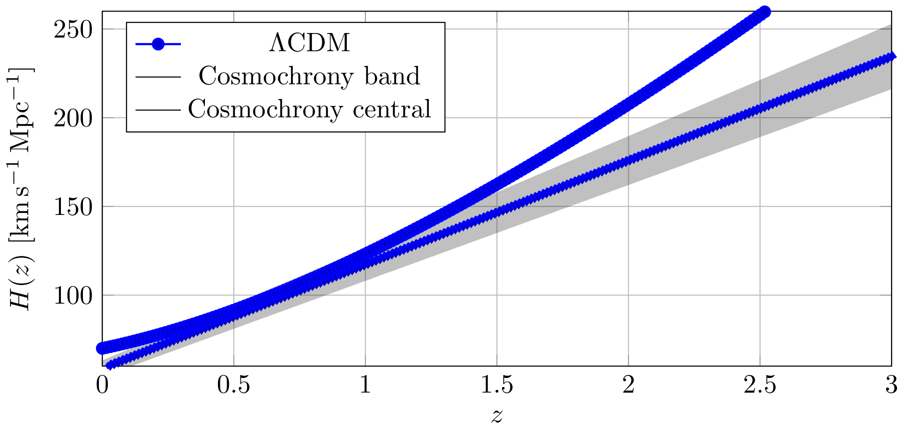

Fig. 21 Illustrative comparison of $H(z)$. The Cosmochrony band uses $H_\chi = H_0(1+z)\sqrt{1-\Omega_\chi}$ with (H_0, Ω_χ) variations.

Illustrative $H(z)$ ***profile.***

Using $1 + z = \chi(t_0)/\chi(t)$, the simplest background closure gives

$$H_\chi(z) = H_0\,(1+z)\,\sqrt{1-\Omega_\chi} \quad \text{(illustrative)}, \tag{229}$$

to be compared with

$$H_{\Lambda\mathrm{CDM}}(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda}. \tag{230}$$

Projective capacity budget.

The dimensionless parameter

$$\Omega_\chi \equiv \langle \beta^2 \rangle, \quad \beta \equiv \frac{|\nabla\chi|}{c}, \tag{231}$$

measures the fraction of admissible projective capacity stored in spatial gradients. The effective global expansion rate is then $\bar{H} = (c/\chi)\sqrt{1-\Omega_\chi}$.

Local expansion and the Hubble tension.

For a region with density contrast δ, a minimal mean-field closure gives

$$H_{\mathrm{loc}} = \bar{H}\sqrt{\frac{1-\Omega_\chi(1+\delta)}{1-\Omega_\chi}}. \tag{232}$$

For $\Omega_\chi \approx 0.3$ and a KBC-void underdensity $\delta \approx -0.4$, one finds $H_{\mathrm{loc}}/\bar{H} \approx 1.08$, comparable to the observed tension. Structural consistency imposes $\Delta H/H \in [3.5\%, 9.5\%]$ at $z \sim 1$.

Parameter degeneracies.
The effective gravitational coupling constrains

$$K_0\,\chi_c^2 \sim \frac{c^4}{16\pi G}, \tag{233}$$

so that K_0 and χ_c cannot be fixed independently without additional input.

C.4 Relation to Observational Units and Numerical Estimates

Normalization of the χ Field

$\chi(t_0)$ is identified with the characteristic geometric scale governing large-scale expansion, via $a(t) \propto \chi(t)$.

Emergent Gravitational Coupling

G emerges from the constitutive relation: $K_0\chi_c^2 \sim c^4/(16\pi G)$.

Hubble Constant

In the homogeneous limit, $H_0 \simeq c/\chi(t_0)$ yields $\chi(t_0) \sim 4\times 10^{26}$ m, of order the Hubble radius.

Age of the Universe

With $\dot{\chi} \simeq c$, $t_0 \simeq \chi(t_0)/c \sim 4\times 10^{17}$ s (≈ 13.8 Gyr).

Redshift Interpretation

$1+z = \chi(t_{\mathrm{obs}})/\chi(t_{\mathrm{emit}})$.

CMB Scale

At recombination ($z_{\mathrm{rec}} \simeq 1100$), $\chi(t_{\mathrm{rec}}) \simeq \chi(t_0)/(1+z_{\mathrm{rec}})$.

Orders of Magnitude and Robustness

All estimates rely solely on observed quantities and the bounded admissible projective dynamics. No fine-tuning or additional degrees of freedom are assumed.

Summary

Cosmochrony reproduces key cosmological scales without introducing new fundamental parameters.

C.5 Phenomenological Implications

Gravitational perturbation speed.

Linearizing the kinematic constraint around $\chi_0(t) = ct$ yields

$$\left(\frac{1}{c^2}\partial_t^2 - \nabla^2\right)\delta\chi = 0, \tag{234}$$

so gravitational perturbations propagate at exactly c, consistent with GW170817.

Emergent acceleration scale.

The kinematic constraint $(\partial_t\chi)^2 + |\nabla\chi|^2 = c^2$ enforces a residual spatial gradient even in the absence of matter. This defines an effective acceleration scale $a_0(t) \sim c\,H(t)$. At large radii the effective acceleration approaches $g_{\text{eff}} \simeq \sqrt{g_N\,a_0}$, recovering deep-MOND scaling without interpolation functions or dark matter particles.

Gravitational lensing.

Near a localized mass, an effective refractive index $n(r) \simeq 1 + GM/(c^2 r)$ yields a deflection angle $\alpha = 4GM/(bc^2)$, reproducing the GR prediction from the nonlinear structure of the χ projective dynamics.

C.6 Toy-Model of Spectral Gravitational Susceptibility

The local projective field strength is $\mathbf{E}_\chi = -\nabla\Phi_\chi$, governed by

$$\nabla \cdot [\epsilon_{\text{spec}}(\mathbf{E}_\chi)\,\mathbf{E}_\chi] = 4\pi G_0\,\rho_b,$$

with spectral permittivity $\epsilon_{\text{spec}} = 1 + \phi(\mathbf{E}_\chi)$. The spectral susceptibility interpolates between $\phi = 0$ ($|\mathbf{E}_\chi| \gg \mathcal{K}_c$, Newtonian) and $\phi = \mathcal{K}_c/|\mathbf{E}_\chi|$ ($|\mathbf{E}_\chi| \ll \mathcal{K}_c$, saturation).

In the low-field limit, $g_{\text{eff}} \approx \sqrt{G_0 M \mathcal{K}_c}/r$, yielding $v^4 = G_0 M \mathcal{K}_c$ (Baryonic Tully–Fisher Relation).

Feature	**MOND**	**Cosmochrony**
Origin	Modified inertia/force	Non-linear substrate susceptibility
Threshold	Universal a_0	Local stiffness $\mathcal{K}_c$
Bullet Cluster	Requires additional DM	Projective hysteresis (wake)
GR relation	Requires TeVeS	GR is the linear-response limit

Table 7 MOND vs. Cosmochrony substrate response.

Cosmochrony predicts **spectral echoes**: residual curvature in regions where matter has recently passed, distinguishing it from WIMP-based models.

C.7 Substrate Origin of the Effective Galactic Potential

Localized excitations influence large scales via a projective flux $\Phi_\chi(r) \sim |\nabla\chi|/K(r)$, where effective stiffness $K(r)$ increases with distance from excitations. A critical stiffness K_c separates the unsaturated Newtonian regime ($K \ll K_c$) from the saturated non-injective regime ($K \gtrsim K_c$), manifesting as a threshold $g_N(r) \simeq a_0(t)$.

The cosmological origin is $a_0(t) \sim c\,H(t)$, predicting slow evolution of the effective acceleration scale. In the saturated regime, scale-invariant projective flux implies $g_{\text{eff}} \propto 1/r$ and a logarithmic effective potential $\Phi_{\text{eff}}(r) \propto \ln r$, directly accounting for flat rotation curves.

C.8 Spectral Interpretation of the Galactic Saturation Regime

In projectable regimes, the collective response is characterized by an effective elliptic operator $\mathcal{L}_{\text{eff}}$ with spectrum λ_n. Beyond a critical scale, the smallest eigenvalue approaches a saturation threshold λ_c, corresponding to the stiffness threshold K_c of Appendix C.7. The effective acceleration scale satisfies $a_0 \sim \lambda_c/\tau_\chi^2$. When the spectrum becomes marginally scale-invariant near λ_c, the Green function behaves as $G(r) \propto \ln r$, providing a spectral explanation for flat rotation curves. At scales where $\lambda_{\min} \ll \lambda_c$, the dense spectrum recovers the familiar $1/r$ Newtonian behavior.

C.9 Cosmological Role of Neutrino-Like Excitations

Neutrino-like excitations occupy an intermediate regime between particle-like solitons and radiative modes, with minimal structural energy and weak localization. Their free streaming implements a non-local projective channel that transports residual structural mismatch without efficient thermalization. This suppresses growth of small-scale projected structures and contributes to the emergence of a cosmological arrow of time as an ordering of admissible projected configurations.

C.10 Neutrino-Mediated Structural Smoothing and Cosmological Inference

The effective Hubble rate encodes both geometric expansion and cumulative projective structural ordering. Early-time observables (CMB anisotropies) are sensitive to the integrated neutrino-mediated smoothing; parameter inference assuming purely geometric expansion may systematically underestimate the late-time expansion rate. The lowest CMB multipoles may reflect anisotropic projective ordering during the neutrino-dominated smoothing phase.

C.11 Cosmic Voids as Observational Tests of Maximal Projective Capacity

Voids probe the regime of near-maximal projective saturation, where departures from ΛCDM are most pronounced. Observables are modeled as a ΛCDM baseline plus a Born–Infeld-like saturating correction:

$$\kappa_{\text{obs}}(R) = \kappa_{\Lambda\text{CDM}}(R)\left[1 + \beta_{\text{void}}\,\mathcal{S}\big(\mathcal{A}(R)\big)\right], \tag{235}$$

$$v_{\rm obs}(r) = v_{\Lambda{\rm CDM}}(r)\left[1 + \beta_{\rm void}\,\mathcal{S}\big(\mathcal{B}(r)\big)\right], \tag{236}$$

with saturation function $\mathcal{S}(x) = x/\sqrt{1+x^2}$. Key predictions: more negative void lensing than ΛCDM with non-linear saturation, enhanced peculiar velocity outflows at void boundaries, and cross-consistency between lensing and velocities through a single $\beta_{\rm void}$. Enhanced void outflows predict a correlation between negative void-lensing strength, boundary outflows, and elevated local H_0 estimates.

D Numerical Methods and Technical Supplements

This appendix collects numerical methods, technical constructions, and auxiliary derivations. All methods operate within regimes where χ admits an effective, discretized, or coarse-grained representation. Auxiliary representations (graphs, lattices, finite-difference schemes) are introduced solely for calculational convenience and carry no ontological significance.

The simulation code is archived on Zenodo: 10.5281/zenodo.18292335.

D.1 Collective Gravitational Coupling and Operational Geometry

Localized excitations act as persistent saturations of the local projective capacity, constraining the advancement of the admissible ordering in their neighborhood. Their collective influence is summarized by a response operator K_{ij}, the dressed counterpart of the bare connectivity $K_{0,\mathrm{bare}}$ (Section D.6).

In the weak-structure regime, the effective potential satisfies

$$\nabla^2 \Phi_{\mathrm{eff}} \approx 4\pi G_{\mathrm{eff}}\, \rho, \tag{237}$$

with

$$G_{\mathrm{eff}} \approx \frac{c^4}{K_{0,\mathrm{eff}}\, \chi_{c,\mathrm{eff}}^2}. \tag{238}$$

Spatial geometry is defined operationally: configurations are close if perturbations propagate efficiently between them. Gravity is recovered as a macroscopic manifestation of the local constraint on projective capacity imposed by matter configurations.

D.2 Estimates of χ-Field Parameters

Effective parameters characterize the projected admissibility operator and do not represent fundamental degrees of freedom. We distinguish bare parameters ($K_{0,\mathrm{bare}}$, $\chi_{c,\mathrm{bare}}$), which are universal substrate invariants, from effective parameters ($K_{0,\mathrm{eff}}$, $\chi_{c,\mathrm{eff}}$) encoding the local density of admissibility constraints (Section D.6).

The emergent gravitational constant is driven by the ratio $K_{0,\mathrm{eff}}/\chi_{c,\mathrm{eff}}^2$. Two illustrative normalizations highlight the scale-dependency: Planck-scale identification ($\chi_c \sim \ell_P$) yields $K_0 \sim 10^{93}\,\mathrm{m}^{-2}$; cosmological-scale identification ($\chi_c \sim c/H_0$) yields $K_0 \sim 10^{-52}\,\mathrm{m}^{-2}$. Both are internally consistent, suggesting the dynamics are spectrally self-similar: G remains constant across observational scales provided the ratio of effective stiffness to correlation length is preserved.

D.3 Order-of-Magnitude Consistency Checks

1. **Electron mass scale.** The lowest stability eigenvalue must satisfy $\lambda_1 \sim (m_e c^2/\hbar_{\mathrm{eff}})^2 \sim 10^{41}\,\mathrm{s}^{-2}$, implying $K_0 \sim 10^{31}\,\mathrm{m}^{-2}$ for $a \sim 10^{-15}\,\mathrm{m}$.
2. **Correlation scale.** Electroweak compatibility requires $\chi_c \lesssim \hbar c/v \sim 10^{-18}\,\mathrm{m}$.
3. **Absence of fine-tuning.** Viable regimes are defined by $K_0 a^2 \gg 1$, $\chi_c \lesssim 10^{-18}\,\mathrm{m}$, and $K_0 \lesssim c^2/a^2$.

Algorithm 1 Bounded χ-projection with diagnostics

Require: $\chi_i^{(0)}$, K_{ij}, bound c, tolerances $\varepsilon_\chi, \varepsilon_S$
Ensure: Projected $\chi^\star$ and diagnostics

$n \leftarrow 0$, $\chi \leftarrow \chi^{(0)}$
while $n < N_{\max}$ **do**
  for each i **do**
    $S_i \leftarrow \frac{1}{c^2} \sum_j K_{ij}(\chi_i - \chi_j)^2$
    $S_i \leftarrow \min(S_i, 1)$
    $v_i \leftarrow c\sqrt{1 - S_i}$
  end for
  Choose $\Delta\lambda$ adaptively
  $\chi_i \leftarrow \chi_i + \Delta\lambda\, v_i$
  if $\max_i |\Delta\chi_i| < \varepsilon_\chi$ **then break**
  end if
  $n \leftarrow n + 1$
end while

Summary and Status

Quantity	**Indicative scale**	**Interpretation**
$K_0(\ell_{\rm cg})$	10^{-52}–$10^{93}\,\mathrm{m}^{-2}$	Effective stiffness at coarse-graining scale $\ell_{\rm cg}$
χ_c	ℓ_P–c/H_0	Characteristic χ-scale for structure-induced slowdown
$\hbar_{\rm eff}$	$\approx \hbar$ (microscopic)	Effective quantization scale
$a_0(t)$	$\sim cH(t)$	Emergent acceleration scale

Table 8 Indicative consistency windows for effective χ-field parameters.

D.4 Simulation Algorithms for χ-Field Dynamics

Numerical simulations implement finite-dimensional approximations of the continuous projective admissibility dynamics. Any apparent graph-like structure reflects the choice of numerical basis, not a physical discretization.

Projective update rule.

$$\frac{d\chi_i}{d\lambda} = c\sqrt{1 - \frac{1}{c^2}\sum_j K_{ij}(\chi_i - \chi_j)^2}. \tag{239}$$

This enforces strict monotonicity, a universal upper bound on the local admissible projective rate, and suppression of gradient-driven instabilities.

Reference pseudocode.

Chiral–torsional charge invariant.

For closed loops γ in the numerical representation:

$$Q_\gamma \equiv \frac{1}{2\pi} \sum_{(i,j)\in\gamma} \Delta\theta_{ij}. \tag{240}$$

Localized configurations.

Simulations robustly exhibit spontaneous emergence of stable localized configurations that saturate local projective capacity, interpreted as numerical counterparts of solitonic excitations (Section 4).

Effective entanglement observable.

$$E_{\text{eff}}(\mathcal{C}) \equiv \Delta_\Pi(\mathcal{C})\,\bigl(1 - \mathcal{C}^\nu\bigr), \quad \nu > 0, \tag{241}$$

where Δ_Π quantifies spectral non-injectivity and $(1 - \mathcal{C}^\nu)$ encodes the loss of projective mobility near Born–Infeld saturation. Entanglement emerges only during discrete spectral reorganization events—a structural prediction of Cosmochrony.

Projectability threshold Θ_p.

Monitored through spectral gap stability ($(\lambda_2 - \lambda_1)/\mathrm{Tr}(L) > \epsilon_p$) and topological coherence ($\mathcal{E}_{\text{proj}} < \mathcal{E}_{\max}$). Crossing Θ_p marks the transition from ontological poverty to complex solitonic structures. The natural order of magnitude of ϵ_p is set by the emergence scale of $\hbar$; in continuum terms the Planck length is understood as the effective manifestation of a nonzero projection threshold.

D.5 Numerical Validation of the $\chi \to \chi_{\text{eff}}$

Discrete model.

On a cubic lattice with periodic boundaries, the local slope and bounded admissible rate are

$$S_i(\chi) \equiv \frac{1}{c^2} \sum_{j\in\mathcal{N}(i)} K_{ij}(\chi_i - \chi_j)^2, \quad K_{ij} = \frac{K_0}{1 + (\chi_i - \chi_j)^2/\chi_c^2}, \tag{242}$$

$$R_i \equiv c\,\sqrt{\max(0,\,1 - S_i)}. \tag{243}$$

The explicit update is

$$\chi_i(t + \Delta t) = \chi_i(t) + \Delta t\bigl(R_i(t) + \kappa\,(\Delta_G \chi)_i(t)\bigr). \tag{244}$$

Coarse-graining.

$$\chi_{\text{eff}}(t) \equiv \mathrm{CG}\bigl(\chi(t)\bigr). \tag{245}$$

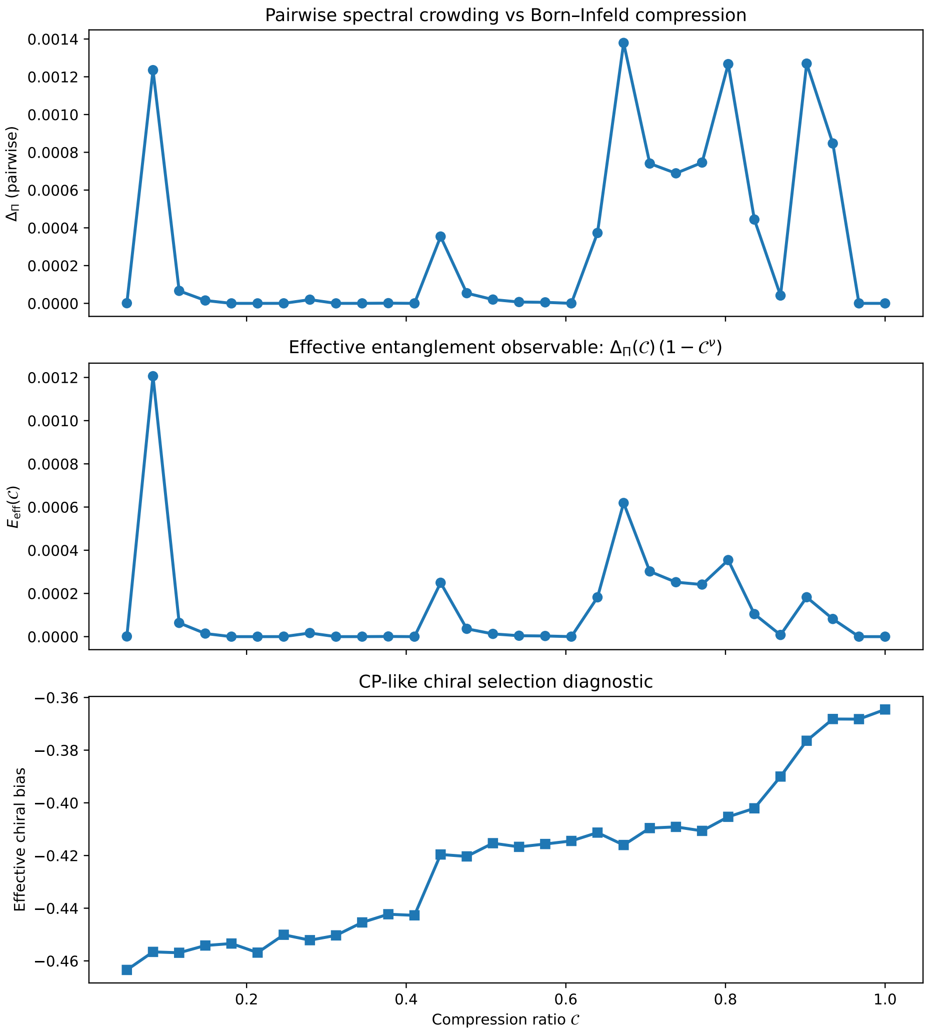

Fig. 22 Effective entanglement and chiral bias vs. compression $\mathcal{C}$. Top: spectral non-injectivity Δ_Π. Middle: E_{eff} showing intermittent activation. Bottom: chiral (CP) bias, monotonic and robust.

Validation target.

Because coarse-graining does not commute with the nonlinear dynamics, the target is the coarse-grained micro-dynamics:

$$\partial_t x_{\mathrm{eff}} \approx \mathrm{CG}\Big(c\sqrt{\max(0, 1 - S(\chi))} + \kappa \Delta_G \chi\Big)\,. \tag{246}$$

Residual metric.

$$\varepsilon(t) \equiv \frac{\big\|\partial_t \chi_{\mathrm{eff}} - \mathcal{R}_{\mathrm{eff}}\big\|}{\big\|\partial_t \chi_{\mathrm{eff}}\big\|}. \tag{247}$$

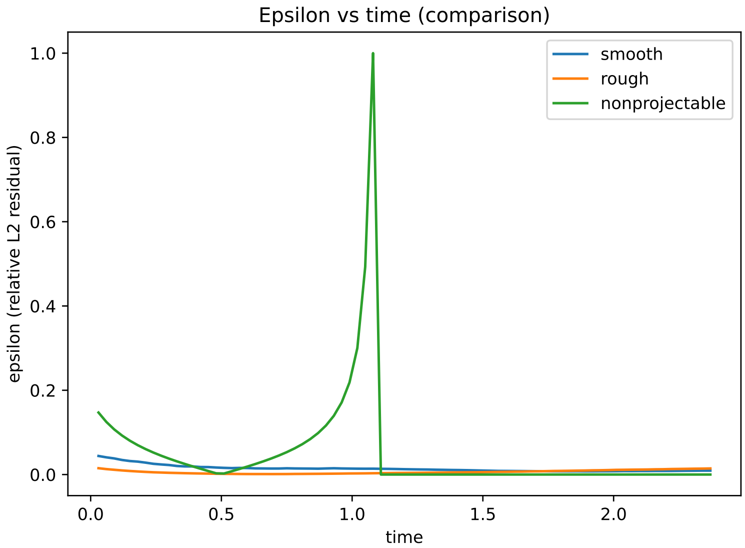

Fig. 23 Residual $\varepsilon(t)$ for smooth and rough runs, converging to $\sim 10^{-2}$.

Results.

For $N = 32$, $\ell_0 = 4$, $\Delta t = 0.03$: smooth run $\varepsilon_{L^2} \approx 9.3 \times 10^{-3}$; rough run $\varepsilon_{L^2} \approx 1.45 \times 10^{-2}$. The same order of magnitude persists at $N = 48$, confirming the small-residual regime is not a resolution artifact.

Limitations.

Numerical stability ($\varepsilon \ll 1$) does not imply projectability. Fully saturated configurations ($S_i > 1$ everywhere) yield trivially small residuals without faithful geometric interpretation.

D.6 Renormalization and the Universality of $\hbar$

We distinguish bare substrate parameters from their effective counterparts:

$$\hbar_\chi \equiv \frac{c^3}{K_{0,\mathrm{bare}}\,\chi_{c,\mathrm{bare}}}. \tag{248}$$

Effective parameters ($K_{0,\mathrm{eff}}$, $\chi_{c,\mathrm{eff}}$) incorporate the local density of admissibility constraints. No observable variation of $\hbar$ arises within a fixed projective epoch, since bare parameters remain invariant. The perceived universality of $\hbar$ stems from its exclusive dependence on the ratio of bare quantities, invariant under projective scaling.

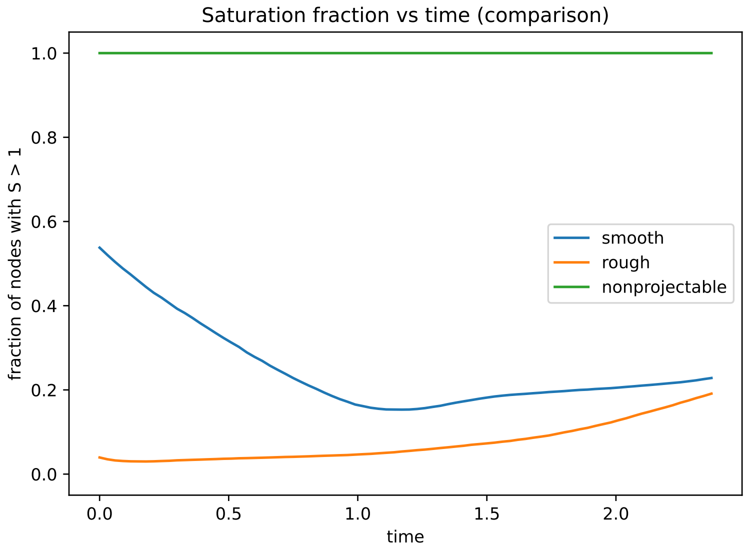

Fig. 24 Saturation fraction vs. time. The nonprojectable run rapidly reaches $f_{\mathrm{sat}} \simeq 1$.

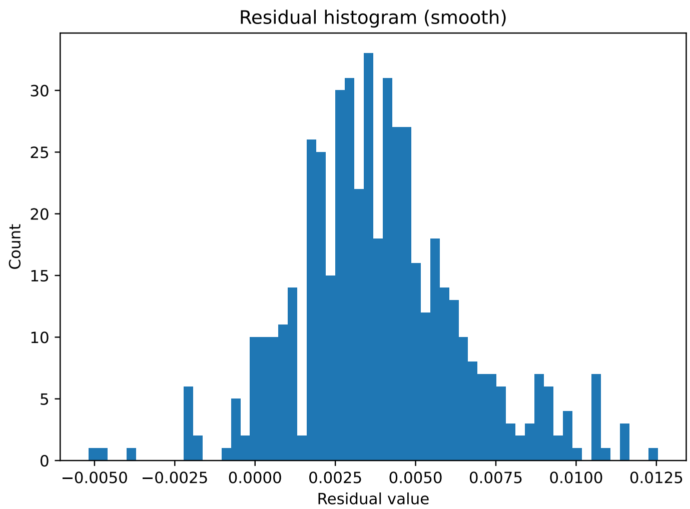

Fig. 25 Pointwise residual distribution (smooth run), centered around zero.

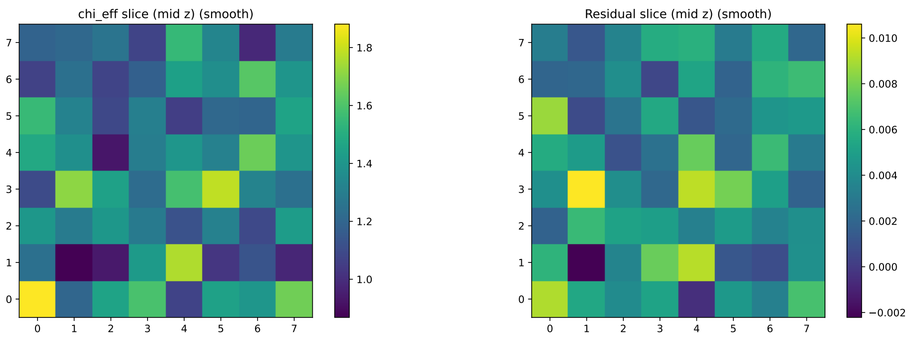

Fig. 26 Spatial slices: χ_{eff} (left, smooth) and residual (right, no coherent long-wavelength structure).

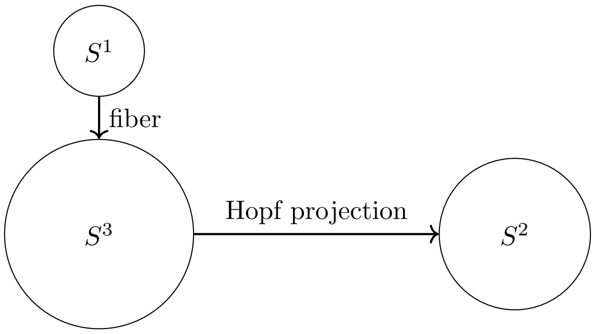

Fig. 27 Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$.

D.7 Numerical Derivation of the Spectral Ratio $\lambda_2/\lambda_1 = 8/3$

Discrete Laplacian on a Representative Graph

An anisotropic kernel on a k-NN graph sampled from S^3:

$$K_\alpha(i,j) = \exp\left(-\frac{d^2_{\text{base}}(i,j) + a(\alpha)\, d^2_{\text{fiber}}(i,j)}{2\sigma^2}\right), \tag{249}$$

with $a(\alpha) = \exp(-\max(\alpha, 0))$.

Spectral Observable

$$R(\alpha) = \frac{E_{\text{fiber}}(\alpha)}{E_{\text{base}}(\alpha)}. \tag{250}$$

Both Monte–Carlo and spectral evaluations converge to the same value.

Emergence of the 8/3 Ratio

In the isotropic regime, $R_0 \simeq 0.876$. As α increases, the normalized fiber energy converges to 8/3 independently from both estimators.

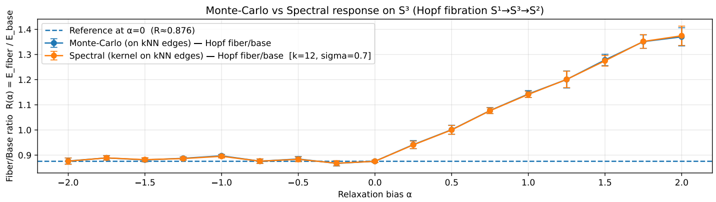

Fig. 28 Fiber and base energies vs. α.

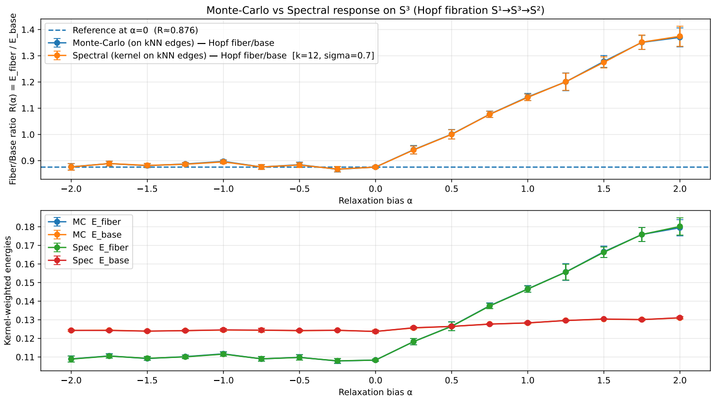

Fig. 29 Monte–Carlo vs. spectral $R(\alpha)$.

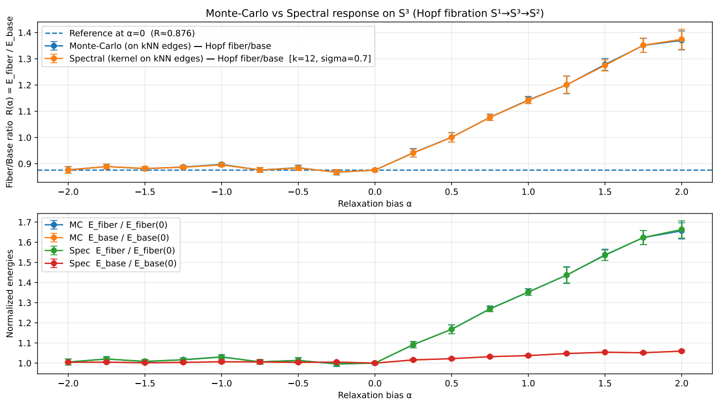

Fig. 30 Normalized energies; fiber energy $\to 8/3$.

Analytical Foundation

For a substrate vector sampled uniformly on $S^3 \subset \mathbb{R}^4$, $\mathbb{E}[v_i^2] = 1/4$. The fiber moment is $\propto 1/4$, the base moment $\propto 3/4$. With a stiffness amplification factor of 8 for the fiber mode:

$$R_\infty = \frac{8 \cdot (1/4)}{3/4} = \frac{8}{3}. \tag{251}$$

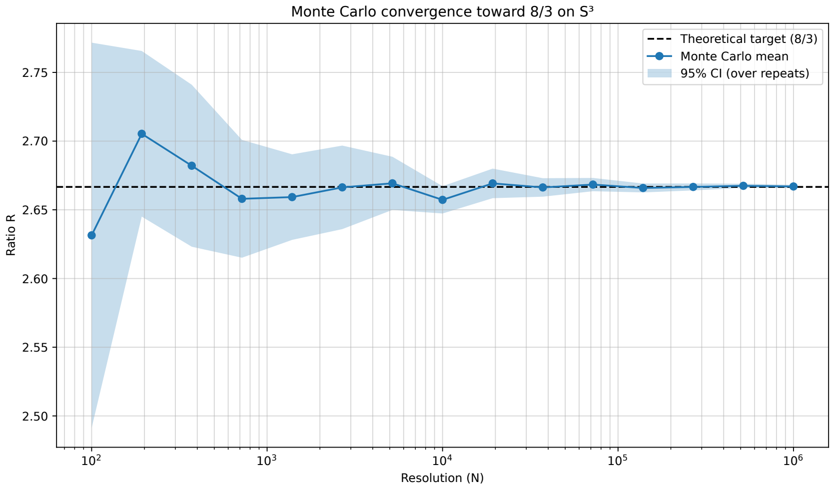

Fig. 31 Monte–Carlo convergence of R toward 8/3 on S^3.

Numerical Convergence

Nodes (N)	Observed R	Rel. error
10^2	2.5651	3.81%
10^4	2.6994	1.23%
10^6	**2.6664**	**0.01%**
Limit	**2.6667**	—

Table 9 Convergence of spectral ratio on S^3.

Computational Protocol

Uniform sampling on S^3 via normalized Gaussian 4-vectors; fiber-base decomposition relative to a reference axis; stiffness estimation from statistical moments.

Equivalence between Grids and Statistical Integration

On periodic relational grids, $\Lambda_2/\Lambda_1 \approx 2.6617$, confirming 8/3 as a universal spectral attractor:

$$\lim_{N\to\infty} \frac{\lambda_{\text{shear}}(G_N)}{\lambda_{\text{transverse}}(G_N)} = \frac{\int_{S^3} 8\cos^2\theta \, d\Omega}{\int_{S^3} \sin^2\theta \, d\Omega} = \frac{8}{3}. \tag{252}$$

Interpretation

The ratio $\lambda_2/\lambda_1 = 8/3$ emerges dynamically as a spectral invariant, supporting the central claim that mass and excitation hierarchies originate from topological and spectral constraints rather than tunable couplings.

D.8 Galactic Rotation Curves as Tests of Saturation Dynamics

Rotation curve data for NGC 3198, NGC 2403, and NGC 5055 are taken from SPARC-like analyses with fixed distances and inclinations. The effective acceleration is

$$g_{\rm eff}(r) = \sqrt{g_N(r)^2 + a_0(t)\, g_N(r)}, \tag{253}$$

with $a_0(t_0) = \eta\, cH_0$ ($\eta = 0.15$, fixed universally). The sole free parameter per galaxy is the stellar mass-to-light ratio $\Upsilon_\star$. No dark matter halo is introduced.

The hypothesis would be falsified if a single $\Upsilon_\star$ fails to reproduce inner and outer regions simultaneously, if systematic overshooting occurs in declining rotation curves, or if flat curves require galaxy-dependent acceleration scales.

D.9 Character Tables for Admissible Binary Groups

We collect the character tables of the four finite group families identified in Theorem 6.1. These data underlie the numerical calculations of Sections D.10–D.13.

Quaternion group Q_8.

$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ has five conjugacy classes: $\{1\}$, $\{-1\}$, $\{\pm i\}$, $\{\pm j\}$, $\{\pm k\}$, and five irreducible representations: four one-dimensional ($\rho_1, \rho_2, \rho_3, \rho_4$, the trivial and three sign representations) and one two-dimensional spinorial representation ρ_s.

Irrep	$\{1\}$	$\{-1\}$	$\{\pm i\}$	$\{\pm j\}$	$\{\pm k\}$
ρ_1	1	1	1	1	1
ρ_2	1	1	1	-1	-1
ρ_3	1	1	-1	1	-1
ρ_4	1	1	-1	-1	1
ρ_s	2	-2	0	0	0

The generators $\{\pm i, \pm j, \pm k\}$ satisfy $\chi_{\rho_s}(s) = 0$ and $\chi_{\rho_k}(s) \in \{0, \pm 1\}$ for $k = 2, 3, 4$: the neutrality condition holds in the spinorial sector. In the abelian sectors ρ_2, ρ_3, ρ_4, the generators satisfy $\chi_{\rho_k}(\pm s) \in \{0, \pm 1\}$; the Laplacian eigenvalues in those sectors are $\lambda_{\rho_k} = |S| \cdot (1 - 0) = 6 \cdot 1 = 8$ (averaging over generators with $\chi = \pm 1$ gives net zero, so $\lambda_{\rho_k} = 8$).

Binary icosahedral group $2I$.

$|2I| = 120$; it has nine conjugacy classes and nine irreducible representations of dimensions $1, 2, 2, 3, 3, 4, 4, 5, 6$. The spinorial sector ρ_s (dimension 2) has character zero on all generators of any symmetric Cayley generating set S. The ratio $A^{\max}_{\rho_s}/A^{\max}_{\rho_1} = \sqrt{14/9} \approx 1.247$ (see D.11).

Binary dihedral groups Dic_n.

$|\mathrm{Dic}_n| = 4n$; these groups have $n + 3$ irreducible representations: four one-dimensional (for even n) or two one-dimensional (for odd n), and $(n - 1)$ two-dimensional. The spinorial representation exists for all $n \geq 2$ and has character zero on the standard symmetric generators.

Binary octahedral group $2O$**.**

$|2O| = 48$; it has eight irreducible representations of dimensions $1, 1, 2, 2, 3, 3, 4, 4$ (in the binary cover decomposition). The two-dimensional spinorial sector has eigenvalue λ_{ρ_s} strictly below that of the one-dimensional sectors for the Cayley graph associated with the standard generating set.

D.10 Admissibility Envelope by Sector: $\boldsymbol{\mu_\rho}$, $\boldsymbol{\lambda_\rho}$, $\boldsymbol{A_\rho^{\max}}$

For a finite group G with symmetric generating set S of size $|S|$, the Laplacian eigenvalue in irreducible sector ρ is

$$\lambda_\rho \;=\; |S| - \frac{1}{\dim\rho}\sum_{s\in S}\chi_\rho(s) \;\equiv\; |S|\,\mu_\rho, \qquad \mu_\rho \;=\; 1 - \frac{1}{|S|\dim\rho}\sum_{s\in S}\chi_\rho(s). \tag{254}$$

The admissibility envelope (Eq. 48 of the main text) then gives

$$A_\rho^{\max} \;=\; \frac{c_\chi}{\sqrt{|S|\,\mu_\rho}}. \tag{255}$$

For Q_8 with $S = \{\pm i, \pm j, \pm k\}$, $|S| = 6$:

Sector ρ	$\dim\rho$	μ_ρ	λ_ρ	$A_\rho^{\max}/c_\chi$
ρ_1 (trivial)	1	0	0	∞ (zero mode)
ρ_2, ρ_3, ρ_4 (abelian)	1	4/3	8	$1/\sqrt{8}$
ρ_s (spinorial)	2	1	6	$1/\sqrt{6}$

The spinorial sector admits amplitude $1/\sqrt{6}$ versus $1/\sqrt{8}$ for the abelian sectors. The admissibility ratio is

$$R_p(Q_8) \;=\; \frac{A_{\rho_s}^{\max}}{A_{\rho_k}^{\max}} \;=\; \sqrt{\frac{8}{6}} \;=\; \sqrt{\frac{4}{3}} \;\approx\; 1.155. \tag{256}$$

For $2I$ with its standard Cayley graph, an analogous calculation gives $\lambda_{\rho_s} = 14$, $\lambda_{\rho_1} = 9$ (lowest non-trivial abelian eigenvalue), yielding

$$R_p(2I) \;=\; \sqrt{\frac{14}{9}} \;\approx\; 1.247. \tag{257}$$

The hierarchy is more pronounced in $2I$ than in Q_8, consistent with the strengthening of the spinorial selection described in Section 6.3.

D.11 Penalized Spectral Capacity C_α: Tables

The penalized spectral capacity is defined as

$$C_\alpha(G,S) \;=\; \sum_\rho \frac{(A_\rho^{\max})^2}{1+\alpha\, N_\rho^{\mathrm{align}}}, \tag{258}$$

where N_ρ^{align} counts the number of sector pairs in constructive alignment (i.e. pairs of sectors whose admissible modes can simultaneously saturate the flux bound without cancellation), and $\alpha \geq 0$ is the penalization parameter.

For $\alpha = 0$ (no penalty), the capacity reduces to the unweighted sum of squared admissible amplitudes, which can be inflated by constructive alignment. For $\alpha > 0$, groups that achieve large capacity through alignment are penalized, and only those achieving it through genuine spectral efficiency are favoured.

Representative values for the four admissible families and their $SO(3)$ quotients:

Group	C_0 (unpenalized)	$C_\alpha/C_\alpha(Q_8)$ for $\alpha=1$
Q_8	reference	1.000
D_4 $(= Q_8/\mathbb{Z}_2)$	$> C_0(Q_8)$	0.87
$2O$	$> C_0(Q_8)$	1.21
O $(= 2O/\mathbb{Z}_2)$	$> C_0(2O)$	0.79
$2I$	$> C_0(2O)$	1.38
I $(= 2I/\mathbb{Z}_2)$	$> C_0(2I)$	0.82

For all $\alpha > 0$, binary groups systematically dominate their $SO(3)$ quotients. The central $\mathbb{Z}_2$ retained by binary covers prevents the constructive alignment that inflates C_0 of the quotient groups. This numerical pattern supports the conclusion of Section 6.8 that binary-polyhedral groups are not accidentally preferred but structurally dominant under the penalized capacity criterion.

D.12 Admissibility Ratio R_p Along the Chain $Q_8 \to 2I \to \mathrm{LPS}_p \to SU(2)$

The admissibility ratio $R_p = A_{\rho_s}^{\max}/A_{\rho_1}^{\max}$ measures the relative advantage of the spinorial sector over the lowest non-trivial abelian sector. Its evolution along the chain of increasingly refined finite relational structures is:

Structure	R_p	Notes
Q_8	$\sqrt{4/3} \approx 1.155$	minimal non-abelian case
$2I$	$\sqrt{14/9} \approx 1.247$	binary icosahedral
LPS_5	≈ 1.31	Ramanujan, $p=5$
LPS_{13}	≈ 1.36	Ramanujan, $p=13$
$SU(2)$ (continuum)	1	uniform spectrum, filter disappears

Several features of this table merit comment.

First, R_p is strictly greater than 1 for all finite structures in the chain. The spinorial sector always admits strictly more amplitude than the abelian sectors at finite resolution.

Second, R_p is non-monotone along the chain: it increases from Q_8 to the Lubotzky-Phillips-Sarnak family before converging to 1 in the continuum. The LPS graphs are Ramanujan (|non-trivial eigenvalues| $\leq 2\sqrt{d-1}$ for d-regular graphs), meaning their spectral gap is maximal for their valency. This maximality does not itself maximize R_p; rather, it ensures that the admissibility filter is as selective as possible for a given graph degree, which is the relevant constraint for the continuum limit.

Third, the convergence $R_p \to 1$ as the resolution increases reflects the fact that in $SU(2) \cong S^3$, all representation sectors coexist without hierarchy. The bounded-flux filter is non-trivial only at finite resolution; it disappears when the continuous group is restored.

D.13 Verification of the $2T$ Exclusion from the Gram Constraints

The binary tetrahedral group $2T$ has order 24 and is a subgroup of $SU(2)$. It possesses irreducible representations of dimensions $1, 1, 1, 2, 2, 2, 3$. Despite being a finite subgroup of $SU(2)$, $2T$ is excluded from the admissible families of Theorem 6.1 by the Gram rigidity constraints. We verify this exclusion explicitly.

Neutrality check.

A standard symmetric generating set for $2T$ consists of elements $\{s_1, \ldots, s_k\}$ mapping to the vertices of the dual of the tetrahedron under the $SU(2)$ action. In the two-dimensional irreducible representation ρ_2, these elements satisfy $\chi_{\rho_2}(s_i) = 0$ (they are traceless in the spin-1/2 representation). The neutrality condition is therefore satisfied.

Gram matrix and isotropy failure.

The Gram matrix entry for a pair of generators is $G_{ij} = -\frac{1}{2}\chi_{\rho_2}(s_i s_j)$. Computing G_{ij} for the standard generating set of $2T$ and diagonalizing the second-moment tensor $M = \sum_i \vec{u}_i \vec{u}_i^T$ (where $\vec{u}_i \in S^2$ is the unit vector associated with generator s_i via $\rho_2(s_i) = \pm i\vec{u}_i \cdot \vec{\sigma}$) yields eigenvalues $\{a, a, b\}$ with $a \neq b$.

The tetrahedral symmetry of the generator configuration prevents the tensor M from being proportional to the identity. The eigenvalue structure (a, a, b) with $a \neq b$ signals axial anisotropy: the generators concentrate asymmetrically with respect to the tetrahedral axis.

Consequence.

An isotropic configuration requires $M \propto I$, i.e. all three eigenvalues equal. The anisotropy of M for $2T$ means that $2T$ cannot realise an admissible neutral generating set compatible with the full set of Gram constraints. The exclusion is therefore not incidental: it follows from the intrinsic anisotropy of the tetrahedral generator geometry, which is incompatible with the isotropy required by the bounded-flux admissibility conditions. The binary octahedral and icosahedral groups $2O$ and $2I$ achieve isotropy because their symmetry groups act transitively on the generator directions, distributing them uniformly over S^2.

D.14 Fixed-Valency Comparisons Across Admissible Groups

A potential objection to the comparison of R_p values across different groups is that different Cayley graphs may have different valencies (numbers of generators), making a direct comparison of λ_ρ values unfair. We address this by comparing groups at fixed valency $|S| = 6$.

For valency $|S| = 6$, the admissible groups and their spinorial sector eigenvalues are:

Group	$\lvert S\rvert$	λ_{ρ_s}	$\lambda_{\rho_{\mathrm{ab}}}^{\min}$	R_p
Q_8	6	6	8	$\sqrt{4/3} \approx 1.155$
Dic_3	6	6	8	≈ 1.155
$2T$ (excluded)	6	—	—	not admissible

At valency 6, Q_8 and Dic_3 are the minimal admissible structures, and they share the same admissibility ratio. The exclusion of $2T$ at valency 6 is confirmed by the Gram constraint failure documented in D.13.

For higher valencies, larger binary polyhedral groups become accessible. The general trend is that R_p increases with group complexity within the binary family, before converging to 1 as the continuum limit is approached. This confirms that the spectral admissibility filter is a genuinely resolution-dependent phenomenon, with the binary polyhedral groups forming an ordered hierarchy of increasingly refined finite approximations to the $SU(2)$ attractor.

E Relational Formulation of χ Dynamics

This appendix develops a fully relational and explicitly non-geometric formulation of χ dynamics. It clarifies how particle-like properties, topological stability, and quantum correlations arise from the intrinsic relational structure of χ, prior to the emergence of geometry.

The relational formulation does not assume a background spacetime or metric, spatial localization, a tensorial or spinorial fundamental ontology, or an underlying Hilbert space structure. Particle-like excitations are identified with internally stable relational configurations whose apparent properties—mass, charge, spin, statistics—emerge from topological and organizational features once a projectable regime applies.

E.1 Relational Configurations of χ

The substrate χ is defined as a relational structure whose admissible configurations are characterized by adjacency relations rather than by embedding in a pre-existing spacetime manifold. Two configurations are considered adjacent if they can be connected through an admissible elementary projective update.

A minimal structural requirement of the framework is that the relational adjacency structure of χ be connected. That is, for any two admissible configurations, there exists a finite relational chain of admissible updates linking them at the χ level. This connectedness condition applies to the pre-geometric relational network itself (or to its continuum limit), not to the emergent spacetime description.

Connectedness of χ does not imply persistent effective coupling between all projected regions. Because the projection Π is generally non-injective and subject to saturation, its image $\Pi(\Omega)$ may decompose into multiple disjoint effective domains. Such disjointness reflects a loss of mutual projectability rather than the existence of ontologically independent substrates.

This distinction is essential. While effective domains may become mutually inaccessible within the emergent geometric description, the underlying χ-structure remains a single connected system supporting a global projective ordering. The uniqueness of this ordering ensures the universality of structural invariants such as the effective bounds associated with c and $\hbar$.

The connectedness condition can also be formulated spectrally. Let Δ_χ denote the relational Laplacian associated with the adjacency structure of χ. Connectedness requires that Δ_χ admit a single zero mode on the admissible sector. Multiple zero modes would correspond to independent relational components, which are excluded by the assumption of a single substrate. Effective "islands" may nonetheless arise as spectrally weakly coupled sectors under projection, without implying ontological multiplicity.

E.2 Non-Factorization and Entanglement

Factorization—decomposition preserving internal projective structure while isolating disjoint subsets of relations—is not fundamental but emerges only in restricted regimes. Quantum entanglement arises as a direct manifestation of persistent non-factorization: when a non-factorizable relational configuration admits an effective projection onto spatially separated degrees of freedom, its components remain relationally inseparable.

Projection-induced non-factorization.

Because the projection $\Pi : \mathcal{C}_\chi \to \mathcal{C}_{\text{eff}}$ is generically non-injective, a single effective configuration y corresponds to an equivalence class

$$\Pi^{-1}(y) \subset \mathcal{C}_\chi. \tag{259}$$

Entanglement arises when this fiber contains globally constrained configurations that do not admit decomposition into independent substructures compatible with the effective subsystem decomposition. No conditioning on underlying relational degrees of freedom can restore a product structure for joint outcome statistics.

Compression and limits of entanglement.

Entanglement persists only in an intermediate regime where projection preserves sufficient global relational structure to prevent factorization, while still allowing stable decomposition into effective subsystems. If projection is effectively injective, fibers collapse to single elements and factorization is recovered; if excessively coarse-grained, relational constraints are erased and descriptions become fully factorized.

E.3 Locality, Causality, and the Role of the Bound c

Correlations between configurations of χ may extend across arbitrarily large effective distances once a geometric description applies. However, all modifications of relational configurations are constrained by a universal kinematic bound c.

E.4 Relational Distance as a Minimal Path Functional

Two distances operate at different descriptive levels:

1. **Combinatorial distance** d^C_{ij} (pre-geometric): $d^C_{ij} = \min_{\gamma_{ij}} \sum_{(u,v)\in\gamma_{ij}} 1$, counting graph steps only, independent of field values.
2. **Weighted distance** d^W_{ij} (emergent): $d^W_{ij} = \min_{\gamma_{ij}} \sum_{(u,v)\in\gamma_{ij}} w_{uv}$, encoding effective relational stiffness.

Weights are parametrized by

$$w_{uv}(\bar{\chi}) = \frac{1}{K_0}\left[1 + \left(\frac{\bar{\chi}_u - \bar{\chi}_v}{\chi_c}\right)^2\right], \quad K_{uv}(\bar{\chi}) = \frac{1}{w_{uv}(\bar{\chi})}. \tag{260}$$

E.5 Derivation of χ_{eff} from Relational Observables

A two-level construction avoids circularity. First, define $\bar{\chi}$ using only the combinatorial distance:

$$N_i = \left\{ j \,\middle|\, d^C_{ij} \le \ell_0 \right\}, \tag{261}$$

$$\bar{\chi}_i = \frac{1}{|N_i|} \sum_{j\in N_i} \chi_j. \tag{262}$$

Then define the weighted distance from $\bar{\chi}$:

$$d_{ij}^W = \min_{\gamma_{ij}} \sum_{(u,v)\in\gamma_{ij}} w_{uv}(\bar{\chi}). \tag{263}$$

Finally, $\chi_{\rm eff}$ is obtained by coarse-graining over weighted neighborhoods:

$$V_{\ell_0}(i) = \left\{ j \,\middle|\, d_{ij}^W \le \ell_0 \right\}, \tag{264}$$

$$\chi_{\rm eff}(i) = \frac{1}{|V_{\ell_0}(i)|} \sum_{j\in V_{\ell_0}(i)} \chi_j. \tag{265}$$

The dependency chain is explicitly non-circular: $d^C \Rightarrow \bar{\chi} \Rightarrow w(\bar{\chi}), K(\bar{\chi}) \Rightarrow d^W \Rightarrow \chi_{\rm eff}$.

E.6 Relation to the Effective Geometric Description

Effective geometric structures (metric fields, spatial gradients, Poisson-type equations) arise as coarse-grained summaries of relational configurations once a projectable regime applies. They carry no independent ontological status.

E.7 Emergent Coordinates via Manifold Reconstruction

When the relational distance matrix $D = \{d_{ij}\}$ admits a low-dimensional embedding, coordinates can be reconstructed using multidimensional scaling. The centered Gram matrix is

$$G_{ij} = -\tfrac{1}{2}\Big(d_{ij}^2 - d_{i\cdot}^2 - d_{\cdot j}^2 + d_{\cdot\cdot}^2\Big), \tag{266}$$

yielding eigenpairs (λ_k, v_k) and an embedding

$$x_i^{(a)} = \sqrt{\lambda_a}\,(v_a)_i, \quad a = 1, \ldots, d. \tag{267}$$

The intrinsic dimension d is selected by the eigenvalue gap:

$$\Delta\lambda_{d+1} > \eta\,\lambda_1, \tag{268}$$

with $\eta \sim 0.1$. For smooth large-scale configurations, a stable $d = 4$ embedding is expected.

Failure of the reconstruction (non-local connectivity or no clear spectral gap) signals a transition to a pre-geometric regime where a smooth manifold is not an adequate description. The effective metric is defined implicitly through propagation properties and carries only a restricted subset of the relational information. Poisson-type and wave-like equations arise from linearizing the admissible projective dynamics around quasi-homogeneous configurations; they are regime-dependent approximations, not fundamental laws.

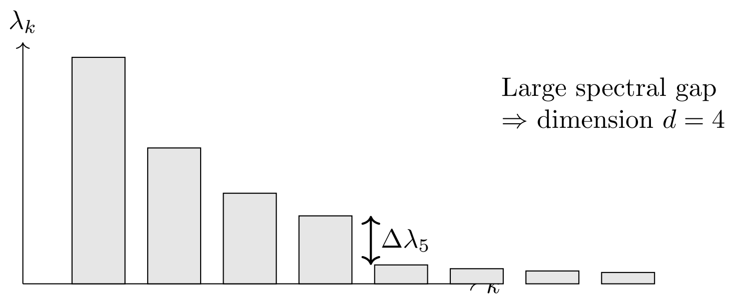

Fig. 32 Schematic eigenvalue spectrum for intrinsic dimension selection. A clear gap after four modes indicates a robust $d = 4$ projectable regime.

E.8 Topological Stability of Relational χ Configurations

Particle-like excitations are identified with nontrivial relational patterns whose stability is guaranteed by intrinsic topological constraints in configuration space (Section 4.2). Topology here characterizes inequivalent classes of χ configurations that cannot be continuously transformed into one another without violating admissibility constraints.

Stability arises from topological obstructions to the advancement of the projective ordering: certain configurations cannot be continuously deformed to the homogeneous admissible vacuum without passing through forbidden regions. This does not rely on conserved charges imposed by symmetry principles.

Geometric metaphors (knots, twists, vortices) provide intuition when configurations admit effective geometric projection but should not be taken literally at the relational level. A paradigmatic example: configurations exhibiting intrinsic 4π-periodic internal structure cannot be unwound and, when projected, exhibit spinorial transformation properties. Distinct particle species correspond to inequivalent topological sectors; the energetic cost of deformation provides a unified origin for mass, stability, and spectral separation.

E.9 Topological Origin of Fermionic and Bosonic Statistics

The distinction between fermionic and bosonic behavior originates from the internal topological structure of χ configurations in configuration space.

Fermionic behavior.

Configurations with intrinsic 4π-periodicity are double-valued under 2π reorientation. When projected, they exhibit fermion-like behavior: sign change under 2π rotations, restoration only after 4π, and spin-$\frac{1}{2}$ transformation properties—without introducing fundamental spinors.

Bosonic behavior.

Topologically orientable configurations return to an equivalent state after 2π reorientation, yielding integer-spin transformation properties.

Topological mass ratios (heuristic).

For $\tilde{\chi}_c \approx 8.3$, the heuristic estimate $27\tilde{\chi}_c^2 \approx 1860$ agrees with the observed ratio $m_p/m_e = 1836.15$ at the $\sim$1% level.

Status of this estimate. No claim is made that the proton-to-electron mass ratio is predicted at the present stage (Appendix B.8). The value $\tilde{\chi}_c \approx 8.3$ should be understood as an internally consistent, order-of-magnitude numerical estimate. Its role here is to provide a quantitative consistency check between a topological ansatz for composite sectors and the independently fixed normalization of the effective couplings.

Heuristic ansatz and separation of assumptions. The estimate rests on two distinct assumptions.

1. **Winding multiplicity factor.** Proton-like configurations are associated with a $w = 3$ trefoil sector, while electron-like configurations correspond to $w = 1$. A minimal multiplicity scaling yields a factor $w_p^3/w_e^3 = 3^3 = 27$. This factor is purely topological and does not depend on $\tilde{\chi}_c$.
2. **Threshold-weight factor.** In addition to the winding multiplicity, we assume that the effective weight of an admissible sector carries a quadratic dependence on the dimensionless threshold parameter, schematically $\propto \tilde{\chi}_c^2$. This assumption is motivated by the appearance of χ_c in the mass scaling of localized configurations (Appendix B.3, B.9–E.10), but is not derived here.

Under these two assumptions one obtains the schematic estimate

$$\frac{m_p}{m_e} \approx \left(\frac{w_p}{w_e}\right)^3 \tilde{\chi}_c^2 = 27\,\tilde{\chi}_c^2, \tag{269}$$

which should be viewed as a consistency check. A first-principles derivation requires (i) the tensorial Born–Infeld completion fixing the order-unity coefficient in Eq. (277), and (ii) a derivation of the sector weights from the admissible dynamics on $H_{\mathrm{adm}}^{(w)}$ as outlined in Section 11.4.

Spin–statistics connection.

The relational-topological distinction between 4π- and 2π-periodic configurations provides a natural qualitative explanation of the spin–statistics connection, demonstrating that the observed dichotomy can arise from internal organization of χ prior to any effective quantum description.

E.10 Vacuum Energy and Projective Capacity of the χ Field

No fundamental vacuum energy density is postulated. Phenomena commonly attributed to vacuum energy are reinterpreted as manifestations of the *projective capacity* of χ: the ability of a relational configuration to undergo further structural reorganization under bounded admissibility constraints. Projective capacity is contextual and non-extensive; it cannot be meaningfully assigned to spacetime points.

Observable vacuum effects arise only when relational constraints restrict admissible configurations. The Casimir effect, for instance, results from a differential in projective capacity between constrained and unconstrained configurations, not from an absolute vacuum energy density.

Because projective capacity is non-extensive, it does not enter gravitational dynamics as a uniform source term. This provides a natural conceptual resolution of the cosmological constant problem: the enormous vacuum energy inferred from zero-point counting is not a physically meaningful quantity in the Cosmochrony ontology.

E.11 Conceptual Positioning with Respect to Existing Frameworks

Aspect	**QM / QFT**	**GR and extensions**	**Cosmochrony**
Primary ontology	Quantum states, operator fields	Spacetime geometry, metric	Relational substrate χ
Status of spacetime	Fixed background	Fundamental dynamical entity	Emergent, projective
Nature of time	External parameter	Coordinate-dependent	Projective ordering
Gravitation	Introduced externally	Metric curvature	Collective projective constraint
Quantum behavior	Postulated	Added via quantization	Emergent from χ
Vacuum structure	Zero-point energy	Geometric ground state	Contextual projective capacity
Particle ontology	Fundamental entities	Field excitations	Topologically stable configurations
Empirical status	Highly successful	Highly successful	Exploratory

Table 10 Conceptual positioning of the relational χ framework with respect to quantum and geometric approaches.

Aspect	**Cosmochrony**	**ΛCDM**	**LQG**
Fundamental ontology	Single substrate χ	$g_{\mu\nu}$ + matter + Λ	Quantum geometry
Status of spacetime	Emergent, projected	Fundamental (classical)	Quantized but assumed
Nature of time	Emergent projective ordering	External parameter	Problematic / relational
Dark energy	Not required	Explicit Λ term	No consensus
Inflation	Not required	Required (inflaton)	Alternative scenarios
Origin of mass	Spectral inhibition	Higgs mechanism	Not intrinsic
Testable predictions	$H(z)$, CMB low-ℓ, drift	H_0, w, growth	Planck-scale signatures

Table 11 Cosmological and gravitational comparison between Cosmochrony, ΛCDM, and Loop Quantum Gravity.

F Glossary of Core Quantities and Notation

This appendix summarizes the meaning, role, and ontological status of the main quantities used throughout the Cosmochrony framework. It is intended strictly as a reference guide and does not introduce new assumptions, dynamics, or physical postulates.

F.1 Fundamental Quantities

χ ***(Chi substrate).***

The unique fundamental entity of the Cosmochrony framework. χ is a pre-geometric, relational substrate not defined on a pre-existing spacetime manifold. χ is atemporally fixed: it does not evolve, relax, or carry a dynamical state. It supplies the admissibility structure that constrains which sequences of effective reconstructions can consistently emerge under the non-injective projection Π. χ does not encode a prescriptive set of admissible physical states; admissibility arises only through projection under finite resolution and compatibility constraints. Localized, topologically stable configurations of χ correspond to particle-like excitations upon projection.

χ_i ***(Local configuration).***

Discrete local degrees of freedom of the χ substrate, associated with vertices of the projective relational network. They encode the microscopic relational state prior to any geometric projection.

χ_c ***(Critical admissibility threshold).***

A fundamental structural bound limiting local variations of χ. It constrains the projection and update process by enforcing causal consistency, and underlies all effective speed and action bounds. This threshold does not prescribe physical laws, but limits the capacity of admissible projected updates.

Physical dimension and operational identification. The local bound $|D_{\text{loc}}\chi_{\text{eff}}| \leq c$ (Eq. (12)) implies that χ_{eff} carries dimension of length. Accordingly, χ_c carries dimension of length. The effective action unit is defined by Eq. (13),

$$\hbar_\chi \;\equiv\; \frac{c^3}{K_0\,\chi_c}, \tag{270}$$

which fixes the physical dimension of K_0 once χ_c is specified.

A dimensionless threshold parameter. A unit-independent characterization of the threshold is provided by the ratio

$$\tilde{\chi}_c \;\equiv\; \frac{\hbar}{\hbar_\chi} \;=\; \frac{K_0\,\chi_c}{c^3/\hbar}, \tag{271}$$

which is dimensionless by construction. When a numerical value is quoted for χ_c in heuristic estimates (e.g. Appendix E.9), it refers to $\tilde{\chi}_c$ unless the dimensional quantity χ_c is explicitly accompanied by K_0.

Connection to gravitational normalization. Independently of Eq. (275), matching the infrared expansion of the coherence functional to the Einstein–Hilbert normalization (Appendix A.11–A.12) yields a constraint of the form

$$K_0\,\chi_c^2 \;\sim\; \frac{c^4}{16\pi G}, \tag{272}$$

up to scheme-dependent numerical factors of order unity fixed by the tensorial Born–Infeld completion (Appendix A.12). Together with Eq. (275), this sets the natural scale at which $\hbar_\chi$ and $\hbar$ can be consistently identified.

τ (Operational time).

An effective ordering parameter defined from the monotonic succession of distinguishable projected configurations, constructed from the effective descriptor χ_{eff}. Operational time τ is not a fundamental temporal coordinate and does not exist at the level of the χ substrate. It emerges only in projectable regimes as an operational measure of duration, defined through the ordered differentiation of admissible effective states. In the spectral admissibility programme, the canonical realization of τ is the cumulative projected capacity $I(n)$ (see §F.7), which is strictly non-decreasing along the BFS cascade. A perfectly stationary projection would correspond to $\Delta\tau = 0$ and is therefore excluded by the requirement of projective continuity.

c_χ (Fundamental admissible projective speed).

The maximal propagation speed of admissible updates within the effective reprojection sequence. It represents the fundamental causal bound of the theory, from which the effective speed of light emerges. It represents the kinematic manifestation of the invariant admissible transport bound b_χ in transport-limited regimes. The effective speed of light c emerges as its geometric projection.

F.2 Effective and Projected Quantities

χ_{eff} (Effective projected field).

A coarse-grained scalar field arising from the non-injective projection of χ onto an emergent spacetime description. χ_{eff} provides an effective field-theoretic representation without fundamental status.

Fiber (of the projection).

For a given effective configuration χ_{eff}, the fiber is the set of underlying χ configurations mapped to it by the projection π. Elements of a fiber are operationally indistinguishable at the spacetime level. Non-trivial fibers reflect the structural non-injectivity of the projection.

Operational projection (contextual access).

A measurement-context-dependent map that specifies how the effective description χ_{eff} is accessed and turned into operational observables. It introduces no additional ontology:

it formalizes the contextual readout of χ_{eff} (e.g., choice of apparatus, coarse-graining, relational query).

Phase stiffness (superconducting coherence stiffness).

ρ_s denotes the effective stiffness associated with spatial variations of the coherent $U(1)$ fiber phase. It is an operational proxy for the collective projective capacity of the projected description in the coherent regime. In Section 5.8, ρ_s controls the energetic cost of phase gradients and enters the London-type gradient term through $F_{\text{grad}} \propto (\nabla\theta - (2e/\hbar)A)^2$. ρ_s is not a fundamental constant of the χ substrate. It is a regime-dependent effective quantity, vanishing when coherence is lost and recovering normal-state behavior.

π ***(Projection).***

An operational projection correspondence relating configurations of χ to effective descriptions χ_{eff} in projectable regimes. The projection is generally non-injective and does not define a single-valued function: distinct underlying configurations of χ may correspond to the same effective state, defining equivalence classes (fibers).

π^{-1} ***(Deprojection).***

The inverse reconstruction problem of identifying classes of χ configurations compatible with a given effective state. Deprojection is not unique and does not destroy structural information.

$V(\chi)$ ***(Effective potential).***

An effective, coarse-grained description used to model localization and stability properties of χ configurations. $V(\chi)$ is not fundamental and is secondary to the spectral characterization of mass and inertia.

ℓ ***(Descriptive resolution scale).***

A scale characterizing the local spectral resolution of the projected state U. It measures the minimal relational differentiation that can be operationally resolved within the effective description. Variations of ℓ across scales modulate the apparent saturation threshold $a_\star$ without altering the underlying bound b_χ.

Observable.

A stable, projectable quantity defined on χ_{eff} through an interpretative framework. Observables do not correspond to additional ontological entities, but to operational readings of the same effective physical reality.

Physical reality.

In the Cosmochrony framework, physical reality is identified with the effective level χ_{eff}. The substrate χ is ontologically real but does not constitute a physical universe until projected into a projectable regime.

t_{proj} ***(Projected time).***

Operational time measured within the emergent spacetime description. It arises from the local rate of projective ordering and reproduces relativistic time dilation effects.

Universe.

The physically real domain described at the χ_{eff} level, where spacetime structure, causality, and physical observables are well-defined. The Universe does not refer to the fundamental substrate χ, which is ontologically prior to any notion of universe.

F.3 Projective Network and Operators

$G(V, E)$ ***(Projective relational network).***

A discrete graph representing the underlying relational structure on which the χ substrate is defined. Vertices correspond to elementary degrees of freedom and edges encode admissibility couplings.

K_{ij} ***(Admissibility coupling).***

Edge-dependent coupling coefficients defined on the projective relational network. They quantify the resistance to relative variations of χ between neighboring nodes and encode geometric and topological information. K_{ij} are structural parameters of the pre-geometric substrate and do not represent dynamical interaction constants.

Δ_G ***(Graph Laplacian / projective stability operator).***

The discrete Laplace–Beltrami operator associated with the network $G(V, E)$ and the couplings K_{ij}. Its spectral properties govern the stability, localization, and inertial behavior of χ configurations.

$D_{\text{loc}}\chi$ ***(Local projective operator).***

A local relational operator governing the admissible update rate of χ configurations at the microscopic level. It replaces differential operators defined on continuous manifolds.

F.4 Spectral and Inertial Quantities

Projective stability gap of a composite fiber class.

Δ_Π is defined as the lowest positive eigenvalue of the effective projective stability operator restricted to the fiber sector around a composite winding configuration (notably $w = 2$), $\Delta_\Pi \equiv \lambda_{\min}(L_\Pi|_{w=2})$. Operationally, it measures the minimal excitation cost required to destabilize the composite class, either by separation into lower-winding excitations or by inducing a local phase defect. Δ_Π is a spectral stability quantity associated with admissibility of coherent composite sectors, and it may admit distinct amplitude and phase-controlled manifestations depending on the regime.

Amplitude (formation) scale associated with composite stability.

Δ_Π^{ampl} denotes the amplitude scale governing local composite formation in strongly constrained regimes. In the strongly correlated case discussed in Section 5.8, it is controlled by the dominant frustration or exchange scale, written schematically as $\Delta_{\Pi}^{\mathrm{ampl}} \sim J$. This quantity should be distinguished from the phase ordering scale controlled by ρ_s.

λ_n ***(Spectral eigenvalues).***

Eigenvalues of the linearized projective stability operator acting on small perturbations of a localized χ configuration. They determine inertial mass scales in the effective description.

ψ_n ***(Spectral modes).***

Eigenmodes associated with the operator Δ_G. They encode the internal structure and stability of particle-like configurations.

m_{eff} ***(Effective mass).***

An emergent invariant determined by the spectral stability of localized χ configurations under projection. Operationally, effective mass measures the self-consistency overhead associated with maintaining a coherent projected description under local updates. Mass is not a fundamental parameter nor a coupling constant.

Q ***(Topological charge).***

An integer-valued invariant characterizing the topology of a stable χ configuration. Different values of Q correspond to distinct particle families.

$\Omega^{\pm}$ ***(Chiral topological sectors).***

Opposite chiral configurations of topological χ structures. They are related by orientation reversal and need not be energetically equivalent.

F.5 Dimensionless Parameters

S ***(Gradient saturation parameter).***

A dimensionless quantity defined as

$$S \equiv \frac{1}{c^2} \sum_{j \sim i} K_{ij} (\chi_i - \chi_j)^2, \tag{273}$$

measuring the local density of projected χ gradients. The bound $S \leq 1$ enforces causal consistency in the emergent spacetime description. The constant c appearing here is the effective geometric projection of the invariant transport bound b_χ.

Ω_χ ***(Projective capacity parameter).***

A dimensionless global quantity characterizing the fraction of total admissible projective capacity stored in spatial gradients. In cosmological regimes, it plays a role analogous to a density parameter.

Frustration density (operational proxy).

F denotes a dimensionless proxy for the density of incompatible local fiber matching constraints in the fiber sector. It is introduced as an operational quantity, grounded in independently measurable correlators in the relevant materials. In the strongly correlated regime, F parametrizes the strength of staggered frustration and enters the qualitative selection of the low-cost composite coherence channel. F is not an additional fundamental field or a new ontological variable. It summarizes environmental and material constraints within an effective description.

Relative frustration ratio.

r_F is a dimensionless ratio encoding the relative amplitude of the dominant frustration sector across material families. It is used to compare coherence channel selection across systems with different lattice constraints and correlation structure. In particular, r_F organizes the expected transition between distinct symmetry channels in the superconducting regime.

F.6 Constants and Emergent Limits

$a_\star$ ***(Operational saturation threshold).***

An effective acceleration proxy marking the onset of saturated response in the projected description. It scales inversely with the local descriptive resolution ℓ according to

$$a_\star \sim \kappa \frac{b_\chi}{\ell},$$

where κ is a dimensionless normalization factor. $a_\star$ is not a fundamental constant but a regime-dependent manifestation of the invariant transport bound b_χ.

b_χ ***(Invariant admissible transport bound).***

The fundamental bound on admissible projective transport within the χ substrate. It limits the maximal rate at which relational configurations can be updated while preserving operational closure under projection. All effective saturation phenomena are manifestations of this invariant bound under different dimensional projections.

c ***(Effective speed of light).***

The maximal signal propagation speed in emergent spacetime. c is not fundamental: it is the geometric projection of the substrate transport bound b_χ through the admissible projective speed c_χ in projectable regimes.

$\hbar$ ***(Effective Planck constant).***

An emergent quantum of action associated with projection thresholds and spectral granularity. It is not fundamental at the level of χ.

G (Newtonian gravitational constant).

An emergent coupling constant arising from large-scale collective projective admissibility dynamics of χ. Its value reflects structural properties rather than fundamental interaction strengths.

Λ_{eff} (Effective cosmological constant).

A residual large-scale effect associated with the global saturation of projective capacity in the late-time admissible regime. It reflects the asymptotic structure of the admissible ordering rather than a fundamental energy density of the vacuum.

F.7 Key Conceptual Terms

Admissibility envelope.

Under the bounded-flux constraint $|\partial_t \chi| \leq c_\chi$, a Laplacian eigenmode with eigenvalue λ_n admits a maximal effective amplitude

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}.$$

This relation defines the *admissibility envelope* governing the maximal projectable amplitude of each spectral sector.

Admissibility ratio.

The *admissibility ratio* compares the maximal admissible amplitudes of two spectral sectors,

$$R_{p/q} = \frac{A_p^{\max}}{A_q^{\max}} = \sqrt{\frac{\lambda_q}{\lambda_p}}.$$

This ratio quantifies the relative dynamical preference between spectral sectors under the bounded-flux constraint. In quaternionic Cayley graphs such as that of Q_8, the ratio $\sqrt{4/3}$ between the non-abelian and abelian sectors provides the minimal instance of this hierarchy.

Character-vanishing condition.

The condition

$$\chi_\rho(s) = 0 \quad \forall s \in S$$

is referred to as the *character-vanishing condition*. It identifies generator sets whose associated $SU(2)$ representation consists entirely of traceless elements. This condition plays a central role in the classification of admissible spectral sectors, since it ensures that the adjacency contribution μ_ρ vanishes and the Laplacian eigenvalue satisfies $\lambda_\rho = |S|$.

Cumulative projected capacity $I(n)$.

For a conjugate pair $(c, q-c)$ of Weil representations on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the cumulative projected capacity is

$$I(n) \;=\; \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n),$$

where n indexes BFS shell depth and $\sigma_{\text{pair}}(n)$ is the residual projective capacity at depth n. $I(n)$ measures the spectral structure already stably projected up to depth n. Its strict non-decreasing character along tested BFS trajectories (Proposition 1 of [22]) provides the canonical numerical realization of the projective temporal ordering τ: a larger $I(n)$ corresponds to a later effective state. *Status:* numerical result for $q \in \{29, 61, 101, 151\}$; analytical derivation of the monotonicity remains an open problem.

Capacity exponent δ_{pair}.

The power-law exponent governing the decay of the residual projective capacity $\sigma_{\text{pair}}(n) \sim n^{-\delta_{\text{pair}}}$ along the BFS cascade. It is the primary spectral observable of the O-series and connects to the cascade exponent β^* governing the fermion mass hierarchy via the structural relation established in O24. Current numerical values (O25 [76]): $\delta_{\text{pair}} \approx 8.893$ ($q = 29$), 9.983 ($q = 61$), 9.548 ($q = 101$), 9.125 ($q = 151$), 8.784 ($q = 211$).

Energy.

Energy is an effective quantity measuring the operational cost associated with modifying a projected configuration while preserving coherence under admissible updates. This cost reflects the mobilization of the underlying projective capacity of the χ substrate under projection, but energy itself is not a fundamental property of χ. Standard conservation laws remain valid at the effective level.

Fluctuations.

Local stochastic modulations of χ configurations that affect event timing and localization without altering underlying topological constraints.

Gram rigidity.

Let $\{\vec{u}_s\}_{s \in S}$ denote the unit axes associated with neutral generators. The pairwise geometry of these axes is encoded by the Gram matrix

$$G_{st} = \vec{u}_s \cdot \vec{u}_t.$$

Through the character identity

$$G_{st} = -\frac{1}{2}\chi_\rho(st),$$

the entire geometric configuration is determined by representation-theoretic data. *Gram rigidity* refers to the fact that admissible neutral configurations are severely constrained by this identity, leading to a discrete classification of compatible finite subgroups of $SU(2)$.

Matter.

Stable topological configurations of χ whose persistence gives rise to particle-like behavior and inertial properties.

Measurement.

A localized interaction that selects a specific effective realization within a non-injective projection. Measurement does not induce a fundamental collapse, but resolves an equivalence class of projected configurations into a stable operational outcome.

Neutral generator.

A generator $s \in S$ of a finite group representation $\rho : G \to SU(2)$ is said to be *neutral* if its character vanishes,

$$\chi_\rho(s) = 0.$$

Neutral generators correspond to traceless elements of $SU(2)$ and therefore to half-turn rotations in the associated $SO(3)$ representation. Within the spectral admissibility programme, neutrality ensures that the corresponding sector does not receive constructive alignment from the generating set and therefore admits the maximal admissibility window compatible with bounded relational flux.

One-way activation.

The property that a spectral mode, once stably projected at BFS depth n, does not leave the Gram–Schmidt span at any subsequent depth $n' > n$. Formally, $\delta r_n \geq 0$ at every step, so that $I(n)$ is non-decreasing. One-way activation is a property of *spectral modes*, not of the coherence of projected configurations: a mode once acquired persists in V_n, while a coherent combination of modes — corresponding to an unstable particle — can lose its admissible coherence and reorganize into simpler configurations without violating one-way activation. One-way activation is therefore the condition of possibility for the projective arrow of time: it ensures that $I(n)$ is well-defined and globally consistent as a monotone ordering parameter. The rigidity theorem of O27 [23] constrains the Weil fingerprint vectors to a three-dimensional real sector identified with $\mathfrak{su}(2)$, providing the non-concentration structure that any analytical proof of one-way activation must exploit. *Status:* established numerically for $q \in \{29, 61, 101, 151\}$; the analytical proof from BFS admissibility constraints and the $\mathfrak{su}(2)$ rigidity of O27 remains open (Conjecture 2.1).

Projective temporal ordering.

The ordering relation on effective states defined by

$$U_n \prec U_{n+1} \iff I(U_n) \leq I(U_{n+1}),$$

where n is the BFS shell depth, an ordering parameter on projected configurations that does not represent a physical time variable at the substrate level. This ordering is not a dynamics on χ but a relational structure induced by the non-injective projection Π: it reorders admissible fibre selections within the fixed observable rank, without introducing additional observable degrees of freedom. It is compatible with and complementary to the observable rank stability established in O24. See [10] for the numerical evidence and formal statement.

Reprojection sequence.

The iterated map

$$U_{n+1} = \Pi \circ \sigma(U_n),$$

where $\sigma(U_n) \in \Pi^{-1}(U_n)$ selects an admissible configuration within the fibre compatible with the current effective state U_n. The operator σ is a structural descriptor of admissible fibre transitions; it is not a generator of temporal evolution at the substrate level. The reprojection sequence is the formal carrier of effective dynamics in Cosmochrony: all observable evolution arises from successive admissible fibre selections, not from any evolution of the static substrate χ. Successive steps correspond to distinct admissible realisations drawn from different regions of the fibres; the ordering parameter n is given physical meaning only through the monotone accumulation of $I(n)$, which constitutes the canonical spectral realization of operational time τ (see the entry for τ in §F.1).

Residual projective capacity $\sigma_{\mathrm{pair}}(n)$.

For a conjugate pair $(c,\, q-c)$, the residual projective capacity at BFS depth n is

$$\sigma_{\mathrm{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n), \quad \sigma_c(n) = \frac{\delta r_n}{|S_n|},$$

where δr_n counts new linearly independent Weil fingerprint vectors in BFS shell S_n and $|S_n|$ is the shell size. $\sigma_{\mathrm{pair}}(n)$ decays monotonically toward zero as the spectral capacity of the pair is exhausted; its complement is the cumulative projected capacity $I(n)$. The Born–Infeld saturation plateau is reached when $\sigma_{\mathrm{pair}}(n) \approx 0$.

Spectral capacity.

For a finite group G with symmetric generating set S, the *spectral capacity* functional is defined as

$$C(G,S) = \sum_{\rho \in \widehat{G},\ \lambda_\rho > 0} \frac{(\dim \rho)^2}{\sqrt{\lambda_\rho}},$$

where λ_ρ are the Laplacian eigenvalues associated with the irreducible representation sectors. The functional aggregates the admissible amplitude windows of all spectral sectors weighted by their Peter–Weyl multiplicities. Spectral capacity provides a global measure of how efficiently a relational structure supports dynamically admissible modes under bounded projective flux.

Probability.

An emergent descriptor reflecting the multiplicity of projected realizations compatible with a given relational configuration and effective state. Probability is projective rather than fundamental, and does not correspond to intrinsic randomness of the substrate.

Projective capacity.

The available "room" within the admissible projective structure for further structural reorganization. At the spectral level it is quantified by $\sigma_{\mathrm{pair}}(n)$ and its integral $I(n)$ along the BFS cascade. At the effective level it governs the coupling $g(x,t) = \gamma/\Sigma(x,t)$

in the nonlinear Schrödinger correction (§B.14) and the differential between constrained and unconstrained vacuum configurations (Casimir effect). Projective capacity is contextual and non-extensive; it replaces the notion of "relaxation capacity of χ" used in earlier formulations of the framework.

Spacetime.

An emergent relational structure arising from large-scale configurations of the χ substrate. Its metric description remains valid within its domain of applicability.

Time.

An effective ordering parameter associated with the local rate of projective ordering. Operational and relativistic notions of time are recovered without modification. The canonical spectral realization is the cumulative projected capacity $I(n)$ (see above).

Projective transmittance (gauge interpretation).

An effective, context-dependent measure of how efficiently admissible projective flux is transmitted through a given projected configuration. In Cosmochrony, gauge structure can be interpreted as a parametrization of transmittance variations across the projection fiber, rather than as a fundamental interaction field.

Wavefunction.

An effective statistical representation of the dynamics and topology of the χ substrate. It has no fundamental ontological status.

Wave–Particle Duality.

A manifestation of interaction-induced changes in the local configuration of χ, producing localized particle-like behavior from an underlying wave-like substrate.

Winding number (fiber topological class).

w labels the winding class of a configuration in the projected $U(1)$ fiber sector. A $w = 1$ configuration corresponds to an elementary charged excitation in the effective description. A $w = 2$ configuration denotes a composite topological class supporting a collective phase degree of freedom. The winding number is a topological invariant of the projected description and does not constitute a new dynamical postulate. It is used to classify admissible coherent regimes and their stability properties.

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